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FRANCO MODIGLIANI

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Franco Modigliani in 1992.

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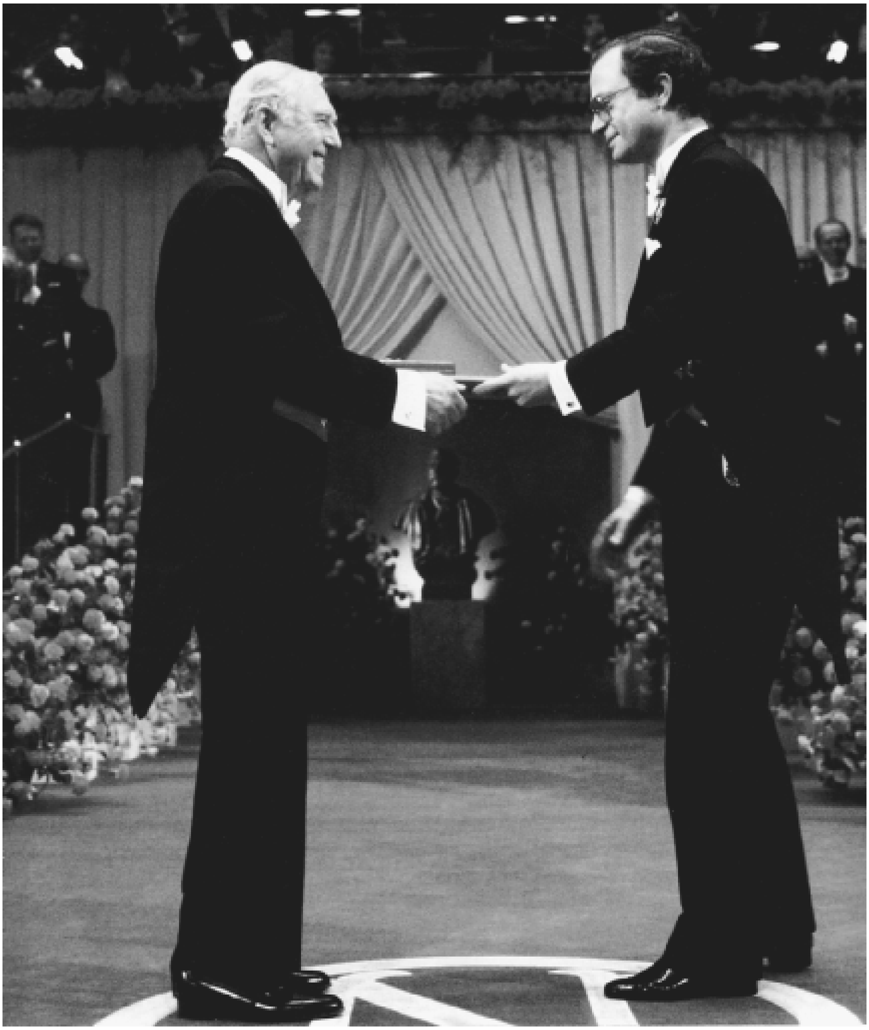

In Stockholm in 1985, receiving the Nobel Prize from King Gustav Adolph of Sweden.

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PREFACE

The essays collected in this volume represent the last decade of Franco Modigliani's scientific contribution to economics. This volume has been on the drawing board for many years, but Modigliani kept postponing it on the grounds that there were one or more nearly finished papers whose inclusion in the collection seemed essential to him. He completed the last paper, "The Chinese Saving Puzzle and the Life-Cycle Hypothesis" on September 24, 2003. On September 25, he did not wake up.

A couple of years ago, Franco Modigliani asked me to edit the collection. I thought the job would take no more than a few months. I did not know Franco Modigliani: while he drew up the list of papers to include in the final volume in 2001, Franco never stopped challenging his indefatigable mind, and in the following three years wrote three more papers which have since become the backbone of the present collection. In these three years I have spent reading and editing his research, Franco Modigliani taught me a lesson he learned a long time ago: "there is no such thing as the final word."

Most of the articles in this volume are the fruit of an intense collaboration with a multitude of coauthors: Richard Brumberg, Albert Ando, Larry Cao, Jean Paul Fitoussi, Tullio Japelli, Beniamino Moro, Denis Snower, Robert Solow, Alfred Steinherr, Paolo Sylos Labini, and Leah Modigliani.

Many people have been involved in the preparation and the editing of this volume, especially Thomas Lissey and Gwendolyn Swasey. Thobias Adrian, Pol Antras, Roberto Benelli, Guido Lorenzoni, and Thomas Philippon read and provided comments on a portion of the papers. But this volume would not exist without the dedicated effort of Serena Modigliani to ensure the completion of her husband's work.

Francesco Franco

INTRODUCTION

The papers are arranged by topic, as in the earlier volumes of Franco Modigliani's collected papers. Within topics, the order is chronological. The first theme is the Life-Cycle Hypothesis, which comprises the entirety of part I. Franco Modigliani wanted to include in this section three older articles originally included in the second volume but now out of print. I interpret this decision as a final attempt to both revisit his research on the life-cycle and to present it to the reader as a coherent whole. This interpretation is supported by the editor's note in the last published article by Franco Modigliani, "The Chinese Saving Puzzle and the Life-Cycle Hypothesis," which is the last essay of part I of this volume.[1]

The first essay of part I, "Utility Analysis and the Consumption Function: An Interpretation of Cross-Section Data," written with Richard Brumberg, lays down the foundation of the Life-Cycle model, which applies the classical postulate of utility maximization to the household's life-cycle: a working period followed by a non-productive phase of retirement. Combining this simple model with a few more assumptions provided a unified explanation for empirical regularities in saving behavior. These included the high frequency cyclical volatility of the saving rate (rather than low frequency), the lack of significant association between aggregate private saving and real per capita income, and the strong relationship between the saving rate and both the rate of growth of per capita income and the age distribution of the population. The second essay, "The Life-Cycle Hypothesis of Saving, Aggregate Implications and Tests," written with Albert Ando, is an empirical verification of the Life-Cycle Hypothesis (LCH). It explores the implications for the short and long run propensity to consume out of income. The third essay, "Long Run Implications of Alternative Fiscal Policies and the Burden of the National Debt," explores and tests policy implications of the LCH. In particular it studies issues related to the burden of the national debt, and attributes this burden to the "crowding out" of productive capital. The fourth essay, "Recent Declines in the Saving Rate: a Life Cycle Perspective," is an empirical test of the LCH applied to developed countries during the last three decades of the twentieth century. It shows that the LCH can successfully account for the decline in the saving rate experienced in the OECD countries during that period of time. The fifth essay, "The Age-Saving Profile and the Life-Cycle Hypothesis," written with Tullio Jappelli, addresses the empirical observation that dissaving by the elderly is limited or absent, a fact contrary to the mechanism beneath the LCH. It shows that measures of saving generally used in previous tests are based on a measure of income that does not properly account for the role of mandated public pension arrangements. It is shown that when correctly adjusted, saving and wealth exhibit the characteristic hump shape over the life-cycle that is implied by the LCH. The sixth essay, "The Chinese Saving Puzzle and the Life Cycle Hypothesis," written with Larry Cao, is the last published work of Franco Modigliani. The essay uses the LCH paradigm to explain the large increase in private saving experienced in China in the last two decades of the twentieth century.

The second section of the volume includes Franco Modigliani's work on the European economy during the last decade of the twentieth century. The first essay, "An Economists' Manifesto on Unemployment in the European Union," written with a plethora of fellow economists, addresses the high level of unemployment that has gripped Europe in the 1980s and 1990s. The manifesto suggests actions aimed at the revival of aggregate demand (demand policies), as well as the reform of the labor and product markets and the system of benefits for the unemployed (supply policies). The salient aspect is the complementary of demand and supply policies. The main focus of the second essay, "The Shameful Rate of Unemployment in the EMS: Causes and Cures," contains a lesson Franco Modigliani taught his students for 50 years: high long-term interest rates resulting from an insufficient supply of real money, combined with too-tight fiscal policy, give rise to insufficient aggregate demand, and therefore an insufficient number of jobs. The author identifies the Maastricht fixed exchange rate regime as the cause of the contractionary monetary stance in Europe. Two popular essays originally published in the Financial Times conclude part II.

Part III is really a collection of topics. The first essay, "Long-term Financing in an Inflationary Environment," is a reflection on inflation and the problems which arise with mortgages in the in the presence of inflation. The second essay, "Risk-Adjusted Performance: How to Measure it and Why," written with Modigliani's granddaughter Leah Modigliani, proposes a new measure for risk to evaluate performance of financial portfolios. The measure is now known as the M^2 measure and has been adopted by many institutions to evaluate mutual funds. The third essay, "The Rules of the Game and the Development of Security Markets," written with Enrico Perotti, attempts to characterize the underdeveloped institutional (legal) framework prevalent in many developing countries, and offers an analysis of the economic consequences for the emergence of a developed security market. The fourth essay, "Emerging Issues in the World Economy," is based on two lectures Modigliani gave at the beginning of the 1990s in Singapore on his visions at that time on the future of the world economy. The fifth essay, "The Keynesian Gospel According to Modigliani," is the last scientific paper of the present volume. In this paper Franco Modigliani, 60 years after his first path-breaking article "Liquidity Preference and The Theory of Interest and Money," revisits the General Theory and its implications for mass unemployment.

The volume concludes with an interview of Franco Modigliani conducted by William Barnett and Robert Solow.

Note

1. In submitting this paper to the Journal of Economic Literature, Franco Modigliani wrote: "This is a paper which has a special meaning for me, as I see it as a fitting conclusion to my life's work on saving."

I

THE LIFE-CYCLE HYPOTHESIS

1

UTILITY ANALYSIS AND THE CONSUMPTION FUNCTION: AN INTERPRETATION OF CROSS-SECTION DATA

Franco Modigliani and Richard Brumberg

Introduction

Of John Maynard Keynes' many hypotheses, the one that has been subject to the most intensive empirical study is the relation between income and consumption. By now, his generalization is familiar to all:

> The fundamental psychological law, upon which we are entitled to depend with great confidence both a *priori* from our knowledge of human nature and from the detailed facts of experience, is that men are disposed, as a rule and on the average, to increase their consumption as their income increases, but not by as much as the increase in their income.[1]

The study of the consumption function has undoubtedly yielded some of the highest correlations as well as some of the most embarrassing forecasts in the history of economics. Yet the interest in the subject continues unabated since, if it were possible to establish the existence of a stable relation between consumption, income, and other relevant variables and to estimate its parameters, such a relation would represent an invaluable tool for economic policy and forecasting.

The work done in this area during the last few years has taken two directions.[2] One has consisted in extensive correlations of data on aggregate consumption, or saving, with income and a large collection of additional miscellaneous variables. The second direction has been the exploitation of cross-section data. Old material has been reworked and new information collected. The most elaborate of the new studies have been those of the Survey Research Center of the University of Michigan. As in the time series analysis, more and more variables have been included, or are proposed for inclusion, in order to discover stable relations.[3]

By now the amount of empirical facts that has been collected is truly impressive; if anything, we seem to be in imminent danger of being smothered under them. What is, however, still conspicuously missing is a general analytical framework which will link together these facts, reconcile the apparent contradictions, and provide a satisfactory bridge between cross-sectional findings and the findings of aggregative time series analysis.

It is our purpose to attempt to provide such an analytical framework through the use of the well-developed tools of marginal utility analysis. We have shown elsewhere[4] that the application of this instrument proves of great help in integrating and reconciling most of the known findings of aggregative time series analysis. In this paper, we shall attempt to show how the same model of individual behavior can be applied to the analysis of cross-section data. We hope to demonstrate that this model provides a consistent, if somewhat novel, interpretation of the existing data and suggests promising directions for further empirical work.

Part I Theoretical Foundations

I.1 Utility Analysis and the Motives for Saving

Our starting point will be the accepted theory of consumer's choice. The implications of this theory have been so incompletely recognized in the empirically-oriented literature of recent years, that it will be useful to retrace briefly the received doctrine.[5]

Consider the following variables:

- c_t consumption of the individual during the t-th year (or other specified interval) of his life, where t is measured from the beginning of the earning span;
- y_t income (other than interest) in the t-th year (for an individual of age t, y_t and c_t denote current income and consumption, while y_τ and c_τ, for $\tau > t$, denote expected income and planned consumption in the τ-th year):
- s_t saving in the t-th year;
- a_t assets at beginning of age period t;
- r the rate of interest;
- N the earning span;
- M the retirement span; and
- L the life span of economic significance in this context, that is, $N + M$.

It is assumed that the individual receives utility only from present and prospective consumption and from assets to be bequeathed. If we assume further that the price level of consumables is not expected to change appreciably over the balance of the life span, so that the volume of consumption is uniquely related to its value, then for an individual of age t, the utility function can be written as

$$U = U(c_t, c_{t+1}, \dots c_L, a_{L+1}). \tag{1.1}$$

This function is to be maximized subject to the budget constraint, which, if the rate of interest, r, is not expected to change appreciably over the balance of the life span, can be expressed by means of the equation

$$a_t + \sum_{\tau=t}^{N} \frac{y_\tau}{(1+r)^{\tau+1-t}} = \frac{a_{L+1}}{(1+r)^{L+1-t}} + \sum_{\tau=t}^{L} \frac{c_\tau}{(1+r)^{\tau+1-t}}.[6] \tag{1.2}$$

For the utility function (1.1) to be maximized, the quantities c_τ and a_{L+1} must be such as to satisfy the first order conditions:

$$\begin{cases} \dfrac{\partial U}{\partial c_\tau} = \dfrac{\lambda}{(1+r)^{\tau+1-t}}; \quad \tau = t, t+1, \dots, L \\[2ex] \dfrac{\partial U}{\partial a_{L+1}} = \dfrac{\lambda}{(1+r)^{L+1-t}} \end{cases} \tag{1.3}$$

where λ represents a Lagrange multiplier. The equation (1.3), together with (1.2), yields a system of $L - t + 3$ simultaneous equations to determine $L - t + 1$ $\bar{c}_\tau$'s, $\bar{a}_{L+1}$ and $\bar{\lambda}$, the barred symbols being used to characterize the maximizing value of the corresponding variable.

If current income, $y_t + ra_t$, is unequal to c_t, the individual will be currently saving (or dissaving); and similarly, if $y_\tau + ra_\tau$ is not equal to $\bar{c}_\tau$, the individual will be planning to save (or dissave) at age τ. The traditional model suggests that we may usefully distinguish two separate reasons for such inequalities to arise. We refer to these reasons as the "motives for saving."[7]

I The first of these motives is the desire to add to the estate for the benefit of one's heirs; it arises when $\bar{a}_{L+1}$ is greater than a_t. Under this condition $y_\tau + ra_\tau$ must exceed $\bar{c}_\tau$ for at least some $\tau \geqq t$.

II The second motive arises out of the fact that the pattern of current and prospective income receipts will generally not coincide with the preferred consumption, $\bar{c}_\tau$, for all $\tau \geqq t$. This clearly represents an independent motive in that it can account for positive (or negative) saving in any subinterval of the life span, even in the absence of the first motive.

It is precisely on this point that a really important lesson can be learned by taking a fresh look at the traditional theory of the household; according to this theory there need not be any close and simple relation between consumption in a given short period and income in the same period. The rate of consumption in any given period is a facet of a plan which extends over the balance of the individual's life, while the income accruing within the same period is but one element which contributes to the shaping of such a plan. This lesson seems to have been largely lost in much of the empirically-oriented discussion of recent years, in the course of which an overwhelming stress has been placed on the role of current income, or of income during a short interval centering on the corresponding consumption interval, almost to the exclusion of any other variable.

Before proceeding further with the implications of our model, it is necessary to devote brief attention to one conceivably important element that we have neglected so far, namely, the phenomenon of uncertainty. No attempt will be made in this paper to introduce uncertainty in the analysis in a really rigorous fashion. The reason for this procedure is simple; we believe that for the purposes in which we are interested, a satisfactory theory can be developed without seriously coming to grips with this rather formidable problem. An examination of the considerations that support this conclusion, however, is best postponed until we have fully explored the implications of our model under certainty. We may simply note at this point that the presence of uncertainty might be expected to give rise to two additional motives for saving:

III The precautionary motive, *i.e.*, the desire to accumulate assets through saving to meet possible emergencies, whose occurrence, nature, and timing cannot be perfectly foreseen. Such emergencies might take the form of a temporary fall in income below the planned level or of temporary consumption requirements over and above the anticipated level. In both cases the achievement of the optimum consumption level might depend on the availability of previously acquired assets.

IV Finally, as a result of the presence of uncertainty, it is necessary, or at least cheaper, to have an equity in certain kinds of assets before an individual can receive services from them. These assets are consumers' durable goods. If there were no uncertainty, a person could borrow the whole sum necessary to purchase the assets (the debt cancelling the increase in real asset holdings), and pay off the loans as the assets are consumed. In the real world, however, the uncertainty as to the individual's ability to pay forces the individual to hold at least a partial equity in these assets.

While we have thus come to distinguish four separate motives for saving, we should not forget that any one asset in the individual's "portfolio" may, and usually will, satisfy more than one motive simultaneously. For example, the ownership of a house is a source of current services; it may be used to satisfy part of the consumption planned for after retirement; it may be bequeathed; and, finally, it is a source of funds in emergencies. It follows that any possession which can be turned into cash will serve at least one of the four motives and should accordingly be treated as an asset. These possessions include, in particular, equities in unconsumed durable goods.

Saving and dissaving can then usefully be defined as the positive, or negative, change in the net worth of an individual during a specified time period. Correspondingly, consumption will be defined as the expenditure on nondurable goods and services—adjusted for changes in consumers' inventories—plus current depreciations of direct-service-yielding durable goods.[8]

As we shall see, the fact that assets are capable of satisfying more than one motive simultaneously provides the foundation for our earlier statement that it should be possible to neglect the phenomenon of uncertainty without too serious consequences. But a fuller development of this point will have to wait until later.

I.2 Some Further Assumptions and Their Implications

The currently accepted theory of the household, even in the very general formulation used so far, has begun to broaden our view of saving behavior. It is, however, too general to be really useful in empirical research. If we are to derive from it some propositions specific enough to be amenable to at least indirect empirical tests, it will be necessary to narrow it down by introducing some further assumptions about the nature of the utility function (see Assumption II below). For convenience of exposition, however, we shall also find it useful to introduce several additional assumptions whose only purpose is to simplify the problem by reducing it to its essentials. These assumptions will enable us to derive some very simple relations between saving, income, and other relevant variables, and it will appear that the implications of these relations are consistent with much of the available empirical evidence. While some of the simplifying assumptions we are about to introduce are obviously unrealistic, the reader should not be unduly disturbed by them. In the first place we have shown elsewhere[9] that most of these assumptions (except Assumption II) can be greatly relaxed or eliminated altogether, complicating the algebra but without significantly affecting the conclusions. In the second place, the question of just which aspects of reality are essential to the construction of a theory is primarily a pragmatic one. If the theory proves useful in explaining the essential features of the phenomena under consideration in spite of the simplifications assumed, then these simplifications are thereby justified.

It may be well to recall, first, that one simplifying assumption has already been made use of in developing our basic model of equation (I.1) to (I.3); this is the assumption that (on the average) the price level of consumables is not expected to change appreciably over time. The first of our remaining assumptions will consist in disregarding altogether what we have called the "estate motive." Specifically, we shall assume that the typical household, whose behavior is described by equations (I.1) to (I.3), does not inherit assets to any significant extent and in turn does not plan on leaving assets to its heirs. These conditions can be formally stated as:

Assumption I $a_1 = 0, \quad \bar{a}_{L+1} = 0.$[10]

Assumption I, together with equation (I.2), implies that our household, in addition to having no inherited assets at the beginning of its life, also does not receive any gift or inheritance at any other point of its life; it can only accumulate assets through its own saving.

From Assumption I and equation (I.2), it follows immediately that current and future planned consumption must be functions of current and expected (discounted) income plus initial assets, *i.e.*,

$$\bar{c}_\tau = f(v_t, t, \tau), \quad \tau = t, t+1, \ldots L;$$

where

$$v_t = \sum_{\tau=t}^{N} \frac{y_\tau}{(1+r)^{\tau+1-t}} + a_t,$$

and t denotes again the present age of the individual.[11]

Now what can be said about the nature of the function f? Or, to reformulate the question, suppose that, on the expectation that his total resources would amount to v_t, and with a given interest rate, our individual had decided to distribute his consumption according to the pattern represented by $\bar{c}_\tau$. Suppose further that, before carrying out his plan, he is led to expect that this resources will amount not to v_t but, say, to $v_t + \Delta v_t$. Should we then expect him to allocate the additional income to increase consumption in any specific period of his remaining life (*e.g.*, his early years, or his late years, or his middle years), relative to all other periods, or can we expect him to allocate it so as to increase all consumptions roughly in the same proportion?

We are inclined to feel that the second alternative is fairly reasonable; or, at any rate, we are unable to think of any systematic factor that would tend to favor any particular period relative to any other. And for this reason we are willing to assume that the second answer, even if it is not true for every individual, is true on the average.[12] This gives rise to our second assumption, the only one that is really fundamental for our entire construction, namely:

Assumption II The utility function is such that the *proportion* of his total resources that an individual plans to devote to consumption in any given year τ of his remaining life is determined only by his tastes and not by the size of his resources. Symbolically, this assumption can be represented by the following equation:

$$\bar{c}_\tau = \gamma_\tau^t v_t, \quad \tau = t, t+1, \ldots, L \tag{I.4}$$

where, for given t and τ the quantity γ_τ^t depends on the specific form of the function U and on the rate of interest r, but *is independent of "total resources,"* v_t.

As a result of well-known properties of homogeneous functions, it can readily be shown that a sufficient condition for Assumption II to hold is that the utility function U be homogeneous (of any positive degree) in the variables c_t, c_{t-1}, . . . , c_L.[13]

The remaining two assumptions are not essential to the argument, but are introduced for convenience of exposition.[14]

Assumption III The interest rate is zero, *i.e.*, $r = 0$.
As a result of this assumption, the expression

$$v_t = \sum_{\tau=t}^{N} \frac{y_\tau}{(1+r)^{\tau+1-t}} + a_t$$

can be rewritten as $y_t + (N - t)y_t^e + a_t$, where

$$y_t^e = \left(\sum_{\tau=t+1}^{N} y_t \right) \Big/ (N-t),$$

represents the average income expected over the balance of the earning span.

Equation (I.4) now reduces to:

$$\bar{c}_\tau = \gamma_\tau^t [y_t + (N-t)y_t^e + a_t] \tag{I.4$'$}$$

which implies

$$\sum_{\tau=t}^{L} c_\tau = [y_t + (N-t)y_t^e + a_t] \sum_{\tau=t}^{L} \gamma_\tau^t.$$

Furthermore, taking into account Assumption I ($\bar{a}_{L+1} = 0$) we also have

$$\sum_{\tau=t}^{L} c_\tau = y_t + (N-t)y_t^e + a_t. \tag{I.5}$$

From (I.4$'$) and (I.5) it then follows that

$$\sum_{\tau=t}^{L} \gamma_\tau^t = 1. \tag{I.6}$$

Assumption IV All the γ_τ^t are equal; *i.e.*, our hypothetical prototype plans to consume his income at an even rate throughout the balance of his life.

Let γ_t denote the common values of the γ_τ^t for an individual of age t. From (I.6) we then have

$$\sum_{\tau=t}^{L} \gamma_\tau^t = (L+1-t)\gamma_t = 1; \tag{I.7}$$

or,

$$\gamma_\tau^t = \gamma_t = \frac{1}{L+1-t} \equiv \frac{1}{L_t},$$

where $L_t \equiv L + 1 - t$ denotes the remaining life span at age t.

Part II: Implications of the Theory and the Empirical Evidence

II.1 The Individual Consumption Function and the Cross-section Consumption-Income Relation

Substituting for γ_τ^t in equations (I.4′) the value given by (I.7), we establish immediately the individual consumption function, *i.e.*, the relation between *current* consumption and the factors determining it:

$$c = c(y, y^e, a, t) = \frac{1}{L_t} y + \frac{(N-t)}{L_t} y^e + \frac{1}{L_t} a; \tag{II.1}$$

where the undated variables are understood to relate to the current period.[15] According to equation (II.1), current consumption is a linear and homogeneous function of current income, expected average income, and initial assets, with coefficients depending on the age of the household.

$$s = y - c = \frac{L-t}{L_t} y - \frac{N-t}{L_t} y^e - \frac{1}{L_t} a. \tag{II.2}$$

In principle, equations (II.1) and (II.2) could be directly tested; but they cannot easily be checked against existing published data because, to our knowledge, data providing joint information on age, assets (which here means *net worth and not just liquid assets*), and average expected income, do not exist.[16] We must, therefore, see whether we can derive from our model some further implications of a type suitable for at least indirect testing in terms of the available data. Since most of these data give us information about the relation between consumption in a given short interval and income over the same interval (or some small neighborhood thereof) we must seek what implications can be deduced as to the relations between these variables.

If the marginal propensity to consume is defined literally as the increment in the current consumption of the household accompanying an increment in its current income, divided by the increment in income, keeping other things constant, then, according to equation (II.1), this quantity would be

$$\frac{\partial c}{\partial y} = \frac{1}{L_t},$$

which is independent of income but dependent on age. The consumption function (II.1) would be represented by a straight line with the above slope and an intercept

$$\frac{(N-t)y^e + a}{L_t};$$

and, since this intercept can be assumed positive,[17] the proportion of income saved should tend to rise with income.

In order to get some feeling as to the quantitative implications of our results, let us say that the earning span, N, is of the order of 40 years, the retirement span, M, 10 years, and therefore the total active life span, L, of the order of 50 years. These figures are not supposed to be anything more than a very rough guess and their only purpose is to give us some notion of the magnitudes involved. On the basis of these figures, the marginal propensity to consume would lie somewhere between a minimum of 1/50, or 2 percent, and a maximum of 1/11, or 9 percent, depending on age.

These figures seem unreasonably small. This is because the above definition of the marginal propensity to consume is clearly not a very reasonable one. A change in the current income of the household will generally tend to be accompanied by a change in its expected income, y^e, so that there is little sense in including y^e among the things that are supposed to be constant as y changes. Note, however, that the same objection does not apply to a, for a denotes initial assets and, for a given household, assets at the beginning of the current period necessarily represent a constant.

Once we recognize that y^e is generally a function of y, the marginal propensity to consume at age t may be defined as

$$\frac{dc}{dy} = \frac{1}{L_t} + \frac{N-t}{L_t}\frac{dy^e}{dy}. \tag{II.3}$$

Since

$$\frac{dy^e}{dy}$$

would generally tend to lie between 0 and 1,[18] the marginal propensity to consume would fall for different individuals between a minimum of 1/50 and a maximum of 4/5, depending both on age and on the value of

$$\frac{dy^e}{dy}.$$

Unfortunately, the empirical validity of these statements cannot be tested from observations of actual individual behavior. The reason is that consumption and income can only have a *single* value for a *given individual at a given age*. To be sure, we might be able to observe the behavior of an individual whose income had changed in time; but, even if we could control the value of y^e, we could not keep constant his age nor probably his initial assets (*i.e.*, assets at the beginning of each consumption period). The only way we could possibly check these conclusions is by observing the behavior of (average) consumption of *different* households at different income levels, *i.e.*, by observing the "cross-section" average and marginal rate of consumption with respect to income.[19]

Suppose we make these observations and, for the sake of simplicity, suppose further that all the households we examine have approximately the same age and in every case $y = y^e$. Should we then expect the marginal rate of consumption to be

$$\frac{N-t+1}{L_t},$$

as equation (II.3) would seem to imply? The answer is no; the individual marginal propensity to consume cannot be simply identified with the cross-section marginal rate of consumption. Turning back to equation (II.1), we can easily see that (if all individuals behave according to this equation) the cross-section marginal rate of consumption should be

$$\frac{d'c}{d'y} = \frac{N+1-t}{L_t} + \frac{1}{L_t}\frac{d'a}{d'y}, \tag{II.4}$$

where the differential operator d' is used to denote cross-section differentials. Although

$$\frac{da}{dy}$$

must be zero for an individual, there is no reason why the cross-section rate of change,

$$\frac{d'a}{d'y},$$

should also be zero. Quite the contrary. Our model leads us to expect a very definite relation between the (average) net worth of house-holds at a given income level and the income level itself, which relation we now proceed to explore.

II.2 The Equilibrium Income-Asset-Age Relation and the Consumption-Income Relation in a Stationary and Nonstationary Cross Section

To see clearly the implications of our model it will be useful to examine at first the nature of the cross-section relation between consumption and income in a special case which we shall call a cross section of "stationary" households. A household will be said to be in stationary position if it satisfies the following two conditions: (a) at the beginning of its active life it expects a constant income throughout its earning span; and (b) at every point of its life cycle it finds that its original expectations are completely fulfilled in the sense that its past and current income are as originally expected and its expectations for the future also coincide with its original expectations.[20] From equations (I.4) and (I.7) (since by assumption $y_1^e = y_1$ and $a_1 = 0$) we see that for such a household the consumption plan at age 1, which we denote by $\bar{c}_\tau^1$, must be

$$\bar{c}_\tau^1 = \frac{N}{L} y_1, \quad \tau = 1, 2, \ldots, L; \tag{II.5$'$}$$

and the saving plan

$$\bar{s}_\tau^1 = \begin{cases} \dfrac{M}{L} y_1, & \tau = 1, 2, \ldots, N. \\[2ex] -\dfrac{N}{L} y_1, & \tau = N+1, N+2, \ldots, L. \end{cases} \tag{II.5$''$}$$

Finally, the asset plan, which is the sum of planned savings, must be

$$\bar{a}_\tau^1 = \begin{cases} \dfrac{(\tau-1)M}{L} y_1, & \tau = 1, 2, \ldots, N \\[2ex] \dfrac{N(L+1-\tau)}{L} y_1, & \tau = N+1, N+2, \ldots, L. \end{cases} \tag{II.5$'''$}$$

We will make use of equation (II.5‴) to define the notion of "stationary equilibrium" assets. We say that initial asset holdings at age t, a_t, are in equilibrium relative to any given level of income, y, if

$$a_t = a(y,t) = \begin{cases} \dfrac{(t-1)M}{L}y, & t = 1, 2, \ldots, N \\ \dfrac{N(L+1-t)}{L}y, & t = N+1, N+2, \ldots, L. \end{cases} \qquad \text{(II.6)}$$

Now it can readily be shown that for households fulfilling the stationary conditions and behaving according to equation (II.1), assets, at any age, will coincide precisely with those planned at age 1.[21] But, by definition, for a household of age $t \leqq N$ in stationary position, current income, y, equals y_1; it follows that its assets at the beginning of the current period must be

$$a = \bar{a}_t^1 = a(y_1,t) = a(y,t) = \frac{(t-1)M}{L}y, \quad t \leqq N \qquad \text{(II.8)}$$

which exhibits explicitly the relation between current initial assets, income, and age. Substituting from (II.8) into (II.1) and (II.2) and remembering that, by assumption, income expectations coincide with current income, we find for any age $t \leqq N$,

$$c = \frac{N}{L}y, \quad s = \frac{M}{L}y; \qquad \text{(II.9)}$$

i.e., for households fulfilling the stationary conditions and within the earning span, current consumption and saving are proportional to the current income.[22]

At first sight this conclusion may appear to have little empirical meaning, since the notion of a stationary household is a theoretical limiting concept with little operational content. But our result need not be interpreted literally. Clearly our model has the following very significant implication: if we take a cross section of households within their earning span, which are reasonably well adjusted to the current level of income (in the sense, that, for each household, current income is close to the level the household has received in the past and it expects to receive in the future), then we should find that the proportion of income saved is substantially the same at all levels of income.[23] Even this more general conclusion is not easy to test from available data. Yet we shall be able to show presently that some rather striking evidence in support of this implication of our model is provided by certain recent studies.

From the result just established it follows directly that, if our sample consisted primarily of households in stationary position, then the cross-section rate of change of consumption with respect to income,

$$\frac{d'c}{d'y},$$

is entirely different from the individual marginal propensity to consume defined by equation (II.3). According to equation (II.9) the cross-section rate of change must be

$$\frac{N}{L},$$

a result which can also be derived from equation (II.4), by observing that, for stationary households, equation (II.8) holds so that, for given age t,

$$\frac{d'a}{d'y} = \frac{(t-1)M}{L}.$$

Note, in particular, that (a) the individual marginal propensity varies with age, whereas the cross-section rate of change is independent of the age composition of the sample (provided all households are within the earning span); and (b) the slope of the cross-section line could not even be expected to represent some average of the marginal propensities at various ages. Indeed, even under unity elasticity of expectations, the marginal propensity at any age (except age 1) is less than N/L.[24]

In general, however, a random sample would not consist entirely, or even primarily, of households in stationary position. Let us therefore proceed to the more general case and see what we can learn from our model about the behavior of households who are not in stationary equilibrium.

Making use of the definition of $a(y,t)$ given in (II.6), the individual saving function (II.2) can be rewritten in the following useful form:

$$\begin{aligned} s &= \frac{M}{L} y^e + \frac{L-t}{L_t}(y - y^e) - \frac{1}{L_t}[a - a(y^e,t)] \\ &= \frac{M}{L} y + \frac{N(L-t)-M}{LL_t}(y - y^e) - \frac{1}{L_t}[a - a(y^e,t)]. \end{aligned} \tag{II.2$'$}$$

The quantity $(y - y^e)$, representing the excess of current income over the average level expected in the future, may be called the "nonpermanent component of income"[25] (which may be positive or negative). Similarly, the quantity $[a - a(y^e,t)]$, representing the difference between actual initial assets and the volume of assets which would be carried by an individual fully adjusted to the "permanent" component of income y^e, may be called the "imbalance in initial assets" or also "excess assets" relative to the permanent component of income.

The first form of equation (II.2′) states the proposition that saving is equal to: (1) a constant fraction of the permanent component of income (independent of both age and income) which fraction is precisely the stationary equilibrium saving ratio; plus (2) a fraction of the nonpermanent component of income [this fraction is independent of income but depends on age and is larger, in fact much larger, than the fraction under (1)]; minus (3) a fraction, depending only on age, of excess assets. A similar interpretation can be given to the second form of (II.2′).

Equation (II.2′) is useful for examining the behavior of an individual, who, after having been in stationary equilibrium up to age $t-1$, experiences an unexpected increase in income at age t so that $y_t > y_{t-1}^e = y_{t-1}$. Here we must distinguish two possibilities. Suppose, first, the increase is viewed as being strictly temporary so that $y_t^e = y_{t-1}^e = y_{t-1}$. In this case $a_t = a(y_t^e, t)$.[26] There is no imbalance in assets and, therefore, the third term is zero. But the second term will be positive since a share of current income, amounting to $y_t - y_{t-1}$, represents a nonpermanent component. Because our individual will be saving an abnormally large share of this portion of his income, his saving ratio will rise above the normal figure

$$\frac{M}{L}.$$

This ratio will in fact be higher, the higher the share of current income which is nonpermanent, or, which is equivalent in this case, the higher the percentage increase in income.[27] Let us next suppose that the current increase in income causes him to raise his expectations; and consider the limiting case where $y_t^e = y_t$, *i.e.*, the elasticity of expectations is unity.

In this case the transitory component is, of course, zero; but now the third term becomes positive, reflecting an insufficiency of assets relative to the new and higher income expectation. Accordingly, the saving ratio rises again above the normal level

$$\frac{M}{L}$$

by an extent which is greater the greater the percentage increase in income. Moreover, as we might expect, the fact that expectations have followed income causes the increase in the saving ratio to be somewhat smaller than in the previous case.[28]

Our model implies, then, that a household whose current income unexpectedly rises above the previous "accustomed" level (where the term "accustomed" refers to the average expected income to which the household was adjusted), will save a proportion of its income larger than it was saving before the change and also larger than is presently saved by the permanent inhabitants of the income bracket into which the household now enters. The statement, of course, holds in reverse for a fall in income.[29]

II.3 A Reinterpretation of the Cross-section Relation Between Consumption and Income

The propositions we have just established are not easy to test directly with existing data, since such data do not usually provide information on the "accustomed" level but, at best, only on the level of income during a previous short period. However, even this type of information should be useful, at least for an indirect test of our conclusions. For, suppose we divide all the households into three groups: (*1*) those whose income has increased in the current year; (*2*) those whose income is approximately unchanged; and (*3*) those whose income has fallen. Then, unless most of the income changes just happen to be such as to return the recipients to an accustomed level from which they had earlier departed,[30] group (*1*) should contain a greater proportion of people whose income is above the accustomed level than group (*2*) and, a fortiori, than group (*3*). Hence, according to our model, the households in group (*1*) whose income has risen should save, on the average, a larger proportion of income than those in group (*2*), which in turn should save a larger proportion than those in group (*3*). It is well known that this proposition is overwhelmingly supported by empirical evidence.[31] Even where some apparent exceptions have been reported, these have occurred largely because of the inclusion in expenditure of current outlays for durable goods which we do not include in consumption as far as they result in an increase in net worth.[32]

We will readily recognize that the proposition we have just derived is far from novel; but notice that our model suggests an explanation that is quite different from the one usually advanced. According to the usual explanation, which is already to be found in the *General Theory* (p. 97), consumer expenditure habits are sticky and only adjust with a lag to the changed circumstances; in the meantime, savings, which are considered as a passive residual, absorb a large share of the changed income. In our model, on the other hand, savings tend to go up either because the new level of income is regarded as (partly or wholly) transitory or, to the extent that it is regarded as permanent, because the initial asset holdings are now out of line with the revised outlook. If the outlook has improved, assets are too low to enable the household to live for the rest of its expected life on a scale commensurate with the new level of income; if the income outlook has deteriorated, then, in order for the household to achieve the optimum consumption plan consistent with the new outlook, it is not necessary to add to assets at the same rate as before, and perhaps even an immediate drawing-down of assets to support consumption may be called for.

We feel that this alternative interpretation has merits. While not denying that the conventional explanation in terms of habit persistence may have some validity, we feel that it has been made to bear too heavy a weight. We all know that there are hundreds of things that we would be eager to buy, or do, "if only we could afford them," and nearly as many places where we could "cut corners" if it were really necessary. Therefore, as long as we are dealing with moderate variations in income (variations whose possibility we had already envisaged at least in our day-dreams), there is not likely to be any significant lag in the adjustment of total expenditure. Of course, there may be significant lags (and leads) in certain types of expenditures: moving to a more exclusive neighborhood may take years to be realized, but meanwhile that dreamed-of vacation trip may come off at once.[33]

Our discussion of the effect of income changes enables us to proceed to an analysis of the cross-section relation between current consumption and current income that we should expect to find, and actually do find, in a random sample which does not consist primarily of households in stationary position.

As is well known, budget studies typically show that the proportion of income saved, far from being constant, tends to rise from a very low or even negative figure at low levels of income to a large positive figure in the highest brackets. These findings are by no means inconsistent with our earlier results concerning the saving-income relation for a stationary cross section. Quite the contrary; the observed relation is precisely what we should expect on the basis of our model when we remember that, in the type of economies to which these budget data mostly refer, individual incomes are subject to short-term fluctuations, so that current income generally will tend to differ more or less markedly from the previous accustomed level and from current income expectations. Such fluctuations may vary in intensity according to time, place, occupation, and other characteristics of the sample covered, but they will never be entirely absent and frequently will be substantial.

The very same reasoning we have used in the discussion of the effect of income changes leads to the conclusion that, in the presence of short-term fluctuations, the highest income brackets may be expected to contain the largest proportion of households whose current income is above the accustomed level and whose saving is, therefore, abnormally large. Conversely, the lowest income brackets may be expected to contain the largest proportion of people whose current income is below the accustomed level and whose saving is, therefore, abnormally low. As a result of the presence of these groups, which are not fully adjusted to the current level of income, in the lowest brackets the proportion of income saved will be lower, and in the highest brackets it will be higher than the normal figure (M/L in our model) which we should expect to be saved by the permanent

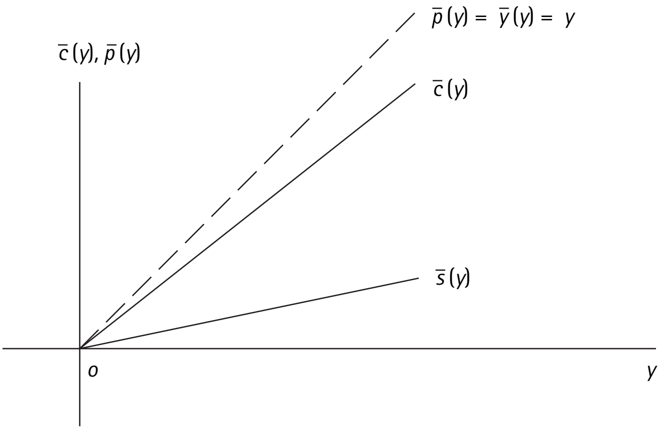

Figure 1.1
Consumption-Income relation in the case of a stationary cross-section

inhabitants of these respective brackets. Thus, the proportion of income saved will tend to rise with income, and the cross-section relation between consumption and income will tend to be represented by a line obtained by rotating the stationary line clockwise around a fixed point whose x and y coordinates coincide approximately with the average value of income and consumption respectively.

While the general line of argument developed above is not new,[34] it may be useful to clarify it by means of a graphical illustration developed in figures 1.1 and 1.2. We will start out by analyzing a cross section of households all belonging to a single age group within the earning span, and will examine first the consumption-income relation; once we have established this relation, the saving-income relation can easily be derived from it.

Our consumption function (II.1) can be rewritten in a form analogous to the saving function (II.2′), namely:

$$\begin{aligned} c &= \frac{N}{L} y^e + \frac{1}{L_t}(y - y^e) + \frac{1}{L_t}[a - a(y^e, t)] \\ &= \frac{N}{L}\left\{ y^e + \frac{L}{NL_t}(y - y^e) + \frac{L}{NL_t}[a - a(y^e, t)] \right\} \end{aligned} \tag{II.1′}$$

In the construction of our figures, we shall find it convenient to have a symbol to represent the expression in braces; let us denote it by $p = p(y, y^e, t, a)$. This expression may be regarded as the stationary equivalent income of the current set of values y, y^e, t and a, for the household, in the sense that, if the household were fully adjusted to a level of income $p = p(y, y^e, t, a)$, then it would behave in the same

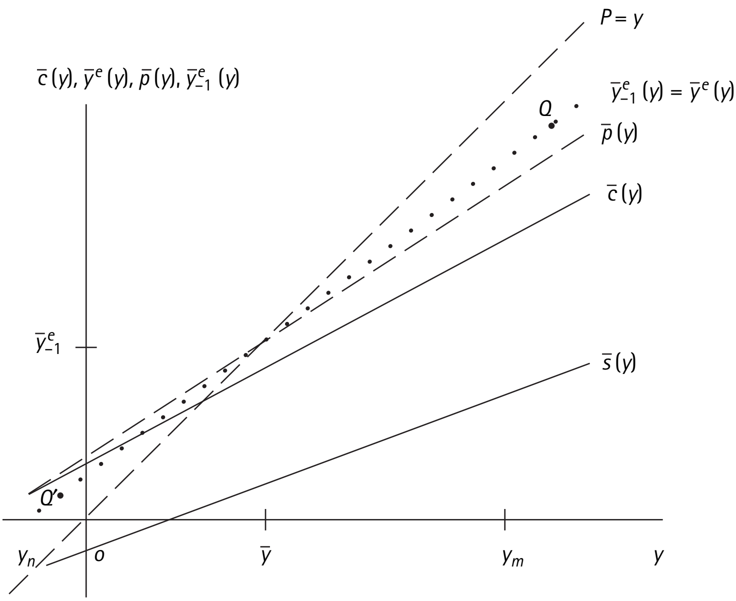

Figure 1.2
Consumption-Income relation and short-term fluctuations in income

way as it currently does. Let us further denote by the symbol $\bar{x}(y)$ the average value of any variable x for all the members of a given income bracket, y. Then the proportion of income consumed by the aggregate of all households whose current income is y is, clearly, $\bar{c}(y)/y$. Our problem is, therefore, that of establishing the relation between $\bar{c}(y)$ and y. But, according to (II.1′), $\bar{c}(y)$ is pr portional to $\bar{p}(y)$ whose behavior, in turn, as we see, depends on that of $\bar{y}^e(y)$ and $\bar{a}(y)$. We must therefore fix our attention on the behavior of these last two quantities.

In the case of a *stationary* cross section, illustrated in figure 1.1, we know that for *every* household, $y^e = y$ and also $a = a(y^e,t)$. It follows that, for *every* household, $p = y$, and therefore the average value of p in any income bracket y, $\bar{p}(y)$, is also equal to y, *i.e.*, $\bar{p}(y) = y$. Thus, the cross-section relation between $\bar{p}(y)$ and y is represented by a line of slope one through the origin—the dashed line of our figure 1.1. The consumption-income relation is now obtained by multiplying each ordinate of this line by the constant N/L, with the result represented by the upper solid line. Because the $\bar{p}(y)$ line goes through the origin, so does the consumption-income relation, $\bar{c}(y)$, and the elasticity of consumption with respect to income is unity. These same propositions hold equally for the saving-income relation, $\bar{s}(y)$ (lower solid line of figure 1.1), obtained by subtracting $\bar{c}(y)$ from y. This merely illustrates a result established in the preceding section.

Let us consider now what we may expect to happen if income is subject to short-term fluctuations, a case illustrated in figure 1.2. We may assume for expository convenience that on the average these fluctuations cancel out so that average current income, $\bar{y}$, is the same as the average future income expected in the year before by the sample as a whole, $\bar{y}^e_{-1}$. But, because of the presence of short-term variations, this equality will not hold for individual households; for some households current income will be higher than y^e_{-1}, while for others it will be lower. As a result of these fluctuations, as we have already argued, in the highest income brackets there will be a predominance of households whose current income is above y^e_{-1}. This, in turn, means that in these brackets the average value of $y^e_{-1}, \bar{y}^e_{-1}(y)$, will be less than y itself;[35] in terms of our graph, $\bar{y}^e_{-1}(y)$ will fall below the dashed line, which represents the line of slope one through the origin. For instance, corresponding to the highest income bracket shown, $y_m, \bar{y}^e_{-1}(\bar{y}_m)$ may be represented by a point such as q in our graph. Conversely, in the lowest income bracket shown, y_n, there will tend to be a preponderance of people whose current income is below y^e_{-1}, and therefore the average value of y^e_{-1} in this bracket, $\bar{y}^e_{-1}(y_n)$ will be greater than y_n and above the dashed line, as shown by the point q', in the figure. Extending this reasoning to all values of y, we conclude that the relation between $\bar{y}^e_{-1}(y)$ and y will tend to be represented by a curve having the following essential properties: (a) it will intercept the dashed line in the neighborhood of a point with abscissa $\bar{y}$; and (b) to the right of this point, it will fall progressively below the dashed line, while to the left of it, it will stand increasingly above this line. In our graph this relation is represented by the dotted straight line joining the points q' and q; in general, the relation need not be a linear one, although the assumption of linearity may not be altogether unrealistic. What is essential, however, is that the $\bar{y}^e_{-1}(y)$ curve may be expected to have everywhere a slope smaller than unity and to exhibit a positive intercept; and that its slope will tend to be smaller, and its intercept larger, the greater the short-term variability of income.

From the behavior of $\bar{y}^e_{-1}(y)$, We can now readily derive that of $\bar{y}^e(y)$, which is the quantity we are really interested in. The latter variable is related to $\bar{y}^e_{-1}(y)$ and to y itself through the elasticity of income expectations. The elasticity of expectations relevant to our analysis can be defined as the percentage change in income expectation over the two years in question,

$$\frac{y^e - y^e_{-1}}{y^e_{-1}},$$

divided by the corresponding percentage difference between current income and the previous year's expectation,

$$\frac{y - y^e_{-1}}{y^e_{-1}}.$$

If we denote this elasticity by the symbol E, we have

$$E = \frac{y^e - y^e_{-1}}{y - y^e_{-1}},^{36} \tag{II.11}$$

which in turn implies:

$$y^e = y^e_{-1} + E\left(y - y^e_{-1}\right) = (1 - E)y^e_{-1} + Ey; \tag{II.11$'$}$$

i.e., the current income expectation is a weighted average of the previous expectation and current income, with weights depending on the elasticity of expectation. If E is close to zero, current income will have little influence in reshaping expectations and y^e will be close to y^e_{-1}; if, at the other extreme, E is close to unity then current expectations will be determined primarily by the recent behavior of income. From (II.11′) we readily deduce the relation between $\bar{y}^e(y)$ and $\bar{y}^e_{-1}(y)$, namely:

$$\bar{y}^e(y) = \bar{y}^e_{-1}(y) + E\left[y - \bar{y}^e_{-1}(y)\right].^{37} \tag{II.12}$$

Equation (II.12) admits of a very simple graphical interpretation. Suppose, first, that E is zero: then $\bar{y}^e(y)$ coincides with $\bar{y}^e_{-1}(y)$, the dotted line of figure 2. Next, suppose E is positive but less than one; then, for any value of y, $\bar{y}^e(y)$, if it were drawn, would lie between y and $\bar{y}^e_{-1}(y)$, *i.e.*, between the dashed and the dotted line and precisely E percent of the way from the dotted to the dashed line. Finally, if E is greater than one, or less than zero, then $\bar{y}^e(y)$ will fall outside the band marked off by our two lines. The last two cases, however, may be regarded as extremely unlikely, when one remembers that y^e and y^e_{-1} are defined as expectations about the average level of income over the entire balance of the earning span.[38] In our graph we have assumed a zero value for E so that $\bar{y}^e(y)$ coincides with $\bar{y}^e_{-1}(y)$; this assumption has been chosen not because we think it is realistic but only because it eliminates the necessity of showing a separate curve in our figure. In general, we should rather expect E to be positive but less than one, so that the $\bar{y}^e(y)$ curve would fall between the dotted and the dashed line. *The slope and intercept of this curve will thus depend on the degree of short-term variability of income* [which determines the shape of $\bar{y}^e_{-1}(y)$] *and on the elasticity of expectations*. But note that where short-run fluctuations play a large role, as might be the case, say, for a sample of farmers, the elasticity of expectations may, itself, be expected to be small on the average, since current income will contain little new reliable information on the basis of which to reshape the previous expectation about average future income, y^e_{-1}. Hence, a large short-term variability of income will tend to depress the slope, and raise the intercept, of $\bar{y}^e(y)$ for two reasons: (1) because it pulls the $\bar{y}^e(y)$ closer to $\bar{y}^e_{-1}(y)$, and (2) because it diminishes the slope of $\bar{y}^e_{-1}(y)$.

We have thus exhibited the behavior of the first component of $\bar{p}(y)$. As we see from (II.1′), the second component is very simple to handle; it represents again a fraction,

$$\frac{L}{NL_t},$$

of the difference between y and $\bar{y}^e(y)$, *i.e.*, of the distance from the dotted to the dashed line. Furthermore, for any reasonable assumption about the values of L and N, this fraction is quite small within the earning span. Thus, the sum of the first two terms of $\bar{p}(y)$ could be represented by a new line falling somewhere between the dashed and the dotted line but very close to the latter. Since it would be so close, this new line is not shown separately in our figure and we will assume that the line $\overline{q'q}$ represents already the sum of the first two components.

We may now proceed to the last component of $\bar{p}(y)$ which measures the (average) unbalance in assets at each current income level. A repetition of the familiar reasoning suggests once more that in the highest income brackets there should tend to be an abnormally large proportion of households whose current income expectation is higher than it had been in the past and for whom, accordingly, current assets are low relative to the current expectation y^e; and that the opposite should be true in the lowest income brackets. Hence, the last term will again be negative for sufficiently large values of y and positive for sufficiently small values. The zero point is again likely to occur for a value of y close to $\bar{y}$, although its location may vary depending on the previous history of the economy (especially the history of the last t years).[39] In any event $\bar{p}(y)$, which is obtained by adding this last term to the previous two, will tend to have a slope everywhere smaller than our dotted line and a larger intercept; its graph may therefore look something like the dashed-dotted line of figure 1.2.

We are now ready to exhibit the consumption-income relation $\bar{c}(y)$, which is obtained simply by multiplying the ordinates of $\bar{p}(y)$ by the constant, N/L. The result is represented by the upper solid line. As we intended to show, it is a line much flatter than the stationary cross-section line of figure 1.1 (from which it can be obtained by rotation around its point $(\bar{y},\bar{y}^e)$) and it exhibits the large positive intercept characteristic of budget data. In other words, in the presence of short-term fluctuations in income, the proportion of income consumed will tend to fall with income and the elasticity of consumption with respect to income will be less than one. Another important result we have established is that the elasticity of consumption with respect to income should depend, according to our theory, on three major factors which are capable of variation in different samples. This elasticity depends, inversely, on (a) the degree of short-term variability of income; (b) the magnitude of variations over time in the permanent component of income that give rise to unbalances in the relation between assets holdings and the permanent component itself; and (c) directly, on the elasticity of income expectations.[40] Given the magnitude of these three factors, the elasticity should depend also on the age of the households in the sample since, as we see from equation (II.1′), the coefficient of the second and third components of p vary with age. This point, however, will receive proper attention in the next section.

In our discussion so far we have been concerned with a cross section of households within the earning span. The effect of the inclusion of retired households in the sample can readily be established. According to our model, these households should have, on the average, levels of consumption not very different from the over-all average,[41] while, at the same time, earning incomes well below average (in our present simplified version, no income at all). The inclusion of these households, therefore, will have the effect of raising still further the $\bar{c}(y)$ curves for low levels of y, thereby increasing its general tilting and possibly making it convex to the income axis.[42]

Summing up, the typical findings of budget studies as to the relation between consumption, saving, and income in a given year, or some other short interval of time, are precisely what they should be according to our model. But, as we see, the interpretation of these findings that follows from our model is radically different from the one generally accepted. According to the usual interpretation, these findings show that the proportion of income saved rises with the economic status of the household. According to our model, on the other hand, they only show that households whose income is above the level to which they are adjusted save an abnormally large proportion and those whose income is below this level save an abnormally low proportion, or even dissave.

To be sure, up to this point, we have done little more than to show that the findings of budget data are consistent with either interpretation. It may be objected that this demonstration is an insufficient ground for discarding the old and widely accepted explanation for our new and radically different one. It would seem that, to support our claims, we should be able to produce some crucial tests, the result of which will enable us to discriminate between the two competing explanations. We believe that the remarkable piece of research recently reported on by Margaret G. Reid, discussed in the next section, comes as close to providing such a test as seems feasible with the kind of data that are presently available.[43]

II.4 Some Evidence on the Constancy of the Saving Ratio for a Stationary Cross Section

In order to understand the foundations of Reid's highly ingenious techniques and the meaning of her results, we must turn back to our figures 1.1 and 1.2, and to the reasoning underlying their construction. Suppose that, somehow, we had been able to locate a sample of households, within their earning span, each of which fulfilled completely our stationary specifications. For *every* member of this sample, because of the complete absence of short-term income fluctuations, we would have the chain of equalities $y = y^e = y^e_{-1} = y_{-1}$ so that the correlation between current and previous income, r_{yy-1}, would be unity. Furthermore, if our theory is correct, the elasticity of consumption with respect to current income, η_{cy}, would also be unity. Moreover, this conclusion would continue to hold if the above chain of equalities were replaced by "proportionality,"

$$i.e., \quad y = \frac{y^e}{k_1} = \frac{y^e_{-1}}{k_2} = \frac{y_{-1}}{k_3}.$$

In this case, all the households are out of stationary equilibrium but, so to speak, by the same proportion; this would affect the slope of the $\bar{c}(y)$ curve of figure 1, but not its intercept, so that the consumption-income elasticity would remain unity.[44]

Now, since information on the permanent component of income and the degree of adjustment to it is not available, it is impossible to locate a sample fulfilling exactly our specifications. On the other hand, it may not be impossible to find one for which short-term fluctuations are of relatively minor importance and current income is relatively close to the level to which the household is adjusted. These conditions might be satisfied, for instance, by a sample of government employees or of college professors. For such a sample we would expect to find a correlation between current and previous income, r_{yy-1}, close to unity and, *if our model is correct*, an elasticity of consumption with respect to income, η_{cy}, close to unity. At the other extreme, for a group of households for which random short-term fluctuations play a dominant role, say, a sample of farmers over a wide geographical region, we would have a low value of r_{yy-1} and, as we know from the analysis of figure 1.2, a low elasticity of consumption with respect to income. This implication of our model clearly forms the basis for a crucial experiment; this experiment has not been carried out as such, although we look forward to its being performed in the near future by anyone having the resources and interest. Meanwhile, Reid's method is very similar to the comparison we have just proposed. The discussion that has led us to the formulation of our crucial experiment suggests that the correlation between current and previous income can be taken as an indirect, approximate measure of the degree to which the current income of each household is close to the level to which the household is adjusted (or to a constant multiple of this level—see previous paragraph). In the first place, this correlation is a very good direct measure of strictly short-term fluctuations which, as we have seen, control the relation between $\bar{y}^e(y)$ and y. Secondly, it seems reasonable to expect that the gap between $\bar{p}(y)$ and $\bar{y}^e(y)$ will, itself, tend to be smaller the smaller the short-run variability of income; for, when incomes are basically stable, there should also be less opportunity for significant discrepancies between the permanent component of income and assets to develop. Thus, when the correlation between current and previous income is high, $\bar{p}(y)$ itself should tend to be close to y. The $\bar{p}(y)$ and $\bar{c}(y)$ curves should, therefore, be similar to those of figure 1.1, and the elasticity of consumption with respect to income should be close to unity. Conversely, when the correlation between current and previous income is small, evidencing the presence of pronounced short-term fluctuations, the $\bar{y}^e(y)$, $\bar{p}(y)$, and $\bar{c}(y)$ curves should be similar to those shown in figure 1.2, and the elasticity of consumption with respect to income should be well below unity.

Hence, if we take several samples of households and, for each of these samples, we compute (a) the correlation between current and previous income, r_{yy-1} and (b) the elasticity of consumption with respect to income, we should find a clear positive rank correlation between the two above-mentioned statistics and, furthermore, η_{cy} should approach unity as r_{yy-1} approaches unity. In essence, this is precisely the test Reid has carried out and her results clearly confirm our inference. The fact that for none of her groups r_{yy-1} is, or could be expected to be, as high as unity prevents η_{cy} from ever reaching the theoretical stationary value of unity; but her data do show that, in the few cases where r comes close to one, η_{cy} is also impressively close to unity. According to Reid's results η_{cy} ranges from a value as low as .1 for r in the order of .2 to values as high as .8 for the highest values of r, which are in the order of .8.[45]

The above discussion is, of course, not intended as an exhaustive account of Reid's techniques and many valuable results; for this purpose the reader is referred to her forthcoming publications. Nevertheless, the brief sketch presented should be sufficient to indicate why her results seem to us to represent as impressive a confirmation of one important implication of our theory, at the microeconomic level, as one may hope to find at this time.

In addition to the test just described, there are several other, partly older, findings that support our model and acquire new meaning in the light of it.

According to our model the typical findings of budget studies as to the relation between consumption and income are basically due to the fact that, in the presence of short-term fluctuations, income over a short interval is a poor and *seriously biased* measure of the accustomed level. In the previous test the extent of this bias was measured by the correlation r_{yy-1}; the higher the correlation, the smaller the bias and, therefore, the higher the elasticity of consumption with respect to income. Margaret Reid and a few others have suggested and tested alternative methods of getting a more reliable index of the accustomed level than current income and, invariably, it is found that when consumption is related to such a measure, the elasticity of consumption with respect to it rises markedly above the consumption-current income elasticity, and comes close, frequently remarkably close, to unity.[46]

Another set of results that supports our model is that reported in the classical contribution of Dorothy S. Brady and Rose D. Friedman, "Savings and the Income Distribution."[47] As is well known, their major finding is that the saving ratio appears to bear a much more stable relation to the position of the income recipient in the income distribution than to the absolute level of income itself. In other words, the proportion of income saved in a *given decile* varies much less over time and space than the proportion of income saved at a *given level of income*. It is not difficult to see that these results are what one would expect if our model is correct. As should be clear from the reasoning we have followed in developing our figure 1.2, the relative frequency of households in a given income bracket whose income is below or above their accustomed level, depends, not on the absolute level of income, but, rather, on the position of the income bracket relative to the average income. For example, in a given income bracket, say \$10,000, we should expect to find a large proportion of people whose accustomed level is less than \$10,000 if, say, the average income is \$2,000 and the level \$10,000 is in the top decile; while, in this same bracket, we should expect to find a small proportion of people whose accustomed level is below \$10,000 if the community average income were, say, \$50,000 so that the \$10,000 bracket is in the lowest income decile. More generally, it can be shown that provided, as seems likely, there is a fairly stable relation between average income in a given decile and the over-all average income, then the saving ratio, in a given decile, would depend primarily on the (relative) short-term variability of income.[48] Thus, if we compare, over time or space, groups for which the (relative) variability of income is not very different, the proportion of income saved in any given decile should be roughly the same for every group. As an example, we can compare the behavior of nonfarm families in different regions and at different points in time (see Brady and Friedman, charts 1 through 4) and we find our inference confirmed. Furthermore, within a group, the greater the variability in income, the greater should be the variation in the saving ratio as between the lower and the upper deciles. This inference, too, is supported by comparison of nonfarm and farm groups (compare chart 2 with chart 5).

It will be recognized that our theory offers an explanation for the Brady-Friedman findings that is fundamentally different from the social-psychological explanation that has been advanced heretofore.[49] Although our current interpretation is much simpler and integrates these findings with many others without recourse to additional postulates, we do not wish to deny that the earlier explanation may have some validity.

One more finding of some significance has to do with the relation between the elasticity of consumption with respect to income on the one hand and age on the other, to which reference has been made earlier. It can be shown that the slope of $\bar{c}(y)$ should tend to fall, and its intercept to rise, with age, unless, on the average, the elasticity of income expectations is extremely low, say, in the neighborhood of zero.[50] Since such a low average elasticity of expectations is rather unlikely to prevail, especially in a basically growing economy, we should generally expect the elasticity of consumption With respect to income to fall with age. This conclusion finds empirical support in the findings reported by Janet Fisher.[51] Our theory provides a common-sense explanation for her empirical finding. The increase in expected income, y^e, which accompanies the change in current income, produces a relatively larger increase in the anticipated total resources of a younger than of an older household, if E is above the minimal value, because of the greater number of years over which the higher level of income will be received. To give a specific illustration, suppose that $E = 1$, and that y, and therefore also y^e, increase by one unit. For a household of age $N - 1$, which has only one year to go before retirement, total resources, v_{N-1}, increase by only two units, and these two units have to be spread over the remaining two years of earning and M years of retirement. Therefore consumption rises by only

$$\frac{2}{M+1}.$$

At the other extreme, if the household had age 1, its total anticipated resources rise by N units, to be spread over N + M years. Hence, current consumption rises by

$$\frac{N}{N+M}$$

units. By a similar reasoning, one can conclude that the depressing effect on current consumption of the unbalance in assets that has been created by the change in income is greater the older the household, because of the smaller number of years available to the household to redress the unbalance.

In conclusion then, there is already ample evidence, and from widely different sources, which is consistent with the most distinctive implications we have derived so far from our theory and which is not equally consistent or, at any rate, readily explainable in terms of any other single set of hypotheses that, to our knowledge, has been advanced so far.

II.5 Individual Saving, Assets, and Age

In recent years a good deal of attention has been devoted to the influence of assets on consumption, and attempts have been made at estimating the cross-section relation between these variables on the expectation that the parameters obtained would yield information on the time-series relation. Our theory has something to contribute as to the pitfalls of such an attempt. To begin with, it suggests that the relevant concept of assets is *net worth*; unfortunately, most of the recent empirical work has concentrated instead on liquid assets, a variable which, according to our model, bears no definite relation to consumption, except perhaps as a very imperfect proxy for net worth.

But even if information on net worth were available, knowledge of the variation in consumption as between different households having different asset holdings, would give very little (and that little would be biased) information as to how a household would react if its assets were increased unexpectedly by a given amount, say, by an anonymous gift of the usual benevolent millionaire or, to take a more fashionable example, by an unexpected fall in the price level of consumables. This failing occurs because the observed asset holdings do not just happen to be there; instead, they reflect the life plan of the individual, which in turn depends on income and income expectations.

To interpret the positive correlation between assets and consumption as implying that people consume more *because* they have more assets would be only slightly less inaccurate than the inverse inference that people have higher assets because they consume more. The point is that both consumption and assets are greatly affected by the other variables: income, income expectations,[52] and age.

It may be objected that we are destroying a straw man; anyone studying the effect of assets would have sense enough to control the effect of, say, income. This may well be true, yet we feel that the above paragraph contains a useful lesson as to the relation between assets and consumption plans that one may too easily forget. For instance, the statement which has been repeated *ad nauseam* that in the early postwar years people bought durable goods lavishly *because* they had such large holdings of liquid assets (which may even find some apparent confirmation in survey results) might well stand a good deal of re-examination along the lines indicated above.

But our model tells us that it is not enough to control income or even income expectations. As it is brought out clearly by our analysis of the stationary case, assets depend also on age (see equation II.8); and this implication of our model is one for which scattered supporting empirical information is available.[53] Furthermore, as we have seen, our hypothetical household goes on consuming a constant fraction of income (equation II.9) even though its assets continue to rise, and reach their peak just before retirement; the rise in assets relative to income does not depress saving because it is part and parcel of the life plan. In other words, higher assets do not necessarily affect saving; they do so only if, on account of unexpected variations, assets turn out to be out of line with income and age: it is only an excess (or shortage) of assets that affects the saving ratio (see equation II.2′).[54]

Finally, we can see from our equation (II.1) that the cross-section marginal rate of change of consumption (or saving) with respect to asset holdings (income and income expectations constant) could not yield a reliable estimate of the marginal propensity to consume with respect to assets. The reason is simple. From (II.1), it follows that this marginal propensity is

$$\frac{\partial c}{\partial a} = -\frac{\partial s}{\partial a} = \frac{1}{L_t}. \tag{II.15}$$

This expression is independent of assets and income but depends on age. We cannot, therefore, properly speak of *the* marginal propensity to consume with respect to assets, as this quantity will vary substantially from age group to age group, tending to increase with age.

Let us finally remember that, in order to compute the individual marginal propensity from cross-section data, it is also not sufficient to control age by introducing this variable linearly in a linear regression of consumption on income and assets,[55] for, according to our model, age does not enter in a linear fashion. The only way of estimating the marginal propensity at various ages and, in the process, test equation (II.15) is to carry out a full stratification by age groups (or some equivalent procedure). It is to be hoped that data for such a test may sometime be available.[56]

II.6 Uncertainty, Saving, and the Composition of Assets

The analysis of the previous sections is helpful in providing a justification for our earlier contention that the phenomenon of uncertainty can be neglected without seriously affecting the usefulness of the analysis.

As we have seen, even under the assumption of certainty there are sufficient incentives for the household to accumulate assets at a rapid rate during the early years of its life. Since the assets thus accumulated can be used to acquire durable goods and are also available as a general reserve against emergency, it would appear that the last two motives (p. 6), which are the result of uncertainty, need not affect significantly the saving behavior.[57]

To be sure, though assets can satisfy several purposes, their efficiency will not be the same in this respect. For instance, those assets which are best suited to satisfy the fourth motive are frequently not very well suited to satisfy the third. Accordingly, variations in the relative urgency of our four motives as between different individuals, and at various points of the life cycle for the same individual, will be reflected in the composition of the balance-sheet. This composition will also be affected by the current and prospective total resources, by the nature of the available alternatives, and, last but not least, by "social" pressures. Cautious individuals will hold relatively more liquid assets. Individuals with large means will tend to hold more durable goods. Examples of social pressure on the type of assets held are also not far to seek; witness the scramble for common stock during the 'twenties and the adoption of television after the Second World War.[58]

As to the effect of the life cycle on the composition of the "portfolio," one might expect that during the period of family formation people will put most of their savings into durables. Automobiles, refrigerators, stoves, and other appliances are felt to be essential to the establishment of an American household. After the initial purchase of durables, although depreciated goods are to be replaced and some additional goods are to be bought, savings flow into other kinds of assets. Various liquid assets may be acquired. The acquisition of a house, which requires (except for the recent G.I. housing) a prior large stock of cash, can be expected to occur throughout the life span. These generalizations are borne out by the existing data.[59]

In conclusion, uncertainty as well as many other factors must be recognized as being of great importance if one is interested in developing a satisfactory theory of the composition of the "portfolio" or, which is equivalent, of the rate of addition to the specific assets. They do not seem to be essential, however, for the development of a useful theory of the factors controlling the over-all rate of saving. Needless to say, the final justification of this statement must rest on whether or not our theory proves helpful in explaining facts which are presently known or will be revealed by further empirical work. The results so far are encouraging.

II.7 Summary

On the basis of the received theory of consumer's choice, plus one major assumption as to the properties of the utility function, we have been able to derive a simple model of individual saving behavior which is capable of accounting for the most significant findings of many cross-section studies. Inasmuch as this model has been shown elsewhere to be equally consistent with the major findings of time-series analysis, we seem to be near the ultimate goal of a unified, and yet simple, theory of the consumption function.

The results of our labor basically confirm the propositions put forward by Keynes in *The General Theory.* At the same time, we take some satisfaction in having been able to tie this aspect of his analysis into the mainstream of economic theory by replacing his mysterious psychological law with the principle that men are disposed, as a rule and on the average, to be forward-looking animals.[60] We depart from Keynes, however, on his contention of "a greater *proportion* of income being saved as real income increases" (p. 97, italics his). We claim instead that the *proportion of income saved is essentially independent of income*; and that systematic deviations of the saving ratio from the normal level are largely accounted for by the fact that short-term fluctuations of income around the basic earning capacity of the household, as well as gradual changes in this earning capacity, may cause accumulated saving to get out of line with current income and age. The common sense of our claim rests largely on two propositions: (a) that the major purpose of saving is to provide a cushion against the major variations in income that typically occur during the life cycle of the household as well as against less systematic short-term fluctuations in income and needs; (b) that the provisions the household would wish to make, and can afford to make, for retirement as well as for emergencies, must be basically proportional, on the average, to its basic earning capacity, while the number of years over which these provisions can be made is largely independent of income levels. We have shown that our claim is strongly supported by budget data when these data are properly analyzed.

In *The General Theory*, Keynes did not put much emphasis on the proposition quoted above. But in the literature that followed, the assumption that the proportion of income saved must rise with income has frequently been regarded as one of the essential aspects of his theory and, as such, it has played an important role in Keynesian analysis and its applications to economic policy and forecasting.

The task of developing the policy implications of our analysis falls outside the scope of this paper. We may, nonetheless, point out, as an example, that our new understanding of the determinants of saving behavior casts some doubts on the effectiveness of a policy of income redistribution for the purpose of reducing the average propensity to save.

Finally, we hope our study has proved useful in pointing out the many pitfalls inherent in inferences derived from cross-section data without the guidance of a clear theoretical framework. For instance, we must take a dim view of the attempts that have been made at deriving the time-series average and marginal propensity to save from the cross-section relation between saving and income. *The individual marginal propensity cannot, generally, be identified with the cross-section rate of change*; in fact, as we have seen, these two parameters need not bear any definite and stable relation to one another. We have further shown elsewhere[61] that the individual marginal propensity to save bears, in turn, a very complex relation to the time-series marginal propensity and hardly any relation at all to the time-series average propensity.

Needless to say, many implications of our theory remain to be tested. In fact, it is a merit of our hypothesis that it leads to many deductions that are subject to empirical tests and therefore to contradiction. We would be the first to be surprised if all the implications of the theory turned out to be supported by future tests. We are confident, however, that a sufficient number will find confirmation to show that we have succeeded in isolating a major determinant of a very complex phenomenon.

Appendix: Some Suggestions for the Adaptation of our Model to Quantitative Tests and Some Further Empirical Evidence

Of the many cross-section studies of saving behavior in recent years, the one that comes closest to testing the hypothesis represented by our equations (II.1) and (II.2) is the extensive quantitative analysis reported by Lawrence R. Klein in "Assets, Debts and Economic Behavior," *op. cit.* (see especially pp. 220–227 and the brief but illuminating comment of A. Hart, *ibid.*, p. 228). In this appendix we shall attempt to provide a brief systematic comparison of his quantitative findings with the quantitative implications of our theory.

We have already pointed out the many shortcomings of his analysis for the purpose of testing our theory, both in terms of definitions of variables and of the form of the equation finally tested. Because of these shortcomings, the result of our comparison can have at best only a symptomatic value. In fact, the major justification for what follows is its possible usefulness in indicating the type of adaptations that might be required for the purpose of carrying out statistical tests of our model.

Making use of the identity

$$N - t \equiv \frac{N(L-t)}{L} - \frac{Mt}{L},$$

our equation (II.2) can be altered to the form

$$s_i = \frac{L-t_i}{L_i} y_i - \frac{N(L-t_i)}{LL_i} y_i^e - \frac{a_i}{L_i} + \frac{M}{LL_i} t_i y_i^e, \tag{A.1}$$

where the subscript i denotes the i-th household, and L_i is an abbreviation for $L + 1 - t_i$.

Unfortunately, the variable y_i^e is usually unknown. Some information on it might be gathered by appropriate questions analogous to the question on short-term income expectations which has already been asked in the past. Klein himself, however, has not made use of this possible source of information in the study in progress. We must, therefore, find some way of relating y_i^e to the variables actually used by Klein, which are also those most commonly available.

It is clear that y^e must bear a fairly close and reasonably stable relation to current income, y. In fact, at various points in the text we have suggested that the relation between these two variables maybe expressed by the equation

$$\begin{cases} y^e = (1-E)y^e_{-1} + Ey + A + U \\ \quad = (1-E)(\alpha+\beta y) + Ey + U', \end{cases} \tag{A.2}$$

where A is a constant for a given sample (though subject to variation over time); E is the elasticity of income expectations defined by equation (II.11); U and $U' = (1-E)\varepsilon' + U$ are random errors; α, β are defined by equations (II.10′); and ε' is the random component of that equation.

If we also have information on previous year's income, y_{-1}, we can clearly exploit this information to get a better estimate of y^e_{-1}, and (A.2) then takes the form

$$y^e = \alpha^* + \beta_1 y + \beta_2 y_{-1} + U^*. \tag{A.3}$$

The coefficients of this equation, as well as the random component U^*, depend again on the short-term variability of income as measured by the correlation r_{yy-1}, on the variance of U and on the elasticity of expectations E. It is not worth-while, however, to derive here this relation explicitly.

Substituting for y^e from (A.3) into (A.1), and rearranging terms (and neglecting the error term which is proportional to U^*) we get:

$$\begin{cases} s_i = -\dfrac{(N-t_i)\alpha^*}{L_i} + \dfrac{(L-t_i)(L-N\beta_1) - L(N-t_i)\beta_2}{LL_i} y_i - \dfrac{a_i}{L_i} + \\ \qquad \dfrac{(N-t_i)\beta_2}{L_i}(y_i - y_{-1i}) + \dfrac{M\beta_1}{LL_i} t_i y_i. \end{cases} \tag{A.4}$$

Finally, dividing through by y_i and making use of the identity $L \equiv M + N$, we obtain the result[62]

$$\begin{cases} \dfrac{s_i}{y_i} = \dfrac{(L_i-1)(L-N\beta_1) - L(L_i-M-1)\beta_2}{LL_i} - \dfrac{(L_i-M-1)\alpha^*}{L_i}\dfrac{1}{y_i} - \dfrac{1}{L_i}\dfrac{a_i}{y_i} \\ \qquad + \dfrac{(N-t_i)\beta_2}{L_i}(y_i - y_{-1i}) \Big/ y_i + \dfrac{M\beta_1}{LL_i} t_i. \end{cases} \tag{A.5}$$

If the quantity L_i were a constant, this equation would be identical in form with the equation Klein proposed to test;[63] although, in the actual statistical test he has found it convenient to approximate the first two terms by an expression of the form $\lambda + \mu \log y_i$, and the variable

$$\frac{y_i - y_{-1i}}{y_i} \text{ by } \frac{y_i - y_{-1i}}{y_{-1i}}.$$

If we now look at Klein's results, we can take courage from the fact that all his coefficients have at least the sign required by our model; namely, a positive sign for income, income change, and age (t in our notation, a in Klein's) and negative for assets

$$\left(\frac{a}{y} \text{ in our notation, } \frac{L}{Y} \text{ in Klein's}\right).$$

This result is of some significance, especially in the case of the age variable. According to our model, the positive sign of this coefficient reflects the fact that within the earning span, for a given level of income and assets, the older the household the smaller will tend to be its resources per remaining year of life, and therefore the smaller the consumption (*i.e.*, the higher the saving).

It would be interesting to compare the size of Klein's coefficients with the values implied by our model. At this point, however, we must remember that the analogy between Klein's equation and our equation (A.5) is more formal than real; for Klein treats his coefficients as if they were constant, whereas, according to our model, they are all functions of age since they all involve the quantity L_i. Consideration of the sensitivity of the coefficients of our equation to variations in t suggests that the error involved in treating them as constants might be quite serious. We must further remember that the specific value of the coefficients in (A.5) is based on Assumptions III and IV. As we have repeatedly indicated, these Assumptions are introduced for expository convenience but are not an essential part of our model. With the elimination of Assumption III and the relaxation of IV, along the lines suggested note 23, the form of our equations and the sign of the coefficients are unchanged, but the value of these coefficients is not necessarily that given in equation (A.5), nor is it possible to deduce these values entirely on a prior grounds, except within broad limits.

We might, nonetheless, attempt a comparison, for whatever it is worth, by replacing the variable t_i, in the expression L_i by a constant, say, by its average value in Klein's sample. Unfortunately, this average is not published, but we should not go far wrong by putting it at between 15 and 25 and computing a range for each coefficient using these two values." We must also take a guess at the value of β_1 and β_2. These quantities, it will be noted, are not observable. On the basis of an analysis of factors controlling these coefficients we suggest, however, that a value in the order of .5 to .6 for β_1 and of .2 and .3 for β_2, is not likely to be far from the mark. Using our standard assumption as to the value of L and N, we then get the following comparison:

Coefficient of	*Klein's estimates*[64]	*Our Model (linear approximation)*
Assets	−.21; −.25	−.04 to −.03
Income	(.02) (.03)	
Income change	.03; .07	.1 to .2
	(.05) (.06)	
Age	.0013; .0055	.003 to .0045
	(.0022) (.0024)	

The age coefficient falls squarely within the range of Klein's results and nothing further need be said about it. The coefficient of income change estimated from the sample is lower than we should have expected, although, at least in one case, the difference from our estimate is well within the range of the standard error.[65] In the case of assets, however, the statistical coefficient is clearly far too great. A large part of this disparity is due, we suggest, to the fact that Klein's variable L represents liquid assets whereas our variable a represents net worth, which is clearly, on the average, several times as large as liquid assets. Hence, if liquid assets holdings are a reasonably good index of total net worth, Klein's coefficient would have to be a large multiple of ours.[66] While it is doubtful that this multiple could be as large as 6 or 7, there seems little doubt that this correction would cut the excess of Klein's coefficient over the theoretical value very substantially, probably to well within one half.[67]

Unfortunately, no definite statement can be made about the coefficient of income and the constant term on account of the logarithmic transformation introduced by Klein. However, approximate computations we have made suggest that these coefficients are in line with our theory in the case of his first equation and somewhat too large (in absolute terms) for the second.

It would thus appear that, at least in terms of orders of magnitude, Klein's findings agree with the implications of our model. We hasten to repeat that the comparison has very limited significance and its results must be taken with a good deal of salt; yet Klein's estimates, just as the many other empirical data we have been able to locate, would seem to warrant the feeling that we are on the right track.

Notes

1. John Maynard Keynes, *The General Theory of Employment, Interest, and Money*, 1936, p. 96.

2. For two excellent bibliographies in this field, see: G. H. Orcutt and A. D. Roy, "A Bibliography of the Consumption Function," University of Cambridge, Department of Applied Economics, mimeo-

graphed release, 1949; and *Bibliography on Income and Wealth, 1937–1947*, edited by Daniel Creamer, International Association for Research in Income and Wealth, 1952.

3. For example, see Lawrence R. Klein, "Savings Concepts and Their Relation to Economic Policy: The Viewpoint of a User of Savings Statistics," paper delivered at the Conference on Savings, Inflation, and Economic Progress, University of Minnesota, 1952.

4. Franco Modigliani and Richard Brumberg, "Utility Analysis and Aggregate Consumption Functions: An Attempt at Integration," a forthcoming study.

5. For an extensive application of marginal utility analysis to the theory of saving see the valuable contributions of Umberto Ricci, dating almost thirty years back: "L'offerta del Risparmio," Part I, *Giornale degli Economisti*, Feb. 1926, pp. 73–101; Part II, *ibid.*, March, 1926, pp. 117–147; and "Ancora Sull'Offerta del Risparmio," *ibid.*, Sept. 1927, pp. 481–504.

6. See, for instance, J. Mosak, *General Equilibrium Theory in International Trade*, Ch. VI, especially pp. 116–117.

7. Cf. Keynes, *op. cit.*, p. 107.

8. Quite recently, many others have advocated this definition. See, for instance, Kenneth Boulding, *A Reconstruction of Economics*, Ch. 8; Mary W. Smelker, "The Problem of Estimating Spending and Saving in Long-Range Projections," Conference on Research in Income and Wealth (preliminary), p. 35; Raymond W. Goldsmith, "Trends and Structural Changes in Saving in the 20th Century," Conference on Savings, Inflation, and Economic Progress, University of Minnesota, 1951; James N. Morgan, "The Structure of Aggregate Personal Saving," *Journal of Political Economy*, Dec. 1951; and William Hamburger, "Consumption and Wealth," unpublished Ph.D. dissertation, The University of Chicago, 1951.

9. Modigliani and Brumberg, *op. cit.*

10. The assumption $\bar{a}_{L+1} = 0$ might be stated more elegantly in terms of the following two:

$$(a)\ \frac{\partial U}{\partial a_{L+1}} \equiv 0; \quad (b)\ a_{L+1} \geqq 0.$$

The first of these equations specifies certain properties of the utility function U; the second states an institutional fact the individual must take into account in his planning, namely that our economic, legal, and ethical system is set up so as to make it rather difficult to get away without paying one's debts. The addition of these two equations to our previous system implies $\bar{a}_{L+1} = 0$.

11. The fact that $a_1 = 0$ by Assumption I of course does not imply $a_t = 0$.

12. The expression "on the average" here means that if we aggregate a large number of individuals chosen at random, their aggregate consumption will behave approximately as though each individual behaved in the postulated way.

13. More generally, it is sufficient that the utility index be any monotonic increasing function of a function U homogeneous in $c_t, \ldots, c_L$. This assumption can also be stated in terms of properties of the indifference map relating consumption in different periods. The postulated map has the property that tangents to successive indifference curves through the points where such curves are pierced by any one arbitrary radius vector are parallel to each other.

It may also be worth noting that a simple form of the utility index U satisfying our assumption is the following:

$$U = \log U = \alpha_0 + \sum_{\tau=t}^{L} \alpha_\tau \log c_\tau$$

since U is clearly homogeneous in $c_t, \ldots, c_L$. The above expression in turn has the same form as the well-known Weber Law of psychophysics, if we regard U as a measure of the intensity of the sensation and c as the intensity of the stimulus. One may well speculate whether we have here something of deeper significance than a mere formal analogy.

14. For a discussion of the effects of removing them, see Modigliani and Brumberg, *op. cit.* Also, see various notes below.

15. For an individual of age $t > N$, by assumption, $y = y^e = 0$ and only the last term on the right-hand side of (II.1) remains.

16. The valuable work in progress at the Survey Research Center of the University of Michigan gives hope that the variety of data required for such a test may sometime become available. Clearly, the problem of measuring average expected future income may prove a serious challenge. On this point, however, see text below and the Appendix.

17. Obviously, both y^e and a will generally be nonnegative.

18. See below, section 3, especially note 37.

19. We speak here advisedly of average and marginal rate of consumption, rather than of "cross-section marginal propensity to consume," for, as will become clear, the sue of the latter term is likely only to encourage a serious and already too frequent type of confusion. The word "propensity" denotes a psychological disposition and should refer to the way in which a *given* individual reacts to different stimuli, and not to the way in which *different* individuals react to different stimuli. The differential reaction of different individuals in relation to different stimuli may give us information about the individual propensity, but it is not, in itself, a propensity.

20. For a generalization of the notion of stationary household, see below note 23.

21. This proposition may be established by mathematical induction. Suppose that the proposition holds for t, *i.e.*, that

$$a_t = \bar{a}_t^1 = \frac{(t-1)M}{L} y_1; \tag{II.7}$$

then, since by assumption

$$y_t = y_t^e = y_1, \quad t \leqq N-1,$$

we have, from (II.2),

$$s_t = \frac{M}{L} y_1$$

and

$$a_{t+1} = a_t + s_t = \frac{(t-1)M}{L} y_1 + \frac{M}{L} y_1 = \frac{tM}{L} y_1 = \bar{a}_{t+1}^1.$$

Thus, if equation (II.7) holds for t it holds also for $t + 1$. But equation (II.7) holds for $t = 1$, since

$$a_1 = 0 = \frac{(L-1)M}{L} y_1 = \bar{a}_1^1.$$

Hence, equation (II.7) holds for all $t \leqq N$. By similar reasoning, it can be shown that it holds also for $N + 1 \leqq t \leqq L$.

22. While this result implies that, at zero income, consumption itself would have to be zero, it should be remembered that, under our stationary assumptions, the current level of income coincides with the level received in the past and expected in the future. A household whose income is permanently zero could hardly survive as a separate unit. Or, to put it differently, a household, within its earning span, whose current income is zero or negative cannot possibly be in stationary equilibrium.

23. It can be shown that if we eliminate Assumption III, our model still implies that, for a stationary cross section, the proportion of income saved is independent of income; however, this proportion will tend to rise with age, up to retirement. The conclusion that the proportion of income saved is independent of income, given age, also continues to hold if we relax our Assumption IV and assume only that there exists a typical pattern of allocation of resources to current and future consumption

which does not necessarily involve a constant planned rate of consumption over time. Finally this conclusion also remains valid if we recognize the existence of a typical life pattern of income and redefine a stationary household as one who expects its income not to be constant over time but rather to follow the normal life pattern, and whose expectations are continuously fulfilled in the sense stated in part (b) of the original definition in the text. Just what effects these two relaxations would have on the relation between the proportion of income saved and age depends, of course, on the specific shape of the pattern of allocation of resources to consumption over the life cycle and on the shape of the life pattern of income. Note, however, that since the line of relation between saving and income *for each* age group is supposed, in any event, to go through the origin, even if we fail to stratify by age, the regression of consumption on income should tend to be linear (though not homoscedastic) and a regression line fitted to the data should tend to go approximately through the origin.

24. As we have seen, under the assumption $y^e = y\left(\text{and therefore } \frac{dy^e}{dy} = 1\right)$ the individual marginal propensity to consume is

$$\frac{N-t+1}{L_t},$$

which reaches a maximum for $t = 1$, the maximum being

$$\frac{N}{L}.$$

25. Cf. M. Friedman and S. Kuznets, *Income from Independent Professional Practice*, pp. 325 ff.

26. Since the individual was in stationary equilibrium up to the end of period $t - 1$, we must have $a_t = a(y_{t-1}, t)$ and, by assumption, $y_{t-1} = y_t^e$.

27. From the last equality in equation (II.2′) and with $y_t^e = y_{t-1}$, we derive immediately

$$\frac{s_t}{y_t} = \frac{M}{L} + \frac{N(L-t)-M}{LL_t}\left(\frac{y_t - y_{t-1}}{y_t}\right).$$

28. From equation (II.2′), with $y_t^e = y_{t-1}$ and since

$$a_t = \frac{(t-1)M}{L} y_{t-1},$$

we derive,

$$\frac{s_t}{y_t} = \frac{M}{L} + \frac{(t-1)M}{LL_t}\left(\frac{y_t - y_{t-1}}{y_t}\right).$$

It is easily verified that the right-hand side of this expression is necessarily smaller than the corresponding expression given in the preceding note.

29. This conclusion fails to hold only if the elasticity of expectations is substantially greater than unity.

30. More precisely, unless there is a very strong correlation between the current change in income, $y - y_{-1}$, on the one hand, and the difference between the previous "accustomed" level and the previous year income, $y_{-1}^e - y_{-1}$, on the other.

31. See, for instance, G. Katona and J. Fisher, "Postwar Changes in the Income of Identical Consumer Units," *Studies in Income and Wealth*, Vol. XIII, pp. 62–122; G. Katona, "Effect of Income Changes on the Rate of Saving," *The Review of Economics and Statistics*, May, 1949, pp. 95–103; W. W. Cochrane, "Family Budgets—a Moving Picture," *loc. cit.*, Aug., 1947, pp. 189–198; R. P. Mack, "The Direction of Change in Income and the Consumption Function," *loc. cit.*, Nov., 1948, pp. 239–258.

32. Katona and Fisher, *ibid.*, especially Section D, pp. 97–101; and G. Katona, *ibid.*

33. It is, in principle, possible to design an experiment to test which of the two hypotheses represents a better explanation of the observed behavior. One possible test might be as follows. Select the set of households whose income has changed unexpectedly in the given year T, and who expect the change to be permanent. Consider next the subset whose income in the immediately following years remains at the new level and whose expectations are therefore fulfilled. If the traditional explanation is the correct one, then by the year $T + 1$ saving should revert to the average level prevailing for all households who have been for two or more years in the income brackets into which our households have moved. On the other hand, if our hypothesis is correct, the saving of our subset of households should, in the years following T, continue to remain higher, on the average, than the saving of the households who have been *permanent* inhabitants of the relevant brackets. Furthermore, under our model, the difference between the saving of the new and the original inhabitants should tend to remain greater the more advanced the age of the household in the year T. Needless to say, the data for such a test are not available at this time and the case for our explanation must rest for the moment on the evidence that supports our model as a whole. The purpose of describing a possible experiment is to emphasize the fact that the two alternative hypotheses have implications that are, in principle, observationally distinguishable. Some of these implications are, in fact, of immediate relevance for aggregative time series analysis.

34. See, for instance, the brilliant paper of William Vickrey, "Resource Distribution Patterns and the Classification of Families," *Studies in Income and Wealth*, Vol. X, pp. 260–329; Ruth P. Mack, *op. cit.*, and the contributions of Margaret G. Reid quoted in the next section.

35. The specific "technical" assumption underlying the entire discussion can be formulated thus:

$$y_i = y^e_{-1i} + \varepsilon_i \tag{II.10}$$

where the subscript i denotes the i-th household and the random term ε is assumed to have zero mean and to be independent of y_{-1}. From (II.10), making use of a well-known proposition of correlation theory, we deduce

$$\bar{y}^e_{-1}(y) = \alpha + \beta y; \quad \beta = r^{2e}_{yy-1}, \quad \alpha = \bar{y}^e_{-1} - \beta \bar{y} \tag{II.10$'$}$$

Clearly, β is necessarily less than one and α is necessarily positive (since, by assumption, $\bar{y}^e_{-1} = \bar{y}$). A more general and realistic stochastic assumption than the one just formulated would be the following:

$$y_i = k y^e_{-1i}\left(1 + \varepsilon_i^*\right),$$

ε' having the same properties as ε in (II.10). Since this assumption would complicate the analysis considerably (*e.g.*, it would destroy the linearity of (II.10′)) without affecting the essence of our argument, it has seemed preferable to base our discussion in this paper on equation (II.10).

36. This definition is not altogether satisfactory for an individual household since y may have been expected to be different from y^e_{-1} (which is the average income expected over the entire balance of the earning span); in this case the fact that y differs from y^e_{-1} need not generate any revision of expectations, *i.e.*, y^e may be equal to y^e_{-1}. As an alternative definition that would give more adequate recognition to the causal relation between the behavior of current income and changes in expectations, we may suggest the following:

$$E = \frac{y^e - y^e_{-1}}{y^e_{-1}} \Big/ \frac{y - y^{e(t)}_{t-1}}{y^{e(t)}_{t-1}},$$

where $y^{e(t)}_{t-1}$ denotes the income expected in the previous year for the current year. However, when we aggregate a large number of households, it is not unreasonable to expect that, on the average $y^{e(t)}_{t-1} = y^e_{-1}$, in which case our alternative definition leads back to equation (II.11) in the text.

37. In deriving (II.12) from (II.11), we are implicitly assuming that the average value of E is approximately the same at all levels of income, an assumption that does not seem unreasonable for the specific stochastic model with which we are presently dealing. A more general formulation of (II.12), which may be especially useful in establishing a connection between cross-section and aggregative time-series analysis, might be as follows:

$$\bar{y}^e(y) = \bar{y}^e_{-1}(y) + E_1[y - \bar{y}^e_{-1}(y)] + E_2(\bar{y} - \bar{y}^e_{-1}), \tag{II.12$'$}$$

which states that individual expectations depend not only on the behavior of individual income but also on that of average income for the entire community, $\bar{y}$. This hypothesis is supported by the consideration that changes in aggregate income may well represent a more reliable indicator for the future than just a change in individual income. It follows from (II.12′) that the elasticity of expectation for the community as a whole is

$$\frac{\bar{y}^e - \bar{y}^e_{-1}}{\bar{y} - \bar{y}^e_{-1}} = E_1 + E_2,$$

an expression that could be close to unity even if, as seems likely, E_1 is much smaller than one. On the other hand, it can be easily verified that if (II.12′) holds, with $E_2 \neq 0$, the elasticity of expectation as defined by (II.11) will generally change from income bracket to income bracket unless $\bar{y} = \bar{y}^e_{-1}$, an equality which has been explicitly assumed for the purposes of our discussion in the text and which makes (II.12′) identical with (II.12).

38. Such cases are of course not impossible for an individual household; all we claim is that they are unlikely for the average of all households falling in a given income bracket. (See, however, note 36.)

There is, of course, no opportunity to test the above statement from available data, since no attempt has yet been made, to our knowledge, to secure information on y^e and y^e_{-1}. Data such as those collected by the Survey Research Center on income expectations for the following year, and presented, for example, in G. Katona, *Psychological Analysis of Economic Behavior*, p. 120, have only a rather remote connection with our concepts. In terms of our notation, these data refer to y_{iT}, y_{iT-1}, and

$$y^{e(T+1)}_{1T},$$

where T denotes the calendar year of the survey. If one is willing to make the rather risky assumption that

$$y^{e(T+1)}_{iT} = y^e_{iT},$$

and the even more risky one that $y_{iT-1} = y^e_{iT-1}$, then it can be inferred from Katona's tabulations that the proportion of households with E greater than one has been somewhat under 20 percent, and fairly stable, in the surveys for which data are presented.

39. The average gap between initial assets, a, and the stationary equilibrium value,

$$a(y^e,t) = \frac{(t-1)M}{L} y^e,$$

at any given level of income, can be derived from the stochastic assumption introduced in note 34, with the addition of an analogous stochastic assumption as to the relation between a and y^e_{-1}, namely

$$y^e_{-1i} = \lambda x_i + w_i, \tag{II.13}$$

where $x_i \equiv a_i \Big/ \dfrac{(t-1)M}{L}, \quad E(w_i) = 0, \quad E(w_i x_i) = 0.$

x_i may be regarded as a function of the income expectations held by the household during all previous years of its active life.

From (II.13), we derive

$$\bar{x}(y^e_{-1}) = \mu + v y^e_{-1}; \quad v = r^2 x_{y^e_{-1}} \Big/ \lambda, \quad \mu = \bar{x} - v\bar{y}^e_{-1}. \tag{II.14}$$

Since in a growing economy $\lambda = \bar{y}^e_{-1}/\bar{x}_i$ may be expected to be at least as high as unity, we must have $0 \leqq v \leqq 1$, $\mu \geqq 0$. From (II.14) and the definition of x_i, we obtain

$$\bar{a}(y) = \frac{(t-1)M}{L}\bar{x}(y) = \frac{(t-1)M}{L}[\mu + v\bar{y}^e_{-1}(y)].$$

Finally, taking into consideration the definition of $a(y^e,t)$, and using equation (II.12), we derive

$$E\{[a - a(y^e,t)]|y\} = \bar{a}(y) - \frac{(t-1)M}{L}\bar{y}^e(y) = \frac{(t-1)M}{L}[\mu + (v - 1 + E)\bar{y}^e_{-1}(y) - Ey],$$

which can be expressed entirely in terms of y by making use of equation (II.10′). The above expression must fall as income rises, as stated in the text, since its derivative with respect to y is proportional to $-[E(1-\beta)+\beta(1-v)]$, which is necessarily negative (provided $E > 0$, which is assumed). In fact, we can establish that

$$E\{[a-a(y^e,t)]|y\} \gtreqless 0$$

$$\text{as} \quad y \lesseqgtr \bar{y}\left[1-\frac{\lambda-1}{\lambda[E(1-\beta+\beta(1-v))]}\right],$$

which shows explicitly the level of income at which the average gap between a and $a(y^e,t)$ is zero.

40. We have seen that a high elasticity of expectations tends to pull $\bar{y}^e(y)$ closer to the 45-degree line; at the same time, however, it will tend to flatten the $\bar{p}(y)$ curve relative to the $\bar{y}^e(y)$ curve, for, if assets are approximately adjusted to y^e_{-1}, then the larger the gap between y^e and y_{-1}, the larger will tend to be the unbalance in assets. Since, however, the second effect does not quite offset the first (see the discussion of section II.2, especially page 20), a high value of E will on the whole, increase the slope and reduce the intercept of $\bar{p}(y)$ and thus of $\bar{c}(y)$.

41. They may, in fact, have an average consumption level below the over-all average for two reasons: (a) because it seems likely that the level of consumption planned for after retirement will, on the whole, tend to be smaller than the average level during the earning span; (b) because of the secular rise in income per capita which characterizes most of the economies for which we have budget data. See, on this point, Modigliani and Brumberg, *op. cit.*

42. Our statements about retired households find strong support in the data on the size distribution of income by age, reported by Janet A. Fisher, in "Income, Spending, and Saving Patterns of Consumer Units in Different Age Groups," *Studies in Income and Wealth*, Vol. XV, especially tables 1 and 2, pp. 81 and 82. Actually, these tables are likely to overestimate the income (used in our sense) of elderly people, since income, there, is defined to include such items as pensions and retirement pay (p. 81, note 6). These inclusions also reduce the usefulness of the information on saving and dissaving by age groups presented in table 11 (p. 93). Even with the upward-biased definition of income, the proportion of positive savers is smaller in the age group 65 and over than in any other group except the group 18 to 24. For this latter group, of course, the figures are seriously affected by the inclusion of durable goods purchases in consumption. Presumably, had our definitions of income and saving been adopted, the relative scarcity of savers and predominance of dissavers in the elderly group would be much more pronounced.

43. These results were first reported in a brief communication presented at the Conference on Saving, Inflation and Economic Progress, University of Minnesota, in May, 1952. The authors have also had the opportunity to consult a preliminary draft on "The Relation of the Within-Group Permanent Component of Income to the Income Elasticity of Expenditures," and have had the benefit of extensive discussion and consultations with Miss Reid. The hypothesis tested in the above-mentioned paper had already been partly anticipated in Reid's earlier contribution, "Effect of Income Concept upon Expenditure Curves of Farm Families," *Studies in Income and Wealth*, Vol. XV, especially pp. 133–139.

44. If the equalities are replaced by proportionality, the correlation $r_{yy_{-1}}$ remains unity. Furthermore, replacing in (II.1) y^e by $k_i y$, and a by

$$\frac{(t-1)M}{L} v y^e_{-1} = \frac{(t-1)Mvk_2}{L} y,$$

(since by assumption, up to the beginning of the current year the assets of every household are proportional to the stationary equilibrium value, v being the proportionality factor and having the same meaning as in note 41), we obtain

$$c = \frac{L + L(N-t)k_1 + M(t-1)vk_2}{LL_t} y,$$

so that, for any given age t, the consumption-income relation is still a straight line through the origin and therefore η_{cy} is still unity. Since, however, the slope varies with age, if we fail to stratify by age, we may find that the computed value of η_{cy} is not exactly one, even for a stationary cross section. On the other hand, it can be verified, from the above expression, that the variation in slope with age is likely to be rather small so that, in fact, η_{cy} should be quite close to unity, unless k_2 and ν are substantially different from unity, an unlikely case except in deep depression or at the peak of a boom.

45. The figures just quoted are in part approximate since Reid has not used the statistic r_{yy-1} but, instead, one closely related to it. Although we do not have specific information on the age composition of the samples, we understand that retired households, if any, represent a negligible proportion of the samples.

46. See, for instance, Josephine H Staab, "Income-Expenditure Relations of Farm Families Using Three Bases of Classification," Ph.D. dissertation, The University of Chicago, 1952; Reid, *op. cit.* (several new experiments are also reported in Reid's preliminary draft quoted above); Vickrey, *op. cit.* In essence Vickrey's point is that consumption is more reliable than current income as a measure of the permanent component of income (p. 273) and he suggests, accordingly, that the individual marginal propensity to consume (with respect to the permanent component) can be estimated more reliably by relating consumption (per equivalent adult), c, to $\bar{y}(c)$ than by relating $\bar{c}(y)$ to y, as has been usually done. It can be shown that Vickrey's suggestions receive a good deal of support from our model (with the addition of the stochastic assumptions introduced in various notes above) in that the relation between c and $\bar{y}(c)$ should be very similar to that between c and our quantity p. In particular, c should be nearly proportional to $\bar{y}(c)$, a conclusion that Vickrey himself did not reach but which is well supported by his own tabulations. A double logarithmic plot of c against $\bar{y}(c)$, based on his data, reveals an extremely close linear relationship with a slope remarkably close to unity. (We have estimated this slope, by graphical methods, at .97.) On the other hand, using the conventional plot, $\bar{c}(y)$ against y, the slope for the same data can be estimated at somewhat below .85, and, in addition, the scatter around the line of relationship is distinctly wider than for the first mentioned plot.

In the contribution under discussion Vickrey has also been very much concerned with the influence of the size composition of the household on saving behavior. This is a point which, because of limitations of space, we have been forced to neglect in the present paper. We will merely indicate, at this point, that our central hypothesis (that the essential purpose of saving is the smoothing of the major and minor variations that occur in the income stream in the course of the life cycle) provides a framework within which the influence of family size can be readily analyzed. We hope to develop this point in later contributions.

47. *Studies in Income and Wealth*, Vol. X, pp. 247–265.

48. Making use of equations (II.10′), (II.12) and the expression for $\bar{a}(y)$ derived in note 39, our cross-section income-consumption relation can be reduced to the form

$$\bar{c}(y) = A + By = A^*\bar{y} + By,$$

where A^* and B depend on the coefficients E, α, β, λ, μ, ν we have introduced earlier, and on age. These coefficients, in turn, depend primarily on the variability of income as measured by r^e_{yy-1} and $r_x y^e_{-1}$, and, probably to a lesser extent, on the long-term trend of income (which affects λ, μ and ν) and on the cyclical position of the economy (which affects α and β and possibly E). Hence, if we have various samples of households for each of which the variability of income is approximately the same, the coefficients A^* and B should also be approximately the same for each sample, especially if the samples in question do not differ too markedly with respect to age, composition and the cyclical position of total income. Denoting by $\bar{c}_i$ and $\bar{y}_i$ the average value of consumption and income for all households falling in the i-th quantile of a given sample, we must have

$$\bar{c}_i = A^*\bar{y} + B\bar{y}_i.$$

If, furthermore, for each of the samples compared $\bar{y}_i/\bar{y}$ is approximately constant, so that we can write $\bar{y}_i = k_i\bar{y}$, we obtain

$$\bar{c}_i/\bar{y}_i = (\bar{c}/\bar{y})_i = \frac{A^*}{k_i} + B,$$

i.e., the proportion of income consumed in a given quantile, i, should be approximately the same for all samples compared, as stated in the text. We may add that, if we replace our simple stochastic assumption (II.10) by the more realistic one suggested at the end of note 37, the conclusion stated in the text would still stand, although the relation between $(\bar{c}/\bar{y})_i$ and i would be more complex than indicated by the right-hand side of the above equation.

49. For example, see James Duesenberry, *Income, Saving and the Theory of Consumer Behavior*.

50. Making use of equations (II.10′), (II.12) and of the expression for $\bar{a}(y)$ of note 39, it can be verified that

$$\frac{\partial^2 \bar{c}(y)}{\partial y \partial t} < 0,$$

i.e., the slope of the cross-section consumption-income relation falls with age, provided

$$E > \frac{1}{M+1} - \frac{M\beta(1-v)}{(M+1)(1-\beta)}.$$

A very similar condition on E must be satisfied in order for the constant term to rise with age. Since $1/(M + 1)$ should be in the order of 0.1, and β and v are typically smaller than unity, the right-hand side of the above inequality cannot significantly exceed zero and is, in fact, likely to be negative.

51. *Op. cit.*, p. 90 and p. 99.

52. There is ample evidence of a very pronounced correlation between assets and income, at least as far as liquid assets are concerned. See, for example, Lawrence Klein, "Assets, Debts, and Economic Behavior," *Studies in Income and Wealth*, Vol. XIV, Table 1, p. 209; and Fisher, *loc. cit.*, table 6, p. 86.

53. Gustave Cassel, *The Theory of Social Economy*, 1932 ed., p. 244; Horst Mender-hausen and Raymond W. Goldsmith, "Measuring Estate Tax Wealth," *Studies in Income and Wealth*, Vol. XIV, table 4, p. 140; and Fisher, *loc. cit.* The most systematic information we have found on this subject is that presented by Janet Fisher, although, unfortunately, her most interesting tabulations relate to liquid assets only. For this reason we will not attempt an extensive comparison of her findings with our theory. We may point out, however, that in terms of liquid assets, her data agree remarkably well with our theory in several respects. For example, a comparison of the mean income ratio (table 1, p. 81) with the mean ratio of liquid asset holdings (table 4, p. 84) by age groups, reveals that the ratio of assets per spending unit to income per spending unit rises steadily with age. For the age group 45–64, which is her oldest group wherein active households presumably predominate, this ratio is nearly three times as large as for her youngest group, 18–24. For the age group 65-and-over, wherein, presumably, retired people predominate, the ratio nearly doubles again. This very high ratio is precisely what should be expected from our model where assets are drawn down slowly but income falls much faster with retirement. In terms of assets, rather than of the income/asset ratio, the peak should be reached just before retirement. Confirmation of this inference is found in the tabulations of table 5 (p. 85).

Also, suggestive of this relation is the complaint by Lawrence Klein, "Assets, Debts, and Economic Behavior," that: "There is some indication that the influence of age on savings may be obscured by a significant positive correlation between (L/Y) are a" (where Y = income, L = liquid assets, and a = age of the spending-unit head in years)."

54. All this has, of course, some significant implications about the "Pigou effect" which are developed in Modigliani and Brumberg, *op. cit.*

55. Cf. Klein, *ibid.*, pp. 220 ff.

56. More generally, the essential implication of our theory that should be tested is that the (average) relation between consumption and net worth, given income and income expectations, is linear and with a slope which tends to grow with the age of the household. The specific value of the slope given by (II.15) depends on Assumptions III and IV which are convenient for exposition but are not essential to the theory.

57. In the very early years these motives might possibly lead to a somewhat faster rate of accumulation than might occur otherwise. On the other hand, in his early years, an individual may feel less

of a need for precautionary assets since, being a better risk, he will he in a better position to borrow, and may also be able to rely, for emergencies, on his relatives.

58. Cf. Boulding, *op. cit.*, Ch. 3 and Ch. 5 for a novel approach to the choice of asset combinations. See also J. Marshack, "Money and the Theory of Assets," *Econometrica*, Oct. 1938, pp. 311–325.

Because durable goods generally seem more vulnerable to social pressure than the forms of saving that make up the ordinary (*e.g.*, Department of Commerce) definition, it is easy to see why they have been thought of as consumption. Keynes enforced the definition only by his interest in the modes of saving not necessarily matched by investment. However, the idea of ostentatious durables typifying "conspicuous consumption" preceded the *General Theory* by many years.

59. See: Dorothy S. Brady, "An Analysis of Saving on the Basis of Consumer Expenditure Data," *Saving and Capital Market Study*, Section 3, R. W. Goldsmith, Director, preliminary; and Fisher, *op. cit.*, tables 4, 5, and 6, pp. 85–89.

60. Our conclusions are also in complete agreement with B. Ohlin's brief but illuminating remarks and criticism of Keynes developed in "Some Notes on the Stockholm Theory of Savings and Investment," reprinted in *Readings in Business Cycle Theory*; see especially pp. 98–100.

61. Modigliani and Brumberg, *op. cit*.

62. The division by y_i creates certain statistical problems in connection with the random term; this difficulty can be handled by an appropriate modification of equation (A.2) or (A.3) which need not be discussed here since it would greatly complicate the presentation without basically affecting the conclusions.

63. p. 220.

64. Figures in parentheses represent the standard errors. The two figures in this column represent the parameters of the equation for "Home Owners" and "Renters," respectively (*op. cit.*, p. 221). Since the completion of this Appendix, Mr. Klein has kindly informed us that the average age for the two samples combined is 46 years; this implies an average age, since entering the labor force, in the order of 20 to 25 years, which is within the assumed range.

65. We suspect that expressing income change in terms of y_{-1} instead of y may also contribute to the discrepancy.

66. If this explanation is correct, it would also follow that Klein's coefficient greatly overestimates the effect on consumption of an increase in assets due, say, to an unanticipated fall in the price level of consumables. In any event, as already pointed out, the relation between the cross section and the time-series marginal propensity is a complex one. In the companion paper quoted earlier, we have shown that the time-series marginal propensity to consume with respect to net worth should be in the order of 0.1.

67. There is also reason to believe that failure to take into account properly the age variable may lead to an appreciable upward bias in the asset coefficient if the cross section includes retired people. Since we do not know whether Klein's sample does have a significant representation of retired households we cannot say whether this explanation is relevant.

2

THE "LIFE-CYCLE" HYPOTHESIS OF SAVING: AGGREGATE IMPLICATIONS AND TESTS

Albert Ando and Franco Modigliani

The recent literature on the theory of the consumption function abounds with discussions of the permanent income hypothesis of Friedman and other related theories and attempts at their empirical verification. Friedman's formulation of the hypothesis is fairly well suited for testing against cross-section data, though numerous difficulties are associated with this task, and there is now a rapidly growing body of literature on this subject [5] [8] [11] [12] [14] [16] [17] [21] [25]. Friedman's model, on the other hand, does not generate the type of hypotheses that can be easily tested against time series data.

More or less contemporaneously with Friedman's work on the permanent income hypothesis, Modigliani and Brumberg developed a theory of consumer expenditure based on considerations relating to the life cycle of income and of consumption "needs" of households [34] [35]. Several tests of the Modigliani-Brumberg theory using cross-section data have been reported in the past including a comparative analysis of the cross-section implications of this hypothesis as against the Friedman model [8] [12] [32] [33].

Modigliani and Brumberg have also attempted to derive time series implications of their hypothesis in an as yet unpublished paper [34], and their theory appears to generate a more promising aggregative consumption function than does Friedman's. However, at the time of their writing the unavailability of data on net worth of consumers made empirical verification exceedingly difficult and indirect [6] [7] [20]. Since then, this difficulty has been partially eliminated as a result of the work of Goldsmith [18] [19].

In part I of this paper, we give a brief summary of the major aggregative implications of the Modigliani-Brumberg life cycle hypothesis of saving. In part II, we present the results of a number of empirical tests for the United States which appear to support the hypothesis.[1] The reader who is not interested in the derivation and statistical testing of the aggregate Modigliani-Brumberg consumption

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function may proceed directly to part III, where we develop some features of the model which, in our view, make it particularly suitable for the analysis of economic growth and fluctuations, as indicated in our past and forthcoming contributions [1] [2] [3] [4] [29] [31].

I Theory

A Derivation of the Aggregate Consumption Function[2]

The Modigliani and Brumberg model starts from the utility function of the individual consumer: his utility is assumed to be a function of his own aggregate consumption in current and future periods. The individual is then assumed to maximize his utility subject to the resources available to him, his resources being the sum of current and discounted future earnings over his lifetime and his current net worth. As a result of this maximization the current consumption of the individual can be expressed as a function of his resources and the rate of return on capital with parameters depending on age. The individual consumption functions thus obtained are then aggregated to arrive at the aggregate consumption function for the community.

From the above brief description, it is quite apparent that the most crucial assumptions in deriving the aggregate consumption function must be those relating to the characteristics of the individual's utility function, and the age structure of the population. The basic assumptions underlying the shape of the utility function are:

Assumption I The utility function is homogeneous with respect to consumption at different points in time; or, equivalently, if the individual receives an additional dollar's worth of resources, he will allocate it to consumption at different times in the same proportion in which he had allocated his total resources prior to the addition.[3]

Assumption II The individual neither expects to receive nor desires to leave any inheritance. (This assumption can be relaxed in either of two ways. First, we may assume that the utility over life depends on planned bequests but assume that it is a homogeneous function of this variable as well as of planned consumption. Alternatively, we may assume that the resources an individual earmarks for bequests are an increasing function of the individual's resources relative to the average level of resources of his age group, and that the relative size distribution of resources within each age group is stable over time. It can be shown that either of these generalized assumptions implies an aggregate consumption function similar in all essential characteristics to the one obtained from the stricter assumption stated here.)

These two assumptions can be shown to imply (cf. [35, pp. 390 ff.]) that, in any given year *t*, total consumption of a person of age *T* (or, more generally, of a household headed by such a person) will be proportional to the present value of total resources accruing to him over the rest of his life, or:

$$c_t^T = \Omega_t^T v_t^T . \tag{2.1}$$

In this equation[4] Ω_t^T is a proportionality factor which will depend on the specific form of the utility function, the rate of return on assets, and the present age of the person, *but not on total resources*,

v_t^T.

The symbol

c_t^T

stands for total consumption (rather than for consumer's expenditure) in the year *t*. It consists of current outlays for nondurable goods and services (net of changes if any in the stock of nondurables) plus the rental value of the stock of service-yielding consumer durable goods. This rental value in turn can be equated with the loss in value of the stock in the course of the period plus the lost return on the capital tied up. Finally the present value of resources at age

T, v_t^T,

can be expressed as the sum of net worth carried over from the previous period,

a_{t-1}^T,

and the present value of nonproperty income the person expects to earn over the remainder of his earning life; i.e.,

$$v_t^T = a_{t-1}^T + y_t^T + \sum_{\tau=T+1}^{N} \frac{y_t^{e^{T\tau}}}{(1+r_t)^{\tau-T}} \tag{2.2}$$

where

y_t^T

denotes current nonproperty income;

$y_t^{eT\tau}$

is the nonproperty income an individual of age T expects to earn in the τth year of his life; N stands for the earning span and r_t for the rate of return on assets.[5]

In order to proceed further, it is convenient to introduce the notion of "average annual expected income,"

$$y_t^{eT},$$

defined as follows:

$$y_t^{eT} = \frac{1}{N-T} \sum_{\tau=T+1}^{N} \frac{y_t^{eT\tau}}{(1+r_t)^{\tau-T}}. \tag{2.3}$$

Making use of this definition and of (2.2) we can rewrite equation (2.1) as:

$$c_t^T = \Omega_t^T y_t^T + \Omega_t^T (N-T) y_t^{eT} + \Omega_t^T a_{t-1}^T. \tag{2.4}$$

To obtain an expression for aggregate consumption we proceed to aggregate equation (2.4) in two steps, first within each age group and then over the age groups.

If the value of Ω_t^T is identical for all individuals in a given age group T, then it is a simple matter to aggregate equation (2.4) over an age group, obtaining:

$$C_t^T = \Omega_t^T Y_t^T + (N-T)\Omega_t^T Y_t^{eT} + \Omega_t^T A_{t-1}^T \tag{2.5}$$

where C_t^T, Y_t^T, Y^{eT}, and A_{t-1}^T are corresponding aggregates for the age group T of c_t^T, y_t^T, y_t^{eT}, and a_{t-1}^T. If Ω_t^T is not identical for all individuals in the age group, however, the meaning of the coefficients in equation (2.5) must be reinterpreted. It has been shown by Theil [41] that under a certain set of conditions the coefficients of (2.5) can be considered as weighted averages of the corresponding coefficients of (2.4).[6]

Next, taking equation (2.5) as a true representation of the relationship between consumption and total resources for various age groups, we wish to aggregate them over all age groups to get the consumption function for the whole community. Consider the equation:

$$C_t = \alpha_1' Y_t + \alpha_2' Y_t^e + \alpha_3' A_{t-1} \tag{2.6}$$

where

$$C_t, Y_t, Y_t^e \text{ and } A_{t-1}$$

are obtained by summing respectively

$$C_t^T, Y_t^T, Y_t^{eT} \text{ and } A_{t-1}^T$$

over all age groups T, and represent therefore aggregate consumption, current nonproperty income, "expected annual nonproperty income," and net worth.

The theorems given by Theil again specify the conditions under which the coefficients in equation (2.6) are weighted averages of the corresponding coefficients of equation (2.5). In this case, it is likely that the conditions specified by Theil are not satisfied, because both net worth and its coefficient in equation (2.5) are positively correlated with age up to the time of retirement. However, a much weaker set of conditions can be specified which are sufficient to insure stability over time of parameters in equation (2.6). In particular one such set of conditions is the constancy in time of (i) the parameters of equation (2.5) for every age group, (ii) the age structure of population, and (iii) the relative distribution of income, of expected income, and of net worth over the age groups.

B A Priori Estimates of the Coefficients of the Aggregate Consumption Function

Modigliani and Brumberg [34], in order to obtain *a priori* estimates of the order of the magnitude of the coefficients of equation (2.6) implied by their model, introduced a number of rather drastic simplifying assumptions about the form of the utility function and life pattern of earnings, to wit:

Assumption III The consumer at any age plans to consume his total resources evenly over the remainder of his life span.

Assumption IV (a) Every age group within the earning span has the same average income in any given year t. (b) In a given year t, the average income expected by any age group T for any later period τ, within their earning span, is the same. (c) Every household has the same (expected and actual) total life and earning spans, assumed to be 50 and 40 respectively for the purpose of numerica1 computation.

Assumption V The rate of return on assets is constant and is expected to remain constant.

Under these assumptions, if aggregate real income follows an exponential growth trend—whether due to population or to productivity growth—the sufficient conditions for the constancy in time of the parameters of (2.6) are satisfied. The value of these parameters depends then only on the rate of return on assets and on the over-all rate of growth of income, which in turn is the sum of population growth and the rate of increase of productivity.[7]

Table 2.1 gives some examples of the numerical value of the coefficients under the assumptions described above.

It should be emphasized that assumptions III to V have been introduced only for the sake of numerical estimation of the coefficients and are by no means necessary to insure the approximate constancy in time of the parameters in (2.6). A

Table 2.1
Coefficients of the Consumption Function (1.6) under Stated Assumptions[a]

Yield on Assets (percent)	0	0	0	3	5	5	5
Annual Rate of Growth of Aggregate Income (percent)	0	3	4	0	0	3	4
$\alpha_1 + \alpha_2$	0.61	0.64	—	0.69	0.73	—	—
α_3	0.08	0.07	0.07	0.11	0.13	0.12	0.12

[a] Missing values have not been computed because of the complexity of calculation.

change in the assumptions would lead to somewhat different values of the parameters. But both *a priori* considerations and rough numerical calculations suggest that these values would not be drastically affected, and that it is generally possible to infer the direction in which these values would move when a specific assumption is changed. The recognition of the estate motive would tend to yield lower values for both coefficients, especially that of assets.[8]

On the whole, then, the values shown in table 2.1 should be regarded as a rough guide to the order of magnitude of the coefficients consistent with the basic model; i.e., radically different values would cast serious doubts on the adequancy of the life cycle hypothesis.

C The Measurement of Expected Income

The last point that must be clarified before we proceed to the discussion of the empirical tests is the measurement of expected nonproperty income, Y^e, which, at least at present, is not directly observable. A "naive" hypothesis is to assume that expected nonproperty income is the same as actual current income, except for a possible scale factor. Thus, we have:

$$y_t^e = \beta' Y_t; \quad \beta' \simeq 1.$$

Substituting the above expression into (2.6), we obtain the aggregate consumption function

$$C_t = (\alpha_1' + \beta'\alpha_2')Y_t + \alpha_3' A_{t-1} = \alpha_1 Y_t + \alpha_3 A_{t-1}$$

$$\alpha_1 = \alpha_1' + \beta'\alpha_2' \simeq \alpha_1' + \alpha_2'.$$

We designate this formulation as hypothesis I.

A similar but somewhat more sophisticated formulation is to assume that expected income is an exponentially weighted average of past income, weights adding up to one, or slightly more than one in order to reflect the expected growth [15] [16]. But it is quite difficult to determine the weights from the data we have at our disposal, and Friedman, who favors this formulation, has acknowledged its shortcomings [15].

The third possible formulation is a slight modification of the first. Under our definitions, Y, and expected income, Y^e, are nonproperty or labor income, excluding, for instance, profits. We may hypothesize that for these currently employed, average expected income,

$$y_t^e,$$

is current income adjusted for a possible scale factor, i.e.,

$$y_t^e = \beta_1 \frac{Y_t}{E_t} \tag{2.7}$$

where E_t is the number of persons engaged in production. We should expect β_1 to be quite close to unity.

For those individuals who are currently unemployed, we hypothesize that expected income is proportional to the average current income of those who are employed. The proportionality constant in this case represents three factors. First, as before, there may be some influence from expected growth. Second, and probably most important, the incidence of unemployment is likely to be smaller for higher-paid occupations than for lower-paid, less-skilled workers; hence, the average earnings the unemployed can look forward to, if reemployed, are likely to be lower than the average earnings of those currently employed. Third, it seems reasonable to suppose that some of the currently unemployed persons would expect their current unemployment status to continue for some time and, possibly, to recur. We shall therefore assume:

$$y_t^{eu} = \beta_2 \frac{Y_t}{E_t} \tag{2.8}$$

where y_t^{eu} is the average expected income of unemployed persons; and, for the reasons given above, we expect the constant β_2 to be substantially smaller than β_1. The aggregate expected income is then given by:

$$\begin{aligned} Y_t^e &= E_t y_t^e + (L_t - E_t) y_t^{eu} = E_t \beta_1 \frac{Y_t}{E_t} + (L_t - E_t)\beta_2 \frac{Y_t}{E_t} \\ &= (\beta_1 - \beta_2) Y_t + \beta_2 \frac{L_t}{E_t} Y_t \end{aligned} \tag{2.9}$$

where L_t denotes the total labor force.[9]

Substituting (2.9) into (2.6), we obtain the following variant of hypothesis I,

$$C_t = \alpha_1 Y_t + \alpha_2 \frac{L_t}{E_t} Y_t + \alpha_3 A_{t-1} \tag{2.10}$$

where

$$\alpha_1 = \alpha_1' + \alpha_2'(\beta_1 - \beta_2)$$

$$\alpha_2 = \alpha_2' \beta_2; \quad \alpha_3 = \alpha_3'.$$

We designate the formulation embodied in equation (2.10) above as hypothesis II.

Since β_1 is thought to be close to unity, we have

$$\alpha_1 + \alpha_2 = \alpha_1' + \beta_1 \alpha_2' \simeq \alpha_1' + \alpha_2'. \tag{2.11}$$

The individual values of the observable coefficients α_1 and α_2 are, however, dependent on the nonobservable value of β_2, about which there is little we can say *a priori*.

II Empirical Verification and Estimation

In this section we report results of a number of tests of our model for the United States.[10] Unless otherwise stated, the period of observation is 1929 through 1959 excluding the Second World War years 1941–46.[11] Consumption, C, labor income net of taxes, Y, and net worth, A, are all measured in billions of current dollars as called for by our hypothesis.[12]

In recent years, economists have become increasingly aware of the many sources of bias, inconsistency, and inefficiency that beset prevailing estimation procedures, e.g., the existence of simultaneous relations, errors of observations in the "independent" variables, spurious correlation, multicollinearity, and heteroscedascity.[13] As a result, the simple-minded and straightforward least-squares approach is being replaced by a host of alternative procedures. Unfortunately most of these alternative procedures are designed to cope with one specific source of difficulty, and they often do so at the cost of increasing the difficulties arising from other sources. Under these conditions, we feel that the best course is to utilize a variety of procedures, exploiting our knowledge of the structure of the model and the nature of data to devise methods whose biases are likely to go in opposite directions. By following such a procedure, we can at least have some confidence that the estimates obtained by different methods will bracket the value of the unknown parameters being estimated.

The main alternative procedures used and the estimates obtained are summarized in table 2.2. Row (1) shows the results of a straightfoward least-squares fit of hypothesis I.[14] The coefficients of both independent variables are highly significant and R^2 extremely high. But in other respects, the results are not altogether satisfactory. The coefficient of Y, which is an estimate of $\alpha_1 + \alpha_2$, is somewhat higher and that of A appreciably lower than our model would lead us to expect. Furthermore, the Durbin-Watson statistic [10] falls considerably short of 2, suggesting the presence of pronounced serial correlation in the residuals.

As can be seen from row (2), the results do not change appreciably if we replace hypothesis I with II by introducing an additional variable

$$Y\frac{L}{E}.$$

Although the coefficient of

$$Y\frac{L}{E}$$

has the right sign it does not appear to contribute significantly to the explanation of C. Meanwhile, it reduces still further the estimate of the coefficient of net worth, and increases the estimate of $\alpha_1 + \alpha_2$ which, it will be recalled, is approximately given by the sum of the coefficients of Y and

$$Y\frac{L}{E}.$$

Also, the serial correlation of the residuals does not change at all. As will soon become apparent, much of the difficulty with hypothesis II can be traced back to multicollinearity, which makes it rather hard to obtain reliable estimates of the individual coefficients.

Note also that in both (1) and (2) the constant term is very significantly different from zero by customary standards, a result which would seem inconsistent with the hypothesis tested. In our view, however, this result is not as serious as might appear at first glance. The constant term is numerically rather small, amounting to only about three percent of the mean value of the dependent variable. Furthermore, we know that the least-squares estimate of the constant term is upward-biased in the present instance because of the simultaneous-equations bias as well as because of errors of measurement in the independent variables.[15] While the size of these biases cannot be directly estimated, we suspect it to be appreciable. Accordingly, on the basis of presently available evidence, we see no compelling reason to reject the hypothesis that consumption is in fact roughly homogeneous in income and assets. Under these circumstances, a more reliable estimate of the coefficients of these variables might be obtained by suppressing the constant term in accordance with the specification of our model.

Table 2.2
Estimates of the Coefficients of the Consumption Function

			Coefficients and Their Standard Errors of Estimates[b]										
Rows	Hypothesis Tested	Mode of Regression[a]	Constant	Y (α_1)	XY (α_{1x})	$Y\left(\frac{L}{E}\right)$ (α_2)	A (α_3)	XA (α_{3x})	$\alpha_1 + \alpha_2$	Standard Deviation of Dependent Variable	Standard Error of Estimate	R^{2c}	Durbin Watson Statistics
(1)	I	A	5.33 (1.46)	0.767 (0.047)			0.047 (0.010)		0.767	88.289	2.352	0.999	1.29
(2)	II	A	4.69 (1.51)	0.633 (0.112)		0.163 (0.124)	0.040 (0.012)		0.796	88.289	2.314	0.999	1.17
(3)	I	A		0.640 (0.039)			0.077 (0.008)		0.640	88.289	2.860	0.999	0.89
(4)	I	A		0.787 (0.086)			0.058 (0.013)	–0.010 (0.005)	0.787	88.289	2.756	0.999	1.17
(5)	II	A		0.430 (0.108)		0.287 (0.139)	0.058 (0.012)		0.717	88.289	2.690	0.999	0.85
(6)	I	B		0.550 (0.116)			0.079 (0.021)		0.55	8.292	2.335	0.921	2.01
(7)	I	B		0.577 (0.170)	–0.030 (0.138)		0.079 (0.022)		0.577[d]	8.292	2.385	0.918	2.03
(8)	I	B		0.550 (0.119)			0.082 (0.031)	–0.004 (0.025)	0.550	8.292	2.385	0.918	2.03
(9)	II	B		0.444 (0.124)		0.274 (0.147)	0.051 (0.025)		0.718	8.292	2.215	0.929	1.82
(10)	II	B		0.411 (0.127)		0.353 (0.163)	0.068 (0.029)	–0.028 (0.026)	0.764	8.292	2.211	0.929	1.97

(11)	I	C	0.634			0.080		0.634	0.092	0.018	0.962	1.00
			(0.020)			(0.003)						
(12)	I	C	0.644	–0.004		0.078		0.644[d]	0.092	0.019	0.958	1.02
			(0.056)	(0.018)		(0.008)						
(13)	I	C	0.654			0.077	–0.001	0.654	0.092	0.019	0.958	1.05
			(0.054)			(0.008)	(0.004)					
(14)	II	C	0.639	–0.003	0.007	0.077		0.646[d]	0.092	0.019	0.958	1.03
			(0.081)	(0.019)	(0.070)	(0.009)						
(15)	II	C	0.649		0.006	0.076	–0.001	0.655	0.092	0.019	0.958	1.05
			(0.079)		(0.070)	(0.009)	(0.004)					

[a] A: Regressions in which variables are used in the original form.
B: Regressions in which variables are used in the first-difference form.
C: Regressions in which variables are used in the form of ratios to labor income.
[b] Figures in parentheses underneath the estimates are estimated standard errors of respective estimates. Number of observations = 25. Where no estimate is shown, the variable is excluded from the equation.
[c] One minus the ratio of the variance of the residual to the variance of the dependent variable.
[d] Figures given do not include the coefficient of *XY*. In other words, the estimate of the coefficient pertains to the period 1929–40.

The constrained estimation results in the equations reported in rows (3) and (5) of table 2.2. A comparison of row (1) and row (3) shows that this procedure leads to estimates which are more nearly of the order of magnitude suggested by our model. Unfortunately the serial correlation is still so high that the reliability of the estimate is open to serious question. From row (5) it also appears that the addition of the variable

$$Y\frac{L}{E}$$

is again not very helpful. Though its contribution is somewhat more significant, it again lowers the coefficient of A, and the serial correlation remains high.

A common procedure in time-series analysis when serial correlation of errors is high is to work with first differences. In the present instance this procedure also serves to reduce drastically the degree of multicollinearity and provides a more meaningful test for the adequacy of the hypothesis as a causal explanation of consumption. The results, reported in rows (6) and (9) and in figure 2.1, appear quite favorable to the hypothesis. The multiple correlation remains quite high and the coefficient of net worth is highly significant. Also the Durbin-Watson statistic improves considerably and there is no longer any reason to suspect that the reliability of the estimate is seriously affected by serial correlation of residuals. A comparison of row (6) with row (3) reveals that the coefficient of A remains essentially unchanged, while the coefficient of Y is reduced from 0.64 to 0.55. On the other hand, comparison of rows (5) and (9) shows that, for hypothesis II, estimates of all parameters remain essentially unchanged under the two estimation methods.

These results seem to be readily explainable. When we deal with actual values the movements of all variables are dominated by their trend. On the other hand, when dealing with first differences as in (6) and (9), we are primarily focusing on short-run cyclical variations. In this light, the results of row (6) suggest that consumption is less responsive to purely cyclical and temporary fluctuations in labor income than the estimate of row (3) would imply. The close agreement between (5) and (9) tends to support this interpretation, since the presence of the variable $Y(L/E)$—which is cyclically more stable than Y—accounts for the relative stability of expected labor income over current labor income. It should be remembered in this connection that according to our model consumption depends largely on expected rather than current labor income. At the same time the fact that both $Y(L/E)$ and A perform a similar function in stabilizing consumption with respect to short-run variations in Y helps to explain why the addition of the latter variable generally tends to reduce not only the coefficient of Y but also that of A.[16]

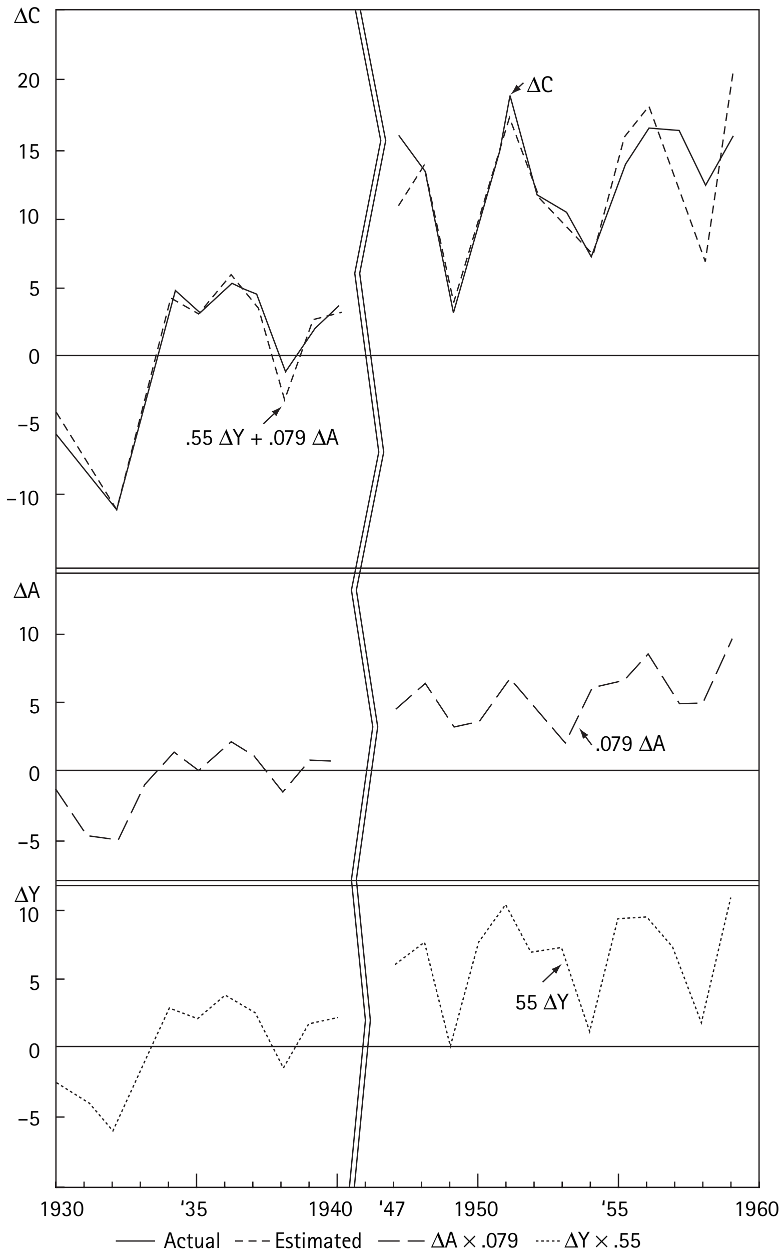

Figure 2.1
Time profile of (6) and its components

All the estimates reported so far are based on the least-squares method applied to a single equation. As is well known this method leads to estimates which are biased, even in the limit, when one or more of the "independent" variables are related to the dependent variable by other simultaneous relations. In the present instance the variable A can be taken as predetermined, but the same is not true of labor income which is related to consumption via total income. That is, the true error component of the consumption function cannot be assumed to be uncorrelated with Y, and hence the least-squares estimates of its parameters are not consistent. Specifically, it can be shown that, asymptotically, the estimator of the coefficient of Y is upward-biased and that of the coefficient of A downward-biased.[17]

The only really adequate way of resolving this difficulty would be to construct a complete model of the U.S. economy and then apply an appropriate simultaneous-equations estimation procedure. This approach would lead at least to consistent estimates, except to the extent that the model was incomplete or misspecified and the exogenous variables were subject to errors of measurement. Furthermore, the efficiency of the estimates might be reduced, particularly if the theory and data relating to other sectors of the economy were less reliable than those relating to the consumption sector. Whatever the merits of this approach, however, we regard the specification of a complete model beyond the scope of this paper.

A compromise followed by some authors is to introduce an accounting identity relating consumption, saving, and income, note that saving is equal to investment, assume that investment is autonomous, and estimate the parameters of the consumption function from the regression of consumption on saving [39] [43]. Now in our view this procedure is likely to lead to bias in the estimates of the parameters which is more serious than that resulting from the conventional regression on income. The arguments supporting this conclusion are developed formally in Appendix section B and can be summarized here as follows:

1. When the "independent" variable is subject to errors of measurement, the resulting estimate of the regression coefficient is biased toward zero, the more so the greater the variance of the error of measurement of the independent variable relative to its true variance. Now since personal saving is in the order of one-tenth of disposable income, it is reasonable to suppose that the variance of true (as distinguished from measured) personal saving is a good deal smaller than the variance of disposable income. At the same time, personal saving, as actually measured in the national income accounts, represents the difference between largely independent estimates of disposable income and of personal consumption. Hence the error of measurement of personal saving is likely to be even larger than that of disposable income. We can therefore be rather confident that the bias toward zero due to errors of measurement will be a good deal more serious when

consumption is regressed on personal saving than when it is regressed on disposable income. Furthermore, given the estimating procedure, the error of measurement of consumption is likely to be negatively correlated with the error of measurement of saving, and this negative correlation will produce a further downward bias in the estimate of the true regression coefficient.

2. Personal saving is *not* identically equal to investment either conceptually or in terms of actual measurement, and investment is *not* exogenous, or independent of consumption, even in the short run, especially when account is taken of investment in inventories. At least over a very short period of time, such as a quarter or less, it is quite likely that random variations in consumption behavior will be accompanied by random variations in personal saving in the opposite direction, and, to the extent that this is true, an estimate of the propensity to consume based on the regression of consumption on saving will be seriously downward-biased, even in the absence of errors of measurement.

For the problem at hand, there exists an alternative and relatively simple way of securing consistent estimates of the parameters in hypothesis I based essentially on the two stage least-squares method. Relying on the specification that the constant term is zero and that the true error is uncorrelated with A, this procedure consists in (*i*) regressing income on assets, and obtaining computed values of Y from this regression, say Y_c; (*ii*) then regressing C on Y_c and on A. The coefficients of Y_c and A so obtained can be shown to be consistent estimators of the coefficients of Y and A respectively.[18]

The application of this procedure to our problem yields estimates of 0.54 and 0.09 for α_1 and α_3, respectively. While these values are reasonably close to those given in row (6) of table 2.2, the loss of efficiency involved in this indirect procedure cautions us against attaching too much significance to these estimates.

Another possible way of coping with the problem of bias in the estimates resulting from a known cause but of unknown magnitude is to construct an alternative estimation procedure in which the same cause may be expected to produce a bias in the opposite direction. If this can be done, then the unknown parameters may be bracketed by the estimates generated by the two alternative procedures, and if they are close together, then we may conclude with some confidence that the bias in either procedure is not too serious. For this purpose, suppose that we divide both sides of equation (2.6) by Y, obtaining

$$\frac{C}{Y} = \alpha_1 + \alpha_3 \frac{A}{Y}$$

and then proceed to estimate the parameters of this equation by the conventional least-squares method. It can be shown that, in so far as the bias due to the positive correlation between Y and the true error of the consumption function is concerned, the least-squares estimate of α_3 will be upward-biased, and hence, that of α_1 will be downward-biased. Thus, for both coefficients, the bias is in the direction opposite to that resulting from other procedures reported so far. This approach has also other desirable properties: it eliminates the difficulties arising from the presence of strong multicollinearity, and the homoscedasticity condition is more likely to be satisfied. Furthermore, it eliminates altogether the common trend in the variables, thus providing a rather stringent test of the relevance of net worth as a determinant of consumption. Its main drawback is that the above-mentioned bias in the estimates may be appreciably reinforced by error of measurement in the variable Y, although error of measurement in A will tend to work in the opposite direction.[19]

The results of this test reported in row (11) are remarkably encouraging. As expected, this procedure yields a higher estimate of α_3 and a lower estimate of α_1 relative to corresponding estimates in row (3). But these estimates are so close to each other that, if our interpretation of these estimating procedures is correct, they provide very narrow limits within which the true values of these coefficients may be expected to lie. However, the very high serial correlation of errors in both (3) and (11) may cast some doubt on the reliability of the estimates.

The high serial correlation in the estimated errors, in this and to a lesser extent in other procedures, suggests the desirability of testing for evidence of a significant change in the parameters of our consumption function as between the prewar and the postwar period. Individual tastes as well as the demographic structure and the rate of return on assets, on which the theoretical values of the coefficients depend, may have changed sufficiently over the two periods to cause a measurable change in the parameters. In addition, there exists a statistical problem arising from the fact that the data, particularly the net worth estimates, are based on somewhat different estimating procedures for the two periods. To test the hypothesis of a shift in parameters we have computed a number of regressions involving a "dummy" variable X, with value zero for the years 1929–40 and value one for the years 1947–59.

The results given in rows (4), (7), (8), (10), and (12) to (14), which constitute a representative sample of the tests we have carried out, show that the coefficient of the dummy variable is quite small in absolute value and statistically insignificant in all cases. Since its sign is consistently negative, we cannot rule out the possibility that there is very slight downward shift of the consumption function from the pre-war period to the postwar period.[20]

On the whole, known statistical properties of various estimation methods utilized appear to explain the pattern of most of the results reported in table 2.2,[21] which we may now endeavor to summarize.

In the first place, all of the tests seem, by and large, to support the basic hypothesis advanced in this paper, and in particular, the importance of net worth as a determinant of consumption. Unfortunately, some difficulties arise in the attempt to secure reliable estimates of the coefficients of the "independent" variables, although some tentative conclusions seem to stand out. First, the different estimation procedures when applied to hypothesis I generally yield a similar estimate of the coefficient of net worth for the period as a whole, very close to 0.08. [Cf. rows (3), (6), (7), (11), and (12).]

At the same time, the estimate of the coefficients of income is somewhat unstable [see rows (3), (4), (6), (8), (10), and (12)]. This instability is appreciably reduced under hypothesis II, where the third variable, $Y(L/E)$ apparently helps to disentangle the effect of purely cyclical and transitory changes in nonproperty income from that of long-fun or permanent changes. The various estimates for the long-run marginal propensity, $\alpha_1 + \alpha_2$, are fairly—consistent—0.72 for the first difference and straight estimates with constant suppressed [rows (5) and (9)] and moderately lower for the ratio estimate [row (15)]. There is however much less consistency in the values α_1 and α_2 separately and hence in the estimates of the short-run marginal propensity to consume with respect to labor income, α_1. While the ratio estimate yields the value of 0.65 [row (15)], all other methods produce much lower estimates, between 0.41 and 0.44 [rows (5), (9), (10)]. On the whole, we are inclined to regard the latter estimates as somewhat more reliable, in part because of the high serial correlation present in rows (5) and (15), but no firm conclusion seems warranted with the available data and methods. At the same time we observe that the introduction of the third variable, $Y(L/E)$ tends to reduce somewhat the estimate of the coefficient of net worth. It would appear that the value 0.08 obtained for hypothesis I, where $Y(L/E)$ is not present, may be somewhat too high—since the cyclically sluggish variable A acts partly as a proxy for expected nonlabor income—and the true value may be closer to 0.06 or even somewhat lower.

As indicated earlier, a few tests of hypothesis I have also been carried out for the period 1900–28. Because the data for this period are mostly obtained from different sources and are subject to very wide margin of error, we have seen little point in combining them with the series relating to the period since 1929. In fact, we are inclined to attach rather little significance to the results of these tests, which are accordingly confined to Appendix section A. For whatever they are worth, these results do not appear grossly inconsistent with those for the period after 1929, especially when account is taken of error of measurement and its likely effects on different estimation procedures. In particular, the contribution of net worth appears again to be significant, its coefficient being of the same order of magnitude as in the later period except in the first-difference test which is obviously most seriously affected by the error of measurement.

Finally, our empirical results are also roughly consistent with the *a priori* numerical predictions reported in table 2.1. The fact that the coefficients of both variables, especially that of net worth, are on the low side may be accounted for by reference to the estate motive which was ignored in the numerical calculations for table 2.1, while it probably plays a nonnegligible role at least for the high-income and/or self-employed groups.[22]

III Some Implications

A Relation to the Standard Keynesian Consumption Functions

The standard Keynesian consumption function [23] is usually written in the form:

$$C = \gamma Y^{*} + \gamma_0 \tag{2.12}$$

where Y^* denotes personal income net of taxes or disposable income and the γ's are constants.[23] A more sophisticated variant of this hypothesis, which has become quite popular of late, consists in separating income into two parts, disposable labor income Y, and disposable nonlabor or property income, which we shall denote by P. Thus,

$$C = \gamma_1 Y + \gamma_2 P + \gamma_0. \tag{2.13}$$

This variant, which reduces to (2.12) when $\gamma_1 = \gamma_2$, is usually advocated on the ground that property income accrues mostly to higher-income and/or entrepreneurial groups who may be expected to have a lower marginal propensity to consume. Accordingly, γ_2 is supposed to be smaller than γ_1 and this supposition appears to be supported by empirical findings.

It is immediately apparent that (2.13) bears considerable similarity to hypothesis I discussed in this paper, i.e.,

$$C = (\alpha_1 + \alpha_2)Y + \alpha_3 A. \tag{2.14}$$

The main difference lies in the constant term which appears in (2.13) but not in (2.14), and in the fact that the wealth variable A in (2.14) is replaced in (2.13) by a closely related variable, income from wealth, P. We can avoid dealing with the first source of discrepancy by working with both hypotheses in first-difference form,

$$\Delta C = \gamma_1 \Delta Y + \gamma_2 \Delta P \tag{2.13a}$$

$$\Delta C = (\alpha_1 + \alpha_2)\Delta Y + \alpha_3 \Delta A. \tag{2.14a}$$

Equations (2.13a) and (2.14a) are quite useful since they allow a straightforward test of the usefulness of the Modigliani-Brumberg hypothesis as compared with the standard Keynesian one. We have already exhibited in table 2.2, row (6), the results obtained by fitting (2.14a) to the data. In order to complete the test we need to estimate the parameters of (2.13a). If the standard Keynesian version is correct, the net worth variable in (2.14) and (2.14a) is merely a proxy variable for the return from wealth, P, and hence substitution of ΔP for ΔA should improve the fit. On the other hand, if (2.14) and (2.14a) are closer to the truth than (2.13), then the substitution of ΔA by a proxy variable ΔP should reduce the correlation.

The estimate of P needed for this test is given in [3].[24] The definition of consumption on which we rely, however, is somewhat different from that customarily used in the standard Keynesian formulation in that it includes the current consumption—depreciation—of the stock of consumer durables, while excluding expenditure for the purchase of such goods.[25] The results obtained for hypothesis (2.13a) are as follows:

$$\underset{}{\Delta C} = \underset{(0.07)}{0.93\Delta Y} + \underset{(0.29)}{0.07\Delta P} \quad R^2 = 0.86. \tag{2.15}$$

Comparison of this result with those reported in table 2.2, row (6), strongly suggests that net worth is definitely not a mere proxy for current property income. While the coefficient of P is positive and smaller than that of Y as expected, this variable is much less useful than A in explaining the behavior of consumption. In fact, its contribution is not significantly different from zero.[26]

These results, besides supporting our hypothesis, serve also to cast serious doubts on the conventional interpretation of the empirical coefficients of ΔY and ΔP in (2.13); namely, that incremental labor income is largely consumed while incremental property income is largely saved. For our tests of the Modigliani-Brumberg model indicate that consumption is quite responsive to variations in the market value of wealth, which, in turn, must largely reflect the capitalization of property income. Note, however, that the market valuation of assets will be controlled by expected long-run returns, say $\bar{P}$, which will tend to be a good deal more stable than *current* property income, P. We suggest therefore that the coefficient of P in (2.13) is small not because property income is largely saved but because short-run changes in P are dominated by transitory phenomena and hence are a poor measure of changes in the relevant long-run, or permanent, property income, which will be reflected far more reliably in the market valuation of assets. Put somewhat differently, the low coefficient of P does not imply a low marginal propensity to consume out of property income but merely a low propensity to consume out of transitory income. Correspondingly the extremely high coefficient of labor income in (2.14) is equally misleading, reflecting the fact that Y acts partly as a proxy for the permanent component of property income, $\bar{P}$.

One might be tempted to estimate the marginal propensity to consume with respect to permanent property income $\bar{P}$ by relying on the estimates of the coefficient of net worth in (2.14a) provided in table 2.2, and on the relation

$$\bar{P} \simeq rA \text{ or } A \simeq \frac{\bar{P}}{r}.$$

Following this reasoning, the coefficient of $\bar{P}$ in the consumption function would be given by

$$\frac{\alpha_3}{r},$$

where r is the rate at which the market capitalizes the return from assets. If we are willing to approximate r with the average realized rate of return on assets, then, from the figures given in [3], we find that r was about 0.04 in the prewar period and somewhat lower (around 0.03) in more recent years.[27] Combining this estimate with our estimate of α_3, which is in the order of 0.06, we seem to be led to the conclusion that the marginal propensity to consume with respect to permanent property income

$$\frac{\alpha_3}{r},$$

far from being low, is actually well above unity.

This result may appear preposterous if judged in terms of the standard Keynesian framework underlying (2.13), with its emphasis on the relation between flows. It is, however, possible to interpret this result in terms of the Modigliani-Brumberg framework. For, in this model, wealth affects consumption not only through the stream of income it generates but also directly through its market value which provides a source of purchasing power to iron out variations in income arising from transitory developments as well as from the normal life cycle. It is therefore not surprising that this model implies a marginal propensity to consume with respect to assets, α_3, larger than the rate of return r (cf. table 2.1), an inference which, as we have just seen, is supported by empirical tests. It should be noted however that

$$\frac{\alpha_3}{r}$$

should not be interpreted as the marginal propensity to consume with respect to permanent property income in the same sense in which $(\alpha_1 + \alpha_2)$ can be said to measure the propensity with respect to permanent nonproperty income, for it measures the *joint* effect on consumption of a change in property income, r constant, and of the accompanying change in assets. It is not possible to infer the two effects separately from knowledge of α_3 and of the average value of r. Although we cannot pursue this subject here, we wish to point out that the effect on C of a change in $\bar{P}$ will be quite different depending on the behavior of A and hence r, as $\bar{P}$ changes.

B Cyclical versus Long-Run Behavior of the Consumption-Income Ratio—Relation to the Duesenberry-Modigliani Consumption Function

As is well known, one of the major difficulties encountered with the standard Keynesian consumption functions (2.12) or (2.13) lies in the constant term γ_0. This constant term is needed to account for the observed cyclical variability in the saving-income ratio, but it also implies a long-run tendency for the saving ratio to rise with income, which is contradicted by empirical findings. The lack of any positive association between income and the saving-income ratio in the long run, at least for the U.S. economy, was first uncovered by Kuznets, and has more recently been confirmed by the extensive investigation of Goldsmith [19], focusing on the years 1896–1949. In his summary recapitulation, he lists as the first item: "Long-term stability of aggregate personal saving at approximately one-eighth of income, and of national saving at approximately one-seventh."[28]

The consumption function proposed here is capable of accounting both for the long-run stability and the cyclical variability of the saving-income ratio. In order to exhibit its long-run properties, let us suppose that Y were to grow at a constant rate n, in which case Y^e can be taken as equal or proportional to Y. Suppose further that the rate of return on assets r is reasonably stable in time. Then the consumption function (2.6) implies that the income-net worth ratio, Y_t^*/A_{t-1} will tend to a constant h, related to the parameters of the consumption function by the equation:[29]

$$h = \frac{n + \alpha_3 - \alpha r}{1 - \alpha}; \quad \alpha = \alpha_1 + \alpha_2. \tag{2.16}$$

When the ratio

$$\frac{Y_t^*}{A_{t-1}}$$

is in fact equal to h, then income and net worth grow at the same rate, n, and the saving-income ratio will be a constant given by:

$$\frac{S_t}{Y_t^*} = \frac{A_t - A_{t-1}}{A_{t-1}} \frac{A_{t-1}}{Y_t^*} = n\frac{1}{h}. \tag{2.17}$$

Similarly, we find:

$$\frac{Y_t}{A_{t-1}} = \frac{Y_t^* - rA_{t-1}}{A_{t-1}} = h - r \tag{2.18}$$

$$\frac{C_t}{Y_t} = \frac{Y_t^* - S_t}{A_{t-1}} \frac{A_{t-1}}{Y_t} = \frac{h-n}{h-r}. \tag{2.19}$$

Thus the model implies that if income fluctuates around an exponential trend the income-net worth ratio will tend to fluctuate around a constant level h, and the saving-income ratio around a constant

$$\frac{n}{h}.$$

The empirical estimates reported in section II suggest that α is around 0.7, and α_3 close to 0.06. The average rate of return, r, is much more difficult to guess. If we are willing to rely on the ratio

$$\frac{P}{A_{t-1}}$$

for this purpose, then r would be around or somewhat lower than 0.04. But this ratio is very likely to under-state the true value for r, since the estimate of P given in [3] corresponds to the conventional definition of personal income and omits a number of items whose exclusion is appropriate for the standard Keynesian model but not for the Modigliani-Brumberg model. Among those items, the more important are imputed net rent on consumer durables and undistributed corporate profits.[30] These adjustments suggest an average value for r slightly over 0.04. If we further take for n, the rate of growth of income, a value in the order of 0.03, then from (3.6), (3.7), and (3.8) we obtain the following estimates for the various ratios under discussion: (*i*) total income to net worth, $h \simeq 0.2$; (*ii*) non-property income to net worth, $h - r \simeq 0.16$; (*iii*) saving to income,

$$\frac{n}{h} \simeq 0.15.$$

It can be seen that the first two of the above figures are in fact close to the values around which the ratios

$$\frac{Y_t^*}{A_{t-1}} \text{ and } \frac{Y_t}{A_{t-1}}$$

fluctuate according to the data given in [3] while the third, the saving-income ratio, is consistent with the findings of Goldsmith reported earlier.

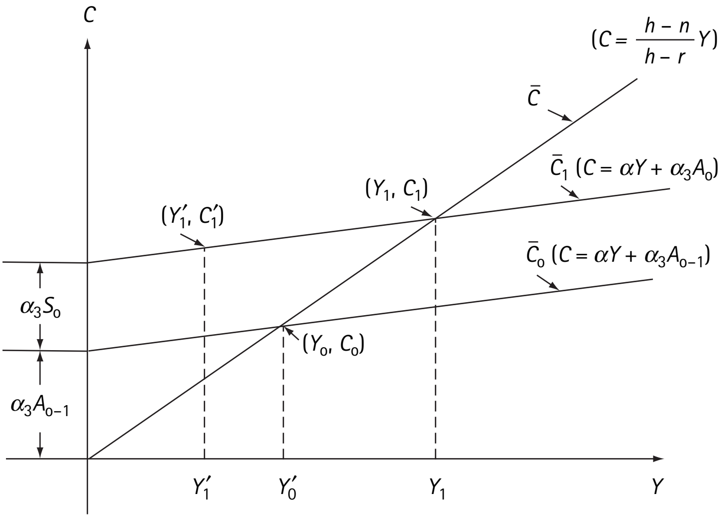

Figure 2.2
Consumption income relations: long-run and short-run

Needless to say, these calculations are very crude and are given here primarily to bring out certain interesting testable implications of the consumption function discussed in this paper. Among these implications the long-run stability of the ratio of net worth to income is particularly significant, for it paves the way for an explanation of the historical stability of the capital-output ratio in terms of the supply of capital, thereby challenging the prevailing notion that the behavior of this ratio is explained by technological requirements [1] [2] [5] [29].

As for the cyclical implications of our model, we need only observe that at any given point in time net worth A_{t-1} is a given initial condition. Hence, retaining for the moment the assumption that $Y^e \simeq Y$, (2.6) implies that the aggregate consumption function for any given year is a straight line in the C-Y plane with slope α and intercept $\alpha_3 A_{t-1}$. It is shown in figure 2.2 for the year 0 as the line labeled $\bar{C}_0$, and looks like the orthodox Keynesian version. Yet, it differs from the latter in one essential respect, namely, that its intercept will change in time as a result of the accumulation (or decumulation) of wealth through saving. As we have shown in preceding paragraphs, so long as income keeps rising on its exponential trend, the growth in net worth will shift the function in such a way that the observed consumption-income points will trace out the long-run consumption function (2.19) represented in our graph by the line $\bar{C}$ through the origin. This point is illustrated in figure 2.2 for two years, 0 and 1. However, suppose that a cyclical disturbance caused income to fall short of Y_1, say to the level Y_1'. Then the consumption C_1' given by the short-rub consumption function $\bar{C}_1$ implies a higher consumption-income ratio and a lower saving-income ratio.

Thus, cyclical swings in income from its long-run trend will cause swings in the saving-income ratio in the same direction,[31] especially since the position of the function will not change appreciably when income is cyclically depressed below its previous peak due to the small or negative saving that would prevail.[32] After income has recovered beyond the previous peak, it may for a while rise rapidly as it catches up with its trend, running ahead of the slowly adjusting wealth. In this phase we may observe points to the right of $\bar{C}$, and the corresponding high saving will tend to make A catch up with Y.

Thus the model advanced here may be expected to generate a behavior of consumption and saving which is very similar to that implied by the earlier Duesenberry-Modigliani type of hypothesis, in which consumption was expressed as a function of current income and the highest previous peak income (or consumption) [9] [30]. If we interpret the role of the highest previous income as that of a proxy for net worth, then the Duesenberry-Modigliani consumption function can be considered as providing a good empirical approximation to the consumption function discussed in this paper, and to this extent the empirical support provided for the Duesenberry-Modigliani type of hypothesis can also be considered as empirical support for the consumption function advanced here, and vice versa. At the same time the present model has the advantage that the hypotheses on which it rests are explicitly stated as specifications of the consumer's utility function. It is also analytically more convenient as a building block in models of economic growth and fluctuations, as we have endeavored to demonstrate in various contributions [1] [2] [3] [29] [31].[33]

Appendix

A Some Statistical Results for Earlier Years

The following are the estimates obtained using data for 1900–28, excluding years 1917, 1918, and 1919. As stated in the text, the data used are very rough, and may not be compatible with the data for the period since 1929. The data and their derivation are described in Ando [2, App.], except for the adjustments needed for different treatments of the government sector. This adjustment is self-explanatory from the description given in [2]. The data presented in [2] are in turn based largely on [18] [19] [22] [27] and [42].

$$\begin{aligned} C = {} & \underset{(0.134)}{0.755Y} + \underset{(0.020)}{0.073A} \qquad R^2 = 0.995 \\ & \qquad\qquad\qquad\qquad DW = 1.63 \end{aligned} \tag{a}$$

$$\Delta C = 0.731\Delta Y + 0.047\Delta A \qquad R^2 = 0.44 \tag{b}$$
$$\qquad (0.180) \quad (0.037) \qquad DW = 2.48$$

$$\frac{C}{Y} = 0.505 + 0.112\frac{A}{Y} \qquad R^2 = 0.51 \tag{c}$$
$$(0.144) \ (0.021) \qquad DW = 1.05$$

B Biases in Estimating the Consumption Function by Regression on Saving

Suppose that true consumption c^* and true income y^* are related by a linear function (all variables being measured from their means),

$$c^* = \alpha y^* + \varepsilon \tag{a}$$

and measured income and measured consumption are related to their respective true values by

$$c = c^* + \eta \tag{b}$$

$$y = y^* + \xi \tag{c}$$

where ε, η, and ξ are random variables. For simplicity, let us assume that ε is uncorrelated with η, and ξ; η and ξ are uncorrelated with c^* and y^*. We also have the definitions

$$s^* = y^* - c^* \tag{d}$$

$$s = y - c = y^* - c^* + \xi - \eta \tag{d$'$}$$

Using (a) and (d), We have

$$c^* = \frac{\alpha}{1-\alpha}s^* + \frac{\varepsilon}{1-\alpha} \equiv \beta s^* + \varepsilon'. \tag{e}$$

In order to concentrate first on the effect of errors of measurements, let us momentarily accept the unwarranted assumption that saving is equal to investment which in turn is truly exogenous. Under this assumption s^* can be taken as independent of ε' and therefore, if we could actually observe s^* and c^*, by regressing c^* on s^* we could secure an unbiased estimate of β from which we could in turn derive a consistent estimate of α. If, however, we estimate β by regressing c on s, then remembering that s is obtained as a residual from y and c, we obtain the estimate

$$\hat{\beta} = \frac{\sum cs}{\sum s^2} = \frac{\sum(\beta s^* + \varepsilon' + \eta)(s^* + \xi - \eta)}{\sum(s^* + \xi - \eta)^2} = \frac{\beta\sum s^{*2} + \sum \eta(\xi - \eta)}{\sum s^{*2} + \sum(\xi - \eta)^2}. \tag{f}$$

The term $\Sigma_\eta(\xi - \eta)$ arises from the fact that when there is a statistical error η in measuring consumption, there will be an error $-\eta$ in measuring saving, except in so far as this is offset by an error ξ in measuring income. This term will tend to be negative and introduces a downward bias into β. The term $\Sigma(\xi - \eta)^2$ in the denominator is the well-known result of an error of measurement of the independent variable, and it introduces an unambiguous bias towards zero into β.

However, as pointed out in the text, the assumption that personal saving is exogenous is completely unwarranted. In order to bring out the nature of the bias resulting from this misspecification, let us make the other extreme and equally unwarranted assumption that disposable income is truly exogenous. We can then regard y^* as independent of ε and by substituting (a) and (d) into (d′) we find

$$s = s^* + \xi - \eta = (1-\alpha)y^* - \varepsilon + \xi - \eta. \tag{g}$$

Equation (f) is then replaced by

$$\beta = \frac{\alpha(1-\alpha)\sum y^{*2} - \sum \varepsilon^2 + \sum \eta(\xi - \eta)}{(1-\alpha)^2 \sum y^{*2} + \sum \varepsilon^2 + \sum (\xi - \eta)^2}. \tag{h}$$

The presence of the term Σ_{ε^2} in both numerator and denominator is the result of the fact, discussed in the text, that when there is residual error ε in the consumption-income relationship, there will be residual error $-\varepsilon$ in the saving-income relationship. Because of the signs, this effect, too, will bias β downward.

Since all the biases are downward and there is no offsetting upward bias of any significance, it is not surprising that recent applications of this approach [39] [43] lead in a number of cases to a negative estimate of the marginal propensity to consume.

Notes

1. Since this paper was submitted and accepted for publication, a model that bears some similarity to the one proposed here has been presented by Alan Spiro in "Wealth and Consumption Function," *Jour. Pol. Econ.*, Vol. 70 (August, 1962), pp. 339–54.

2. The theory summarized here is essentially the same as that developed by Modigliani and Brumberg in [7] and [34]. Because of the untimely death of Richard Brumberg in August, 1954, the original paper by Modigliani and Brumberg [34] has never been published, and it is not likely to be published in the near future. Because of this, we present here a summary. However, the aggregation procedure developed here is different from that followed by Modigliani and Brumberg. This is because they are largely concerned with numerical prediction of parameters while we are interested only in exhibiting the conditions for existence of a particular form of the aggregate consumption function.

3. This equivalence holds on the assumption that consumers deal in perfect markets.

4. For the sake of simplicity, we shall not display the stochastic component of these relations explicitly in this section.

5. To be precise

$y_t^{eT\tau}$

is the income the person expects to earn at age τ, measured in prices prevailing in the year t, and r is the "real" rate of return on assets. In (2.2) the expected real rate is assumed to remain constant over time, but the formula can be generalized to allow for changing rate expectations.

6. See Theil [41, pp. 10–26]. More precisely, the least-squares estimates of the parameters of equation (2.5) will be weighted averages of the least-squares estimates of the corresponding parameters of equations (2.4) only if the set of conditions specified by Theil in the reference cited above is satisfied. Roughly speaking, these conditions require that there be no systematic relations between parameters and variables of equation (2.4) over individuals.

7. Strictly speaking the values of the parameters would vary somewhat depending on whether the growth of income results from population or from productivity growth. However, for rates of growth within the relevant range, say 0 to 4 percent per year, the variation turns out so small that it can be ignored for present purposes.

8. On the other hand, if we assume (*i*) that the preferred pattern of allocation of consumption and the pattern of income over life are the type suggested by the available cross-section data, (*ii*) that income expectation is consistent with the prevailing pattern of income, again suggested by the cross-section data over age groups, then the resulting coefficients of income and assets in equation (2.6) would be somewhat higher than those reported in table 2.1.

9. See Ando [1]. Mincer in [28] relied on a similar device, except that he used population in place of labor force.

10. The data and the procedure by which they have been obtained will be found in Ando, Brown, Kareken, and Solow [3, Data App.]. The derivation of labor income after taxes is particularly troublesome and is based in part on methods suggested by [13] and [38].

11. A few experiments were made using data including the Second World War years, and equation (2.6) appears to explain consumption behavior during these years better than any other consumption function to our knowledge. However, the fit is still not very good, and, at any rate, we do not feel that these years are relevant because of their obviously special characteristics.

12. In this section, the time subscript will be omitted whenever there is no danger of confusion.

13. See, for instance, Theil [40].

14. In this section, we shall refer to equations by the rows in table 2.2.

15. See note 17.

16. The simple correlation between ΔA and $\Delta\left(Y\frac{L}{E}\right)$ is 0.93, higher than that between ΔY and $\Delta\left(Y\frac{L}{E}\right)$, 0.89.

17. Let us denote the *true* error term in equation (2.6) by ε. Then, under the assumption that the correlation between Y and A and that between Y and ε are positive, while A and ε are uncorrelated with each other, it can be readily shown that

$$\text{plim}\,\hat{\alpha}_1 = \alpha_1 + \text{plim}\frac{\sum A^2 \sum Y\varepsilon}{\sum A^2 \sum Y^2 - \left(\sum AY\right)^2} \geq \alpha_1$$

$$\text{plim}\,\hat{\alpha}_3 = \alpha_3 - \text{plim}\frac{\sum Y\varepsilon \sum AY}{\sum A^2 \sum Y^2 - \left(\sum AY\right)^2} \leq \alpha_3$$

where $\hat{\alpha}_1$ and $\hat{\alpha}_3$ are the least-squares estimates of α_1 and α_3, respectively.

The above formulae are stated for the case in which the constant is suppressed, but for the case where the constant is not suppressed, it is only necessary to reinterpret the symbols as deviations from respective means. The asymptotic bias for the constant term can then be expressed in terms of the above limits and the means of the variables involved. Denoting the constant term and its least-squares estimate by γ and $\hat{\gamma}$ respectively, it can be shown that

$$\operatorname{plim}(\hat{\gamma}-\gamma)=\operatorname{plim}\frac{\sum Y'\varepsilon}{\sum A'^2\sum Y'^2-\left(\sum A'Y'\right)^2}\left(\bar{A}\sum A'Y'-\bar{Y}\sum A'^2\right)$$

where the primed symbols denote the deviations from the mean. The estimate from our data of the probability limit of the expression inside the parentheses is positive and fairly large, so that the asymptotic bias of the least-squares estimate of the constant term is most likely to be positive.

18. Suppose that we estimate the parameters of the following equation by the method of least squares:

$$Y=b_yA+a_y+\eta_y \tag{a}$$

where η_y is the error term.

The substitution of the value of Y estimated by (a) into equation (2.6) yields

$$C=\alpha_1(b_yA+a_y)=\alpha_3A+\varepsilon=(\alpha_1b_y+\alpha_3)A+\alpha_1a_y+\varepsilon. \tag{b}$$

Comparison of equation (b) with the regression of C on A

$$C=b_cA+a_c+\eta_c \tag{c}$$

results in

$$\alpha_1b_y+\alpha_3=b_c \tag{d}$$

$$\alpha_1a_y=a_c \tag{e}$$

(d) and (e) can then be solved to give the estimated values of α_1 and α_3. These estimates are identical with those resulting from the two-stage least-squares procedure described in the text.

19. For possible problems arising from the use of ratios, see [26] and [37]. Note that the ratio estimates described above can be regarded as the Aitkin's generalized least-squares estimates of hypothesis I and II under the assumption that the standard deviation of the residual errors is proportional to Y_t.

20. Our model provides one possible clue to this apparent downward shift, An examination of the figures reported in [3] reveals a distinct decline in the ratio of nonlabor income to net worth, which can be taken as a measure of the rate of return on capital. This is presumably attributable in large measure to increases in corporate taxes and in the extent and progressiveness of personal income taxation. An examination of the values of the coefficients of income and net worth implied by our model, reported in table 2.1, suggests that both coefficients should tend to decline as the rate of return on assets declines. However, we do not wish to press this point, especially since we cannot even be sure whether the apparent decline in the coefficients reflects anything more than error of measurement.

21. One uncomfortable feature of table 2.2 is the high serial correlation of residuals in a number of tests. One possible explanation for the high serial correlation in some of the tests is that consumption does not adjust fully to changes in income and assets within the arbitrary time unit of one year. To allow for this possibility hypothesis I might be written as

$$C_t-C_{t-1}=\delta(\alpha_1Y_t+\alpha_3A_{t-1}-C_{t-1}),\text{ or } C_t=(1-\delta)C_{t-1}+\delta\alpha_1Y_t+\delta\alpha_3A_{t-1},$$

where the constant δ, with the dimension 1/time, measures the speed of adjustment of consumption and may be expected to approach unity as the time unit increases. This hypothesis was tested in ratio form (to avoid increasing further the already extremely high multicollinearity prevailing in the direct form) with the following results:

$$\frac{C_t}{Y_t}=0.46+\underset{(0.077)}{0.177}\frac{C_{t-1}}{Y_t}+\underset{(0.016)}{0.072}\frac{A_{t-1}}{Y_t}$$

implying $\delta=0.82$, $\alpha_1=0.56$, $\alpha_3=0.087$.

The relatively low coefficient of C_{t-1} (which is only moderately significant) suggests that a span one year is long enough for most of the adjustment to take place. Also the estimates of the remaining coefficients are not greatly affected and move closer to the first difference estimates. However, somewhat surprisingly, the serial correlation is increased still further.

22. See for instance the results of cross-section studies reported in [24] and [32].

23. Keynes' own formulation (See [23, Book 3]) was considerably more general than that contained in equation (2.12).

24. Our estimates of Y and P do not add up exactly to disposable personal income as usually defined because we include in disposable personal income contributions to, instead of benefits from, the social security system. However, this discrepency is quite minor.

25. Also, our data are in current dollars, while the standard Keynesian version of the consumption function is usually stated in terms of constant dollars.

26. For the sake of completeness several other variants of (2.13a) were tested by adding variables that were included in the test of our hypothesis and which are consistent with the spirit of the Keynesian model. The addition of the variable

$$\Delta\left(Y\frac{L}{E}\right)$$

which might help to sort out the effect of long run from that of purely cyclical variations in income, yields

$$\underset{(0.18)}{\Delta C = 0.47\Delta Y} + \underset{(0.13)}{0.49\Delta\left(Y\frac{L}{E}\right)} + \underset{(0.23)}{0.17\Delta P} \quad R^2 = 0.921. \tag{2.15a}$$

If we also include the dummy variable X to allow for possible shifts from the prewar to the postwar period, its coefficient is uniformly less than its standard error, and in general hypothesis (2.13) does not fare any better. This conclusion can be illustrated by the following result which is the most favorable to that hypothesis among the battery we have run:

$$\underset{(0.14)}{\Delta C = 0.46\Delta Y} + \underset{(0.14)}{0.51\Delta\left(Y\frac{L}{E}\right)} + \underset{(0.28)}{0.26\Delta P} - \underset{(0.39)}{0.21X\Delta P} \quad R^2 = 0.921. \tag{2.15b}$$

The fact that in (2.15a) and (2.15b) the coefficient of the variable

$$\Delta\left(Y\frac{L}{E}\right)$$

is a good deal higher and statistically more significant than in the corresponding tests reported in table 2.2 is readily accounted for by the high correlation between this variable and $\boldsymbol{A}$ which, in the absence of A, makes this variable act partly as a proxy for A. (The correlation in question is 0.93. See also note 16.)

27. See, however, our comment below on the shortcomings of our estimate of P given in [3] as measure of return on assets.

28. Goldsmith [19, Vol.1, p. 22.].

29. Under the stated assumptions we have $Y_t^* = Y_t + P_t$, and $P_t = rA_{t-1}$. Hence, saving can be expressed as

$$S_t = Y_t^* - C_t = Y_t + P_t - C_t = (1-\alpha)Y_t^* - (\alpha_3 - \alpha r)A_{t-1}. \tag{a}$$

We also have $S_t = A_t - A_{t-1}$. (We disregard capital gains in this context, and in this context alone. This simplification seems to be justified for long-run analysis, but for a fuller discussion of this problem, see Ando [2].) Substituting this definition for S_t in (a), dividing through by A_{t-1}, adding and subtracting n, and then rearranging terms, we obtain

$$\frac{A_t - A_{t-1}}{A_{t-1}} = n + (1-\alpha)\left[\frac{Y_t^*}{A_{t-1}} - \frac{n + \alpha_3 - \alpha r}{1-\alpha}\right]. \tag{b}$$

Comparison of (*b*) above with equation (2.16) shows that if Y_t^*/A_{t-1} were larger than *h*, the second term in the right-hand side would be positive, and hence net worth grow at a rate larger than that of income, *n*, causing Y_t^*/A_{t-1} to fall toward *h*: and conversely if Y_t^*/A_{t-1} were smaller than *h*.

This argument is oversimplified and incomplete, particularly since it ignores the interaction between the behavior of consumers and the production process in the economy. A more complete analysis of this growth process is given in Ando [2].

30. The rationale for including corporate saving in property income and personal saving is given in Modigliani and Miller [36].

31. This phenomenon will be further accentuated when we recognize the possibility that a cyclical fall in Y is likely to bring about a small change in Y^e. Also, because property income may be expected to fluctuate cyclically even more than labor income, the ratio of saving to total income will fluctuate even more than the ratio of saving to labor income. See note 30, equation (*a*).

32. Some downward shift of the consumption function might occur even in the absence of dissaving, if there is some downward revaluation of the market value of assets, as the depressed level of property income tends to bring about less favorable evaluation of the long-run prospects for return from assets.

Because of this dependence of the value of assets on property income, the statement made earlier to the effect that A_{t-1} can be taken as a given initial condition in the year *t* is only approximately true. Also the relation $\Delta A_t = S_t$ does not hold in the presence of capital gains and losses. Note however that what is relevant in the present connection is the change in the value of assets in terms of purchasing power over consumption goods and not the change in terms of money value, which may be considerably more severe.

33. On the other hand, because up-to-date estimates of wealth are not readily available, at least for the present, some variant or the Duensenberry-Modigliani model may well be more useful for short-run forecasting.

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3

LONG-RUN IMPLICATIONS OF ALTERNATIVE FISCAL POLICIES AND THE BURDEN OF THE NATIONAL DEBT

Franco Modigliani

I Introduction

The time-honoured controversy over the burden of the national debt has flared up once more. The view, almost unchallenged a few years back, that the national debt is no burden on the economy and that the real cost of government expenditure, no matter how financed, cannot be shifted to "future generations" has been on the retreat under a powerful counterattack spearheaded by the contributions of J. M. Buchanan,[1] J. E. Meade[2] and R. A. Musgrave.[3] These authors, while relying to a considerable extent on older arguments, have significantly enriched the analysis by blending the traditional approach with the new insights provided by the Keynesian revolution. But even these most recent contributions have failed, in our view, to provide an altogether adequate framework—a failure resulting at least in part from the Keynesian tendency to emphasise flows while paying inadequate attention to stocks. It is the purpose of this paper to propose a fresh approach to this problem, and to show that, unlike its predecessors, it leads to a consistent and yet straightforward answer to all relevant questions.

Unless otherwise noted, the national debt will be defined here as consisting of: (1) all claims against the government held by the private sector of the economy, or by foreigners, whether interest bearing or not (and including therefore bank-held debt and (government currency, if any); less (2) any claims held by the government against the private sector and foreigners.[4]

From a methodological point of view, the central contention of our analysis is that to grasp fully the economic effects of alternative fiscal policies and of the national debt, we must pay proper attention to stocks as well as to the usual flow variables and to the long-run as well as to the impact effects. Among the substantive implications of this line of approach, the following may be mentioned here by way of a rough summary: (1) Given the government purchase of goods

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and services, an increase of the (real) national debt, whether internal or external, is generally advantageous to those present at the time of the increase (or to some subset thereof). (2) Such an increase will generally place a "gross burden" on those living beyond that time through a reduction in the aggregate stock of private capital, which, as long as the (net) marginal productivity of capital is positive, will in turn cause a reduction in the flow of goods and services. Furthermore, this loss (as well as the gain under (1) above) will tend to occur even when lack of effective private demand would prevent the maintenance of full employment in the absence of the deficit, though the relative size of gain and losses may be quite different in these circumstances. (3) These conclusions hold in reverse in the case of a reduction in the real national debt. That is, such a decline is burdensome on those present at the time of the reduction and tends to generate a gross gain for those living beyond. (4) *If* the rate of interest at which the government borrows can be taken as a good approximation to the marginal productivity of private capital, then the gross burden (or gain) to "future generations" referred to under (2) and (3) can be *measured* by the interest charges on the national debt. (5) The gross burden may be offset in part or *in toto*, or may be even more than offset, in so far as the increase in the debt is accompanied by government expenditure which contributes to the real income of future generations, *e.g.*, through productive public capital formation.[5]

This summary is very rough indeed and is subject to numerous qualifications and amendments, many of which will be noted below. In any event, I should like to emphasise that the stress of this paper is on developing a method of analysis rather than on presenting a body of doctrines. For this reason I will try to relate my analysis to earlier points of view whenever this seems helpful in clarifying the issues involved. At the same time I will endeavour to stay clear of many traditional but somewhat sterile controversies, such as whether the analogy between private and public debt is true or false.

II A Bird's Eye View of the Classical and Post-Keynesian No-transfer and No-burden Argument

We begin by reviewing the very persuasive arguments supporting the doctrine that the cost of the current government use of resources cannot be transferred to future generations and that the national debt is no burden on them. Since these arguments have been presented many times in the last couple of centuries and have been extensively restated in recent years, we can afford to recapitulate them very briefly in terms of the three propositions presented below—at the cost of glossing over some of the fine points and of foregoing the pleasure of citing "chapter and verse."[6]

(1) Individuals or sub-groups within an economic system can, by means of borrowing, increase the current flow of goods available to them and pay for this increase out of future output. But they can do so only because their borrowing is "external," *i.e.*, matched by a lender who yields current goods in exchange for later output. But a closed community cannot dispose of more goods and services than it is currently producing. It certainly cannot increase this flow by paying with future output, for there is no way "we can dispose to-day of to-morrow's output." Hence the goods and services acquired by the government must always be "paid for" by those present at the time in the form of a reduction in the flow of goods available to them for private use, and cannot possibly be paid for by later generations, whether the acquisition is financed by taxes or by internal borrowing. Only through external borrowing is it possible to benefit the current generation and to impose a burden on the future.

(2) Although internal borrowing will leave in its wake an obligation for future tax-payers to pay the interest on the national debt and, possibly, to repay the principal, this obligation is not a net burden on the community as a whole, because these payments are but transfers of income between future members of the community. The loss of the tax-payers is offset in the aggregate by the gain of the beneficiary of the payment. These transfers may, of course, occur between people of different ages and hence of different "generations," and in this sense internal borrowing may cause "intergenerations transfers," but it will not cause a net loss to society.

The above two arguments, or some reasonable variant thereof, have provided the cornerstone of the no-transfer, no-burden argument over the last two centuries or so. It was left for Keynesian analysis to provide a third argument, removing thereby a potentially troublesome objection to the first two. If the cost of government expenditure always falls on the current generation, no matter how financed, why not forego altogether the painful activity of levying taxes? Yet our common sense rebels at this conclusion. A partial answer to this puzzle was provided by recognising that taxes, even when paid back in the form of transfers, generate some "frictional loss," because most if not all feasible methods of raising tax revenue tend to interfere with the optimum allocation of resources.[7] Presumably the ever-increasing level of the national debt resulting from full deficit financing of current expenditure would require raising through taxes an ever-growing revenue to pay the interest on the debt. Eventually the ratio of such taxes to national income plus transfers would exceed the ratio of government expenditure to national product, giving rise to frictional tax losses which could have been avoided through a balanced budget. While these considerations do provide a *prima facie* case for a balanced-budget policy, the case is not tight, for could not the interest itself be met by further borrowing?

However, we need not follow these fancy possibilities, for the Keynesian analysis has provided a much more cogent argument to support the need for an "appropriate" amount of taxation, although not necessarily for a balanced budget. This argument, which reaches its most elegant formulation in the so-called principle of "functional finance" enunciated by Lerner,[8] can be roughly summarised as follows.

(3) Given the full employment output, say $\bar{X}$, and given the share of this output which it is appropriate to allocate for government use, say $\bar{G}$, there is a maximum amount of output that is left available for the private sector, say $\bar{P} = \bar{X} - \bar{G}$. Now the private sector demand for output, say P, is a function of income and taxes, say $P = \mathscr{P}(X, T)$, with

$$\frac{\delta P}{\delta T} < 0.$$

Taxes are then to be set at that level, say $\bar{T}$, which satisfies the equation $\mathscr{P}(X, T) = P$. A higher level of taxes would generate unemployment and a lower level would generate inflation, both evils which it is the task of the government to avoid. $\bar{T}$ may turn out to be larger than $\bar{G}$, calling for a surplus, or smaller than $\bar{G}$, or even perchance just equal to $\bar{G}$, implying a balanced budget. But in any event, the purpose of taxes is not to make the current members of the community pay for the government use of goods, which they will do in any event, the real reason we need to put up with the unpleasantness of taxes is to prevent the greater social evil of inflation.

III A Bird's Eye View of the Classical and Post-Keynesian Transfer and Burden Argument

The basic contention of this school of thought, which itself has a long tradition, is that in general—though possibly with some exceptions—a debt-financed public expenditure will place no burden at all on those present at the very moment in which the expenditure takes place, and will instead place a burden on all tax-payers living thereafter. This burden may fall in part on those present at the time of the expenditure, but only in so far as they are present thereafter. The arguments which support this position have also been repeatedly stated and have been thoroughly reviewed quite recently by Buchanan. It will therefore again be sufficient to summarise them very briefly in the following two propositions:

(1) The cost of a tax-financed expenditure is borne currently, for the resources obtained by the government come from a forcible reduction in the resources of current tax-payers. But an expenditure financed by debt, whether internal or external, as a rule places no burden on those present at the time of the expenditure in

that, and in so far as, the resources acquired by the government are surrendered in a voluntary exchange by the savers, who thereby acquire government bonds (in lieu of some other asset).

(2) The burden is imposed instead on all future tax-payers, who will have to pay taxes to service the debt. These taxes are *not* a mere transfer of income, but a net burden on society, for, in the absence of the debt-financed expenditure, the taxes would not have been levied, while the investors in bonds would have received the income just the same, directly or indirectly, from the return on the physical assets in which their savings would have been invested. This argument *does not* imply that a debt-financed expenditure will necessarily affect future generations unfavourably. In order to assess the "net outcome," we must subtract from the gross burden represented by the extra taxes benefits, if any, resulting from the expenditure. Thus the net outcome might even be positive if the expenditure undertaken produced greater benefits than the private capital formation which it replaces. But the argument does imply that, through deficit financing, the expenditure of the government is being "paid for" by future generations.

A careful application of the *reasoning* underlying (1) and (2) will reveal circumstances in which the above conclusions do not hold and the allocation of the burden may be independent of the form of financing used. There are in particular two important cases which are treated at some length by Buchanan and which bring to light the contribution of Keynesian analysis also to this side of the argument. The first is the case of debt-financed expenditure in deep depressions, when private capital formation could not, in any event, provide an adequate offset to full-employment saving. Here, according to Buchanan, not even a gross burden need result to future tax-payers, for the expenditure could in principle be financed by interest-free issuance of currency. The second exception discussed by Buchanan is that of a major war. Unfortunately, the chapter on war financing is one of the least convincing in his book, and what follows may represent more nearly my application of his framework than a faithful summary of his argument. Suppose the war effort is sufficiently severe so that the allocation of resources to various uses, and to capital formation in particular, is completely determined by war necessities. In such a situation the way in which the government finances its expenditure cannot affect private consumption or capital formation. It would seem therefore that the burden of reduced consumption must be borne by the current generation, even if the reduction is achieved not through taxes but through a combination of rationing and voluntary increases in saving and the unspent disposable income is invested in claims against the government. Similarly, the burden of the reduction in useful capital formation is borne by those living after the war, again independently of financing. In this case, as well as in the case of depression financing, the taxes levied to pay the interest on the increased debt would indeed seem to result in a pure transfer, for the income associated with the bonds would *not* have come to exist had the government decided respectively to tax, or to print money, instead of borrowing.

J. E. Meade has also lately associated himself with those maintaining that the national debt is a burden,[9] but his argument is quite different from the classical one, and bears instead all the marks of post-Keynesian analysis. He is not concerned with the differential effect of deficit versus tax financing, but asserts none the less that government debt in excess of government-owned physical capital—the so-called deadweight debt—is a burden on the economy. Unfortunately his contribution, which is so stimulating in analysing the effects of a major capital levy, is less than convincing in its attempt to establish that the deadweight debt is a burden. For his demonstration seems to rely entirely on the proposition that elimination of the debt would be a blessing for the economy in that it would encourage saving through a "Pigou type" effect, besides reducing the frictional costs of transfers. Now the tax-friction proposition, though valid, is not new,[10] and had already been generally accepted as a second-order amendment to the no-burden argument. On the other hand, the first and central argument is rather unconvincing. For, as Meade himself recognises, a reduction in national debt, be it through a capital levy, budget surplus or inflation, would spur saving whether or not the debt reduced thereby was "deadweight" debt. In fact, at least the first two means would tend to increase saving, even if they were applied in a situation where the national debt was zero to begin with, and the outcome would be that of driving the economy into a position of net indebtedness *vis-à-vis* the government. Finally, Meade's analysis throws no light on whether the increase in saving following the capital levy is a permanent or a purely transitory phenomenon, nor on who, if anyone, bears the burden of a debt reduction. In spite of these apparent shortcomings, I am encouraged to think that Professor Meade's views are fundamentally quite close to those advanced here. I hope this will become gradually apparent, even without much further explicit reference to his argument.

IV Fallacies in the No-transfer No-burden Argument

The classical argument summarised in the last section appears so far rather convincing, and if so we should be able to pinpoint the fallacies in one or more of the three propositions of section II.

The fallacy in proposition (1) is not difficult to uncover. It is quite true that a closed community cannot increase its current resources by relying on tomorrow's unproduced output. None the less, the way in which we use today's resources can affect in three major ways the output that will result tomorrow from tomorrow's labour input: (i) by affecting the natural resources available to the future; (ii) by improving technological knowledge; and (iii) by affecting the stock of man-made means of production, or capital, available to future generations. Hence government expenditure, and the way it is financed, *can* affect the economy in the future if it affects any of the above three items.

The argument in (3) is also seriously inadequate, and the post-war experience has brought this home sharply. For the demand of the private sector consists of consumption C and capital formation I, and at least the latter component depends not only on income and taxes but also on monetary policy. Once we acknowledge this point, the principle of Functional Finance no longer implies a unique level of taxes. To demonstrate this point and its implications, it will be convenient—though it is not essential—to suppose that monetary policy affects P exclusively through the intermediary of the rate of interest r, *i.e.*, that $P = \mathscr{P}(X, T, r)$ with

$$\frac{\delta P}{\delta T} < 0;$$

and that r in turn depends on X and the quantity of money M. But once we admit that r enters not vacuously in $\mathscr{P}$ we must also recognise that the equation

$$\mathscr{P}(\bar{X}, T, r) = P \tag{3.1}$$

will be satisfied not by one but by many possible values of T, each value being accompanied by the appropriate value of r, say $r(T)$. Now in most circumstances—that is, except possibly in very deep depression—there will be a range of values of T such that the corresponding $r(T)$ is achievable by an appropriate monetary policy. There is therefore not one but a whole schedule of values of T which are consistent with the maintenance of full employment and price stability, each value of T being accompanied by an appropriate monetary policy. Furthermore, within this range there will tend to be a direct connection between T and the capital formation component of $\bar{P}$. If, starting from a correct combination of T, r and M, we lower taxes we will increase consumption, and to offset this we must reduce capital formation by appropriately tightening monetary policy. Conversely, by increasing taxes we can afford to have a larger capital formation. Thus, given the level of government expenditure, the level of taxes, and hence of budget deficit, does affect "future generations" through the stock of capital inherited by them.

Having thus brought to light the weaknesses in arguments (1) and (3), it is an easy matter to establish that, at least under certain conditions, the Keynesian framework is perfectly consistent with the classical conclusion stated in section III. Suppose we take as a starting-point a given G, and some given combination of T and r consistent with full employment. Suppose further, as is generally assumed in Keynesian analysis, that to a first approximation consumption responds to taxes but not to interest rates. Now let the government increase its expenditure by dG while keeping taxes constant. Then the deficit will increase precisely by $dD = dG$. What will happen to capital formation? If we are to maintain full employment without inflation we must have

$$dG + dC + dI = 0$$

and since, by assumption, taxes are constant and hence $dC = 0$, we must have

$$dG = dD = -dI$$

i.e., the debt-financed expenditure must be accompanied by an equal reduction in capital formation (with the help of the appropriate monetary policy).

This outcome is illustrated by a numerical example in table 3.1. Row (*a*) shows the behaviour of the relevant flows in the initial situation, taken as a standard of comparison: here the budget is assumed to be balanced, although this is of no particular relevance. In row (*b*) we examine the consequence of an increase of expenditure by 100, with unchanged taxes, and hence consumption. The amount of resources available for, and hence the level or private capital formation, is cut down precisely by the amount of the deficit-financed expenditure. It is also apparent that this expenditure puts no burden on the "current" members of the community. Their (real) disposable income is unchanged by the expenditure, and consequently so is their consumption, as well as the net current addition to their personal wealth or private net worth. But because capital formation has been cut by the amount of the deficit, the community will thereafter dispose of a stock of private capital curtailed to a corresponding extent.

Thus the deficit-financed expenditure will leave in its wake an overall burden on the economy in the form of a reduced flow of income from the reduced stock of private capital.

V Interest Charges and the "True" Burden of Debt Financing

The analysis of the last section is seen to agree with the classical conclusion that debt financing transfers the burden of the government expenditure to those living beyond the time of the expenditure. At the same time it indicates that this burden consists in *the loss of income from capital* and not in *the taxes levied on later members to pay the interest charges*, as the classical argument contends.

In some respects this amendment to the classical burden position may be regarded as rather minor, for it can be argued that, under reasonable assumptions, the interest charges will provide a good *measure* of the true burden. Indeed, as

Table 3.1

A. Effects of Government Expenditure and Financing on Private Saving and Capital Formation (Full Employment—All variables measured in real terms)

Method of Financing.	Income, X. (1)	Government Expenditure, G. (2)	Taxes, T. (3)	Disposable. Income, Y. $(X - T)$ (4)	Consumption, C. $(c_0 + cY)$ $(c = 0.6)$ (5)	Saving $S = \Delta W$ $(Y - C)$ (6)	Deficit, D. $(G - T)$ (7)	Private capital formation $I = \Delta k$ $(S - D)$ (8)
(a) Initial situation	2,000	300 (G_0)	300	1,700	1,500 (C_0)	200 (S_0)	0	200 (S_0)
(b) Increased expenditure–deficit financed	2,000	400 $(G_0 + dG)$	300	1,700	1,500 (C_0)	200 (S_0)	100 (dG)	100 $(S_0 - dG)$
(c) Increased expenditure—tax financed	2,000	400 $(G_0 + dG)$	400	1,600	1,440 $(C_0 - cdG)$	160 $(S_0 - sdG)$	0	160 $(S_0 - sdG)$

B. Comparative "Burden" Effects of Alternative Budgetary Policies

Budgetary policy.	Private capital formation.	Effect on	"Burden."
1. Joint effect of increased expenditure *and* deficit financing	$I(b) - I(a)$[1] $= (S_0 - dG) - S_0$	$= -dG$	$r^*(dG)$
2. Joint effect of increased expenditure and taxes	$I(c) - I(a) = (S_0 - sdG) - S_0$	$= -sdG$	$r^*s(dG)$
3. Differential effect of deficit financing	$I(b) - I(c) = (S_0 - dG) - (S_0 - dG) = -(1 - s)dG$	$= -cdG$	$r^*c(dG)$

[1] $I(a)$ means investment in situation (a), and similarly for $I(b)$ and $I(c)$.

long as the amount *dD* is not large in relation to the stock of capital (and the flow of saving), the loss in the future stream of output will be adequately approximated by $r^*(dD)$, where r^* denotes the marginal productivity of capital. Now if the government borrows in a competitive market, bidding against other seekers of capital, then the (long-term) interest rate r at which it borrows will also be a reasonable approximation to r^*. Hence the annual interest charges $r(dD)$ will also be a good approximation to the true social yearly loss, or opportunity cost, $r(dD)$[11]—provided we can also establish that the initial reduction in the stock of capital will not be recouped as long as the debt is not repaid.

One can, however, think of many reasons why the interest bill might not even provide a good *measure* of the true loss. To begin with, if the government operation is of sizeable proportions it may significantly drive up interest rates, say from r_0 to r_1, since the reduction in private capital will tend to increase its marginal product. In this case the interest on the debt will overstate the true burden, which will lie somewhere between $r_0(dD)$ and $r_1(dD)$. More serious are the problems arising from various kinds of imperfections in the commodities as well as in the capital markets. In particular, the government may succeed in borrowing at rates well below r^* by forcing banks and other intermediaries to acquire and hold bonds with yields below the market, or, as in war-time, by effectively eliminating the competition of private borrowers. In the first-mentioned case, *e.g.*, we should add to the interest bill the lost income accruing to the bank depositors (or at least to the bank's stockholders). There is also the rather nasty problem that, because of uncertainty, the rate of interest may not be a good measure of the productivity of physical capital. To put it very roughly, r is at best a measure of return net of a risk premium, and the actual return on capital is better thought of as a random variable whose average value is, in general, rather higher than r.[12]

Besides the relation of r to r^* there is another problem that needs to be recognised. In our discussion so far, and in our table, we have assumed that consumption, and hence private saving, were unaffected because taxes were unchanged. But once we recognise that the borrowing may increase the interest rate, we must also recognise that it may, through this route, affect consumption even with unchanged taxes. This problem might well be quickly dismissed under the principle of "*de minimis.*" For, though economists may still argue as to whether an increase in interest rates will increase or decrease saving, they generally agree that the effect, if any, will be negligible.[13] But even if the rate of saving were to rise from, say, S_0 to $S_0 + e$ and the level of capital formation were accordingly reduced only by $dD - e$, one could still argue that r^*dD and not $r^*(dD - e)$ is the relevant measure of the true loss to society. This is because, as suggested by Bowen *et al.*,[14] the income generated by the extra capital formation e may be quite appropriately regarded as just necessary to offset the extra sacrifice of current consumption undertaken by those who responded to the change in r.

Thus it would appear that the classical-burden position needs to be modified to the extent of recognising that the burden of deficit financing consists not in the increased taxes as such, but rather in the fall in income generated by the reduction in the stock of capital. But this modification would seem rather innocuous, since, admittedly, rdD will generally provide a reasonable approximate *measure* of the true burden. In fact, however, the amendment we have suggested turns out to have rather far-reaching implications as we will show presently.

VI Shortcomings of the Classical Transfer and Burden Argument: The Differential Effect of Deficit Versus Tax Financing

The classical conclusion that deficit financing of an expenditure places the burden on the future seems to imply that, if the expenditure were financed by taxes, there would be no burden in the future. Interestingly enough, Buchanan's book provides nowhere a systematic treatment of the temporal distribution of the burden from a tax-financed expenditure. Nor is this really surprising, for if the burden were in fact the interest of the debt, then tax financing could generate no burden on the future.[15] But if the relevant criterion is instead the loss of capital formation, then in order to find the true differential effect of debt financing versus tax financing, we must inquire about the effects of tax financing on private saving and capital formation. Only if this effect were nil or negligible would the classical conclusion be strictly valid.

Now, to an economist steeped in the Keynesian tradition, it is at once obvious that raising taxes to finance the government expenditure cannot fail to affect significantly private saving and capital formation. While tax financing will reduce disposable income by the amount of the expenditure, it will reduce consumption only by an amount $cdT = cdG$, where c is the marginal propensity to consume. The rest of the tax will be borne by a reduction in saving by sdT, where $s = 1 - c$ is the marginal propensity to save. Accordingly, if the initial position was one of full employment, as we are assuming, and inflation is to be avoided, private capital formation must itself be reduced by the amount sdG (through the appropriate monetary policy).[16] This outcome is illustrated numerically in row (c) of table 3.1. By comparing the outcomes (a), (b) and (c) as is done in part B of the table, we find that the differential effect of the deficit versus tax financing is to decrease capital formation by $dG - sdG = cdG$. The balance of the reduction, namely sdG, must be attributed to the expenditure as such, independently of how financed.[17] Hence, even if we are willing to regard the interest rate paid by the government as a good approximation to r^*, the differential burden of debt financing on the future generations is not rdG but only $rcdG$.

It can readily be seen that the above result is not limited to the case we have explicitly discussed so far, in which the deficit arises from an increase in expenditure. If, for whatever reason, a portion dD of the government expenditure is not financed by taxes but by deficit, capital formation tends to be reduced by approximately $c(dD)$. This conclusion is, however, subject to one important qualification, namely that for $T = \overline{G}$, *i.e.*, with a level of taxation balancing the budget, there exists a monetary policy capable of achieving full employment—or, in terms of our previous notation, of enforcing the required rate of interest $r(\overline{T})$. When this condition is satisfied we shall say that there is a "potentially adequate private demand," or more briefly, an "adequate demand." We shall for the moment concentrate attention on this case, reserving the task of examining the implications of a lack of adequate demand to a later point.

Our result so far, then, is that even with an adequate demand, the net or differential burden placed on the future by debt financing is not nearly as large as suggested by the classical conclusion. But note that the implied error is poor consolation for the no-transfer proponents, for they maintained that the burden is always "paid as you go." The error we have uncovered would seem to lie instead in not recognising that a part of the burden of the expenditure is always shifted to the future. This last conclusion, however, is somewhat puzzling and disquieting. And this uneasiness can be easily increased by asking ourselves the following embarrassing question: roughly how large is the coefficient s which determines the unavoidable burden on the future? This question is embarrassing because recent empirical as well as theoretical research on the consumption function suggests that the answer depends very much on the length of time which is allowed for the adjustment. In the long run, the average propensity to save has remained pretty constant in the general order of 0.1, meaning that the marginal propensity is of the same order. But the quarterly increase in saving associated with a quarterly increase in income seems to be of a much larger order of magnitude, with estimates ranging as high as 0.5 and even higher.[18] Which is the relevant figure and why? Or does the answer depend on whether we look only at the impact effect of taxation or also at its delayed and ultimate effects? We will argue that this is indeed the case, and that in so far as we are interested in the distribution of the burden over time and between generations, the total effects are paramount.

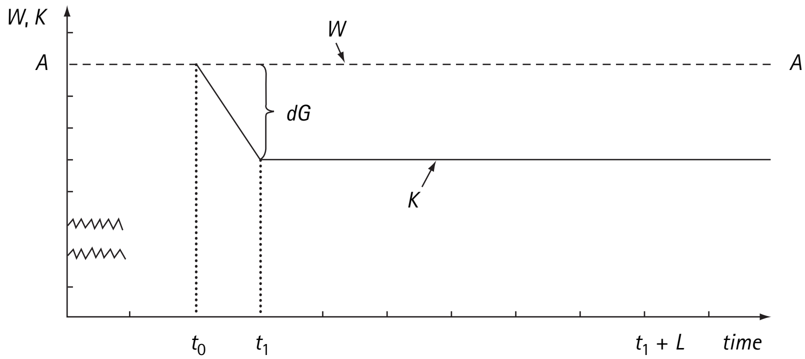
VII Impact Versus Total Effects of Deficit and Tax Financing

Let us come back to a comparison of rows (b) and (c) in table 3.1, but this time let us concentrate on the effect of taxation on the terminal net worth position of the households. We can see that if the expenditure is debt-financed this terminal position is (at least to a first approximation) the same as if the expenditure had not been undertaken. On the other hand, in the case of tax financing, in addition to the concomitant fall in consumption, we find that saving, and hence the increase in net worth, was also cut down from 200 to 160. What effect will this have on later consumption and saving behaviour?

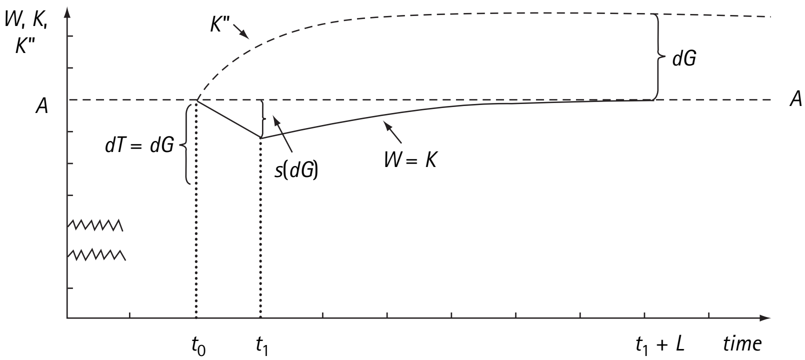
In order to answer this question we need to go beyond the standard Keynesian emphasis on current flows and try to understand why consumers wanted to add 200 to their net worth in the first place and how they will react when this goal has to be scaled down in response to higher taxes. I have elsewhere proposed some answer to these questions by advancing the hypothesis that saving (and dissaving) is not a passive reaction to income but represents instead a purposive endeavour by the households to achieve a desirable allocation of resources to consumption over their lifetime.[19] However, in what follows we do not need to rely, to any significant extent, on that model or any other specific theory of saving behaviour. All that we need to keep before our eyes is the logical proposition that there are, in the final analysis, only two ways in which households can dispose of any addition to their net worth achieved through current saving: namely, either through later consumption or through a bequest to their heirs.

Now let us suppose at first that the bequest motive is small enough to be neglected and that, as a rule and on the average, each household tends to consume all of its income over its lifetime. This assumption can be conveniently restated in terms of the notion of the "over-life average propensity to consume" (*oac*), defined as the ratio of (the present value of) life consumption to (the present value of) life resources, and of the "over-life marginal propensity to consume" (*omc*), defined as the ratio of marginal increments in the same two variables. With this terminology, our current assumption is that both *omc* and *oac* are unity. It should be noted that, under reasonable hypotheses about the typical individual life cycle of earnings and consumption, an *oac* of unity for each household is consistent with a sizeable stock of aggregate assets, in the order of several times national income. With a stationary population and unchanged technology—stationary economy—this aggregate stock would tend to be constant in size, implying zero net saving in the aggregate, but it would be undergoing a continuous reshuffling of ownership from dissavers, such as retired persons, to those in the process of accumulating assets for retirement and short-run contingencies. On the other hand, with a rising population and/or technological progress, there would tend to be positive saving and a rising stock; in fact, the ratio of saving to income and of assets to income would tend to be constant in the long run if the above two forces resulted in an approximately exponential growth trend for aggregate income.[20]

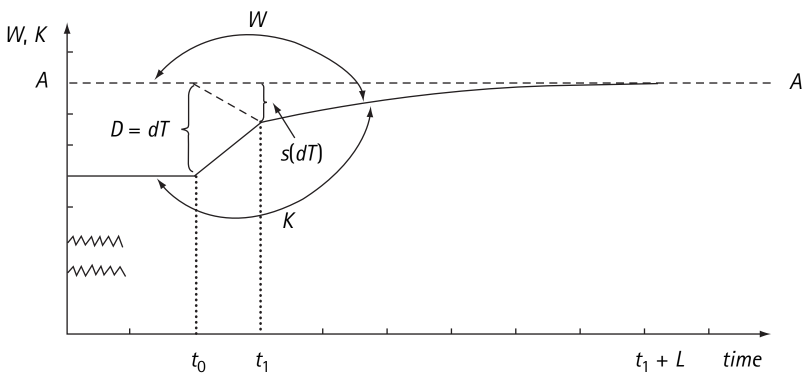
Let us now consider the consequences of a non-repetitive increment in government expenditure *dG*, financed by a deficit, limiting ourselves at first to a stationary economy. Figure 3.1 (*a*) illustrates graphically the effects of this operation



(a) *Deficit Financing*



(b) *Tax Financing*



(c) *Debt Retirement*

Figure 3.1
Effect of deficit and taxes on net worth, *W*, and capital, *K* (unity over-life propensity to consume)

on aggregate private net worth W, and on the net stock of privately owned capital K. The horizontal dashed line AA represents the behaviour of net worth in the absence of dG. It is constant by our assumption of a stationary economy, implying zero net saving, or gross saving and gross investment just sufficient to offset the wear and tear of the capital stock. If we make the further convenient assumption that there is initially no government debt (and ignore non-reproducible tangible wealth), then W coincides also with K. The incremental expenditure dG is supposed to occur in the interval t_0 to t_1 at the constant rate $dG/(t_1 - t_0)$, and is financed by tapping a portion of the gross saving otherwise devoted to capital maintenance. As a result, between t_0 and t_1 K falls, as shown by the solid curve. But the net worth W remains at the same initial level as the fall in K is offset in the consumers' balance sheet by the government debt of dG. By t_1 the gap between W and K amounts to precisely dG, and thereafter the curves remain unchanged until and unless some further disturbance occurs. The final outcome is that the debt-financed expenditure, by generating a permanent wedge $dG = dD$ between W and K,[21] causes the entire cost of the expenditure to be borne by those living beyond t_1 in the form of a reduction in the stock of private capital by dG and in disposable income by r^* (dG).[22] If, in addition, $r^* = r$, then before-tax income will be unaffected and the fall in disposable income will equal the tax collected to pay the interest, as claimed by the classical-burden doctrine.[23]

Consider now the effect of full tax financing of dG, illustrated in figure 3.1 (*b*). The line AA has the same meaning as before. The impact effect of the tax-financed expenditure—*i.e.*, the effect within the interval t_0t_1—is to reduce consumption by cdG and saving and private capital formation by sdG. Hence, as shown by the solid line, by t_1 both W and K have fallen by sdG. As we had already concluded, this fall in K partly shifts the effect of the expenditure to those living beyond t_1. However, by following up the delayed effect of the tax, we can now show that in this case: (*a*) the shift of the burden is only temporary, as W, and hence K, will tend to return gradually to the original pre-expenditure level, and (*b*) the burden transferred to the period following t_1 is borne, at least to a first approximation, entirely by those who were taxed between t_0 and t_1.

To establish this result we need only observe that since those taxed have suffered a loss of over-life (disposable) income amounting to dG as a result of the tax, they must make a commensurate reduction in over-life consumption. If the consumption is cut initially only by $c(dG)$ the balance of the tax, or $s(dG)$, is financed out of a reduction in accumulation—including possibly borrowing from other households—which at first reduces the net worth at time t_1 by $s(dG)$, but eventually must be matched by a corresponding reduction of consumption over the balance of the life span. Let L denote the length of time until the taxed generations die out. In the interval t_1 to $t_1 + L$, then, the consumption of this group will be smaller relative to its income, and hence its rate of saving will be larger, or its rate of dissaving smaller, than it would have been in the absence of the tax. On the other hand, over the same interval, the income consumption and saving of households who have entered the scene after t_1 will be basically unaffected by the operation. Hence, in the L years following t_1 there will arise some positive saving which will gradually die down as the taxed generation disappears. The precise path of aggregate saving will depend on the way the taxed generation chooses to distribute the reduction of consumption over its life. But, in any event, the cumulated net saving over the entire interval t_1 to $t_1 + L$ must come to precisely $s(dG)$, representing the required reduction of consumption relative to income of the taxed generation.[24] This cumulated saving is just sufficient to make up for the initial fall in the stock of $s(dG)$, so that by $t_1 + L$ the stock of capital (as well as W) has returned to the original level, as shown in figure 3.2 (*b*), and we are back in the original stationary state.

The above framework can be readily applied to analyse the effects of deficit or surplus generated under different conditions, *e.g.*, by varying taxes, expenditure constant. Figure 3.1 (*c*), for instance, depicts the outcome of an increase in taxes in the interval t_0 to t_1, utilised to retire the debt D outstanding at t_0. Here again the entire burden of the retirement falls on the taxed generation—although it is spread between t_0 and $t_1 + L$—and the gain accrues to those living after t_1 in the form of an increase in the stock of capital by an amount which eventually approaches the amount of debt retired and reflects the elimination of the wedge between W and K.

It is also easy to verify that our results remain valid for a growing economy, the only difference being that the dashed line AA would turn into an upward-sloping curve. With debt financing the graph of K would, from t_1 on, run at a distance dG below this line, while with tax financing the graph of $K = W$ would initially fall below it by $s(dG)$, but would tend to return to it at $t_1 + L$.

In summary, then, under unit *oac* the cost of an expenditure financed by debt, whether internal or external, tends to fall entirely on those living beyond the time of expenditure, as asserted by the classical-burden position, though it is best measured by r^*dD rather than by the incremental tax bill rdD. This burden may be eliminated at a later date by retiring the debt through a budget surplus, but thereby the full cost of the original expenditure is shifted to the later tax-payer, who financed the surplus. On the other hand, the cost of a tax-financed expenditure will tend to be borne by society as a whole, partly at the time and partly for some finite period thereafter. But the burden beyond t_1 still falls primarily on those who initially paid the tax and reflects the spreading of the burden over their lifetime.[25]

In the analysis so far we have concentrated on examining who bears the cost of the expenditure. To complete the picture we must, of course, also reckon the

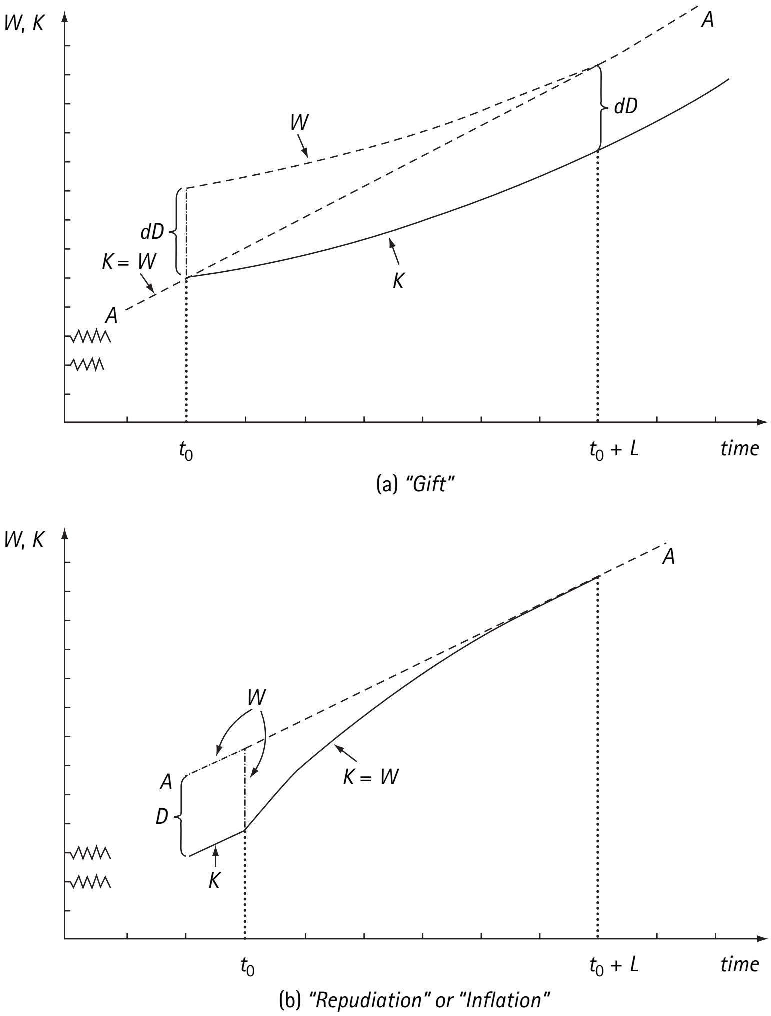


Figure 3.2
Effect of "gratuitous" changes in debt on net worth, *W*, and capital, *K* (unity over-life propensity to consume)

yield, if any, produced by dG beyond t_1. In particular, if dG results in a (permanent) addition to the stock of capital we must distinguish between W, K and K^*, the latter denoting the total stock of private-plus government-owned capital. K^* will exceed K by dG. Thus in the case of a debt-financed capital expenditure, K^* will everywhere tend to coincide with W, the government capital formation simply replacing the private one. For the case of tax financing, the behaviour of K^* is shown by the broken line in figure 3.1 (*b*). Here the burden on the taxed generation results in a permanent gain for those living beyond t_1, which will gradually approach the yield on dG. In this sense one might well say that the cost of current government services can be paid for not only by the current and future generations but even by past generations.

There remains to consider how far our conclusions need to be modified if the *omc* is less than unity. Since a debt-financed expenditure does not affect the behaviour of net worth, our analysis of this case basically stands, although one can conceive of some rather fancy circumstances in which modifications would be necessary.[26] In the case of tax financing, however, an *omc* of less than one implies that part of the burden of the expenditure will fall on later generations, who will receive a smaller bequest. It can be readily seen that the reduction in $K = W$ available to them will be of the order of (*oms*) (dG), where *oms* denotes now the over-life marginal propensity to save. The differential burden of debt versus tax financing on society will correspondingly be of the order of r(*omc*) (dG) instead of rdG.[27] In other words, the propensities to consume and save relevant to the long-run effect are precisely the over-life ones.[28] Unfortunately, these are propensities about which information is currently close to zero, both in terms of order of magnitude and stability, although some attention has begun to be devoted to this question.[29]

Our analysis of the differential burden of tax versus debt financing could stand a great deal of refinement and qualifications to take proper account of the specific nature of the taxes levied to pay for dG or for the interest on the debt. But it is clear that these refinements can in principle be handled by proper application of the framework set out in this section. We shall therefore make no attempt at working out a long list of specific cases,[30] and will proceed instead to point out the implications of our framework for a somewhat different class of problems, namely where the change in debt occurs without any accompanying change either in government purchases or taxation.

VIII "Gratuitous" Increases in Debt, Repudiation, and Inflation

For analytical convenience we may start out by considering a case which has admittedly rather limited empirical relevance: namely, where the government debt is increased at some date t_0 by an amount dD by a "gratuitous" distribution of a corresponding amount of bonds.[31] Presumably, at least in this case, the proponents of the classical burden argument, if they apply their reasoning consistently, would have to agree with the proponents of the classical non-burden argument that the increment in the national debt puts *no burden on the economy as a whole*, either at the time of issuance or thereafter, except for frictional transfer costs. For while it is true that from t_0 on, tax-payers are saddled with extra taxes to defray the interest bill, it is also true that the receipt of interest would not have arisen without the creation of extra debt. Note that this conclusion does not rule out the possibility that the operation may result in some transfer between generations, if by a generation we mean the set of members of the economy born at a particular date: thus the interest accruing to those receiving the gift will very likely be paid, at least partly, by a younger generation. But these are still mere income transfers involving no overall burden.

But once we recognise that the overall burden of the national debt derives from its effects on the private stock of capital, then it becomes clear that, by and large, both classical doctrines agree with the wrong conclusion. This is indeed quite obvious in the case of a unity *oac*, a case illustrated graphically in figure 3.2 (*a*). The solid line *AA* shows as usual the behaviour of $W = K$ in the absence of the gift. For the sake of variety, we deal here with a growing economy with positive saving and rising wealth.[32] If the gift is distributed all in one shot at point t_0, then at that point *W* will rise by *dD*, with no change in *K*. But now between t_0 and $t_0 + L$ the members of the generation that received the gift will gradually dispose of the bonds (or of other assets that they may have exchanged against the bonds) by selling them to current savers and using the proceeds to increase consumption over what it would have been otherwise. As this takes place, the aggregate rate of saving, and hence the accumulation of net worth and capital, is reduced. The result is that *W* gradually approaches the base path *AA*, while *K*, which is always lower by the wedge *dD*, falls below it. By $t_0 + L$ the cumulated rate of saving and physical capital formation will have been reduced by (approximately) *dD*, so that *W* will tend to coincide with *AA* and *K* to be *dD* lower, a position that will tend to be maintained thereafter. Thus an increase *dD* in the national debt, even if it arises from a free gift, will put a burden on the economy as a whole. Under unity *oac*—after a transient period in which *W* is increased as a result of the gift—this burden will approach the level $r^*(dD)$, and hence approximately equal the interest on the debt.[33]

If the *omc* is less than unity, then the burden will be smaller, tending to (*omc*) (r^*dD), because the gift will tend to increase *W* "permanently" by (*oms*) (*dD*) and hence *K* will tend to fall only by (*omc*) (*dD*). As usual, this burden can be removed at any later point by taxation and retirement of the debt, but only at the cost of putting the burden on the taxed generation, which in effect will be paying for the benefits enjoyed by the beneficiaries of the gift.

Our conclusion applies directly, but for an appropriate change of "sign," to the case of a "gratuitous" one-shot reduction in the national debt, as indicated in figure 3.2 (*b*). Such a reduction might be accomplished by repudiation, total or partial, or by a capital levy, or, much more importantly and frequently, by the simple device of (unanticipated) inflation. Thus a (once and for all) doubling of the price level is entirely equivalent to a repudiation of one-half of the national debt at the original price level—although it has, of course, all kinds of other widespread economic effects. As far as the national debt is concerned, this operation puts a burden on the owners of the bonds by reducing the real value of their interest income as well as the real value of the principal. In so far as the first effect is concerned, we have a mere transfer from the bond-holders to the tax-payers, with no overall effect. But the reduction in the principal generates an unmatched reduction in consumption, and hence a *transient* higher rate of saving. The resulting increase in the capital stock will benefit all those living after the inflation—provided, of course, private capital has a positive marginal product and the potentially higher rate of saving is utilised for capital formation rather than being wasted in depressed income and unemployment.

From the content of this section and the previous one it should also be apparent that our analysis basically supports Meade's conclusion concerning the burden of the deadweight debt, although this conclusion is derived by a very different line of reasoning. The deadweight debt is a burden because: (*a*) it generates a corresponding gap between aggregate net worth W and the aggregate stock of capital K^*, and (*b*) we can expect that this gap will result primarily in a fall in K^* rather than in an offsetting rise in W. Thus, if we conceive two communities A and B identical with respect to natural endowments, technical know-how and habits of private thrift, and which differ only in that A has a deadweight debt D' and B has none, community A will be poorer, roughly, by D' times the marginal productivity of capital plus frictional transfer costs.

IX Deficit Financing in War and in Depression

In this concluding section we propose to apply our tools to see what light they shed on the two classical and empirically relevant issues: the pre-Keynesian problem of war financing and the post-Keynesian problem of deficit created as part of a counter-cyclical stabilisation policy.

In order to face squarely the core issue of war financing, let us be concerned with the case of a major war effort of the type outlined earlier in section III, in which the stock of capital in existence at the termination of the war is independent of the methods used to finance it. It follows immediately that, independently of financing, the war will impose a burden on post-war society as a whole to the extent that the stock of capital in existence at its termination—counting only whatever portion is useful to satisfy the post-war requirements—is less than what would have been there in the absence of war. In order to examine the residual effects, if any, of methods of financing, we must suppose that, in spite of the war's severity, we have some choice as to the extent of taxation. Also, to keep separate for the moment the possible role of inflation, we must suppose that untaxed income in excess of the predetermined consumption level is prevented from bidding up the price of goods—whether through voluntary abstention or through a fully successful system of rationing. In these conditions the unspent income must result in an increase in government debt held by the private sector, either directly or through financial intermediaries. Thus the level of taxation versus deficit financing determines essentially the size of the increment in government debt, *dD*. Now suppose, for the sake of the argument, that the war had been entirely financed by taxes, *dD* being accordingly zero. It then follows from our earlier analysis that the burden of the war will be borne almost entirely by members of the war generation. For, in addition to the sacrifice of consumption and other amenities *during* the war, they will also bear the brunt of the reduced capital stock, their accumulation of net worth being limited to the permitted privately financed capital formation. Thus the burden falling on society as a whole after the war will fall primarily directly on the members of the war generation (affecting others only to the extent that the reduction in the stock of capital reduces total income by more than the return on capital). They will be forced after the war to maintain a reduced level of consumption over the rest of their life, tending to save heavily in their remaining earning span and to dissave at a greatly reduced rate thereafter. This behaviour in turn will produce, after the war, an abnormally large rate of aggregate saving, gradually declining with the disappearance of the war generation. As a result, by the time the war generation has disappeared, the war-time reduction in capital formation may have been substantially made up—this being more nearly true the closer the *oac* is to unity and the smaller the initial loss of capital.

If, on the other hand, through lower taxes the war generation is permitted to increase its terminal net worth by an additional amount *dD*, the effect, with respect to the present issue, is essentially the same as though at war's end it had been handed down gratuitously a corresponding amount of government bonds.[34] As usual, this will enable them to maintain a higher post-war consumption, reducing capital formation, by an extent that can range as high as *dD*, if the *oac* is unity. Thus the debt financing will generate both: (i) a transfer from the post-war to the war generation to the extent of taxes levied on the former to pay interest to the latter, and (ii) a permanent burden on society as a whole to the extent that the stock of capital is permanently reduced by *dD*—less any increase in *W* resulting directly from *dD*.[35] In so far as in the immediate post-war period the government, to speed up the reconstruction, pushes capital formation by raising taxes and creating a surplus, the long-run effect is eliminated. But the burden of debt financing is placed to that extent on those living in the relevant post-war period, which may again consist in part of the very same war generation.

If inflation is permitted to develop in the course of the war or immediately following it our analysis remains valid, provided the increment in the debt is measured in real, rather than money, terms. This net real increment can be expressed as

$$\frac{D_0 + dD}{1 + dP} - D_0,$$

where D_0 is the pre-war debt and dP is the relative increase in the price level in the course of the war inflation. The above quantity, it will be noted, may even be negative if

$$dP > \frac{dD}{D_0},$$

i.e., if the increase in prices exceeds the relative increase in the debt. In this case the war generation will be made to carry even more than the cost of the war (unless its plight is improved by post-war transfers of income); and later generations may conceivably end up by benefiting from the war, at least following the transient period of high saving rates and rapid capital accumulation. Perhaps the picture we have been drawing has some relevance for an understanding of the post-war experience of such countries as Germany, Italy, and Japan.

It seems hardly necessary to point out that our analysis in no way implies that in financing a war the use of debt should necessarily be minimised. Quite aside from obvious incentive considerations, there may be perfectly good equity reasons for lightening the burden of the generation that suffered through the war by granting them a more comfortable life after the war, at the expense of later generations.

We come finally to the effects of debt generated as a counter-cyclical measure. In view of the complexity of the problem, we shall have to limit ourselves to a sketchy treatment of a limited class of situations. Our main concern is to show that, even in this case, debt financing, though quite advantageous to the current generation, will generally not be costless to future generations, at least in terms of gross burden.

Consider a situation where, in spite of the easiest possible monetary policy and with the whole structure of interest rates reduced to its lowest feasible level, the demand for private capital formation is inadequate to absorb full-employment saving with a balanced budget. But let us suppose that the situation can be counted upon to be temporary and not to recur for a long time to come. If the government does not step in there will be a temporary contraction of employment accompanied by a contraction of consumption and of addition to net worth, which is limited to the amount of private capital formation. Suppose, on the other hand, the government expands its expenditure to the extent needed to fill the deflationary gap, and thereby runs into a deficit *dD*. Let us also imagine that it succeeds in choosing its action perfectly so as to maintain full employment without inflation. Hence consumption will be kept at the full-employment level and so will the accumulation of net worth; except that this accumulation will now take the form of an addition to the national debt to the extent *dD*. Thus the government action involves a current gain to society which can be measured by the income which would have been lost otherwise. What we wish to know is whether this gain places any cost on later generations (and if so, how it can be valued).

Under the assumed conditions the answer would have to be affirmative at least under unity *oac*. In this case, in fact, we can see that the cost which was spared to society would have fallen entirely on the members of the depression generation. They would have been forced over their lifetime to cut their consumption by an amount (whose present value is) equal to the lost income. This reduction would be distributed partly within the depression but partly *after* the recovery, to an extent equal to the loss in accumulation of net worth in the depression. This reduction of consumption would in turn give rise to a somewhat higher rate of capital formation after the recovery, so that by the time the depression generation disappears the stock of capital would tend to be back to where it would have been without depression. In other words, under the assumed conditions failure of the government to act, though costly to the depression generation, tends to be costless to later generations. On the other hand, if the government acts, the depression generation does not have to maintain a lower consumption after the recovery, and accordingly, the lost private capital formation during the depression is never made up. The creation of *dD* introduces again a corresponding wedge between *W* and *K* which will tend permanently to reduce the amount of physical capital available to future generations. Hence there is a loss to them to the extent that at later points of time an increment to the stock of capital would make any net positive addition to output. If the debt is never meant to be retired, then at least with well-functioning capital markets, the consol rate, being an average of anticipated future short rates, may provide at least a rough measure of the (appropriate time average) annual loss. And in this sense if the government borrows long, the interest bill on the incremental debt may provide a rough measure of the average future (gross) burden placed on society as a whole.[36]

Once more, recognising that the government action may involve a gross cost to future society does not imply that the action should not be taken. In the first place, because of multiplier effects the gain in income to those present is likely to be appreciably larger than the lost stock of capital which approximates the present value of the sacrificed income stream. In the second place, if the government spends on projects which produce a yield in the future, then the gross burden will be offset by the gross yield and the net outcome may even be positive. In the third place, the gross burden can be eliminated if at some future point of time the government runs a surplus and retires the debt. It is true that this will tend to place the burden of the original deficit on those who pay the taxes financing the surplus. But if the surplus follows the deficit in short order these people will be, to a large extent, the very same ones that benefited from the original deficit; thereby the questions of inter-generation equity are minimised. The case for eradicating the deficit with a nearby surplus is, of course, strongest if the government expenditure provides nothing useful for the future, or if the deficit reflects a reduction in the tax bill, expenditure constant, resulting either from built-in flexibility arrangements or from *ad hoc* tax rebates. Thus, our analysis turns out to provide a strong case in favour of what used to be called the cyclically balanced budget.

Although we cannot pursue further here the complex issues raised by the burden aspects of counter-cyclical fiscal operations, we hope to have succeeded in showing how the tools we have developed may provide some insight into the problem. One point should be apparent even from our sketchy treatment: namely, that in so far as counter-cyclical fiscal policy is concerned, our analysis does not require any significant re-evaluation of currently accepted views. Yet, by reminding us that fiscal operations involve considerations of inter-generation equity even when used for stabilisation purposes, it may help to clarify some issues. It does, for example, establish a *prima facie* case, at least with respect to *ad hoc* measures as distinguished from built-in stabilisers, for a course of action that will minimise the "deadweight" deficit and stimulate investment rather than consumption.[37] More generally, considerations of inter-generation equity suggest the desirability of a compromise between the orthodox balanced-budget principle and the principle of functional finance, which might be couched roughly as follows: as a rule, the government should run a "deadweight" deficit only when full-employment saving exceeds the amount of capital formation consistent with the most favourable feasible monetary policy; and it should run a surplus, in so far as this is consistent with full employment, until it has wiped out previous deficits accumulated in the pursuance of this policy.

Notes

1. J. M. Buchanan, *Public Principles of the Public Debt* (Homewood, Illinois: Richard D. Irwin, 1958).

2. J. E. Meade, "Is the National Debt a Burden?" *Oxford Economic Papers*, Vol. 10. No. 2, June 1958, pp. 163–83, and "Is the National Debt A Burden: A Correction," *ibid*., Vol. 11, No. 1, February 1959, pp. 109–10.

3. R. A. Musgrave, *The Theory of Public Finance* (McGraw-Hill, 1959), especially chapter 23. Other recent contributions include: the reviews of Buchanan's book by A. P. Lerner, *Journal of Political Economy*, Vol. 47, April 1959, pp. 203–6; E. R. Rolph, *American Economic Review*, Vol. 49, March 1959, pp. 183–5, and A. H. Hansen, *Review of Economics and Statistics*, Vol. 41, June 1959, pp. 377–8; also "The Public Debt: A Burden on Future Generations?" by W. G. Bowen, R. G. Davis and D. H. Kopf, *American Economic Review*, Vol. 50, September 1960, pp. 701–6; and the forthcoming note by A. P. Lerner, "The Burden of Debt" *Review of Economics and Statistics*, Vol. 54, No. 2, May 1961.

4. This definition implies that the national debt could in principle be negative. Even in this case we shall refer to it by the same name, although its magnitude will be expressed by a negative number. Similarly, we refer to an operation that reduces the algebraic value of the national debt as a "reduction," even if the debt was initially zero or negative.

5. The difference between the increase in the national debt in a given interval and the government expenditure contributing to future income corresponds roughly to the net increase in what Professor Meade has called the "deadweight" debt. Cf. *op. cit*.

6. The reader interested in establishing just who said what will find much useful material in Buchanan, *op. cit*., especially chapters 2 and 8, and in B. Griziotti, "La diversa pressione tributaria del prestito e dell'imposta" in *Studi di scienza delle finanze e diritto finanziario* (Milano: Giuffre, 1956), Vol. II, pp. 193–273.

7. See, *e.g*., J. E. Meade, "Mr. Lerner on 'The Economics of Control,'" *Economic Journal*, Vol. LV, April 1945, pp. 47–70.

8. See, *e.g*., "Functional Finance and the Public Debt," *Social Research*, Vol. 10, No. 1, and "The Burden of the National Debt" in *Income Employment and Public Policy* (New York: Norton & Company, 1948).

9. Meade, *op. cit*.

10. See, *e.g*., the references in note 7 above, and in Buchanan, *op. cit*., p. 14, note 8.

11. This is precisely the position taken by Musgrave, *op. cit*., p. 577.

12. Cf. F. Modigliani and M. H. Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review*, Vol. 58, No. 3, June 1958, pp. 261–97. However, Miller has suggested to me that r may be the more relevant measure of return on capital as it deducts an appropriate allowance for the "cost" of risk bearing.

13. This is especially true if current consumption is appropriately defined to include the rental value and not the gross purchase of consumers' durables.

14. Bowen, Davis and Kopf, *op. cit*., p. 704.

15. See, however note 1, p. 746, for a different explanation of Buchanan's omission.

16. The need to curtail investment when government expenditure is increased at full employment, even though it is fully tax covered, is the counterpart of the so-called multiplier effect of a balanced budget when starting from less than full utilisation of resources. The tax-financed expenditure *per se* increases the aggregate real demand for goods and services by a dollar per dollar of expenditure. But if we start from full employment, this extra demand could only result in inflation. Hence it must be offset by a fall in investment of s dollars per dollar of expenditure, which, taking into account the multiplier effect, will reduce total demand by $s/s = 1$ dollar per dollar, as required.

17. This conclusion has also been reached by W. Vickrey, *op. cit*.

18. See, *e.g*., the following two recent and as yet unpublished studies prepared for the Commission on Money and Credit: D. B. Suits, "The Determinants of Consumer Expenditure: a Review of Present Knowledge"; and E. C. Brown, R. M. Solow, A. K. Ando and J. Kareken, "Lags in Fiscal and Monetary Policy," Part II.

19. F. Modigliani and R. Brumberg, "Utility Analysis and the Consumption Function: An Interpretation of Cross-section Data," in *Post-Keynesian Economics*, K. Kurihara, ed. (Rutgers University Press, 1954).

20. Cf. A. K. Ando and F. Modigliani, "Growth, Fluctuations and Stability," *American Economic Review*, May 1959, pp. 501–24.

21. Permanent in the sense that it persists as long as the debt remains outstanding.

22. Actually the fall in disposable income consequent upon the fall in K is likely to give rise to a further fall in W and hence in K, but this indirect effect will tend to be of secondary magnitude. See on this point note 33.

23. If the reduction in K results in a significant rise in r^* and hence r, then, as pointed out by Vickrey (*op. cit*., p. 135), there will tend to occur a shift in the distribution of *pre-tax* income. Labour income will tend to shrink and property income to increase—and, incidentally, this increase will tend to more than offset the fall in labour's earnings. It does not follow, however, as Vickrey has concluded, that the "primary burden of diminished future income will be felt by future wage earners." For the burden consists in the reduction of *disposable* income, and this reduction will depend on the distribution of the taxes levied to pay the interest as between property and non-property income.

24. It may be noted that the cumulated reduction in consumption will tend to be somewhat larger than $s(dG)$ because the taxed generation will also lose some income as a result of the reduction in their wealth, amounting initially to $s(dG)$. However, the cumulated increase in saving over the interval is still $s(dG)$ because the additional loss in consumption just matches the reduction in income from this source. Actually $s(dG)$ measures essentially the *present value as* of t_1 of the required reduction in consumption.

25. In a stimulating comment to a preliminary draft of this paper, Mr. Buchanan has provided an explanation for his failure to analyse the temporal distribution of the burden of a tax-financed expenditure. He points out that in line with the classic tradition, he defines the burden as the subjective loss of utility suffered by the tax-payer because of the initial loss of resources. The burden in this sense occurs entirely when the tax is levied and the later reduction of consumption cannot be separately counted as burden, as it is merely the embodiment of the original burden. I have serious reservations about the usefulness of this definition. It has, for instance, the peculiar implication that, when as a result of tax financing an heir receives a smaller inheritance or as a result of debt financing he is saddled with a larger tax bill, this cannot be counted as burden on him, as the entire burden is already accounted for in the guise of his father's grief that his heirs will enjoy a smaller net income. It is this peculiar reasoning that underlies Ricardo's famous conclusion that the cost of the government expenditure is always fully borne by those present at the time.

26. It is conceivable that, *e.g*., the tax newly imposed to defray the interest cost might reduce the bequests. To this extent an even greater burden is placed on later generations, which will inherit a smaller K for two reasons: because of the smaller W and because of the wedge dG between W and K. An even fancier possibility is that the new tax might spur the initial generation to increase its bequests to help their heirs pay for the new tax. This would, of course, increase the burden on the current generation and decrease that on posterity.

27. Note that, regardless of the value of *omc*, the current generation must always fare at least as well, and generally better, under debt than under tax financing, even if it capitalised fully the new taxes that must be raised to pay the interest bill on the new debt. For, even in the highly unlikely event that the amount $r(dD)$ per year necessary to pay the interest bill were levied entirely on the initial generations, as long as they lived, this liability is limited by life, and hence represents a finite stream whose present value must be less than the amount dD which would have been taken away in the case of tax financing. See on this point also note 25.

28. Even with an *omc* of less than unity it is likely that the impact of the tax on bequests handed down from one generation to the next would gradually disappear so that W and K would eventually be unaffected by the tax-financed expenditure. But this is in the *very* long run indeed.

29. See, *e.g.*, J. Tobin and H. W. Guthrie, "Intergenerations Transfers of Wealth and The Theory of Saving," Cowles Foundation Discussion Paper No. 98, November 1960.

30. By so doing we are also deliberately by-passing the other major issue of fiscal policy, that of the distribution of the burden between income classes.

31. In order to avoid side issues, we will assume that the coupon rate on these bonds in such as to make their market value also equal to dD, and that no change occurs in the government purchase of goods and services, G.

32. Just for reference, we may note that according to the Modigliani-Brumberg model, if income were growing at approximately exponential rate, W would be growing at the same rate.

33. This conclusion is strictly valid only in so far as the fall in disposable income brought about by the fall in K is matched by an equal fall in consumption. To the extent, however, that consumption falls somewhat less, cumulated saving may fall somewhat more, pushing W and K to a lower position than in our figure: but this extra adjustment will in any event tend to be of a second order of magnitude. The nature and size of this adjustment can be exhibited explicitly with reference to the Modigliani-Brumberg model of consumption behaviour. As indicated earlier, this model implies that, in the long run, the aggregate net worth of consumers tends to be proportional to their (disposable) income, or (1) $W = gY$, where the proportionality constant is a decreasing function of the rate of growth of income. Suppose initially income is stationary as population and technology are both stationary. We also have the identity (2) $W = K + D$, where D denotes the national debt. With population and technology given, the effect of capital on income can be stated by a "production function" (3) $Y = f(K)$. We have stated in the text that a gratuitous increase in D, or more generally an increase in D which does not result in government capital formation or otherwise change the production function, will tend to reduce K by dD and Y by r^*dD: *i.e.*, we have asserted

$$\frac{dK}{dD} \simeq -1 \text{ and } \frac{dY}{dD} \approx -r^*,$$

where

$$r^* = \frac{df}{dK} = f' \simeq r.$$

By means of equations (1)–(3) we can now evaluate these derivatives exactly. Solving (2) for K and using (1) and (3) we have $K = gf(K) - D$. Hence

$$\frac{dK}{dD} = gf'\frac{dK}{dD} - 1 \text{ or } \frac{dK}{dD} = \frac{-1}{1-gf'} = \frac{-1}{1-gr^*}.$$

Similarly,

$$\frac{dY}{dD} = \frac{-r^*}{1-gr^*} \text{ and } \frac{dW}{dD} = \frac{-gr^*}{1-gr^*}.$$

Thus, if $r^* \simeq r = 0.05$ and g is in the order of 4, then

$$\frac{dK}{dD} \text{ is } -1.25 \text{ instead of } -1 \text{ and } \frac{dY}{dD} \text{ is } -0.625 \text{ instead of } -0.05.$$

I am indebted to Ralph Beals, presently a graduate student at Massachusetts Institute of Technology, for pointing out that these formulæ are not entirely general, for, within the Modigliani-Brumberg model, the second-order effect is not independent of the nature of taxes employed to defray the interest bill. In fact, the formulæ derived above are strictly valid only if the revenue is raised entirely through an income tax on non-property income. With other kinds of taxes, one obtains somewhat more complicated formulæ. For instance, if the taxes are levied on property income this will depress the net yield of wealth below r, which in turn will, in principle, affect the proportionality constant g of equation (1). However, exploration of several alternative assumptions suggests to me that the outcome is unlikely to be appreciably different from that derived above, at least in the case of direct taxes.

It can also be shown that the above formulæ will tend to hold, at least asymptotically, for an expanding economy in which population grows at an approximately constant rate and/or so does productivity as a result of technological change which is neutral in Harrod's sense (*cf. Toward a Dynamic Economics* (Macmillan, 1949), p. 23). The main features of such a growth model are discussed in Ando and Modigliani, *op. cit*.

34. If the bonds issued during the war carried an exceptionally low rate of interest because of the monopoly position of the Government in the market the gift in question should be regarded, for present purposes, as represented by the market value of the bonds.

35. Note that the incremental debt dD could be regarded as a burden on society even if the economy tended to suffer from long-run stagnation, *i.e*., a chronic tendency for a very low or zero marginal productivity of capital. For while it is true that the larger consumption bestowed on the war generation would help to sustain consumption, and thus full employment, the same result could be achieved by reducing the taxes and expanding the consumption and saving of whoever was present at the appropriate later time.

36. Of course, under our present assumptions the burden as measured by the opportunity cost will be essentially zero during the period in which the debt is created, regardless of whether it takes the form of long-term debt, short-term debt or currency creation. But in the last two cases the current interest cost will not appropriately reflect the average future burden, unless we also take into account the rate the government will have to pay on bonds sold at later points of time to refinance the short-term debt or to reduce the money supply in order to prevent inflation.

37. These consideration, *e.g*., cast some doubt on the desirability of relying on personal tax cuts explicitly announced to be but temporary. For there is at least some ground for supposing that the temporary nature of the cut will tend to reduce the desirable impact effect on consumption and increase instead short-run saving and the (possibly) undesirable delayed consumption.

4

RECENT DECLINES IN THE SAVINGS RATE: A LIFE CYCLE PERSPECTIVE

Franco Modigliani

Introduction

In the course of the last three decades, the national saving to income ratio, which has traditionally been regarded as fairly stable at least in the medium run, has undergone profound changes, especially among the industrialized countries. These changes have tended to run in the same direction for all countries, namely down. While this development has been very negative for the world economy, as it has reduced the flow of resources available for investment and generally raised interest rates, it offers a unique opportunity for the scientist trying to validate—or reject—theories of aggregate saving behavior. The main purpose of this paper is to rely on this episode to test whether and how far it can be explained by the so-called Life-Cycle Hypothesis (LCH). We propose to show that, relying on that framework, one can account fairly well for the unusual behavior of saving in recent decades.

1 A Review of Theory and Facts

1.1 Recent Trends of National Saving

Table 4.1 presents the basic statistics on the national and private saving ratio for the 21 OECD countries for which the necessary information could be gathered, for the period 1960–1987. Since we are interested in studying long-run, rather than cyclical, behavior, each observation for every variable in each country is an average of many years: the years ’60 to ’70 for the first period (top of table 4.1); ’71 to ’80 for the second period (middle section); and ’81 to ’87 for the last period. The row at the bottom of each section gives the average for all countries in each period. It is seen that the national saving rate declined from an impressive level

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Table 4.1
Regression Variables: 1960–1970

		NS/NNP	APSYT	GROWTH	GLAGGED	SURPLUS/ NNP	NEWSURP/ NNP	SURPYT	NETTAX/ NNP	NEWTAX	LFWORK	DEP	DEBTAD/ NNP
Canada	1960	0.113	0.103	0.047	0.047	0.028	0.033	0.044	0.225	0.231	0.365	0.552	0.006
United States	1960	0.106	0.107	0.035	0.026	0.013	0.023	0.030	0.215	0.225	0.450	0.500	0.010
Japan	1960	0.256	0.219	0.095	0.081	0.078	0.075	0.090	0.173	0.170	0.571	0.394	–0.003
Australia	1960	0.137	0.093	0.050	0.042	0.053	0.064	0.081	0.203	0.214	0.409	0.478	0.010
Austria	1960	0.182	0.167	0.043	0.094	0.068	0.074	0.113	0.345	0.350	0.512	0.368	0.006
Belgium	1960	0.144	0.143	0.045	0.031	0.013	0.029	0.036	0.176	0.193	0.385	0.375	0.016
Denmark	1960	0.174	0.173	0.048	0.039	0.050	0.044	0.059	0.254	0.249	0.526	0.371	–0.006
Finland	1960	0.157	0.101	0.048	0.047	0.086	0.083	0.114	0.275	0.271	0.627	0.215	0.003
France	1960	0.193	0.192	0.050	0.046	0.045	0.048	0.064	0.245	0.248	0.466	0.412	0.004
Germany	1960	0.199	0.190	0.041	0.076	0.064	0.061	0.083	0.276	0.272	0.486	0.344	–0.003
Greenland	1960	0.153	0.132	0.069	0.059	0.040	0.044	0.054	0.173	0.177	0.387	0.391	0.004
Iceland	1960	0.137	0.045	0.040	0.054	0.104	0.103	0.137	0.247	0.246	0.386	0.597	–0.001
Ireland	1960	0.122	0.130	0.042	0.015	0.007	0.015	0.019	0.174	0.183	0.377	0.538	0.009
Italy	1960	0.208	0.245	0.052	0.057	–0.001	0.010	0.012	0.179	0.190	0.320	0.342	0.011
Netherlands	1960	0.199	0.180	0.046	0.053	0.043	0.066	0.089	0.238	0.260	0.283	0.459	0.022
Norway	1960	0.161	0.123	0.038	0.037	0.071	0.076	0.111	0.306	0.311	0.372	0.395	0.005
Portugal	1960	0.198	0.215	0.058	0.043	0.024	0.020	0.025	0.179	0.176	0.232	0.460	–0.003
Spain	1960	0.167	0.161	0.066	0.036	0.045	0.028	0.033	0.153	0.137	0.263	0.436	–0.016
Sweden	1960	0.166	0.112	0.042	0.034	0.032	0.090	0.134	0.327	0.325	0.537	0.322	–0.002
Switzerland	1960	0.212	0.201	0.043	0.043	0.047	0.049	0.061	0.189	0.191	0.394	0.368	0.002
United Kingdom	1960	0.112	0.085	0.026	0.031	0.032	0.051	0.071	0.260	0.279	0.507	0.364	0.019
Average of all countries		0.166	0.148	0.049	0.047	0.048	0.052	0.070	0.229	0.233	0.422	0.413	0.004

Table 4.1 (continued)
Regression Variables: 1971–1980

		NS/NNP	APSYT	GROWTH	GLAGGED	SURPLUS/ NNP	NEWSURP/ NNP	SURPYT	NETTAX/ NNP	NEWTAX	LFWORK	DEP	DEBTAD/ NNP
Canada	1971	0.133	0.156	0.038	0.047	0.010	0.018	0.024	0.253	0.261	0.505	0.403	0.008
United States	1971	0.089	0.104	0.029	0.035	–0.012	0.007	0.010	0.195	0.214	0.541	0.388	0.019
Japan	1971	0.246	0.243	0.049	0.095	0.046	0.046	0.056	0.176	0.176	0.535	0.354	0.000
Australia	1971	0.110	0.089	0.030	0.050	0.015	0.045	0.060	0.233	0.263	0.493	0.424	0.030
Austria	1971	0.180	0.162	0.035	0.043	0.058	0.073	0.110	0.322	0.336	0.484	0.364	0.014
Belgium	1971	0.139	0.139	0.031	0.045	–0.011	0.033	0.044	0.195	0.239	0.445	0.371	0.044
Denmark	1971	0.133	0.133	0.023	0.048	0.052	0.046	0.071	0.354	0.348	0.636	0.347	–0.006
Finland	1971	0.142	0.117	0.038	0.048	0.075	0.064	0.095	0.342	0.331	0.662	0.327	–0.011
France	1971	0.163	0.161	0.034	0.050	0.036	0.048	0.067	0.274	0.286	0.517	0.377	0.012
Germany	1971	0.143	0.157	0.028	0.041	0.033	0.033	0.048	0.303	0.304	0.494	0.325	–0.001
Greenland	1971	0.207	0.217	0.044	0.069	0.020	0.034	0.043	0.192	0.206	0.327	0.374	0.014
Iceland	1971	0.140	0.080	0.044	0.040	0.083	0.085	0.122	0.305	0.307	0.451	0.489	0.002
Ireland	1971	0.131	0.154	0.046	0.042	–0.032	0.020	0.027	0.225	0.277	0.345	0.530	0.052
Italy	1971	0.167	0.175	0.037	0.052	–0.047	0.034	0.045	0.158	0.239	0.374	0.328	0.081
Netherlands	1971	0.164	0.194	0.028	0.046	0.008	0.024	0.033	0.262	0.278	0.326	0.388	0.016
Norway	1971	0.140	0.091	0.048	0.038	0.086	0.085	0.142	0.400	0.399	0.560	0.375	–0.001
Portugal	1971	0.220	0.269	0.046	0.058	–0.007	0.005	0.006	0.188	0.200	0.427	0.439	0.012
Spain	1971	0.167	0.159	0.036	0.066	0.026	0.036	0.043	0.166	0.175	0.323	0.431	0.009
Sweden	1971	0.118	0.128	0.021	0.042	0.062	0.038	0.060	0.396	0.371	0.675	0.318	–0.025
Switzerland	1971	0.194	0.186	0.009	0.043	0.041	0.044	0.054	0.189	0.191	0.529	0.331	0.002
United Kingdom	1971	0.082	–0.015	0.018	0.026	0.005	0.091	0.148	0.295	0.382	0.550	0.363	0.087
Average of all countries		0.153	0.148	0.034	0.049	0.026	0.043	0.062	0.258	0.275	0.486	0.383	0.017

Table 4.1 (continued)
Regression Variables: 1981–1987

		NS/NNP	APSYT	GROWTH	GLAGGED	SURPLUS/ NNP	NEWSURP/ NNP	SURPYT	NETTAX/ NNP	NEWTAX	LFWORK	DEP	DEBTAD/ NNP
Canada	1981	0.094	0.161	0.031	0.038	–0.043	–0.027	–0.036	0.234	0.250	0.617	0.321	0.016
United States	1981	0.039	0.086	0.031	0.029	–0.042	–0.029	–0.037	0.191	0.204	0.630	0.330	0.013
Japan	1981	0.202	0.189	0.037	0.049	0.043	0.048	0.060	0.182	0.187	0.568	0.325	0.006
Australia	1981	0.034	0.034	0.031	0.030	–0.012	0.009	0.013	0.256	0.278	0.539	0.363	0.022
Austria	1981	0.130	0.145	0.017	0.035	0.021	0.034	0.051	0.325	0.338	0.509	0.280	0.013
Belgium	1981	0.065	0.103	0.014	0.031	–0.078	–0.015	–0.020	0.154	0.216	0.503	0.284	0.063
Denmark	1981	0.060	0.083	0.027	0.023	–0.013	0.008	0.013	0.354	0.376	0.742	0.284	0.021
Finland	1981	0.107	0.119	0.032	0.038	0.030	0.030	0.046	0.352	0.351	0.727	0.288	–0.001
France	1981	0.080	0.080	0.018	0.034	0.009	0.026	0.038	0.306	0.323	0.549	0.328	0.017
Germany	1981	0.107	0.129	0.016	0.028	0.013	0.019	0.028	0.311	0.317	0.502	0.232	0.006
Greenland	1981	0.085	0.160	0.012	0.044	–0.067	–0.043	–0.054	0.174	0.197	0.397	0.329	0.024
Iceland	1981	0.067	0.094	0.034	0.044	0.001	0.002	0.003	0.311	0.312	0.442	0.416	0.001
Ireland	1981	0.083	0.147	0.022	0.046	–0.074	–0.024	–0.034	0.220	0.270	0.374	0.494	0.050
Italy	1981	0.109	0.133	0.019	0.037	–0.080	0.010	0.013	0.166	0.256	0.411	0.267	0.090
Netherlands	1981	0.133	0.187	0.014	0.028	–0.016	–0.009	–0.011	0.234	0.241	0.403	0.296	0.007
Norway	1981	0.158	0.039	0.039	0.048	0.097	0.134	0.213	0.335	0.372	0.678	0.320	0.037
Portugal	1981	0.198	0.214	0.020	0.046	–0.018	0.037	0.050	0.194	0.249	0.576	0.364	0.055
Spain	1981	0.096	0.107	0.026	0.036	–0.011	0.012	0.015	0.189	0.212	0.382	0.369	0.022
Sweden	1981	0.060	0.095	0.024	0.021	–0.002	0.005	0.008	0.410	0.417	0.771	0.286	0.006
Switzerland	1981	0.202	0.202	0.017	0.009	0.039	0.041	0.051	0.201	0.202	0.538	0.262	0.002
United Kingdom	1981	0.062	0.068	0.032	0.018	–0.015	0.015	0.022	0.282	0.312	0.593	0.300	0.030
Average of all countries		0.103	0.123	0.024	0.034	–0.010	0.013	0.021	0.256	0.280	0.545	0.321	0.024

of 16.6 percent in the sixties to 15.3 percent in the seventies, and finally crumbled to 10.3 percent in the eighties, an overall decline of some 40 percent in twenty years and of one-third in the last decade. What makes the episode even more impressive is the generalized nature of the decline in saving. This can be seen from table 4.2, which reports the *change* in saving between decades: the top portion gives the change between the '60s and the '80s, the lower panel the change from the decade of the 1970s to that of the '80s. From the top panel we see that saving fell in every single country (except Portugal where there was no change). In the final period, from the '70s to the '80s, the decline again spread to very nearly all countries, with the exception of Switzerland where the change was again near zero, and Norway where it reflected unusual circumstances.

In the case of private saving, the decline, though still notable (from 14.8 percent to 12.3 percent), is less dramatic and less widespread; over the entire period saving declines in 16 countries out of 21, and only one of the remaining five has a pronounced rise in saving. As will be shown presently, the difference between the national and private saving rate reflects of course the large decrease in government surplus, averaging nearly four percentage points.

1.2 A Brief Review of the Life Cycle Model Relevant to Aggregate Saving

The so-called New Theories of the Consumption Function, which include the LCH first developed by Modigliani and Brumberg [15], and the Permament Income Hypothesis (PY) of Milton Friedman [7], are distinguishable from the preexisting Keynesian model, based on crude empiricism, because they rest on a choice theoretic framework and more precisely on the theory of optimum allocation of "income" between alternative "commodities" (theory of consumers' choice). Following the pioneering work of Irving Fisher, the commodities between which the consumer chooses can be identified as consumption at various points of life, and income with the (present value of) resources accruing over life. This formulation has the basic implication that, contrary to the prevailing view, consumption at any point in time, as well as over the lifetime, depends on life resources and not on income at the time of consumption. Thus saving represents the residual between the consumption optimally allocated to a period and whatever the income happens to be in that period.

The essential difference between PY and LCH is that Friedman relies on the approximation that life is infinite and therefore uniform. Friedman's permanent income is the present value of resources over the infinite life times the rate of return, and his main concern is the effect, on the saving of individuals or groups, of random shocks to income away from permanent income, his so-called transitory income. The LCH on the other hand takes explicitly into account that life is

Table 4.2
First Differences of 1980s–1960s

	NS/NNP	APSYT	GROWTH	GLAGGED	SURPLUS/ NNP	NEWSURP/ NNP	SURPYT	NETTAX/ NNP	NEWTAX	LFWORK	DEP	DEBTAD/ NNP
Canada	–0.019	0.058	–0.016	–0.009	–0.079	–0.060	–0.073	0.009	0.019	0.252	–0.231	0.010
United States	–0.067	–0.021	–0.004	0.003	–0.055	–0.052	–0.067	–0.024	–0.021	0.180	–0.170	0.003
Japan	–0.054	–0.030	–0.058	–0.032	–0.035	–0.026	–0.030	0.008	0.017	–0.002	–0.069	0.009
Australia	–0.103	–0.059	–0.019	–0.012	–0.065	–0.054	–0.068	0.053	0.065	0.130	–0.115	0.011
Austria	–0.052	–0.022	–0.026	–0.059	–0.047	–0.040	–0.062	–0.020	–0.012	–0.003	–0.087	0.007
Belgium	–0.079	–0.040	–0.031	0.000	–0.091	–0.044	–0.055	–0.023	0.024	0.118	–0.089	0.047
Denmark	–0.114	–0.090	–0.021	–0.016	–0.063	–0.036	–0.046	0.100	0.127	0.215	–0.086	0.027
Finland	–0.050	0.018	–0.016	–0.009	–0.056	–0.053	–0.069	0.077	0.080	0.100	0.072	0.002
France	–0.113	–0.112	–0.032	–0.036	–0.036	–0.022	–0.026	0.061	0.074	0.083	–0.084	0.013
Germany	–0.092	–0.061	–0.025	–0.048	–0.051	–0.042	–0.055	0.035	0.045	0.016	–0.111	0.010
Greenland	–0.068	0.028	–0.057	–0.015	–0.107	–0.087	–0.108	0.001	0.021	0.010	–0.062	0.020
Iceland	–0.070	0.049	–0.006	–0.010	–0.103	–0.101	–0.134	0.064	0.066	0.056	–0.181	0.003
Ireland	–0.039	0.017	–0.020	0.031	–0.081	–0.039	–0.052	0.046	0.087	–0.003	–0.044	0.041
Italy	–0.099	–0.112	–0.033	–0.020	–0.079	0.000	0.001	–0.013	0.066	0.091	–0.075	0.079
Netherlands	–0.066	0.007	–0.032	–0.025	–0.059	–0.075	–0.100	–0.004	–0.019	0.120	–0.162	–0.015
Norway	–0.003	–0.084	0.001	0.011	0.026	0.058	0.102	0.029	0.061	0.305	–0.075	0.032
Portugal	0.000	–0.001	–0.038	0.003	–0.042	0.017	0.025	0.014	0.073	0.344	–0.097	0.058
Spain	–0.071	–0.054	–0.040	0.000	–0.056	–0.016	–0.018	0.036	0.075	0.120	–0.067	0.039
Sweden	–0.106	–0.017	–0.018	–0.013	–0.094	–0.085	–0.126	0.083	0.091	0.234	–0.036	0.008
Switzerland	–0.010	0.001	–0.026	–0.034	–0.008	–0.008	–0.010	0.012	0.012	0.145	–0.105	0.000
United Kingdom	–0.050	–0.017	0.006	–0.013	–0.047	–0.036	–0.049	0.022	0.032	0.087	–0.063	0.011
Average of all countries	–0.063	–0.026	–0.024	–0.013	–0.058	–0.038	–0.049	0.027	0.047	0.124	–0.092	0.020

Table 4.2 (continued)
First Differences of 1980s–1970s

	NS/NNP	APSYT	GROWTH	GLAGGED	SURPLUS/ NNP	NEWSURP/ NNP	SURPYT	NETTAX/ NNP	NEWTAX	LFWORK	DEP	DEBTAD/ NNP
Canada	–0.039	0.005	–0.007	–0.009	–0.053	–0.045	–0.060	–0.019	0.011	0.112	–0.082	0.008
United States	–0.050	–0.018	0.002	–0.006	–0.030	–0.036	–0.047	–0.004	–0.010	0.089	–0.058	–0.006
Japan	–0.044	–0.054	–0.012	–0.046	–0.003	0.002	0.004	0.006	0.011	0.033	–0.029	0.006
Australia	–0.076	–0.055	0.001	–0.020	–0.027	–0.036	–0.047	0.023	0.015	0.046	–0.061	–0.008
Austria	–0.050	–0.017	–0.018	–0.008	–0.037	–0.039	–0.059	0.003	0.002	0.025	–0.084	–0.001
Belgium	–0.074	–0.036	–0.017	–0.014	–0.067	–0.048	–0.064	–0.041	–0.023	0.058	–0.087	0.019
Denmark	–0.073	–0.050	0.004	–0.025	–0.065	–0.038	–0.058	0.000	0.028	0.106	–0.063	0.027
Finland	–0.035	0.002	–0.006	–0.010	–0.045	–0.034	–0.049	0.010	0.020	0.065	–0.039	0.010
France	–0.083	–0.081	0.016	–0.016	–0.027	–0.022	–0.029	0.032	0.037	0.032	–0.049	0.005
Germany	–0.036	–0.028	–0.012	–0.013	–0.020	–0.014	–0.020	0.008	0.013	0.008	–0.093	0.005
Greenland	–0.122	–0.057	–0.032	–0.025	–0.087	–0.077	–0.037	–0.018	–0.009	0.070	–0.045	0.010
Iceland	–0.073	0.014	–0.010	0.004	–0.082	–0.083	–0.119	0.006	0.005	–0.009	–0.073	–0.001
Ireland	–0.48	–0.007	–0.024	0.004	–0.042	–0.044	–0.061	–0.005	–0.007	0.029	–0.036	–0.002
Italy	–0.058	–0.042	–0.018	–0.015	–0.033	–0.024	–0.032	0.008	0.017	0.037	–0.061	0.009
Netherlands	–0.031	–0.007	–0.014	–0.018	–0.024	–0.033	–0.044	–0.028	–0.037	0.077	–0.092	–0.009
Norway	0.018	–0.052	–0.009	0.010	0.011	0.049	0.071	–0.065	–0.027	0.118	–0.055	0.038
Portugal	–0.022	–0.055	–0.026	–0.012	–0.011	0.032	0.044	0.005	0.049	0.149	–0.075	0.043
Spain	–0.071	–0.052	–0.010	–0.030	–0.037	–0.024	–0.028	0.023	0.037	0.059	–0.062	0.013
Sweden	–0.058	–0.033	0.003	–0.021	–0.064	–0.033	–0.052	0.014	0.046	0.096	–0.032	0.031
Switzerland	0.008	0.016	0.008	–0.034	–0.002	–0.003	–0.003	0.012	0.011	0.009	–0.069	0.000
United Kingdom	0.020	0.083	0.014	–0.008	–0.020	–0.076	–0.126	–0.013	–0.070	0.043	–0.063	–0.057
Average of all countries	–0.049	–0.025	–0.009	–0.015	–0.036	–0.030	–0.042	–0.002	0.005	0.060	–0.062	0.007

Table 4.2 (continued)

Legend for variable abbreviation:

NS/NNP	= net national saving as a percentage of national income.
APSYT	= private saving corrected for inflation as a percentage of national income less net taxes corrected for inflation.
GROWTH	= the current period's average annual rate of real GDP growth.
GLAGGED	= the previous time period's average annual rate of real GDP growth.
SURPLUS/NNP	= the average of aggregate general government current receipts less current disbursements as a percentage of national income.
NEWSURP/NNP	= average inflation-adjusted surplus as a percentage of national income.
NETTAX/NNP	= total general government net taxes as a percentage of national income, where net taxes are defined as current receipts less net transfers less net interest payments on the public debt.
NEWTAX	= NETTAX/NNP corrected for inflation.
LFWORK	= the average of the annual labor force participation ratio for females aged 25–54.
DEP	= the ratio of population under 15 to the entire population.
DEBTAD/NNP	= the inflation correction calculated as: (net public debt/nnp) × (cpi).

finite and structured, and that, therefore, income and consumption exhibit in addition to random shocks also systematic variations arising from the life cycle characterized by the succession of a preworking, a working, and a retired phase. LCH is thus in a position to explore the implications not only of random shocks, which are basically the same as for PY, but also of systematic life cycle variations of saving and wealth. These turn out to be crucial for an understanding of *aggregate* saving whereas the variations of transitory income are not too relevant since under aggregation they tend to cancel out. Not surprisingly therefore, PY, in contrast to LCH, has little to say about aggregate consumption and saving. In what follows we concentrate on the aggregate implications of LCH, since they are the ones that are relevant for the study of the recent drying up of saving.

1.3 The Aggregative Implications of LCH: The Basic Model

To understand the implication of LCH it is useful to rely on two successive approximations. In the first approximation, or basic model, it is assumed that there are no bequests and there is no government—or at least no unbalanced budget. In addition, we rely on one basic assumption about the nature of allocation preferences, namely that behavior can be described as though the representative agent had a homothetic preference map, stable over time (meaning that the ratio of preferred consumption at any age to life resources is independent of the size of life resources).

1.3.1 The Role of Income Growth

From these assumptions one can establish the following propositions (Modigliani-Brumberg [15]): 1) the saving rate of a country is independent of its per capita income; 2) two countries may have different rates of saving even though individuals are equally thrifty, in the sense of having the same life path saving and wealth; 3) the saving rate depends critically on the rate of growth in the following sense: (*i*) with zero growth, the saving will be zero, regardless of income or thrift habits, (*ii*) there can be saving only when there is growth and between economies with equal thrift, the one with the fastest growth can be expected to save the most; 4) an economy can accumulate a very substantial stock of wealth relative to income, even in the absence of bequests.

These propositions are the most elementary example of life cycle model in which income is constant through the earning span, zero thereafter, and consumption is constant through life and national income is stationary. In this case there will be saving till retirement, and dissaving after retirement equal to the amount accumulated till then. It should be clear that, in this economy, the wealth-income ratio is independent of income, national or per capita; it can also be readily shown that it is equal to half the number of retired years. For this simple economy, proposition 1) is a direct implication of homothetic maps. Since each household consumes all his accumulation over his life, proposition 3) (*i*) can be established by noting that in the absence of growth, i.e. stationary population and no other source of growth, national saving must be zero as the saving of the young is offset by the dissaving of the old. But if population is growing steadily, that will mean an increase in the relative number of the young who save, relative to the old who dissave; thus the saving will be positive and increase with population growth. Similarly if income grows through rising productivity, even if population is stationary, saving will be positive (at least for growth rates in the relevant range) because the young who are in their saving phase enjoy higher lifetime resources than do the old who dissave (proposition 3) (*ii*)).

The nature of the relation between saving and trend growth can be confirmed also by marking use of the fundamental dynamic relation that must hold, in steady state, between the saving ratio, s, the wealth income ratio w, and the rate of growth g, namely, $s = gw$ (which is the equivalent of the well-known Harrod-Domar proposition for investment and the capital stock).

This equation directly confirms proposition 3) (*i*)), as $s \gtreqless 0$ if $g \gtreqless 0$. For positive values of s we found that the slope of the relation between s and growth, g, is given by:

$$ds/dg = w + g(dw/dg)$$

Thus, if w were not a function of growth, or $dw/dg = 0$, s would be proportional to growth and with slope equal to the wealth income ratio w, which, as noted earlier, could be a rather large number, say on the order of five. However, in general w will depend on g. In fact there are reasons to believe that it decreases with g, which means that the second term in the above expression will become more and more negative. Thus the slope declines as g rises, or the relation between s and g is concave to the g axis. However, the slope must be positive at the origin and by continuity, in a neighborhood thereof, which may be expected to include the relevant range of g which is empirically small—seldom more than 10%.

One can throw some light on the relation between the wealth-income ratio and growth by a different route. It was shown by Modigliani and Brumberg [15] that under the assumption stated at the beginning of this section, aggregate consumption can be expressed as a linear function of wealth and labor income, or:

$$C = \alpha \cdot YL + \delta W \tag{4.1}$$

where YL is labor income.

Since personal income is $Y = YL + rW$, where r is the rate of return on wealth, saving can be expressed as:

$$S = YL + rW - C = (1-\alpha)Y + (\delta - \alpha r)W$$

or:

$$s = (1-\alpha) - (\delta - \alpha r)w \tag{4.2}$$

where: $s = S/Y$ and $w = W/Y$.

But with steady growth $s = gw$, implying:

$$(1-\alpha) - (\delta - \alpha r)w = gw \tag{4.3}$$

or:

$$w = (1-\alpha)/(g + \delta - \alpha r) \tag{4.3a}$$

$$s = g(1-\alpha)/(g + \delta - \alpha r) \tag{4.3b}$$

It is apparent from (4.3a) that the wealth-income ratio will tend to decline with g while the saving ratio will be an increasing function of g for any finite value of g, at least as long as the parameters α and δ are not very sensitive to g, which is consistent with simulation reported in Modigliani and Brumberg [15].

A comparison of equations (4.2) and (4.3b) shows that there are two alternative ways to express the saving ratio, s; first, in terms of wealth-income ratio (as in (4.2)), and second in terms of the growth rate, g, as in (4.3b). The formulation of equation (4.2) has some advantages in that it may be expected to hold reasonably well even in the short run. The drawback is the very limited availability of data on wealth, whereas g is available as long as there exist national income accounts. For this reason in our test we have used a linear approximation to equation (4.3b).

1.3.2 Other Factors Affecting the Saving Rate

Besides depending on growth, s will also be affected by any factor which affects the wealth-income ratio. Equation (3) suggests that one such factor is the interest rate, and that a rise in r will then raise w and s. However the second inference is not warranted because the parameters α and especially δ are generally themselves a function of r. Results presented in Modigliani and Brumberg [15] suggest that δ rises with r and may rise even faster, in which case s and w will decline with r.

Other factors at work include, in addition to length of retirement, the timing and size of other major expenditures for which provisions must be made, such as for the purchase of a house, children's education, or other major durables. The need for accumulation will be increased in the presence of credit rationing. Another reason for the accumulation of assets is the precautionary motive (Keynes [10], p. 107) which in turn interacts with the availability of public and private insurance against various events and of life annuities. While precaution and liquidity constraints make for higher saving, there may be another factor making for smaller saving, namely myopia or the failure to make adequate preparations for retirement. But there is so far little evidence of myopia being a widespread phenomenon.

Finally, the saving rate will be affected by socio-demographic variables. To begin with, population growth is expected to affect the saving rate through the age structure of the population. But in addition, several other variables have been suggested, e.g., life expectancy, the proportion of retired people, retirement habits (the extent to which people retire and the prevailing age at which they to), and the incidence of female participation in the labor force.

Of all these variables in our empirical test we will limit ourselves to two: first, the proportion of the population below working age (which we take to be 15); we expect this variable to have a negative effect on s, because of the extra load imposed on parents through a larger family size. Second, female participation, which we measure as the ratio of working women age 25–54 to the total number of women in the same age group. This variable is expected to have a negative sign, on the grounds that when a second spouse enters employment, expenditures will rise proportionately more than income because of the need to replace, for a price, some of the activities that the working spouse can no longer perform. Recently, Guiso and Jappelli [8] have stressed a second reason, namely that, when both spouses work, there is less need to accumulate a precautionary balance. The other variables will be omitted because they are more questionable or are not readily measurable, like retirement habits.

1.4 The General Model

1.4.1 The Role of Bequests

We can now move to the second approximation, or general model, by allowing first for the presence of bequests. One must distinguish between bequests resulting from the precautionary motive and those resulting from a true bequest motive. The former arise because of uncertainty as to the time of death. Since such bequests may be expected to be proportional to life resources, they may be lumped with life cycle saving and will not affect the properties of the basic model though they imply a somewhat larger wealth-to-income ratio. As for inheritances resulting from a true bequest motive, they can also be incorporated in the model without affecting its basic properties by postulating that bequest behavior is such that the flow of true inheritances is proportional to the flow of life cycle saving, or equivalently, the stock of inherited wealth at any point in time is proportional to total wealth; this of course implies that for any given growth trend, inherited wealth is also proportional to income. The evidence appears consistent with this assumption at least for the U.S. where the information is available. In fact if inherited wealth rose systematically faster or slower than income, then we should expect that the wealth-income ratio itself would rise or fall over time, whereas the observed ratio appears to have remained remarkably stable for many decades (at least since the beginning of the century).

But why should bequests, like life cycle saving, be a constant fraction of income? It was shown in Modigliani and Brumberg [15] that for the proportionality to hold it is sufficient that the following two conditions be satisfied. First, even though "richer" people may tend to leave a larger fraction of their resources than "poor" people (who may leave none), the proportion left should depend on relative not absolute, life cycle resources. Second, the size distribution of life resources should be reasonably stable over time. There is some empirical evidence that appears consistent with these assumptions.

We can conclude that under the conditions stated above, the presence of bequests, whether precautionary (and therefore unintended) or intended, does not change the basic conclusions: the general model still implies that saving depends on growth and not on income. Of course the ratio of bequests to life cycle saving may reflect differential bequest behavior; and, other things equal, the larger bequeathed wealth, the larger the wealth-income ratio and hence the larger the saving ratio, for given growth (even though it should be realized that bequests can, to some extent, have a depressing effect on life cycle wealth).

This raises the question of how important is bequeathed wealth relative to life cycle wealth. I have concluded from a good deal of evidence that, for the U.S., the ratio of bequeathed to total wealth can be placed at between 1/4 and 1/5, though this estimate has been hotly debated (Kotlikoff [11]). However, this issue is irrelevant for our tests, since our concern is with the widespread fall in the overall saving ratio. In any event information on the share of wealth that is inherited is not available at present for almost any country. But this is not likely to be too serious, since there are no grounds for thinking that the phenomenon under investigation could be, to any significant extent, the result of a worldwide reduction in true bequests whether due to legislation or to a spontaneous change in preferences. And second, bequests or not, the growth rate and the other variables mentioned remain major determinants of the saving ratio.

1.4.2 The Role of Government

In the traditional Keynesian model, introducing a government requires distinguishing between national income and disposable income (income minus taxes) and recognizing that national saving is the sum of government saving and private saving. Private saving is defined as disposable income less private consumption, and it is taken to be a function of disposable income.

In an LCH framework instead, private saving will be affected not only by private income but also by deficits. For if consumers are rational and base their consumption on life resources, they should realize that a current deficit increases the national debt which must eventually increase future interest and thus taxes. If life were infinitely lasting, then the additional burden due to the deficit, the present value of the resulting interest forever, would be just equal to the deficit.

But then in order to maintain a smooth consumption path, the rational agent must prevent the future taxes from cutting into his future consumption by setting aside currently (saving) an additional amount equal to the deficit for which he will have to pay. Thus national saving is unaffected by the deficit (expenditure constant) as the government dissaving is completely offset by additional private saving. If the rise in the deficit is due to government expenditure rather than taxes, the effect on national saving is not zero, but still small, namely equal to the marginal propensity to save (see equation (4.6) below). This is because while resources are not changed by taxes, they are reduced the government preempts their use.

But in LCH, life is very much finite. Therefore the burden arising from the future service of the debt is equal to the present value of the future taxes that will be levied on those currently present and only as long as they live, and pay taxes. This is clearly but a fraction of the deficit. How large a fraction depends on such things as the (real) interest rate, the distribution of life expectancy of those present, and the rate of growth of income. Sterling [18] has estimated the fraction of the deficit that will finally be borne, through future taxes, by the current generation. He reports a range of values depending on the interest rate, growth and life expectancy. A typical result is around 30 percent. To see the implication of this generalization for the impact of fiscal policy on saving, we first note that, once we allow for the Sterling effect in the LCH framework, the consumption function can be written as:

$$C = (1 - f(g))(Y - T + \lambda GS) \tag{4.4}$$

where Y is NNP, T is total net taxes (i.e. net of transfers), $f(g)$ is a function expressing the impact of growth on the saving ratio (equation (4.3b)), GS is government surplus and λ is Sterling's fraction of the deficit falling on the current generation.

From (4.4) we can derive the private saving function:

$$PS = (Y - T) - C = f(g)(Y - T) - (1 - f(g))\lambda GS \tag{4.5}$$

and the national saving function, by adding GS to both sides of the equation, thus:

$$NS = f(g)(Y - T) + [1 - (1 - f(g))\lambda]GS \tag{4.6}$$

To avoid nonlinear estimation we approximate $f(g)$ by a linear function and write the private saving rate in the form:

$$PS/(Y-T) = a + bg + cGS/(Y-T) \tag{4.5a}$$

where $c = -[1 - f(\text{g})]\lambda$ and $[1 - f(g)]$ can be approximated by a constant. This approximation is somewhat crude since $1 - f(g)$ is the consumption ratio which varies appreciably in our sample, between 0.75 and 1.

In the case of the national saving ratio, which should be

$$\begin{aligned} NS/Y &= f(g)(1 - T/Y) + d(GS/Y) \\ &= a + bg - hT/Y + d(GS/Y) \quad \text{where } d = 1 - (1 - (g))\lambda \end{aligned} \tag{4.5b}$$

Recently Barro [4] has proposed an extension of LCH according to which the Sterling coefficient, λ, must be unity. Clearly this conclusion would be consistent with LCH if and only if life were of infinite duration. But he manages to reconcile his claim with a finite life by postulating that the agent is so concerned with the nature of his heirs that the life cycle relevant for the allocation of resources extends beyond his own life to encompass that of his heirs, heirs of heirs, etc. to infinity. Under these conditions the gain from a tax cut financed by a deficit has to be totally saved in order to prevent the government from bringing about a reshuffling of resources from the future lineage to the current generation. This means that consumption and national saving will be totally unaffected by the deficit.

There have been criticisms of Barro's hypothesis or "Ricardian equivalence" (because the hypothesis was first proposed, but also rejected, by Ricardo), a few of which will be mentioned here.

(i) When the maximization of utility over all future generations calls for transferring resources from future generations to the present, there is no way of enforcing it. The actual allocation will then represent a corner solution, and a deficit will provide the means for raising the utility of the current generation by increasing current consumption at the expense of future consumption.

(ii) Similarly, with a corner solution due to credit rationing, a tax cut will be utilized, at least in part, to increase consumption.

(iii) Asymmetries: a tax cut cannot have neutral effects as Barro supposes because it redistributes wealth from current and prospective large families, who are made worse off, to small families.

(iv) Implausibility: the evidence suggests that relatively few people at the top of the distribution of wealth are concerned with bequests. My own estimate (1988) is that the wealth received by bequest is only about 1/4 of the total. Also no evidence has been found that different generations of one family live on anything like the same scale (Altonji, Hayashi, and Kotlikoff [2]).

All this points, to the conclusion that Barro-like behavior cannot by any means be universal, but is at best limited to a small minority. We therefore expect the coefficient λ to be below unity, contrary to Barro's view. It should instead be somewhere around Sterling's estimate of 0.3; it could be somewhat smaller in that not all consumers are likely to be rational enough to save now in view of future taxes, but it could also be somewhat higher in so far as some portion of those present may be responding to the higher taxes that will fall on later generations by saving even more than called for by the Sterling estimate. Fortunately these contrasting views can be tested by relying on the estimate of the coefficient c of $GS/(Y - T)$ in equation (4.5a).
Using the definition of c we find:

$$\lambda = -c/[1 - f(g)] \tag{4.7}$$

and $f(g)$ can be approximated by the mean saving ratio (0.14 in our sample).

2 Empirical Tests for the OECD Countries

2.1 Nature of Tests

The battery of tests reported below covers the last three decades: the sixties, the seventies, and the eighties as far as available, namely 1987. For each country and for each variable we compute an average value for each decade. Since there are 21 countries and three decades, we have a total of 63 observations on each variable. We have performed three types of test: the first is based on a cross-section time-series of the 63 observations; the next test is based on a cross section of the 21 countries for each decade separately; the third relies on a cross section of first differences. We present two versions of this test. In one the observations are the change in the value of each variable between the '60s and '80s; in the other the changes are between the '70s and '80s.

2.2 Results—Level Tests

Table 4.3 reports the results of a number of tests of the life cycle and of the role of fiscal variables. The first two rows test the variable that is the distinctive feature of the life cycle model, namely income growth g. It is seen that the coefficient of this variable is highly significant, as measured by the t-ratio. In line 1 of table 4.3, the estimated coefficient of g is somewhat lower than in line 2, but one may suspect some downward bias in the former estimate because in our sample we find

Table 4.3

	Period & estimation technique	Constant	g	g_1	Surplus (GS/Y)	Tax effect	LFWORK	Dep. ratio	Inflation correction	Adj. R^2 & S.E.
1	1960–1987 OLS	0.08 (4.1)		1.20 (3.1)						0.12 0.053
2	1960–1987 OLS national savings	0.06 (4.2)		1.81 (6.1)						0.37 0.041
3	1960–1987 OLS	0.08 (5.1)		1.76 (4.9)	–0.53 (4.5)	–0.68 (5.8)				0.33 0.046
4	1960–1987 OLS	0.29 (5.4)		1.41 (4.3)	–0.40 (3.7)	–0.55 (5.1)	–0.19 (3.6)	–0.26 (3.1)	–0.56 (3.4)	0.49 0.040
5	1960–1987 WLS (pop.)	0.27 (5.3)		1.73 (6.5)	–0.27 (2.4)	–0.42 (3.7)	–0.23 (4.5)	–0.22 (2.7)	–0.72 (4.4)	0.67 0.035
6	1960–1987 OLS national savings	0.27 (6.7)		1.08 (4.6)	0.77 (6.0)	–0.46 (3.4)	–0.10 (2.2)	–0.19 (3.1)	–0.38 (2.2)	0.67 0.029
7	1960–1970 WLS (pop.)	0.22 (3.3)		2.24 (5.0)	–0.77 (2.3)	–0.92 (2.7)	–0.08 (0.9)	–0.20 (1.7)	–0.55 (0.9)	0.75 0.027
8	1971–1980 WLS (pop.)	0.33 (3.2)		1.81 (5.9)	–0.41 (1.9)	–0.56 (2.6)	–0.22 (2.7)	–0.43 (2.1)	–0.54 (2.4)	0.87 0.026
9	1981–1987 WLS (pop.)	0.20 (2.3)		3.9 (4.0)	–0.04 (0.2)	–0.19 (1.0)	–0.10 (1.0)	–0.47 (2.2)	–0.35 (1.4)	0.55 0.031
10	1st diff national savings OLS 1980s–1960s	no constant		0.72 (2.1)	0.52 (2.5)	0.03 (0.7)	0.05 (0.62)	0.08 (0.64)	–0.43 (1.2)	0.83 0.029
11	1st diff national savings 1980s–1960s OLS	no constant		0.78 (2.6)	0.60 (3.8)	0.18 (0.65)			–0.48 (1.6)	0.85 0.028

Table 4.3 (continued)

	Period & estimation technique	Constant	g	g_1	Surplus (GS/Y)	Tax effect	LFWORK	Dep. ratio	Inflation correction	Adj. R^2 & S.E.
12	1st diff national savings 1980s–1970s OLS	no constant		0.59 (1.6)	0.98 (6.1)	0.71 (2.3)	–0.02 (0.19)	0.06 (0.39)	–0.64 (1.6)	0.88 0.020
13	1st diff national savings 1980s–1970s OLS	no constant	0.93 (2.8)	0.64 (2.7)	0.90 (7.9)	0.61 (3.2)			–0.37 (1.4)	0.92 0.016

Legend:
Absolute values of t-ratios are in parentheses. All regressions were run on period average private savings out of disposable income corrected for inflation except where noted. The data used for the regressions in this table was obtained from the OECD National Income Accounts, OECD Economic Outlook, and International Financial Statistics.
The explanatory variables used in these regression are defined as follows: $g(g_1)$ = current (lagged) real GDP growth taken as the average rate of growth for all years from the previous period surplus and taxes = surplus of the general government defined as "net taxes" current receipts less transfers less interest on the debt, corrected for inflation · less government expenditures on goods and services tax effect = see note #4 LFWORK = the labor participation ratio for women aged 25–54 dep. ratio = the percentage of the population below the age of 15 inflation correction = consumer price deflator · net public debt. For those countries with incomplete net public debt data, the following approximation was used: $(\dot{p}/r) \cdot net$ where $\dot{p}$ = consumer price deflator; r = yield on long term government bonds; net = net interest paid on the public debt nnp.
The tax effect for the equations in which the dependent variable is adjusted private savings is computed from the following formula; tax effect = surplus coefficient –0.15. For the regressions with national savings as the dependent variable, which include a specific tax term, the tax effect is the sum of the coefficients on the surplus and tax variables.

a positive association between growth and government saving, and the government saving in turn should reduce to some extent private saving (equation (4.5)). This suspicion is confirmed by the results in line 3, in which we add the fiscal variable, government surplus. In line 3, the growth coefficient rises to 1.76, a value close to that of line 1. These estimates of g are of interest for several reasons. First, in the estimate, lines 2 and 3 are broadly confirmed by all the other results presented in table 4.3. The estimates fall predominantly between 1.5 and 1.8 with very few, usually explainable, exceptions, as indicated below. Second, it is consistent with the results reported in various other contributions beginning with my 1970 paper. Third, this estimate is broadly consistent with the value suggested by the LCH. As we have seen in equation (4.3b), the LCH implies a non-linear relation between the saving ratio and g. In our tests we have approximated (4.3b) linearly:

$$s = f(g) \cong a + bg$$

The slope b of this approximation might be expected close to $f'(\bar{g})$, the slope of (4.3b) computed around the mean. By differentiation of (4.3b), the slope at any value of g is found to be:

$$f'(g) = \delta'(1-\alpha)\big/(\delta' + g)^2$$

where:

$$\delta' = \delta - r\alpha$$

Modigliani and Brumberg, using simulation, found that $(1 - \alpha)$ could be placed in the neighborhood of 0.3, and δ' at around 0.07. The estimates have since been broadly supported by direct time series estimated, e.g. for the US (Ando and Modigliani [3]; Modigliani and Sterling [17]), and for Italy (Modigliani and Jappelli [16]), though the estimates of δ have sometimes been on the low side. Hence the slope of (4.3b) at the mean value of g, which is around four percent can be placed at 1.7 for $\delta' = 0.07$ (1.8 if $\delta' = 0.06$), quite consistent with our estimates of the slope.

In line 3 of table 4.3 the government surplus, *GS*, appears with a significant negative sign, implying that private saving is reduced by government saving. This result is consistent with Barro and LCH, though it rejects the traditional Keynesian formulation. The difference between LCH and Ricardian equivalence is in the magnitude of the coefficient. If λ were 1 as asserted by Barro, then from (4.5a) the absolute value of the coefficient of *GS* should be $1 - f(g) = 0.86$ in the sample, which measures the long-run propensity to save, or somewhat less, if we adjusted for the effect of deficit on private saving. The actual coefficient of *GS* is only 0.40. With a standard error of the coefficient of 0.11, the actual coefficient is seen to be four standard deviations below the value that should hold if Barro were correct! On the other hand, for the LCH model with Sterling's estimate of 0.3, the coefficient should be 0.26. This is somewhat lower than the observed coefficient but only by one standard deviation. An alternative way of stating the result is to calculate from (4.6a) and line 6 of table 4.3. We fine that λ is below 0.4, as compared with the prediction of LCH of 0.3, and of 1 under the Barro hypothesis.

Line 4 adds the remaining variables. The coefficients of *g* and *GS* are not appreciably changed though the latter coefficient gets a bit more negative. The coefficients of demographic variables have the expected sign and are significant, especially the labor force participation.

Unfortunately we have no prior value to assess conformity with the estimate coefficient. But the estimate is certainly large considering that between the '60s and '80s labor force participation increased on average by 12.4 percent. With our coefficient estimate of 0.2, this implies that the variable would account for a decline in saving of some 2.3 percent out of a total decline of 2.6 percent. The most striking result in this equation, however, is the strong showing of the inflation correction. It implies that a very high fraction of the inflation correction component of interest income is treated, by the private sector, as though it were ordinary income available for consumption. That fraction can be estimated by dividing the coefficient in the last column by the propensity to consume: to obtain 0.56/0.85 = 0.66. Since I have long been convinced that inflation illusion is widespread, I do not find this figure shocking or incredible, though others might. It is worth noting that the only other estimate of the role of the inflation adjustment of which we are aware, based on a time series analysis for Italy for the years 1962–1988, yields a similar, but even higher result (Modigliani [13], tables 3 and 4). Taking that estimate at face value, and considering that the inflation adjustment (divided by NNP) rose between the '60s and '80s by an estimated 2.0 percentage points, would suggest that inflation illusion could account for a decline in saving of 1.1 percentage points.

Line 5 is a rerun of line 4, except that the observations are weighted by population. This is a procedure that has been frequently followed before since the early work of Houthacker [9] and Modigliani [12], on the grounds that there are enormous variations in the size of the countries in the sample, from a few millions to hundreds of millions, and one would expect the observations from a larger country to be more representative and more reliable because they are based on a larger sample. The results present no major surprise—indeed, both the coefficient of *g* and that of *GS* are right on the nose of LCH predications. In line 6 the dependent variable is national saving. Here the coefficient of *g* is somewhat low, but the coefficient of *GS* nearly coincides with LCH prediction, as it implies a value of λ of just about 0.3, and that one dollar of government deficit reduces national saving by 77 cents, when according to Barro [4] the effect should be no more than some 15 cents. The coefficient of the other variables are somewhat reduced, especially that of labor force participation, which is nearly halved.

The next battery of tests presents results for cross-section tests for each of the decades. In these tests, the results are broadly consistent with those reported above, but, presumably because of the much smaller samples, and apparent multicollinearity, the estimates are less stable and tend to be a good deal less significant. We report one single equation corresponding to line 5, i.e. with all the variables and population weighted. Line 7 of table 4.3 is based on the first decade. The problem of instability, and multicollinearity, is clearly visible in the results.

The point estimates of the coefficients other than that of *GS* are not very different from those of line 5 but the last three variables are all insignificant, despite the high R^2, and insignificantly different from the coefficients in line 5. The coefficient of *GS* is the one that is appreciably more negative and thus less consistent with LCH. It implies that a one dollar increase in government deficit would be offset to the extent of 77 cents by higher private saving. However, this result is not a serious challenge to LCH. In the first place it is unique in the battery of tests, and in the second place the estimate is not very reliable: it has a standard error of 0.33. Thus while it is not significantly different from the value of 0.85 called for by Barro, it is also not significantly different (even at the 10 percent level) from the value implied by LCH of around 0,25. The coefficient estimate is just too unreliable to permit a sound conclusion.

In line 8, covering the '70s, most coefficients, including that of *GS*, are back at a level close to those of line 5, expect possibly fior that of Dependency. The remaining regression for the '80s is the least satisfactory. All the coefficients are generally different from (5), but the estimates are subject to such large error that the differences from the estimates of line 5 are not significant.

2.3 First Difference Tests

The purpose of the tests reported in the last four lines of table 4.3 is to inquire how well the LCH can account for the large changes in saving that occurred between the '60s and '80s. In this case there are no solid grounds for expecting that the coefficients will be very close to those obtained from the level cross section like lines 5 and 6 because of the different structure of errors; in particular one can expect that the difference in a given variable between countries will tend to correspond more nearly to a permanent difference than might be the case with differences for one country, between two relatively close points in time.

We found in fact systematic differences between the coefficients of a magnitude that was rather surprising. In particular the coefficients of the two demographic variables were never significant in accounting for the change between the '60s and the '80s, or between the '70s and the '80s. This can be illustrated by means of line 10. Only the coefficient of *GS* is consistent with earlier results and has a respectable *t*-ratio. These disappointing results seem to reflect multicollinearity since in both lines 10 and 12, R^2 is rather high. Dropping the demographic variables in line 11, makes relatively little difference. Line 12, where the variables are first differences between the '70s and '80s, is even more surprising as the only coefficient significant at the one percent level is that of *GS*, and that coefficient is too large even for LCH. However, dropping from line 12 the insignificant variables and allowing for a faster response to growth by adding the current, in addition to the lagged, rate of growth, appears to eliminate or reduces appreciably the unsatisfactory aspects of line 12: the growth effect (the sum of the two growth coefficients) more than doubles to 1.6 and thus returns to the prevailing range; the coeficient of government surplus declines, though remaining somewhat high for LCH and *a fortiori* for the Barro hypothesis. The coefficient of taxes agrees with LCH, and only the coefficient of the inflation correction declines in value and reliability.

In table 4.4, we have utilized the equation of line 13 to examine how well the model can account for the large decline in the saving rate between the '70s and '80s, both for the average of all countries and for each individual country. Each column in the table gives the contribution to the change in saving of the variable indicated in the top row. The last row gives, for each variable, the average of all countries. Column 6 reports the change computed from the equation (the sum of the entries in columns 2 to 5, and 7 gives the actual change. It is seen from column 7 (bottom row) that saving declined almost universally in this period, with an average decline of nearly 5 percentage points. It is also apparent, from column 6, that the equation accounts quite accurately for this average fall.

Column 2 shows the contribution to the change in saving attributable to the change in the rate of growth. It brings out the fact that the almost universal decline in this variable is a major factor in the fall of saving, accounting for some 40 percent of the total. The other major component is the nearly as widespread decline in surplus, accounting on the average for three percent of *NNP*, which is responsible for over half of the decline. The remaining two variables, taxes and inflation correction, also contributed to the fall but to a minor extent (about 10 percent). This reflects the fact that these variables had very little change on the average over the period under review, though they contributed substantially to the changes in saving of a number of countries. Indeed, from the individual rates one can discern a considerable variety of experience, though negative signs predominate in every column. Thus, the UK experienced a rather small decline in saving, despite the appreciable deterioration in the fiscal stance, due mostly to

Table 4.4
The Role of Major Variables Affecting the Change in the Savings Rate, 1970s–1980s
(based on regression equation)
$dns - 0.93 \times dgrowth + 0.64 \times dlag + 0.90 \times dsurp - 0.28 \times dtax - 0.37 \times ddebtad$

Country (1)	Growth (*) (%) (2)	Surplus (%) (3)	Net tax (%) (4)	Inflation correction (%) (5)	Estimated change (%) (6)	Actual change (%) (7)	Difference (%) (8)
Canada	–1.22	–4.02	0.31	–0.29	–5.22	–3.90	–1.32
United States	–1.18	–3.31	0.29	0.25	–2.06	–5.00	2.04
Japan	–4.02	0.24	–0.33	–0.20	–4.31	–4.40	0.09
Australia	–1.17	–3.16	–0.44	0.29	–4.47	–7.60	3.13
Austria	–2.17	–3.49	–0.05	0.05	–5.66	–5.00	–0.66
Belgium	–2.46	–4.37	0.64	–0.71	–6.89	–7.40	0.51
Denmark	–1.20	–3.42	–0.77	–1.02	–6.41	–7.30	0.89
Finland	–1.22	–3.08	–0.57	–0.37	–5.24	–3.50	–1.74
France	–2.54	–1.94	–1.05	–0.18	–5.73	–8.25	2.53
Germany	–1.94	–1.29	–0.38	–0.21	–3.82	–3.60	0.22
Greenland	–4.58	–6.96	0.24	–0.36	–11.67	–12.20	0.53
Iceland	–0.67	–7.41	–0.15	0.02	–8.20	–7.31	–0.89
Ireland	–2.00	–3.96	0.21	0.08	–5.67	–4.80	–0.87
Italy	–2.62	–2.17	–0.47	–0.33	–5.59	–5.80	0.21
Netherlands	–2.45	–2.93	1.06	0.34	–3.97	–3.10	–0.87
Norway	–0.21	4.33	0.78	–1.41	3.49	1.80	1.69
Portugal	–3.17	2.89	–1.38	–1.61	–3.27	–2.20	–1.07
Spain	–2.87	–2.14	–1.05	–0.49	–6.55	–7.10	0.55
Sweden	–1.06	–2.96	–1.29	–1.15	–6.45	–5.80	–0.65
Switzerland	–1.40	–0.29	–0.31	0.03	–1.97	0.80	–2.77

Table 4.4 (continued)

Country (1)	Growth (*) (%) (2)	Surplus (%) (3)	Net tax (%) (4)	Inflation correction (%) (5)	Estimated change (%) (6)	Actual change (%) (7)	Difference (%) (8)
United Kingdom	0.83	–6.86	1.98	2.13	–1.92	–2.00	0.08
Average of all countries	–1.82	–2.68	–0.13	–0.25	–4.88	–4.94	0.06

Legend: *dns* = change in inflation-adjusted national savings as a percentage of national income;
dgrowth = change in the growth rate from 1970s to 1980s;
dlag = change in the growth rate from 1960s to 1970s;
dsurp = change in inflation-adjusted general government surplus as a percentage of national income;
dtax = change in inflation-adjusted general government net taxes as a percentage of national income;
ddebtad = change in the inflation correction, calculated as: (net public *debt*/*nnp*) * (*cpi*).

(*) Sum of the effects of the change in current and lagged growth.

higher expenditure, because of a favorable growth experience (it was the only country with positive growth in the period), and because a reduction in inflation, coupled with a large debt, produced a large reduction in the inflation adjustment. On the other hand, Greece suffered the largest decline in saving (−12.2 percent), well accounted for by the equation, as resulting from the largest decline in growth and the second-largest rise in deficit. And Japan had an about average decline in saving, mostly because of a slowdown in growth.

In general, the equation accounts well for the behavior of saving, though there are exceptions. In the case of France, e.g. with the second-largest contraction in saving (−8.2 percent), the equation estimates a large decline, but not large enough. Even for the U.S., where, somewhat surprisingly, the decline in saving is not more than average, the only significant explanation offered by the equation is the rise in the budget deficit, reflecting both higher expenditure and lower taxes, which however turns out to be inadequate to account for the decline.

On the whole, the evidence appears to be reasonably consistent with the LCH, supporting its implications of a major role for income growth and fiscal policy.

3 A Brief Digression on the Role of Income and Other Variables

3.1 Income

In summarizing the foundations and aggregative implications of LCH, it was pointed out that one of the significant and unusual consequences of that theory is the independence of the aggregate saving ratio from the level of economic well-being of the economy as measured, say, by per capita real income. This independence is an implication of the postulate of homothetic preference. It was also indicated that it has been supported by a number of tests, including some carried out with the data used in the tests reported above. As a result, some recent studies have not felt it necessary to test the possible role of that variable.

It should be noted, however, that the tests have typically relied on samples heavily weighted with developed countries enjoying relatively high income, simply because these are the countries for which reliable data are most likely to be available.

But one may well question whether these results can be extrapolated to very low income countries. For at least that portion of the population that lives at, or near, the starvation level may find it impossible or too burdensome to set aside resources now in order to provide for later consumption. People in that predicament may tend to live more from hand-to-mouth, skipping retirement or being supported by the extended family. It is thus conceivable that, for a sufficiently low value of per capita income, the wealth-income ratio may tend to rise with income; if so, the saving-income ratio for given growth would also tend to rise with income. There have been some tests of this hypothesis based on good-sized samples of developing countries, beginning from a much debated paper by Leff[1] based on a sample of the countries, to the recent contribution of Ram[2] analyzing a sample of 121. All of these studies present results which are consistent with the above hypothesis; for the poor developing countries the saving ratio tends to rise with income, while for developed countries, there is no significant systematic income effect. These results can be checked using newly available information assembled by the International Monetary Fund has assembled information for a large sample of developing countries (85) which was utilized in an article by Aghevli *et al.* [1]. The underlying data have generously been made available to us in the form of averages of each variable in each country for the years 1982–1988.

The variables available include, besides the ratio of national saving to national income and income growth, several variables that the authors deemed especially relevant to underdeveloped countries: per capita real income, inflation measured by a consumer's price index, terms of trade, population structure as measured by the proportion of "active" population, and a 0.1 dummy for whether the country experiencedd a debt crisis. Unfortunately no information is available on fiscal variables.

The data were used by the authors to estimate an equation which is reported on page 59 of the aforementioned publication, and is reproduced on line 1 of table 4.5. The results appear rather disappointing: (*i*) the coefficient of growth is very small, less than $\frac{1}{5}$ as large as for the OECD sample, and it is on the borderline of significance: (*ii*) the root mean square error of the estimated regression is some twice as large as the typical error in table 4.3; (*iii*) income appears to have a positive effect on saving though not significant, even at the 10 percent level; (*iv*) the effect of inflation is positive but quantitatively negligible and insignificant; (*v*) the positive coefficient for the variable "Active" is consistent with a negative coefficient for the variable "Dep" in table 4.3, since the two variables move in opposite directions.

However, after the article was published, the identification of an inconsistency between the IMF and our OECD data led the authors of the IMF article to the realization that the series they had used for the rate of growth of output was faulty and should be replaced by a different series that they kindly made available to us. The complete data set used for the table 4.5 regressions is reproduced in table 4.6.

Substituting the revised for the erroneous growth series produces a number of significant changes, including some in the expected direction, as can be seen from line 2. The standard error of the equation falls sharply and the growth coefficient

Table 4.5
Estimates for Developing Countries (1)

	Sample size & estimation technique	Constant	g	National income	CPI	Terms of trade	Active	Dummy for debt	Adj. R^2 & S.E.
1	85 OLS (2)	10.7 (1.2)	0.275 (2.0)	9.7×10^{-3} (1.52)	2×10^{-3} (1.2)	−0.53 (3.1)	0.40 (2.3)	5.02 (3.05)	0.43 6.83
2	85 OLS	2.7 (0.3)	1.32 (5.2)	1.5×10^{-3} (2.8)	2.35×10^{-3} (−0.8)	−0.56 (−3.9)	0.16 (1.0)	2.73 (1.7)	0.53 5.89
3	85 OLS	6.8 (0.9)	1.30 (5.5)	−1557 (−4.5)		−0.60 (−4.5)	0.17 (1.3)	4.2 (2.9)	0.59 (5.49)
4	82 WLS (pop.)	−16.8 (−1.7)	1.30 (6.5)	−894 (−2.6)		−0.51 (−4.3)	0.58 (3.3)	4.5 (3.1)	0.84 3.73
5*a*	24 OLS (3) (high)	22.3 (1.2)	2.0 (3.9)	11688 (1.0)		−0.91 (−3.2)	−0.13 (−0.5)	2.2 (0.8)	0.44 5.81
5*b*	61 OLS (3) (low)	6.2 (0.7)	1.33 (5.2)	−1090 (−3.1)		−0.50 (−3.4)	0.14 (0.8)	4.0 (2.4)	0.56 4.96
6*a*	42 OLS (4) (positive)	2.9 (0.3)	1.67 (4.7)	−1494 (−3.0)		−0.63 (−2.6)	0.20 (1.2)	4.3 (1.9)	0.58 5.65
6*b*	43 OLS (4) (negative)	3.6 (0.3)	1.37 (1.9)	−1426 (−2.6)		−0.57 (−3.3)	0.23 (0.9)	5.5 (2.3)	0.40 5.60

(1) The data used for the regressions in this table were obtained from the IMF and are defined as follows:
g = real annual GDP per capita growth over the period 1982–1988;
national income = per capita national income except in regressions 3–6 where the reciprocal of per capita national income is used;
cpi = consumer price index;
terms of trade = measure of a country's trade balance;
active = the proportion of the population aged 15–64;
dummy for debt = indicator variable establishing whether or not a country has debt servicing problems.
(2) This regression is the one reported on page 59 of the IMF paper cited as the source for the data in this table.
(3) This pair of regressions was fitted to the data after the observations were partitioned into two groups: 1) high income countries with per capita national income greater than \$1650, and 2) low income countries with per capita national income less than \$1650.
(4) This pan of regressions was fitted to the data after the observations were partitioned into two groups: 1) those with positive growth and 2) those with negative growth.

Table 4.6

Name	Savings rate (in %)	Growth rate (in %)	National income	Change in CPI (in %)	Change in terms of trade	% of pop active	Debt crisis
Algeria	34.7	–1	2.388.3	8.7	–14.3	49	1
Argentina	12.7	–0.9	3.189.6	338.9	–6.2	61	0
Bangladesh	9.1	0.8	140.2	10.9	2.4	52	1
Barbados	16.5	1.5	4.691.3	4.8	–3.6	62	1
Benin	3.9	–1.4	274	6.7	–1.1	52	0
Bolivia	3.5	–3.5	693.1	1.961.3	0.1	52	0
Botswana	35.3	10.9	1.233.7	9.4	4.7	47	1
Brazil	16.9	1.1	2.337.8	249.6	4.5	58	0
Burkina Faso	13.9	2.7	168.5	4.3	–2.9	49	0
Burma	12.7	1.8	191.7	11	–7.1	54	0
Burundi	9.7	0.9	253.6	6.6	5.6	52	1
Cameroon	28.4	0.2	890.7	11.7	–12.4	51	0
Cape Verde	25.8	4.1	501.7	11.7	–0.2	55	0
Cent. Afr. Rep.	2	–0.4	242.6	5.2	0	54	0
Chile	7.7	0.1	144.3	20.2	1.1	63	0
China	34.7	9.8	281.6	6.8	–1.5	62	1
Colombia	15.3	1.6	1.149.2	21.8	2.6	58	1
Comoros	6.2	–0.2	378	6.7	–2.8	49	0
Costa Rica	20.5	–0.3	3.781.6	28.4	1.1	56	0
Côte d'Ivoire	8	–3.3	767.7	5.6	–1.5	56	0
Cyprus	24.6	5.2	4.479.9	4.3	1.6	64	1
Dominican Rep.	16.6	–0.4	880.8	19.7	3.8	56	0
Ecuador	14.6	–1	949	33.5	–9.6	52	0
Egypt	10.8	2.3	472.5	19.8	–8.7	56	0
El Salvador	11.8	–0.8	902.3	19.4	–1.4	51	0
Ethiopia	7.8	–2.8	128.6	6	–1.4	50	1
Fiji	17.2	–2.6	1.823.5	6.3	3.2	58	1
Gabon	29.3	–1.6	2.673.2	6.2	–8.5	55	0
Gambia	14.8	–0.7	360.3	21.2	–3.2	56	0
Ghana	7.3	–0.4	1.219.8	40.2	–0.3	48	0
Greece	15.4	1	4.211.5	19	1.3	64	1
Guatemala	8.4	–2.6	1.158.2	13.5	–0.3	53	0
Haiti	11.5	–2.1	406.5	6.8	1.7	54	0
Honduras	9.3	–1	746.9	5.2	0.5	49	0
Hong Kong	27.5	6.3	5.453.3	6.6	–0.4	68	1
India	22	3.6	293	8.4	2.8	57	1
Indonesia	23.7	1.1	448.7	8.5	–6	56	1
Israel	16.6	0.4	7.251.4	147.1	1.8	58	1
Jamaica	14.3	–0.1	1.185.8	14.7	–6	55	0
Jordan	20	0.2	1.798.4	2.9	6.7	47	1
Kenya	19.8	–0.3	335.8	10.4	–4.2	45	1
Korea	31.9	8.2	2.290.4	4	3	64	1
Lao P.D.R.	4.2	2.5	131.5	48.7	–1.7	52	1

Table 4.6 (continued)

Name	Savings rate (in %)	Growth rate (in %)	National income	Change in CPI (in %)	Change in terms of trade	% of pop active	Debt crisis
Lesotho	21.9	1.1	269	13.9	–8.6	54	1
Liberia	17	–3.6	411.1	3.5	–1.4	53	0
Madagascar	5.8	–1.6	238.4	18.2	–2.9	51	0
Malawi	8.7	–0.3	180.1	17.6	–0.9	50	0
Malaysia	28.9	2	1.784.8	2.6	–0.5	58	1
Maldives	23	6.5	392.7	7	–1.1	53	1
Mali	4.7	1.3	167.2	3.5	0.1	49	0
Malta	24.6	1.9	4.192.3	1.1	1.9	71	1
Mauritania	5.7	–1.3	417.8	8.3	–0.8	50	0
Mauritius	20.7	5.7	1.179.5	6.2	3.4	61	1
Mexico	21.3	–2.4	2.175.6	88	–6.8	53	0
Morocco	15	1.4	654.8	7.3	–0.5	52	0
Nepal	13.8	2.1	153.9	10.7	4.5	52	1
Niger	6.5	–3.2	273.8	1.3	5.2	49	0
Nigeria	5.6	–2.5	792.2	17.8	–10.1	48	0
Pakistan	13.4	3.2	351.9	5.9	2.9	53	1
Panama	15.7	–3.4	1.918.2	1.9	0.4	57	0
P.N. Guinea	15.2	1.3	782	5.5	1	56	1
Peru	18.6	–1.9	1.079.2	182.8	0.2	55	0
Philippines	15.8	–1.4	638.5	15.2	4.4	57	0
Portugal	22.9	1.8	2.831.7	18.2	1	64	1
Rwanda	11.5	0.4	309.3	5	4.1	50	1
Senegal	4.7	1.3	465.9	7.8	–1.2	52	0
Sierra Leone	5.2	–2.9	277.3	76.5	–1	58	0
Singapore	42.5	4.9	7.213.6	1.3	–2.5	67	1
Somalia	12.2	–0.2	174.4	47.9	–2.3	49	0
South Africa	24.1	–0.8	2.309.1	14.9	–1.7	55	0
Sri Lanka	17.2	2.6	399	9.1	2.9	60	1
Syrian A.R.	17.7	–3.1	1.649.2	28.9	–14.8	49	0
Tanzania	12.1	–0.9	268	30.4	–1.7	51	0
Thailand	21.8	4.1	812.8	3.1	0	58	1
Togo	15.8	–2.1	324.6	3.1	–1.5	50	0
Trin. & Tobago	16.1	–5.3	5.514.7	11.1	–10.3	61	1
Tunisia	20.3	3.4	1.230.6	8.6	0.2	55	1
Turkey	19.4	3.5	1.213.5	41.6	3.1	58	1
Uruguay	7.6	–1.3	1.896.4	56.8	0.6	61	0
Vanuatu	5.3	–1.3	1.094.7	7.3	14.1	53	1
Venezuela	19.3	–1.6	2.931.6	15.4	–7.4	55	0
Yemen A. Rep.	10.9	5	1.021.3	16.6	–0.2	47	0
Yugoslavia	27.6	0.3	2.179.6	87.1	–2.7	67	0
Zambia	1.8	–3.3	321.1	36.8	2.2	47	0
Zimbabwe	17.8	–0.7	1.313.3	13.8	–4.4	46	1

increases over four times to a level close to that found for the OECD sample. The coefficient of income nearly doubles and becomes significant. At the same time the inflation coefficient becomes negative but entirely insignificant. Accordingly, in line 3, we have dropped that variable. At the same time we have replaced the hypothesis that the saving ratio is a linear function of income with the more appropriate one that to a first approximation saving itself is a linear function of income, but with a negative intercept. The original specification specifies that saving is a quadratic function of income, so that the marginal propensity to save rises with income, which is the opposite of what our hypothesis suggests. Our specification instead results in the saving ratio being a linear function of the reciprocal of Y:

$$S/Y = a/Y + b; \quad a < 0, b > 0$$

and implies a constant marginal propensity to save but a saving ratio rising with income and approaching an asymptote.

It appears from line 3 that replacing income with its reciprocal produces an appreciable improvement in fit. Also the coefficient of the reciprocal, reported in column 3, has a t-ratio (4.5) which is distinctly higher than that of the income coefficient in line 2, and is larger in absolute terms. This result might be considered as bad news for LCH, (as well as inconsistent with well established earlier results). But we suggest that in reality there is no inconsistency and that the result should be more appropriately considered as a generalization of the LCH which helps to clarify the limits within which the standard model may be expected to hold.

There is no contradiction because the older results were based primarily on high income countries whereas the results cited here rely on samples which include much poorer countries. The poorest country in our sample, Ethiopia, has an estimated real per capita income of 130 dollars, and the average income for the sample is 1,650 dollars, whereas the poorest country in the OECD sample, Portugal, has an income of 2,830 dollars. At the same time the results in line 3 can be shown to be consistent with the older ones because they imply that, for countries as well off as those in the OECD sample, the impact of income is negligible. To establish this inclusion we may note that the estimate of the coefficients of the reciprocal of income (1,560) implies that, for any country whose income is in the range above 1,560, the effect of income on the saving ratio is less than one percentage point, and tending to zero as income grows—in other words, it is of negligible magnitude. But that range is well below even the poorest country in the OECD sample.

Line 3 of table 4.5 has been re-estimated weighing observations by population, in line 4. The standard error of the regression is greatly reduced while the coefficient estimates are substantially the same though the role of income is somewhat downgraded, in both magnitude and significance.

As a further test of the approximate independence of saving ratio from income above some low level, we have estimated separate equations for countries below and above a cut-off represented by mean per capita income (1,650 dollars). The results, given in lines 5*a* and 5*b* confirm that income is unimportant above the cut-off. Indeed, the point estimate of the coefficient of the reciprocal becomes positive, which would imply that the saving ratio declines with income, except that it is insignificant. On the other hand, for countries below the mean, the effect is positive, and quantitatively important. For instance, the coefficient implies that for a poor country such as Ethiopia with per capita income of 130 dollars, the saving ratio might be eight percentage points below what it would be with an average income.

Lines 6*a* and 6*b* report the results of one more test suggested by the observation that the sample contains a surprisingly large number of countries that report a decline in income, namely 42, or just about half. Now the implications of the life cycle are supposed to hold in steady state. Negative growth is unlikely to represent a steady state but more likely to reflect transitory development. This should weaken the association between growth and saving ratio for countries with declining income. To carry out the test we have broken up the sample into two corresponding subsamples, one consisting of the growing countries (6*a*) and the other of the declining ones (6*b*). As expected, for the growing countries the coefficient of income growth is larger, highly significant, and of the same magnitude as for the OECD sample. For the declining countries the significance of the coefficient is borderline and the whole fit is poorer than in line 6*a*.

In conclusion, it would appear that the LCH proposition that the saving ratio does not depend on per capita income is supported, but only above some threshold that included all the OECD countries in our sample. However for poorer countries the saving ratio may be expected to respond significantly to per capita income.

But note that in the generalized LCH, the effect of income on saving, even for poor countries, is not a direct one as in the Keynesian formulation; rather, per capita resources affect the wealth income ratio which, multiplied by the saving rate of growth determines the saving ratio; in particular; it is still true that with zero growth there should be no saving, no matter what the rate of growth.

3.2 Interest Rate and Inflation

There are two further variables that we have barely mentioned in reviewing the LCH, namely the (real) interest rate and inflation, that are frequently regarded as playing a major role in saving decisions though, paradoxically, there is a lot of disagreement about the direction of the effect. We have not included them in the tests of table 4.3 because we do not regard these variables as quantitatively important and we were eager to conserve degrees of freedom. We do not expect the interest rate to be very important because we believe in a strong income effect, reflecting the high wealth income ratio. As for the inflation, aside from the role of the inflation adjustment on government interest which has been included, its effect is the uncertain outcome of two opposing effects. On the one hand inflation typically increases uncertainty and that generally makes for more saving. But on the other the uncertainty of the return from saving discourages saving; and inflation illusion, other than through government interest or some other variables might also make for lower saving. Nevertheless, before concluding, we will summarize the results of some rudimentary tests whose purpose is to make sure that the inclusion of those variables would not prove to be of major significance or radically change the results of table 4.3.

In the first test we added interest rate and inflation to line 4 of table 4.3. In the case of inflation the coefficient is fairly significant (*t*-ratio of 2.4), but rather small; an increase in inflation by one percentage point would decrease the saving rate by 0.3 points. For the interest rate the estimate is even more negative (–0.7). This is puzzling, considering that this coefficient should measure the substitution effect since the income effect is included in the income term. Now the substitution effect should, without question, be positive. However the coefficient of interest is not significant (at the one percent level). Furthermore, when the same variables are added to line 5, in which all variables are weighted by population, the coefficient falls sharply, down to near zero. On the other hand, the coefficient of inflation becomes more negative (–0.60) and more significant (*t*-ratio = 2.9). The other coefficients of the equation are largely unaffected. Similar results are obtained in tests using one variable at a time. The inflation effect is always negative, and become especially significant in the weighted form (–0.6, *t*-ratio of 3). However, the interest effect remains negative in the unweighted form but in the weighted form it turns positive, though not significant.

This admittedly crude experiment supports the conclusion that neither the interest rate nor the inflation are to be regarded as important contributors to the determination of saving, although inflation appears to have but a mildly negative impact, but mild at best. As for interest rate, though we cannot reject the hypothesis that the substitution effect has the expected sign we can say that, in any event, it appears to be small enough so that the total effect of a rise in interest in positive on consumption and negative on saving.

In concluding it may be useful to summarize our results by comparing them briefly with those reported in a recent study by Bosworth (1988). He has tested a model similar to the one underlying our table 4.3 but using time series data for

12 different OECD countries for the period 1969 to 1986. His variables are the same or analogous to ours, except for not allowing for the possibility of consumption out of the inflation correction, and for the inclusion of a time trend which unfortunately turns out to be high (and negative) for a number of countries. This is disturbing, for a trend is a non-explanation.

Income growth is highly significant, generally the most significant (except possibly for trend), with but one exception, though the coefficient is typically smaller than in table 4.1 (range 1.5 to 0.5). The estimated effects of government surplus are consistent in the two studies and therefore reject pure Barro in favor of a partial Barro effect.[3] The interest rate was included for the eight larger countries. For the four countries with the highest *t*-ratios, 2 have positive and 2 negative coefficients. For the remaining countries, 3 are positive and one negative, but none are significant. On the whole, these results do not make a strong case for a clear positive effect.

In the case of inflation, his estimated effect on saving is prevailingly positive (in 9 cases out of 12), and the coefficients are pretty stable across countries (around 0.3). This is contrary to our results, which however are not reliable enough to offer a challenge to the time series estimates.

Notes

1. See Leff, Nathaniel H.: "Dependency Rates and Savings Rates", *American Economic Review*, Vol. LIX, 1969, pp. 886–96.

2. See Ram R, "Dependency Rates and Aggregate Savings: A New International Cross-Section Study", *American Economic Review*, n. 72, 1982, pp. 537–44.

3. There is only one country for which the deficit effect is estimated to be quite large—Italy, with a coefficient of –0.67 (*t*-ratio = 4). But in a detailed study of that country (Modigliani [13]) I found that the effect was quite small, no more than –0.19.

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5

THE AGE-SAVING PROFILE AND THE LIFE-CYCLE HYPOTHESIS

Tullio Jappelli and Franco Modigliani

1 Introduction

The Life-Cycle Hypothesis posits that the main motivation for saving is to accumulate resources for later expenditure and in particular to support consumption at the habitual standard during retirement. According to the model, saving should be positive for households in their working span and negative for the retired ones, and wealth therefore should be hump-shaped (Modigliani 1986). Yet, if one looks at the microeconomic evidence on household saving rates by age, dissaving by the elderly is seldom observed.

To take just one example, in the introductory essay of a collection of country studies on saving, Poterba (1994) reports that in virtually all nations the median saving rate is positive well beyond retirement, concluding that "the country studies provide very little evidence that supports the Life-Cycle model." Based on the country studies, Poterba also reports that the median saving rate in the age class 70–74 is 1.1 percent in the United States and 6 percent in Canada; in Italy and Japan for those aged 65 and older it is even higher.

These figures are inconsistent not only with the elementary version of the LCH, but also with more elaborate versions. In its basic formulation, the LCH posits that saving behavior is forward looking and driven by the desire to prepare for future expenditures above later income throughout life. The main foreseeable event it one's life is old age and retirement. At this time earned income may be expected, on average to dwindle to a level well below active life consumption. This implies that an essential observable implication of the LCH is the existence of phases of life—notably during the retirement period—when consumption tends to exceed earned income financed by negative saving in the form of a reduction in wealth accumulated in the earning span. Refinement of the standard model, allowing for uncertainty, precautionary saving, and accidental bequests may affect the age after which one should start observing wealth decumulation. It does not, however, affect the main implication of the theory that individual wealth should eventually tend to fall with age, with saving becoming prevailingly negative. Thus, the widely reported positive saving rates at old ages are interpreted as a strong contradiction of the LCH, and as consistent only with alternative behavioral models of saving. The following are some examples: models in which irrational consumers make no preparation for retirement and cannot draw on previously accumulated assets or do not need to draw thanks to a government handout; or models in which saving is driven by the widespread imperative to leave a large bequest.

In this paper we demonstrate why these tests throw no light whatsoever on the empirical relevance of the LCH, and are at best a test of a mockery of that hypothesis. A simple explanation for this error, is that individuals have forgotten that there are multiple ways of defining income and consumption, and hence numerous ways of measuring saving, which is essentially the difference between the two.

To illustrate, one may define income as total output, nominal or real, personal, private or national, gross or net of depreciation, gross or net of various kinds of government levies, and so on which produce a dozen measures of a country's saving. Surprisingly enough, the above authors have failed to ask themselves which of these many concepts are relevant for a test of the LCH. They have thus failed to realize that there is only one concept that is appropriate: anyone who understands the spirit of the LCH—consumption smoothing in the face of the life cycle of earned income—will agree that the appropriate concept of income is *earned family income (net of personal taxes) less family consumption.*

This is the measure of income (and saving) that will be used in our tests. In other words for testing the LCH, saving must be defined and measured to include any portion of current earned income that is not consumed, but is used to provide purchasing power for later expenditures, such as retired consumption. Therefore that part of the pay that is allocated to a pension plan—private or public, voluntary or mandated—and contributes to support later consumption must be included in saving in any test of the LCH. On the other hand pensions are not currently earned income but the counterpart for past income; they are paid out of accumulated past contributions and must be excluded from income in computing saving. By contrast the authors we have criticized have routinely relied on the customary measure of saving used in the National Income Accounts, namely Disposable Income minus consumption, labeled Personal Saving. This definition differs from ours in that it endeavors to measure "cash income," and not income earned. It excludes from income and saving all contributions to Public Pension institutions. Although part of the workers pay, they are generally not received in the form of cash. Pensions, rather, are paid in cash and hence are included in disposable income and saving, although they are not part of the current pay, or income produced.

It is obvious that the concept of cash income is not only irrelevant, but also highly misleading. What comes to be subtracted and added from earned income to arrive at disposable income has the effect of "flattening out" the original path of the variables into an NIA path, which *cuts off from the age profiles of earned income, saving and wealth; the very humps* the life cycle paradigm is all about.

In most developed countries, disposable and earned income can thus be expected to exhibit a quite different life path. Using Italy as an example, we will demonstrate these propositions. Italy is admittedly an extreme case with pension contributions in excess of 30 percent and inordinately high replacement rates. But the subtractions and additions are very large in developed countries, in particular in Western Europe. It is not surprising, then, that the other authors find little life cycle left in their NIA measure and that, using the correct measure of family earnings, saving and wealth, the bumps return with a vengeance.

2 Definitions, Measurements, and Relation to Earlier Work

We implement appropriate definitions and measures of income and saving to Italian repeated cross-sectional data. In particular, income will be measured as earned income (labor plus property income) net of personal taxes. By subtracting consumption we get the relevant measure of total family accumulation, which we call *total* (*household*) *saving*. This quantity in turn can be broken down into two components:

- Contributions to pension plans less pensions received, which for practical purposes in Italy consist only in contributions to and from Social Security and are therefore referred to *as mandatory saving*.
- The difference between total and mandated saving, which we label personal or *discretionary saving*. This component coincides with the NIA "Personal Saving" and with the concept erroneously used in earlier tests.

Central to our analysis is the proposition that (net) contributions to pension plans are to be regarded as a component of total saving, because, like any other type of life cycle saving, they constitute a portion of current income that is not consumed, but used to build up reserves for later consumption. One might challenge this point of view on the ground that true saving should result in an increment in national wealth or capital. Yet, in many public pension systems—notably those that rely on pay-as-you-go financing—there is no connection between contributions and national saving. To answer this objection one must understand the relation—and interaction—between various saving flows, private and public.

Consider first the relation between mandated saving, discretionary saving, and their sum, total saving. This interaction has been the subject of pioneering contributions of Munnell (1974) and Feldstein (1976) through the "extended life-cycle model." They pointed out that pension wealth should be counted as part of individuals' resources, and argued forcefully that the transition to a social security regime would affect discretionary saving. In fact, if the LCH is correct in asserting that total saving is controlled by a target accumulation to support retirement, one might conclude that social security and discretionary wealth (or saving) should largely offset each other. This offset is what the above authors call the substitution effect—pension saving crowding out discretionary saving. But they go on to point out that this effect might be well below one-for-one because of the induced retirement effect: the provision of social security pension facilitates earlier, longer retirement, which in turn tends to raise target wealth and saving.

Since these contributions, many authors have estimated pension wealth with microeconomic data and provided age-breakdowns; to name just a few, Blinder, Gordon, and Wise (1983) and Gale (1998) for the US, King and Dycks-Mireaux (1982) for Canada, Alessie, Kapteyn, and Klijn (1997) for the Netherlands, and Jappelli (1995), Attanasio, and Brugiavini (2003) and Jappelli, Padula, and Bottazzi (2003) for Italy. Several studies also tried to estimate the impact of mandated wealth on discretionary wealth, and the prevailing conclusion is that the former does tend to reduce the latter, but the effect is well below one-for-one; in fact the reduction is generally less then one half. This result, however, may reflect not only the retirement-induced effect, but also other effects, such as ignorance or concern that in the end the social security system may be unable to deliver on its promises. Whatever the reason, the implication is that social security contributions may well result in significant increast in total household wealth, and of retirement wealth in particular.

As for the relation between personal saving and national saving, the key point is that national saving is the sum of private and government saving and so the relation depends on what happens to government saving. That, in turn, largely depends on the way pensions are financed. If they are financed by the traditional *funded* system, then the total saving is invested directly or indirectly in productive capital; thus the mandated contribution might well *increase* national saving by some fraction of the contribution. On the other hand, one can readily verify that under an ideal pay-as-you go system the social security contributions will have no effect on national saving. This is because while they increase private saving and wealth, they are offset by an equal government deficit. The situation is similar to what happens when the private sector uses some of its saving to buy newly issued government debt, and the proceeds are then used to cover a current account deficit.

The exercise we perform in this paper is related to the intergenerational accounting framework proposed by Auerbach, Gokhale, and Kotlikoff (1991) who aim to measure how much existing generations can be expected to pay to the government over their remaining lifetimes. Generational accounts provide measures of cohort-specific receipts and payments that can be used to evaluate the intergenerational redistributive impact of fiscal policy. Our focus here is primarily the computation of cohort-adjusted mandatory saving age profiles implied by the current pension arrangements. Other kinds of transfers, such as medical payments, are of course important, but are neglected in the present analysis.

Some researchers have already recognized that total saving should be estimated as the accumulation of total wealth, including pension wealth.[1] The earliest attempt is Bosworth, Burtless, and Sabelhaus (1991), who compute various saving definitions to data from the U.S. Consumer Survey. In one of their calculations, they show that with the pension adjustment the saving rates of households aged 64 plus falls from 11 to –4 percent (their adjustment refers to private pension funds but excludes social security wealth).

More recently, Gokhale, Kotlikoff, and Sabelhaus (1996) report U.S. propensities to save out of two definitions of income. *Conventional* disposable income is the sum of labor income, capital income, and pension income less taxes. *Alternative* disposable income classifies social security contributions as loans to the government, and social security benefits as the repayment of principal plus interest on past social security loans, less an old age tax. They find that the propensity to save is negative in old age, and much lower when using the alternative definition (figure 10, p. 346). They also stress that any test of the LCH should not rely on income flows, which are based on questionable definitions, but rather focus on the propensity to consume out of total resources (the sum of discretionary, pension and human wealth). It is an implication of the LCH that the propensity to consume out of total resources should be increasing with age, an implication that is supported by their figure 9 (p. 345).[2]

In section 3 we present the dataset, which is drawn from the 1989–2000 Survey of Household Income and Wealth (SHIW), a total of six cross-sections, each of which is representative of the Italian population. These data contain information not only on flows of income, consumption, pension contributions, and pension benefits, but also on the stock of wealth held by the respondents. This enables us to construct age profile of measures of assets corresponding to the three saving flows. The survey provides direct information on the wealth accumulated through personal or discretionary saving, which we label *discretionary wealth*. We construct a measure of accumulation through social security (*mandated or pension wealth*) from statutory information about contributions, imputed rate of return (growth), replacement rates, retirement age, and life expectancy. Finally, by summing the two measures, we obtain our estimates of the life profile of total wealth.

The estimate of the life cycle of wealth and its component is of interest in terms of testing whether they are hump shaped. In addition they can be used for an interesting test of the internal consistency of saving and wealth data, based on the consideration that saving should result in a corresponding change in wealth. We can thus compare the life cycle path of each component of saving with the change in the corresponding stock.

In the next two sections we report the two independent estimates of the age-profile of saving implied by our cohort data. In section 4, we first estimate the age-profile of discretionary wealth, pension wealth, and total wealth. From these estimates we derive for each component an estimate of the age profile of saving by taking the increment in wealth from one age to the next. In section 5, for each component we report the age-profile of saving measured, in the standard fashion, as the difference between the relevant measures of income and consumption flows. We find that for two concepts of saving—total and pension saving—the change in wealth and the flow measure are reasonably similar and consistent with the LCH. However, in the case of discretionary accumulation, the two measures show surprising discrepancies.

In section 6, we discuss various reasons why saving estimated from flow data may differ from the change in wealth. By relying on previous studies and other cross-sectional data, we gain significant insight into the puzzle, although we fall short of a full understanding. In section 7, we discuss the implications of the estimated age-saving profiles for the importance of bequest motives and for the evaluation of the research on the saving behavior of the elderly. Section 8 concludes.

3 The Repeated Cross-sectional Data

It is well known that age profiles of individual variables like income, consumption, or wealth cannot be estimated using cross-sectional data in a developing economy. This is because in such data the age effect is confounded with "cohort" effects—older people come from an earlier cohort—and different cohorts frequently have different experiences and resources; e.g., with steady technological progress, older cohorts are life cycle poorer than younger ones.

It is only with panel data that one can track individual saving and wealth trajectories over time. But in Italy and many other countries this tracking is not feasible for lack of long panel data with data on assets, income, and consumption. If they are available, repeated cross-sectional data can partly overcome their absence. Although the same individual is only observed once, a sample from the same cohort is observed in a later survey, so that one can track the income or consumption not of the same individual, but of a representative sample of individuals of the same cohort.

We use a time-series of cross-sections of Italian households spanning the 1989–2000 period to control for cohort effects, and to estimate age-profiles of consumption, income, saving, and wealth. The noteworthy characteristic of our dataset is that it contains separate information on income and consumption flows and on wealth, from which alternative measures of saving can be constructed and compared.

The survey we use is the Bank of Italy SHIW (Survey of Household Income and Wealth). The purpose of this survey is to provide detailed data on demographics, consumption, income, and households' balance sheets. The data set used in this study includes seven independent cross-sections (1989, 1991, 1993, 1995, 1998, and 2000), for a total of almost 50,000 observations. The appendix describes the main features of the survey. Brandolini and Cannari (1994) and D'Alessio and Faiella (2002) report further details.

To gauge the quality of the data, it is useful to compare the SHIW measures of saving and wealth with the national accounts. Table 5.1 indicates that the SHIW measure of saving is substantially higher than the national account measure in all years. This is because income in the SHIW is more accurately reported than consumption. Furthermore, the national accounts saving rate declines substantially over time, while the saving rate estimated with microeconomic data is rather constant between 1989 and 2000.[3] Finally, the SHIW wealth-income ratio is

Table 5.1
A Comparison between Measures of Aggregate Saving and Wealth Derived from National Accounts and Survey Data
The national accounts saving rate and wealth-income ratio are drawn from the Annual Report of the Bank of Italy and the OECD Economic Outlook. The microeconomic estimates of the aggregate saving rate and of the aggregate wealth-income ratio are computed using the SHIW. The microeconomic estimates use sample weights and the entire data set for each survey year.

Year	Household saving rate		Wealth-income ratio		Number of households	
	National accounts	Survey data	Financial accounts	Survey data	Total	Used in the estimation
1989	16.7	26.4	4.41	4.34	8,297	7,938
1991	18.2	24.0	4.55	4.94	8,188	7,884
1993	15.8	25.0	5.09	5.93	8,089	7,908
1995	13.6	23.4	5.00	6.03	8,135	7,993
1998	14.7	28.6	5.15	5.92	7,147	7,061
2000	11.8	27.3	5.40	6.30	8,001	7,868

generally higher than the national accounts estimate, although in this case the difference between the two is less systematic.

Brandolini and Cannari (1994) report that disposable income is under-reported by 25 percent with respect to the national accounts data, while consumption is under-reported by 30 percent. An identifiable measurement error can therefore partly reconcile the level of the aggregate saving rate with the one obtained from the microeconomic data. This should be kept in mind when evaluating the saving profiles in the next sections, especially when we compare the saving profiles obtained as the difference between income and consumption with those obtained by first differencing wealth.

Households headed by persons older than 80 or younger than 25 (regardless of year of birth) are excluded from our analysis. These exclusions are motivated by concern over two sources of potential sample bias. The first arises from the mortality problem. As a rule, surveys only elicit information from individuals who are alive at the time of the survey. Surveys therefore miss information on the fraction of the cohort that was alive at the time of one survey, but died before a later wave. The seriousness of this problem is demonstrated by the fact that we have no ground for believing that the survivors are an unbiased sample of those who were alive at the last interview. It is well known that survival probabilities tend to be positively correlated with wealth, which implies that the non-survivors will tend to have lower income and wealth than the survivor sample. The proportion of non-survivors rises with age: e.g., for Italy, between age 60 and 64, it reaches 5 percent for women and 11 percent for men. However, once we reach the age class 80–84 it rises to some 30–40 percent. Clearly the information obtainable from survivors over 80 cannot be regarded as representative.

The second source of potential bias is a correlation between wealth and young household heads peculiar to our sample. In Italy, young working adults with independent living arrangements tend to be wealthier than average, because most young working adults live with their parents.[4] For instance, the fraction of income recipients below 30 years of age is about 20 percent, while the fraction of household heads in that age bracket is less than 10 percent. Excluding observations whose data was missing for consumption, income or wealth, our final sample consists of 46,945 households.

We use the repeated cross-sections to sort the data by the year of birth of the head of the household. The first cohort includes all households whose head was born in 1910. The second includes those born in 1911, and so on up to the last cohort, which includes those born in 1974. We then create 334 age/year/cohort cells. The average cell size is 136; the minimum is 21 and the maximum 218. As with other survey data, the saving and wealth distributions are skewed, and means may not adequately characterize the age-wealth or the age-saving profiles. We therefore rely on the median as measure of location and on median regressions for econometric analysis.

4 Estimates of Saving Age Profiles with Wealth Data

Figure 5.1 offers fundamental insights into the process of wealth accumulation, plotting the median discretionary wealth of 11 cohorts. Discretionary wealth is the sum of financial and non-financial assets, net of liabilities. To make the graph more readable, we plot only the wealth of selected cohorts. The numbers in the graph refer to the year of birth, extending from 20 (individuals born in 1920) to 70 (individuals born in 1970). Except for the youngest and the oldest generations, each cohort is observed at six different points in times, one for each cross-section. The cross-sections run from 1989 to 2000. Thus, each generation is observed for 11 years with each line being broken (for instance, cohort 20 is sampled six times from age 69 to age 80).

Following retirement, median wealth falls for most cohorts. This fact is more evident in figure 5.2, where we plot the wealth profile for cohorts sampled in old age (again, to make the graphs more readable, we consider only cohorts born in

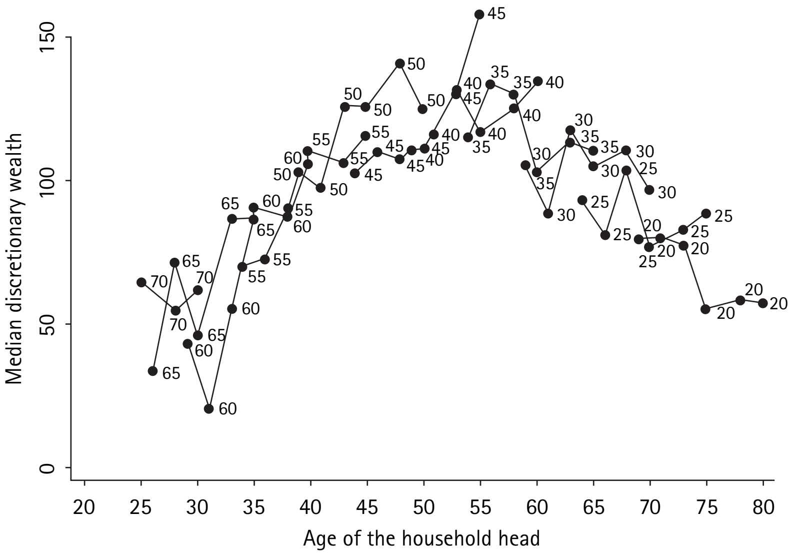

Figure 5.1
Discretionary wealth, by age and cohort
The figure plots median discretionary wealth, by age and cohort. Each number in the graph represents a different cohort, going from 20 (household heads born in 1920) to 70 (heads born in 1970). Wealth is expressed in thousand euro

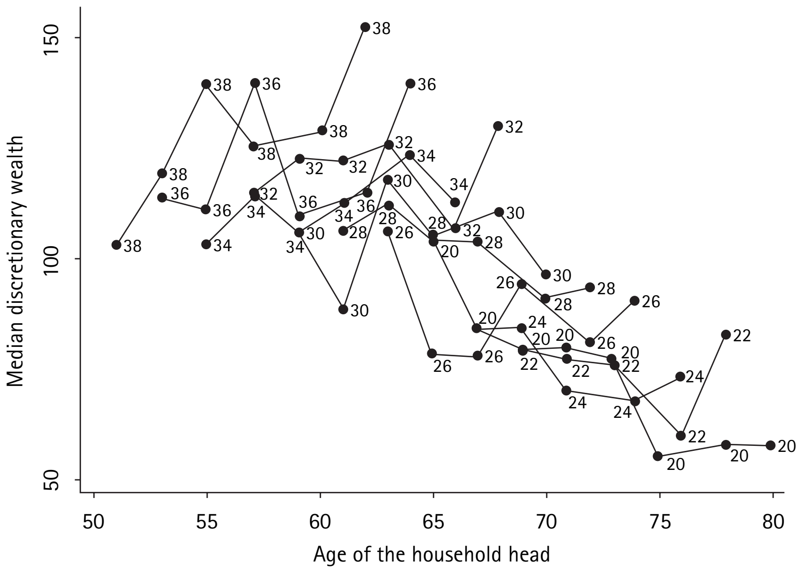

Figure 5.2
Discretionary wealth of older cohorts
The figure plots median discretionary wealth, by age and cohort. Each number in the graph represents a different cohort, going from 20 (household heads born in 1920) to 36 (heads born in 1936). Wealth is expressed in thousand euro

even years). Although some cohorts exhibit an increase in discretionary wealth between age 60 and 65, wealth declines considerably at the oldest ages.

Figures 5.1 and 5.2 also clearly show the presence of cohort effects (the broken lines for younger cohorts tend to lay above those of the older ones). Time effects also affect the data. For instance, the wealth of several cohorts increases in 2000, reflecting either measurement errors or common business cycle shocks.[5]

In the absence of uncertainty the LCH implies that the shape of the wealth profile depends on age, regardless of resources, while lifetime resources, regardless of age, set the position of the profile. Introducing a positive real interest rate or more realistic earnings profiles does not change this basic implication of the model.[6] Therefore, one way to combine the information contained in figure 5.1 in a framework that is consistent with the LCH is to assume that the shape of the age-wealth profile is the same for each cohort, and that its level depends on cohort-specific intercepts, which primarily reflect differences in productivity across generations.

With uncertainty, macroeconomic shocks lead to revisions in households resources, and therefore in assets accumulation. Measurement errors can also generate disturbances to the wealth equation. However, it is well known that the separate effects of age, cohort, and time cannot be identified.

One possibility is to rule out uncertainty and measurement errors and eliminate all year dummies. Deaton and Paxsom (1994) adopt a slightly less restrictive approach and assume that the year dummies sum to zero and are orthogonal to the time trend. This is equivalent to assuming that all trends in the data can be interpreted as a combination of age and cohort effects and are therefore, by definition, predictable. The time effects then reflect additive macroeconomic shocks or the residual influence of non-systematic measurement error. It is important to keep in mind that this normalization of time effects rules out time-age or time-cohort interaction terms.

The wealth equation is estimated on 334 age/year/cohort cells. Given the structure of our sample, the regressors include 55 age dummies, 64 cohort dummies, a set of restricted time dummies, and a constant term. Under the assumptions described above, the estimated dummies can be interpreted as an individual age-wealth profile, purged from cohort effects, while the cohort dummies can be interpreted as the cohort (or generation) effect. All other estimates and profiles reported in the paper are constructed in similar fashion.

For illustrative purposes figure 5.3 plots the estimated coefficients on each of the 55 age dummies and a smoothed profile obtained by fitting a third-order

Figure 5.3
Age profile of discretionary wealth
The age dummies are obtained from a regression of median discretionary wealth on age dummies, cohort dummies, and restricted time dummies. The smoothed profile is obtained by regressing the age dummies on a third order age polynomial. Wealth is expressed in thousand euro

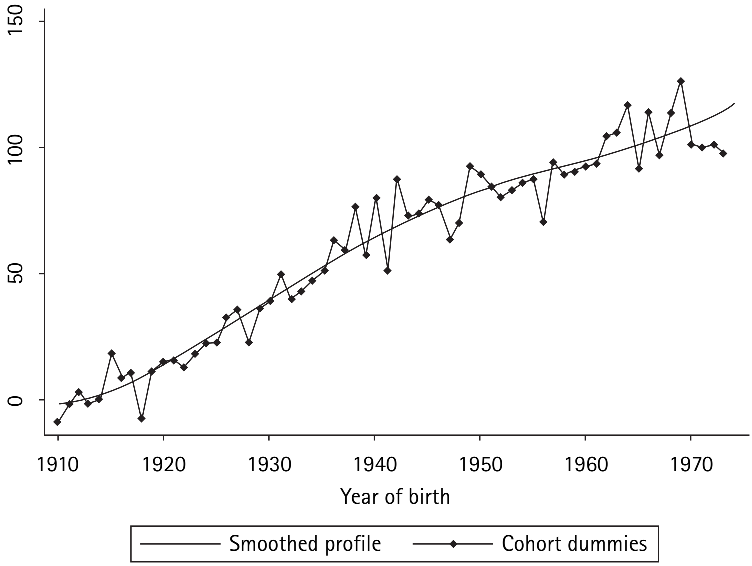

Figure 5.4
Cohort effect of discretionary wealth
The age dummies are obtained from a regression of median discretionary wealth on age dummies, cohort dummies, and restricted time dummies. The smoothed profile is obtained by regressing the age dummies on a third order age polynomial. Wealth is expressed in thousand euro

polynomial to the original coefficient estimates. In figure 5.3 discretionary wealth rises until age 60, the peak retirement age in Italy. It then declines slowly, but remains substantial even in old age.

Figure 5.4 plots the estimated cohort effect in wealth—that is, the coefficients of the 64 cohort dummies. Even though the estimated coefficients exhibit some noise, the smoothed profile is remarkably stable, increasing by about one thousand euro per year of birth. The most natural explanation for the shape of the cohort effect is that it reflects the increase in productivity (and therefore resources) of each successive generation.

In figure 5.5, we plot pension wealth and total wealth, the sum of discretionary and pension wealth. Pension wealth is the difference between the discounted values of social security benefits and contributions.[7] Constructing pension wealth thus requires assumptions about expected benefits and expected contributions, since they depend on projected income, demographic trends, and future legislation. The concave shape of pension wealth is not surprising, since the pension system collection and payment are designed to be an imitation of the LCH saving and dissaving.

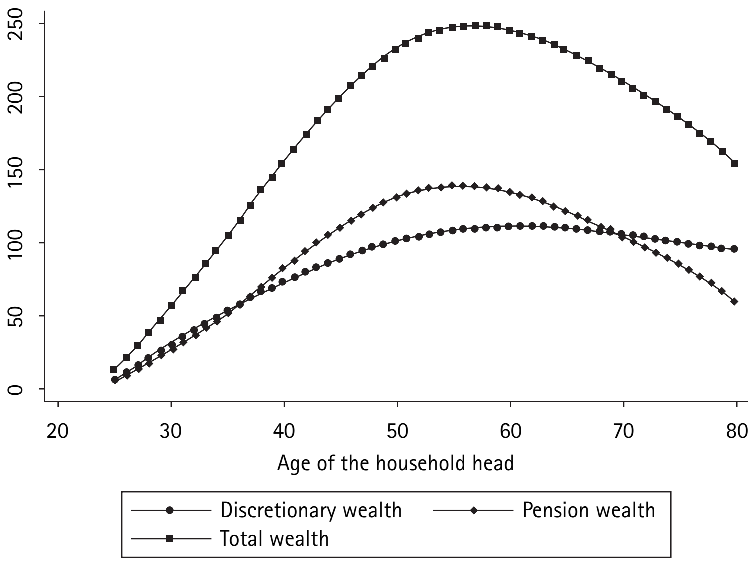

Figure 5.5
Age profile of discretionary, pension, and total wealth
The lines in the figure represent the age dummies estimated from regressions of median discretionary wealth, pension wealth, and total wealth on age dummies, cohort dummies, and restricted time dummies. The age dummies are smoothed with a third order age polynomial. Wealth is expressed in thousand euro

Total wealth is also hump-shaped, rising to a peak of around 250 thousand euro at age 60, and then declining during retirement. By taking first differences of the estimated wealth and of its components at successive ages we can obtain a picture of the age profile of the accumulation of net worth through life, which is shown in figure 5.6. In this figure, discretionary saving is computed as the change in discretionary wealth, and is similar for mandatory saving and pension wealth. Total saving corresponds to the change in total wealth. All three measures of saving derived from wealth are seen to conform to the implications of the LCH. They are positive during the work-span, and negative after the retirement age of 60.[8]

5 Estimates of Saving with Flow Data

In this section we present an alternative estimate of the age profile of saving based on income and consumption flows. We construct two measures of income. One is earned income, which we claim is the appropriate measure, and the other is the conventional disposable income that was used inappropriately in earlier tests.

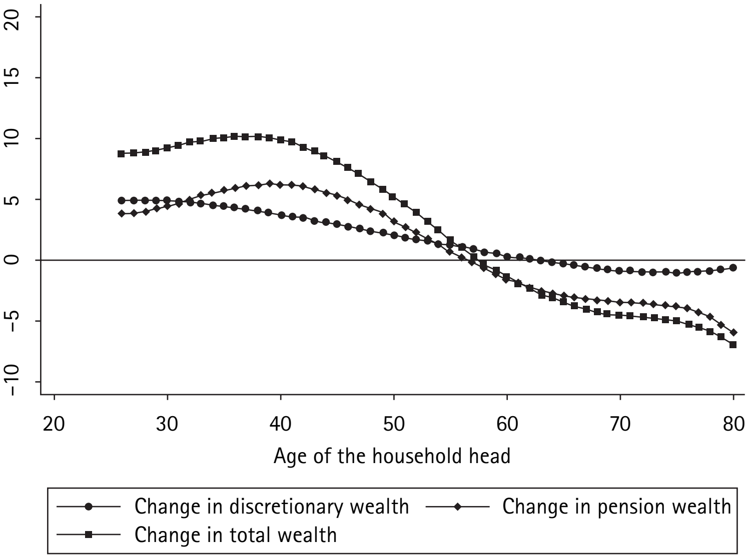

Figure 5.6
Age profile of change in wealth
The lines in the figure represent the increment from one age to the next of discretionary wealth, pension wealth, and total wealth in figure 5.5. Data are expressed in thousand euro

Consumption is the sum of expenditure on durable and non-durable goods and includes imputed rents on owner occupied housing.

Conventional disposable income is obtained directly from the respondent, but then adjusted for imputed rents. Earned income is obtained by adding the mandated contribution to social security, which is taken as an approximation to mandated saving through public pensions, and subtracting pensions, to disposable income. It should be recognized that social security contributions should be counted as part of income and saving to the extent to which contributions actually increase the stock of wealth. We approximate mandatory saving with the total amount of contributions actually paid by each worker less the total amount received as pension benefits. Since contributions are levied at a flat rate on gross earnings, we can estimate earned income by blowing up reported disposable income.[9]

As it is estimated, the mandatory saving need not coincide with the change in pension wealth reported in figure 5.6. There we take into account survival probabilities, and make explicit assumptions about the growth rate of the economy and the real interest rate. Here we simply impute the yearly contributions from net earnings and the benefits from the respondent. In practice we find that the two measures of mandatory saving are broadly consistent.

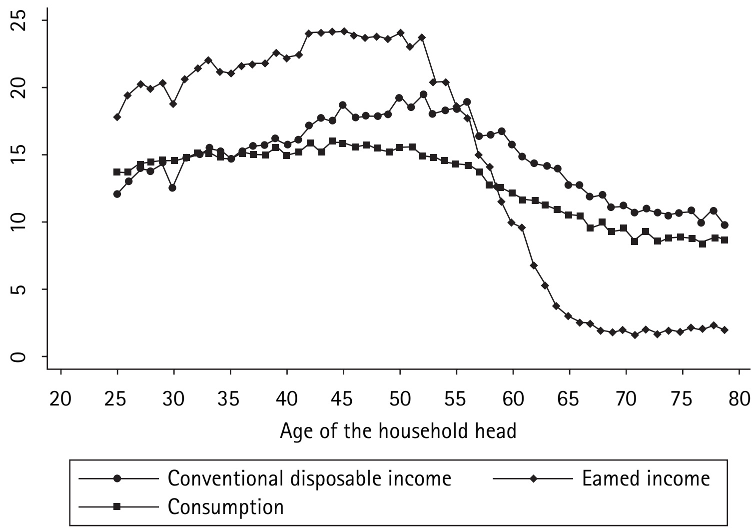

Figure 5.7
Age profile of consumption and income
The age dummies are estimated from regressions of median income and consumption on age dummies, cohort dummies, and restricted time dummies. Income and consumption are expressed in thousand euro

In figure 5.7, we plot the age profile of consumption and of the two income measures. Each profile is obtained using the microeconomic data in the same fashion used for wealth. Median consumption and income in each age/year/cohort cell are regressed on a full set of age dummies, cohort dummies and restricted time dummies. The smoothed coefficients of the age dummies are then plotted in figure 5.7.

As hypothesized by LCH, consumption is remarkably flat. The moderate hump appears to reflect a similar hump in the age profile of family size reported in figure 5.8, which mirrors the entrance and exit of children (and one spouse) from the households. Since it is not essential to our basic argument, we forego any adjustment for family size.

In figure 5.7, the profile of earned income, in contrast to that of consumption, is very hump-shaped. It peaks around age 50, which in some respects is surprisingly early. This peak may reflect the very young age at which some pensions have been awarded in Italy (so called baby pensions). It declines rapidly after age 55, a reflection of the increasing number of retired individuals belonging to older age groups. Retirement earned income consists mainly of capital income (the return to discretionary wealth), much of which is accounted for by imputed rents on owner occupied housing.

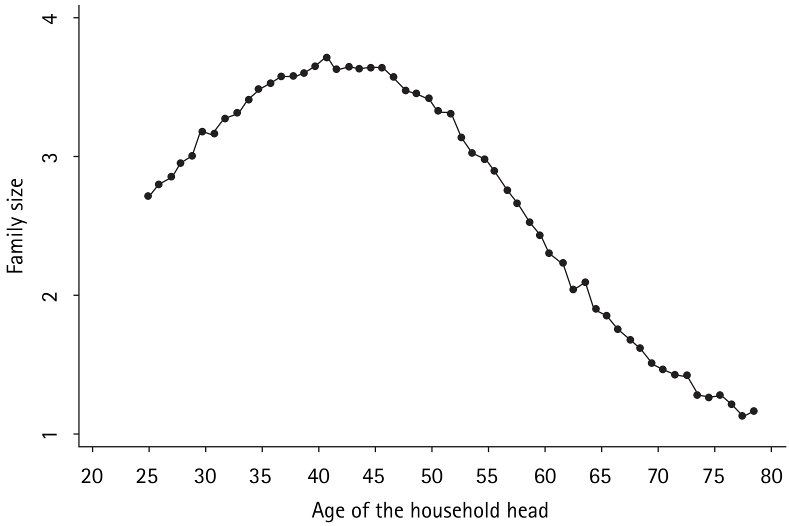

Figure 5.8
The life cycle of family size
The age dummies are obtained from a regression of family size on age dummies, cohort dummies, and restricted time dummies

Figure 5.9 reports the life cycle of saving and its component, implied by figure 5.7. The path of total or family savings is sharply humped, reflecting the hump path of earned income and the flat path of consumption. It becomes increasingly negative after the mid-fifties, which is a pattern largely consistent with the LCH. It is also largely consistent with the pattern implied by the change in total wealth in figure 5.6.

A comparison of the graph of earned and disposable income reported in figure 5.7 brings to light how conspicuous subtractions from, and additions to, earned income for Social Security contributions and pensions have the effect of largely smoothing and eliminating the humps in earned income (which is of course what they were designed for). As a result, the humped life cycle of earned income is turned into a remarkably flat path of disposable income, very similar to the life cycle of consumption. In fact disposable income and consumption stay very close, so that the path of discretionary saving in figure 5.9 is itself quite flat. What is surprising is that disposable income is consistently *above consumption*, which means that discretionary saving *remains positive* throughout the life cycle, or at least until the age of 80.

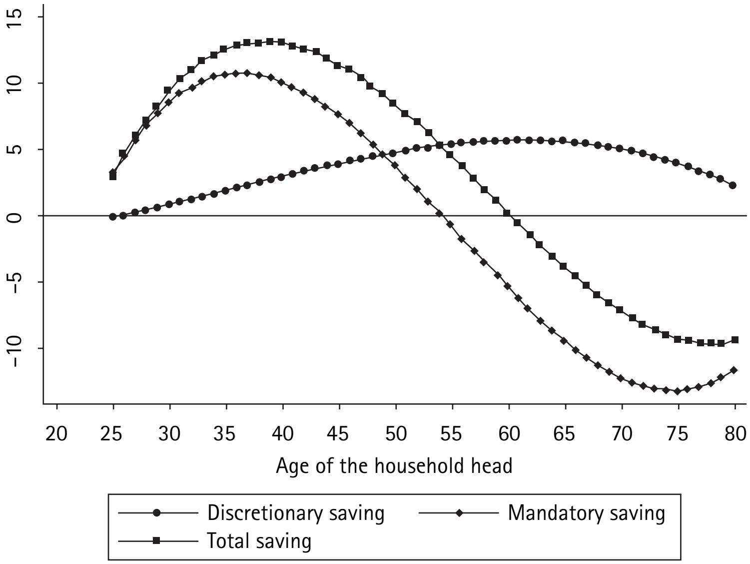

Figure 5.9
The age profile of discretionary, mandatory, and total saving
The lines in the figure represent age profiles of discretionary saving, mandatory saving, and total saving. Discretionary saving is the difference between conventional disposable income and consumption; total saving is the difference between earned income and consumption; mandatory saving is the difference between total and discretionary saving. Each profile is estimated from regressions of saving on age dummies, cohort dummies, and restricted time dummies. The age dummies are smoothed with a third order age polynomial. Saving is expressed in thousand euro

6 Tracking the Sources of Discrepancy Between Saving and Wealth Measures

The finding of a flat profile for discretionary saving is a somewhat surprising result. This is for two reasons. First, it is glaringly inconsistent with the reported behavior of discretionary wealth. The flat profile implies that discretionary saving never declines, even after retirement, whereas according to the wealth data in figures 5.5 and 5.6 discretionary saving declines steadily from around age 60. The inconsistency is illustrated in figure 5.10. Here, the graph compares the two alternative estimates of the life cycle of discretionary saving. Not only do they have opposite signs at advanced age (after 60), but also at the youngest ages (below 30). Indeed the two paths look like mirror images of each other! In addition, the absence of negative saving has implications for the role of bequests, which we discuss below.

To search for the explanations for the discrepancy between saving and wealth measures, we begin by examining the possible role of factors primarily connected

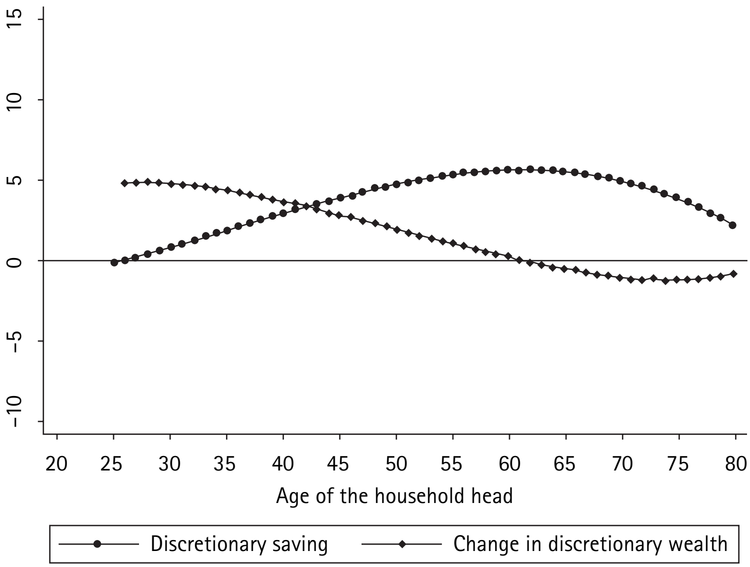

Figure 5.10
Age profile of discretionary saving and of increment in discretionary wealth
The lines in the figure represent the age profiles of the increment in discretionary wealth (from figure 5.6) and of discretionary saving (from figure 5.12). Saving is expressed in thousand euro

with the Italian data or institutions. We then turn to evidence for other countries, and for conceptual differences between saving and changes in wealth.

6.1 Sources Idiosyncratic to Italy

The Role of Smoothing

We began by considering the possibility that the finding of positive saving, even at an advanced age, might be the result of *distortions* induced by smoothing the life cycles of consumption and income by polynomials in age. On the basis of figure 5.11, which shows a plot of the original (not smoothed) data of discretionary saving by age and cohort, we promptly dismiss this hypothesis. Clearly saving is humped, but even by age 80, though it is smaller than at any other age, it is unmistakably positive. This is once again confirmed by the smoothed graph of discretionary saving, exhibited in figure 5.12. Regression analysis—not reported for brevity—indicates that income, wealth, homeownership, education, self-employment, and family size are positively correlated with saving at advanced age. But the most noteworthy feature of saving is that less than 20 percent of the aged households report negative saving, and that this proportion does not increase with age, even conditioning for employment status, education, homeownership, or other household characteristics.

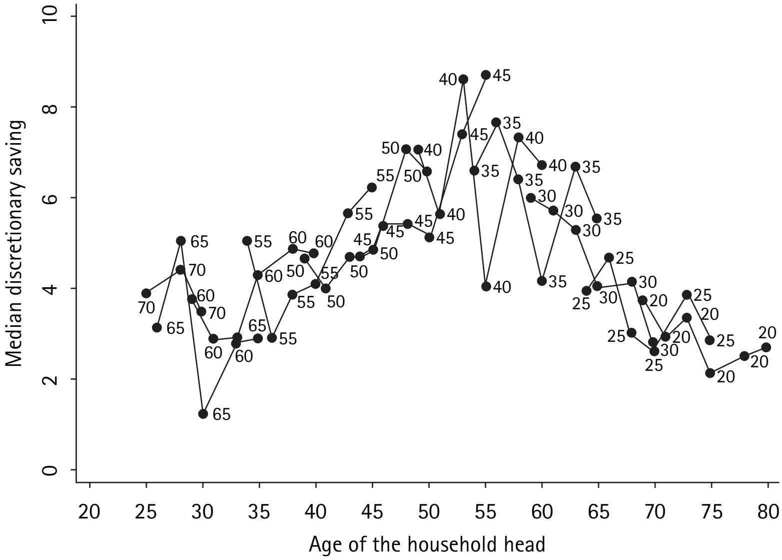

Figure 5.11
Discretionary saving, by age and cohort
The figure plots median discretionary saving, by age and cohort. Each number in the graph represents a different cohort, going from 20 (household heads born in 1920) to 70 (heads born in 1970). Saving is expressed in thousand euro

Figure 5.12
The age profile of discretionary saving
The age dummies are obtained from a regression of median discretionary saving on age dummies, cohort dummies, and restricted time dummies. The smoothed profile is obtained by regressing the age dummies on a third order age polynomial. Saving is expressed in thousand euro

Measurement Error

Next, we have examined a possible role of measurement error, whose presence was already suggested by the comparison of microeconomic data with national accounts data carried on in table 5.1. Cannari and D'Alessio (1994) and D'Alessio and Faiella (2002) report that in comparison with national account data, the SHIW consumption is underestimated by 30 percent, while income is underestimated by 25 percent. For the purposes of this paper we are not so much concerned with bias in the level of income and consumption as we are with whether the *age profile* of saving is systematically biased.[10] For this reason we cannot rely on a comparison with macroeconomic data, and must turn to the only other Italian survey that contains data on family income and consumption, namely a 1991 ISTAT Survey of Household Budgets. This survey represents a large cross-section and contains detailed information on consumption, but has limited income data. We find that the two surveys are generally mutually consistent. Both indicate that saving is rather flat and that there is no negative saving in old age.

The Role of Severance Pay

A third possibility we considered is that the difference between the two measures of saving could be related to one very unique feature of the Italian system: namely, a mandated large severance pay. Each year the employer withholds one months' wage, to be returned to the worker at the time of termination of employment.[11] Such *severance pay* is not included in our definition of income (either disposable or earned). Because survey information on severance pay is scant, we impute contributions paid in the severance fund pay, grossing up labor income in proportion to the contribution rate. The result is that mandatory saving and pension wealth are correspondingly higher before retirement. After the severance pay fund is liquidated, generally at retirement, it should appear as an increment in discretionary wealth without changing discretionary saving. This should generate an increase in discretionary wealth around retirement that is greater than discretionary saving; the opposite of what we see in the data.

6.2 Evidence from Other Countries and Explanations

The International Evidence

We looked to evidence from other countries to see whether it might confirm or reject the findings for Italy. In particular, we were concerned with the apparent inconsistencies between flows and stocks. With respect to discretionary saving, we found that age profiles flat and *uniformly positive* were reported by several studies for those countries where the amount of mandatory saving is substantial. To take just a few recent examples, Poterba (1994) summarizes evidence to this effect for the group of most industrialized countries, Alessie et al. (1997; 1999) for the Netherlands, and Paxson (1996) for the U.S. and the U.K.[12]

At the same time, the decline in wealth at advanced ages is also confirmed by earlier studies. For instance, Diamond and Hausman (1984) find rates of dissaving after retirement of about 5 percent per year in the National Longitudinal Survey of Mature Men. Using the Retirement History Survey, Hurd (1987) finds decumulation rates of about 1.5 percent per year (3 percent excluding housing in the definition of wealth). Furthermore, the inconsistency has been reported in other studies for which both saving and wealth data were available.

Widespread evidence pointing to the apparent inconsistency between stock and flow measures suggests the need to reexamine possible conceptual discrepancies between saving and the change in the value of assets in a given period, as measured in the survey. The search resulted in identifying multiple sources of discrepancies, which fall into two quite distinct types. The first type arises from *conceptual differences* between saving in a period and the change in the value of assets during the period, as measured in standard surveys. The second type results from the fact that as a cohort ages, the individual composing it keeps changing, with young people joining (becoming household heads) at different ages and others exiting (because of disability or death).

Conceptual Differences between Saving and Changes in Wealth

The main conceptual discrepancies arise from the fact that wealth cannot only change through saving, but also through other channels. Two are of major interest: capital gain/losses and inter vivos transfers.

To what extent could capital gains explain the observed discrepancies? Because of formidable obstacles to the estimation of a time series of gains, we cannot offer a precise answer. The best we can do is to estimate the capital gains and losses on *nominal net* assets resulting from the widespread incidence of inflation, especially in the early years. Since in this period the inflation resulted in net capital losses, the adjustment made the change in assets somewhat less negative, but not enough to turn it positive. As for the effect of other capital gains, even in the absence of detailed information we can be fairly certain that there were substantial real gains coming from residential real estate during the period covered. Hence, while one might have expected the increase in wealth to generally exceed saving, these inferences are not supported by the data in figure 5.10. We must conclude that capital gains do not help to explain the discrepancy.

Ando et al. (1994) and Alessie et al. (1999) have suggested a second possible source of conceptual discrepancy, namely inter vivos transfers. They speculated that the two measures of saving could be reconciled by the pattern of intergenerational *net* transfers. These transfers flow prevailingly from the old to the

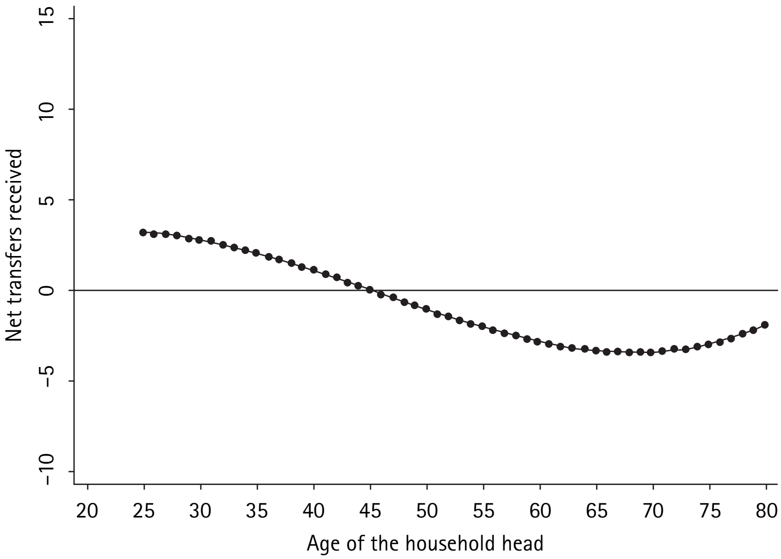

Figure 5.13
Age profile of net inter vivos transfers received by the household
Average net transfers by the household are the difference between transfers received and transfers given. The profile is obtained by regressing net transfers received on third order age polynomial using the pooled 1998–2000 sample. Transfers are expressed in thousand euro

young, and are encouraged by the favorable taxation of gifts. If so, for the younger generations the increase in assets should tend to exceed saving (by net gifts received), while for the old it would tend to fall short (by the net gifts made). This is precisely the nature of the puzzling discrepancy in figure 5.10.

Neither Ando nor Alessie provide direct evidence to support their hunch on the issue. We have been able to carry out a rough test of the hypothesis on the basis of information available in our survey. Unfortunately, the information in consistent form is only available for the years 1998 and 2000. Using the response for inter vivos transfers, we have computed the age profile of average net transfers received by the household.

Because we have only two years' worth of information, we cannot apply the method used for other variables to disentangle the age effect from the cohort effect. To bypass this issue, we simply compute average net transfer by age, and then fit to these values a third degree polynomial in age, in this way smoothing the result. The smoothed curve is reported in figure 5.13. Because figure 5.13 is derived from cross-sectional means and is not adjusted for cohort effects, this profile is not directly comparable with the life cycle of saving presented in figure 5.10. However, the path of net transfers is amazingly similar to the difference between the two inconsistent measures of saving reported in figure 5.10. For the young (who are the net receivers in figure 5.13) the increase in discretionary wealth exceeds saving, whereas for the old, (who are the net givers) the increase in wealth is lower than discretionary saving. Thus the change in wealth tends to overestimate the saving of the young and to underestimate the saving of the old. Still, we are not in position to say whether the explanation is sufficient, or whether it can account for the observed decline in wealth even though saving is positive.

The Effects of Changing Composition of the Cohort

Despite its systematic and potentially important effect on the relationship between the two measures of accumulation, the membership of the cohort being sampled and the fact that it changes between surveys has gone all but unnoticed. One measure is the change in average wealth (W) between the surveys at age t and that at age $t - 1$, say $W(t) - \mathrm{W}(t - 1)$, and the other is the (average) saving rate in the interval between the two surveys, say $S(t)$.

In order to concentrate on the above issue it will be convenient to assume that, for every *individual present*, the change in assets between the survey at time t and the previous one at $t - 1$, is identical with his or her saving over the stated interval. We begin by classifying those living at $t - 1$ into two sets, namely those that are alive and therefore respond at time t, the "survivors," and those dying during the interval, the "deceased." Since at time t only the survivors are responding, the average saving revealed by the survey will be precisely the average saving of the survivors, which we denote by S_s. Thus $S(t) = S_s(t)$, which in turn equals the change in the average wealth of the survivors, $W_s(t) - W_s(t - 1)$. Thus:

$$S(t) = W_s(t) - W_s(t-1), \tag{5.1}$$

gives $S(t)$ in terms of the change in the average wealth of the survivors. We want to express it in terms of the change in the average wealth of the cohort, i.e. $W(t) - W(t - 1)$. Because the survivors are the only respondent at time t, we can conclude that $W_s(t)$ is the same as $W(t)$.

The remaining task is to establish the relation between $W_s(t - 1)$ and $W(t - 1)$. That relation is easily secured by noting that $W(t - 1)$, being average wealth at time $t - 1$, is simply a weighted average of the average wealth of the survivors $W_s(t - 1)$, and of the deceased, $W_d(t - 1)$, weighted by the share of survivors (N_s/N) and of the deceased N_d/N in the cohort at $t - 1$, respectively.

We can use this relation to substitute in (5.1) for $W_s(t - 1)$ in terms of $W(t)$, and rearrange terms to get the following expression for the change in average assets:

$$W(t) - W(t-1) = S(t) + [W_s(t-1) - W_d(t-1)](N_d/N) \tag{5.2}$$

The interpretation of (5.2) should be obvious: if the survivors are richer than the deceased, then the average wealth will rise more than the average saving because it will be boosted by the elimination of a segment poorer than the average. As seen, there exists only one case in which the change in assets coincides with saving; this is when the deceased happened to be, on average, as rich as the survivor.

Unfortunately (5.2) cannot be tested empirically since $W_s(t-1)$ could only be ascertained with a panel, and even with a panel, $W_d(t-1)$ could not be estimated because as a rule, a panel will only interview survivors. Therefore, it cannot provide information about the estate of the deceased. However as suggested earlier, it is well known that the probability of surviving is positively correlated with income and wealth, implying that on average the deceased are poorer than the survivors, i.e. $W_s(t-1) > W_d(t-1)$.

Currently, this conclusion is receiving further support. Through a novel and imaginative undertaking, Michael Hurd and his associates are endeavoring to secure information about the estate at the time of death of deceased members of a US Panel survey. From the information gathered from the Retirement History Survey, Hurd, McFadden, and Merrill (2001) have constructed a table comparing the average wealth in the base year (1993) for the 75–79 age group of those who were still alive in 1995—the survivors—with the wealth of those who subsequently died between 1993 and 1995. For males, the average 1993 wealth of the survivors in dollars thousand was 217, and the average estate of the deceased was 178, which is 18 percent smaller. For females, the corresponding figures are 167 and 145, which is 13 percent smaller.

To the extent that these figures are representative, they suggest an appreciable discrepancy between S and the change in per capita wealth; indeed for the age group 75–79, the ratio of those dying in a year to the survivors can be placed at around 5 percent, and the difference $W_d(t-1) - W_s(t-1)$ at $0{,}85 \times W_s(t-1)$, implying a discrepancy of about 4 percent of wealth, which is large compared with the saving rate (e.g., in Italy, the average wealth holding is not far from 20 times saving). Clearly the size of the correction would vary with age. What is disappointing is that the above reasoning implies that at advanced ages the increase in wealth should *exceed* saving, whereas figure 5.10 shows the very opposite. Of course our deceased effect could be overpowered by others, like the inter vivos transfer effect.

With this background we can also deal quickly with the effect of new entrants in the cohort. Again, we classify those responding to the survey at time t into two sets, those who were present at time $t-1$—the survivors—and those that entered after $t-1$, the entrants. The per capita saving S is the average, of the average saving of each group, weighted by their respective shares of the cohort population. Similarly, the average *terminal* wealth is the weighted average of the wealth of each group, while the average wealth at $t - 1$, $W(t - 1)$, is only the wealth of the survivors, $W_s(t - 1)$. One can then readily verify that the change in average wealth is:

$$W(t) - W(t-1) = S(t) + [W_e(t-1) - W_s(t-1)](N_e/N) \tag{5.3}$$

Thus, the growth of wealth will exceed average saving if the new entrants are wealthier than the survivors. Again, we have no direct (or indirect) information about the average initial wealth of the two groups; but in figure 5.10 we observe that there is a large excess of asset growth over saving. We are lead to wonder whether it might reflect a not implausible, but relatively higher wealth of the entrants together with a high rate of entry.

7 Hump Wealth, LCH, and the Bequest Motive

The search for an explanation of the discrepancies between saving flows and changes in wealth has brought out a number of possible causes. With one exception, which is discussed below, these causes do not seem to go very far in explaining the empirical discrepancies revealed by figure 5.10. Indeed, some suggest a discrepancy opposite to that observed. The one possibly plausible explanation, which could be quantitatively large, is inter vivos transfers. It is substantiated by independent evidence on the timing and importance of intergenerational transfers.

When the two measures are in conflict, which of them provides more useful information? We suggest that there is no unique answer. The two measures focus on somewhat different things, thus which is best depends on what is to be measured. If one is interested in the life path of wealth, one should use the wealth measure and by the available evidence agree that wealth tends to decrease after retirement. If, on the other hand, one is interested in measuring the extent to which in old age, discretionary wealth continues to accumulate through saving—or is decumulated through consumption above income—than the appropriate measure of accumulation is saving. For the case of Italy we must conclude that discretionary saving remains positive, although small.

This conclusion does not have any direct implications for our test of LCH, but it is relevant for an understanding of bequest behavior, and of the relative contribution to national saving of Life Cycle accumulation versus accumulation for intergenerational transfers. It should be remembered in this connection that, contrary to a common perception, some accumulation for bequests is not inconsistent with LCH. The distinctive feature of this model is not the absence of bequests, but rather that some portion of accumulated wealth is drawn down to support retired consumption, which essentially means that the age profile of wealth is hump shaped. In the case of Italy we have seen that this implication is amply verified both in terms of stocks (figure 5.5) and flows (figure 5.9).

The age-profile of discretionary wealth is relevant for bequest behavior because it is the only component of wealth that can be passed to future generations if it is not consumed before the termination of the household. In contrast, annuitized wealth, which includes all public pension rights, disappears when the retired dies: even though part of wealth is transferred through survivors' benefits, the survivors cannot transfer the capital to future generations.

Suppose one observes a hump in total wealth, but not in discretionary wealth, which keeps increasing after retirement. This must imply that part of pension wealth is used to increase discretionary wealth. If retirees continue to accumulate bequeathable wealth, the most natural interpretation is that they plan to leave a bequest.

Beyond this qualitative statement, the age profile of discretionary wealth cannot provide much guidance for estimating the quantitative importance of intergenerational transfers as a source of accumulation. There are several reasons for this conclusion. In the first place there is considerable uncertainty about the path of discretionary wealth in the retirement stretch; we recall that, where data concerning the behavior of wealth are available, they consistently show that it declines in the post retirement period. To be sure, saving estimates, which may conceptually provide a more reliable measure, do not support the decline but still leave room for some uncertainty, especially when one recalls that the survey responses tend to overestimate saving by well over 50 percent.

There is yet a more serious problem to consider, namely the likelihood that, at advanced ages, the saving estimates generated by the survey may be seriously upward biased. It must be remembered that what we are interested in estimating is the average saving rate (disposable income—consumption) during a given age, say the 75th year, for all those members of the cohort that have survived until age 75. What we can draw from the survey (or a panel) is the average saving of those who survived until the next survey, since they are the only ones that can respond as regards their activity in the past period (the 75th year). The overall average we are seeking is clearly the weighted average of the saving rate of the survivors and of the deceased weighted by the size of the two groups. Unfortunately we have no available estimate of the saving rate of the deceased, for the simple reason that they cannot be interviewed. However, we can make some educated guesses about the unknown saving rate by recalling earlier references to a well-established negative correlation between mortality and wealth. It implies that on the average the deceased are poorer than the survivors and should tend to save less. But in addition, the saving of the deceased might be adversely affected by the occurrence of death.

This hypothesis receives some confirmation. Hurd and associates worked to secure information about the estate of deceased members of a panel. These finding make it possible to compare the wealth reported in the last survey preceding death with the estate ascertained as of the time of death; the difference, then, is an estimate of the saving during the interval. According to the figures reported in Hurd and Smith (2001), the average value of the estate at death (95 thousand dollars) was much smaller than that preceding death (130 thousand) by over one-quarter, implying a large rate of dissaving. Since the deceased constitute a rapidly rising share of those surviving, a substantial dissaving on their part, together with a small and dwindling saving rate of the survivors, could easily push the overall average into the negative territory. In other words it is conceivable that the richer, recorded survivor may continue to save, but the poorer unrecorded deceased may dissave enough to push the whole cohort into spending more than income. Because of questions raised by authors regarding treatment of the value of the house in the estate of the surviving spouse, and because we have no idea as to the magnitude of inter vivos transfers, which again should be subtracted from the dissaving, the above estimate should be recognized as very uncertain.

Even if we could obtain a more reliable estimate of the rate of saving (positive or negative) in the retirement phase, it is unlikely that it would be of much help in establishing the quantitative importance of bequests, or even less of the bequest motive. On the one hand, the amount of wealth held at various ages does not represent bequests, and tells us little about them. Clearly bequests will be whatever remains after the survivor spouse dies (intra-family transfers should not be counted as bequests). On the other hand, the amounts bequeathed or transferred include transfers by those that have already died or made transfers.

Some evidence on the importance of bequests and gifts is available for Italy. A special section of the 1991 SHIW asks each member of the household to report the amount of bequests and gifts received in the past from parents or other relatives, and the year of receipt. This information is used in Guiso and Jappelli (2002) to compute the aggregate share of transfers in total wealth. On average, in 1991 each household received about 30,000 euro, 24.3 percent of discretionary wealth (20.2 percent bequests and 4.1 percent gifts). This figure is consistent with the results reported by Modigliani (1988) and Wolff (2002) for the United States, to the effect that the share of transfer wealth does not exceed one-fourth. However, this information is available for recipients, and we have no information on how transfers affect the age saving and wealth profiles of donors.

In closing, it must also be remembered that the amount of bequests left and received cannot be identified with the accumulation dictated by the bequest motive. This is because part of bequests may constitute unintentional bequest resulting from the holding of wealth for precautionary reasons. Given life uncertainty, risk-averse consumers will always find it optimal not to run their assets down to zero.

8 Conclusions

Our analysis, which is based on Italian repeated cross-sectional data, supports the Life-Cycle Hypothesis of humped wealth, or saving turning negative after retirement, once mandated pension saving and pension wealth is duly taken into account. It can be argued that because people cannot choose the amount of mandatory saving, they should be ignored when it comes to understanding the household's behavior. But since people can change discretionary saving in response to changes in mandatory saving, total saving is the relevant measure of the change in assets accumulated for retirement. After all, the existence of mandatory saving programs and the widespread implementation of retirement plans should be interpreted as the social approval of schemes designed to ensure people with adequate reserves to be spend during retirement.

For these reasons, discretionary saving and discretionary wealth (obtained by subtracting mandatory saving and pension wealth, respectively from total saving and total wealth) are not relevant indicators of accumulation for retirement in societies in which a major source of provision for retirement is provided for by mandated pension programs. Thus the shape of one component of total saving, like discretionary saving, cannot be cited as evidence in favor of or against the LCH, as has been done in several writings cited earlier.

As for the age profile of discretionary wealth, the data for Italy (and other countries) leave room for considerable doubt as to whether it is humped shaped, once account is taken of the contribution of the growing number of deceased to the cohort saving. It must be recognized that the decline during retirement is at best slow, which is consistent with non-negligible bequests (partly involuntary, resulting from precautionary motives). Data separately collected by our SHIW survey has resulted in and estimate of transfer wealth of about one-quarter of discretionary wealth, consistent with results for other countries.

Appendix

The Survey of Household Income and Wealth (SHIW). The data set includes the 1989, 1991, 1993, 1995, 1998 and 2000 SHIW. The survey is representative of the Italian population because probability selection is enforced at every stage of sampling. The unit of observation is the family, which is defined to include all persons residing in the same dwelling who are related by blood, marriage, or adoption. Individuals selected as "partners or other common-law relationships" are also treated as families. The interviews are generally conducted between May and September of each year, thus flow variables refer to the previous calendar year, and stock variables are end-of-period values. All statistics reported in this paper use sample weights. Nominal figures are deflated by the CPI and then converted in euro.

Consumption. Sum of durable and non-durable consumption, including imputed rents on owner-occupied housing.

Earned income. Sum of households earnings, transfers, capital income, and income from financial, assets, net of taxes. Earnings are the sum of wages and salaries, self-employment income, less income taxes. Wages and salaries include overtime bonuses, fringe benefits, and payments in kind, and exclude withholding taxes. Self-employment income is net of taxes and includes income from unincorporated businesses, net of depreciation of physical assets. Capital income includes imputed rents on owner-occupied housing.

Disposable income. Earned income plus pension benefits less social security contributions.

Discretionary wealth. Net financial assets and real assets less household debt (mortgage loans, consumer credit, and other personal loans) and business debt. Real assets are the sum of real estate, unincorporated business holdings, and the stock of durable goods.

Pension wealth. Since in Italy very few workers hold private pension funds, pension wealth largely coincides with social security wealth. For current workers we compute pension wealth as the difference between the discounted value of benefits and the discounted value of contributions. Social security contributions are assumed to be a flat rate of 25 percent in all years. At each age, the stream of social security benefits depends on expected retirement, survival probabilities, the expected replacement rate, expected earnings at retirement, the rate of growth of pension benefits during retirement, and the real rate of interest at which people discount future benefits. We do not distinguish between men and women and assume that the household retires at age 60, and that the replacement rate is 70 percent, about the level prevailing in the last decade. We use mortality tables for women to impute the survival probabilities. Expected earnings at retirement are estimated by the fitted value of a regression of log earnings on a fifth-order age polynomial, a set of cohort dummies and a set of restricted time dummies. Using the fitted values from this regression, we project for each age-year-cohort group the earnings at the assumed retirement age of 60. We also assume that the growth rate of earnings equals the real interest rate and that pensions are fully indexed to the cost of living. For the currently retired we compute the present discounted value of pension benefits, based on current pensions (including old age, social, and disability pensions). Survival probabilities refer again to women.

Notes

1. When measuring the aggregate saving rate, defining saving as earned income minus consumption or as disposable income minus consumption does not make a great difference. Gale and Sabelhaus (1999, table 2) adjust NIPA figures for the United States and point out that adding net saving in federal, state and government retirement plans raises the aggregate saving rate by less then 1.5 percentage points over the entire 1960–98 period. The reason is that government pension saving is fairly stable and that in the aggregate mandatory contributions largely offset mandatory benefits.

2. It should be noted that their study is based on an unusual and questionable attribution of resources to individuals, not households, which is an approach that rests on a number of assumptions. First of all, that households are a veil for the individuals who live in them. Second, that there are no public goods within the family. Finally, that there are no economies of scale in consumption.

3. The decline in the aggregate saving rate is not peculiar to the 1989–2000 period, but follows a trend starting just after the period of high and sustained growth of the fifties and sixties. In previous work (Modigliani and Jappelli, 1990) we emphasize that the reduction in productivity growth is the main factor explaining the trend decline in the Italian saving rate, particularly after the 1973 oil shock. Rossi and Visco (1995) argue that the accumulation of social security wealth due to the transition to a pay-as-you-go social security system and the increasing generosity of the system also explain a substantial portion of the fall in household saving. Other explanations focus on the reduced need for precautionary saving due to the increased availability of social insurance schemes.

4. Reasons for such behavior include mortgage market imperfections, which prevent young households from borrowing (Guiso and Jappelli, 2002).

5. Two macroeconomic episodes characterize our sample period. The economy went into a recession in 1991–93. Afterwards the economy began a mild recovery, with the growth rate picking up in 2000.

6. One should keep in mind that wealth accumulation also depends on households' preferences, the interest rate, the life-cycle variation in household size and composition, and the rules governing retirement. Additional control variables (such as education, gender, or region of residence) do not change the qualitative results in the next sections.

7. Since the fraction of Italian households that contributes to private pension funds is tiny, they can be safely neglected. The 1995 pension reform of the social security system implements gradual changes in eligibility rules, accrual rates and pension age. Our estimate of social security wealth abstracts from these institutional details. It provides a rough estimate of expected contributions and benefits whose main purpose is to obtain a benchmark indicator to be compared with discretionary wealth. The assumed replacement rate and contribution rates are, respectively, 70 and 25 percent. See the appendix for details.

8. Sabelhaus and Pence (1999) use a time series of cross-section wealth surveys to measure how wealth accumulation varies across age in the US. They apply a technique similar to ours to the 1989, 1992 and 1995 Survey of Consumer Finance and are therefore able to purge the age-wealth profile from cohort effects. They also compute the change in wealth as a percent of disposable income and find positive saving rates in the order of 20 percent until age 60, and negative saving afterwards, ranging from −20 percent at age 65 to −50 percent at age 75.

9. For employees the contribution rate increases from 26 percent of gross earnings in 1989–93 to 27 percent in 1995–2000. The contributions are split between the employee and the employer. But clearly this is immaterial: in both cases it is the employee who actually bears the burden of paying the contributions (and finally receives the benefit). For the self-employed the contribution rates are set at 15 percent of gross earnings.

10. One could thus compute discretionary saving by "blowing up" disposable income by 25 percent, and consumption by 30 percent. This adjustment reduces discretionary saving at all ages, but does not change the shape of the age-consumption profile, and therefore that of the age-saving profile. In particular, discretionary saving is again flat over the life cycle.

11. Initially, in Italy severance pay was intended to insure the employee against the risk of dismissal, but it gradually evolved into a form of deferred compensation, irrespective of the cause of termination of employment. The employee is entitled to it whether he or she retires, is laid off, or quits.

12. This latter study also presents flat age-saving profiles in Taiwan and Thailand, in which the amount of mandatory saving is much more limited. The evidence in these countries is therefore much less favorable to the LCH than in other developed countries.

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6

THE CHINESE SAVING PUZZLE AND THE LIFE-CYCLE HYPOTHESIS

Franco Modigliani and Shi Larry Cao

1 Introduction

With today's world economy plagued by recession and sluggish performance, China stands out with its sustained high growth, perhaps due to market-oriented economic reforms that began in 1978. The growth has been accompanied by an explosion in China's household saving ratio, which has reached an impressive level in recent years, just when there has been a worldwide reduction in the private saving ratio. In 1994, for example, our estimate of the household saving-to-income ratio in China was close to 34 percent, rivaling the Japanese experience in the 1960s, even though the level of China's per-capita income remained well below those of the industrialized nations (see figure 6.1). More surprisingly, looking back at the postwar history of Chinese household savings, one finds that, from the 1950s through the mid-70s, the "thrifty" Chinese were not so thrifty with an average household saving ratio below 5 percent. In this paper, we attempt to explain the apparent paradox of the sudden spurt in the saving ratio's magnitude by using the framework of the life-cycle hypothesis (LCH) developed by Franco Modigliani and R. Brumberg (1954).

The LCH was initially thought to be relevant for developed market economies only. China's per-capita income in 2002 is still ranked below 100th in the world, according to the World Bank.[1] In the middle of our sample, there was a sharp regime shift from a highly planned economy to a market-oriented economy. Therefore, testing the explanatory power of the LCH for China is not only relevant in endeavoring to offer a plausible explanation of the drastic changes in the personal saving ratio of that country, but also has theoretical implications as to the applicability of the LCH model to a more general environment, including developing countries.

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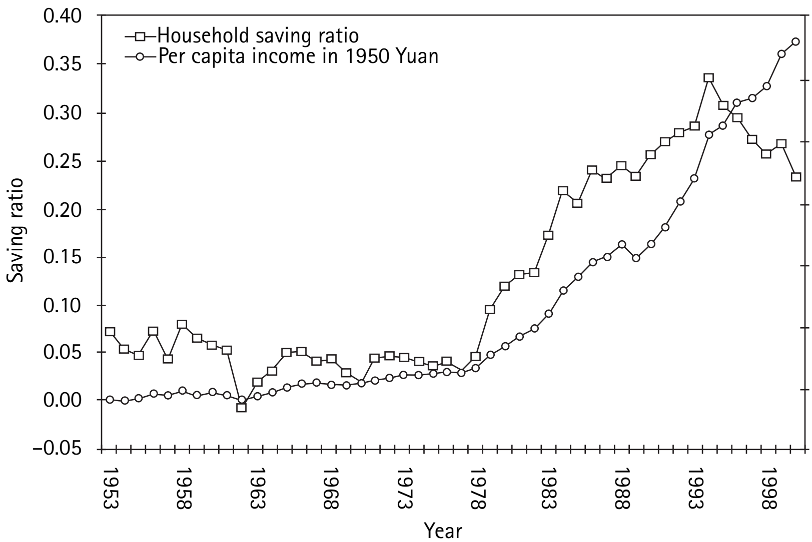

Figure 6.1
China's household saving ratio and per-capita income: 1953–2000

Since it is widely believed that the standard Keynesian theory may play an important part in explaining behavior pertaining to saving in at least the low-income nations, we include a comparison of the results of the LCH and the Keynesian model.

2 Basic Data

Table 6.1 summarizes the income and saving information on China for the period 1953 to 2000. It is important to keep in mind that personal saving (see data appendix) is not measured as (disposable) income minus consumption as is conventional in the national incomes account (NIA). This is due to the unavailability of the required data. In this case, we use an alternative approach that, conceptually, should yield an estimate of NIA saving, except for measurement errors. It consists of measuring the (annual) increase in personal wealth that results from personal saving. Our estimated calculations to show the increase in personal wealth, $W(t) - W(t - 1)$, are the sum of two components:

1. The first is the increase in the holdings of a list of intangible assets, $A(t) - A(t - 1)$. We have endeavored to make the list of relevant assets as comprehensive as possible, a task facilitated by the very limited choices available during

Table 6.1
Chinese Household Income and Saving, 1953–2000

Year	Household Consumption	Household Saving	Household Income	Household Saving Ratio	CPI, 1950 = 100	CPI, Preceeding Year = 100	Real Income	Per capita Real Income	E/M
1953	529.20	41.50	570.70	0.0726	121.4	105.1	470.1	79.95	1.0015
1954	550.00	31.80	581.80	0.0546	123.1	101.4	472.6	78.42	0.9876
1955	602.00	30.10	632.10	0.0476	123.5	100.3	511.8	83.27	0.9796
1956	646.80	51.00	697.80	0.0731	123.4	99.9	565.5	90.01	0.9774
1957	686.60	31.80	718.40	0.0442	126.6	102.6	567.4	87.76	0.9705
1958	724.00	63.00	787.00	0.0801	125.2	98.9	628.6	95.26	1.0528
1959	691.20	48.20	739.40	0.0652	125.6	100.3	588.7	87.60	1.0067
1960	741.70	45.80	787.50	0.0581	128.8	102.5	611.4	92.34	1.0001
1961	816.70	45.90	862.60	0.0532	149.6	116.1	576.6	87.55	0.9840
1962	838.70	–5.30	833.40	–0.0063	155.3	103.8	536.7	79.75	0.9432
1963	844.20	17.20	861.40	0.0199	146.1	94.1	589.6	85.23	0.9341
1964	889.60	29.10	918.70	0.0317	140.7	96.3	652.9	92.62	0.9669
1965	951.50	50.70	1002.20	0.0506	139.0	98.8	721.0	99.40	0.9740
1966	1021.10	54.90	1076.00	0.0510	137.3	98.8	783.7	105.13	0.9925
1967	1081.50	46.90	1128.40	0.0415	136.4	99.4	827.3	108.32	1.0085
1968	1076.60	49.40	1126.00	0.0439	136.5	100.1	824.9	105.04	1.0234
1969	1127.70	34.40	1162.10	0.0296	137.8	101.0	843.3	104.54	1.0394
1970	1206.90	24.90	1231.80	0.0202	137.8	100.0	893.9	107.71	1.0502
1971	1262.00	59.20	1321.20	0.0448	137.7	99.9	959.5	112.57	1.0676
1972	1334.20	66.50	1400.70	0.0475	137.9	100.2	1015.8	116.52	1.0524
1973	1432.50	68.40	1500.90	0.0456	138.0	100.1	1087.6	121.91	1.0507
1974	1467.00	64.00	1531.00	0.0418	138.9	100.7	1102.2	121.31	1.0474
1975	1528.50	57.90	1586.40	0.0365	139.5	100.4	1137.2	123.04	1.0457
1976	1588.50	70.20	1658.70	0.0423	139.9	100.3	1185.6	126.51	1.0704

Table 6.1 (continued)

Year	Household Consumption	Household Saving	Household Income	Household Saving Ratio	CPI, 1950 = 100	CPI, Preceeding Year = 100	Real Income	Per capita Real Income	E/M
1977	1647.80	53.60	1701.40	0.0315	143.7	102.7	1184.0	124.67	1.1034
1978	1759.10	87.60	1846.70	0.0475	144.7	100.7	1276.2	132.58	1.1400
1979	2005.40	213.90	2219.30	0.0964	147.4	101.9	1505.6	154.35	1.1804
1980	2317.10	316.00	2633.10	0.1200	158.5	107.5	1661.3	168.31	1.2317
1981	2604.10	401.30	3005.40	0.1335	162.5	102.5	1849.5	184.81	1.2761
1982	2867.90	449.10	3317.00	0.1354	165.8	102.0	2000.6	196.81	1.3265
1983	3182.50	671.20	3853.70	0.1742	169.1	102.0	2278.9	221.24	1.3990
1984	3674.50	1036.00	4710.50	0.2199	173.7	102.7	2711.9	259.87	1.4964
1985	4589.00	1199.40	5788.40	0.2072	194.4	109.3	2977.6	281.30	1.5773
1986	5175.00	1649.50	6824.50	0.2417	208.0	106.5	3281.0	305.19	1.6393
1987	5961.20	1811.60	7772.80	0.2331	226.3	107.3	3434.7	314.25	1.6889
1988	7633.10	2489.50	10122.60	0.2459	273.1	118.8	3706.6	333.85	1.7620
1989	8523.50	2624.40	11147.90	0.2354	317.6	118.0	3510.0	311.44	1.7722
1990	9113.20	3168.90	12282.10	0.2580	321.7	103.1	3817.9	333.93	2.0185
1991	10315.90	3835.00	14150.90	0.2710	338.1	103.4	4185.4	361.36	2.0081
1992	12459.80	4868.50	17328.30	0.2810	367.2	106.4	4719.0	402.75	2.0048
1993	15682.40	6313.40	21995.80	0.2870	421.2	114.7	5222.2	440.63	2.0000
1994	21230.00	10813.50	32043.50	0.3375	522.7	124.1	6130.4	511.50	2.0063
1995	26944.50	12027.20	38971.70	0.3086	612.1	117.7	6366.9	525.66	2.0553
1996	32152.30	13568.10	45720.40	0.2968	662.9	108.3	6897.0	563.53	2.1347
1997	34854.60	13155.60	48010.20	0.2740	681.4	102.8	7045.8	569.93	2.1885
1998	36921.10	12881.70	49802.80	0.2587	676.0	99.2	7367.3	590.28	2.2374
1999	39334.40	14571.10	53905.50	0.2703	666.5	98.6	8087.4	642.32	2.3342
2000	42911.90	13218.60	56130.50	0.2355	669.2	100.4	8387.7	662.62	2.4556

Notes: Population in 10,000 persons.
Household income, saving, consumption in 100 million current RMB Yuan.
Real income in 100 million constant 1950 RMB Yuan.
E/M = Number of persons employed/Number of persons 14 years and younger.

the period we cover. The assets we include consist basically of nominal assets (money, deposits, and government bonds).
2. The second component is an estimate of the increase in the stock of some major tangibles (e.g. private residences). Here we measure the utilization of saving by the estimated flow of investment I^*.

Thus, $S = [A(t) - A(t-1)] + I^* = I + e$, where I is the aggregate investment. The last equality is meant to emphasize that our measure of saving is logically equivalent to the traditional one, but that the two estimates can differ because of errors of measurement in either or both terms.

Our measure of income is obtained by combining sums for consumption, for which official estimates are available, with sums for saving as estimated above. This approach measures "disposable" income, which is actually very close to personal income, since there were no significant personal income taxes prior to the recent decade. Our saving ratio is, in effect, the ratio of estimated accumulation of assets to the sum of consumption and accumulation.

Table 6.1 shows that the household saving ratio was quite low in the pre-reform period 1953–78. The very low saving rate from 1962 to 1964, negative in the first year, was the result of severe natural disasters in three consecutive years and can be regarded as a transitory phenomenon. But even without these three years, the saving ratio from 1953 to 1978 averaged less than 5 percent. All in all, it is apparent that there was not much household accumulation during this time. Furthermore, the saving rate did not exhibit any clear positive trend even though the real per-capita income increased by 60 percent (see table 6.1 and figure 6.1). It is noteworthy that the growth rate was quite modest during these 25 years. It averaged less than 5 percent annually. In the 22 years subsequent to the economic reforms, the estimated saving ratio has increased consistently from 5 percent in 1978 to an incredible peak of 34 percent in 1994, and still attained the 24-percent level in 2000!

It is worth noting that our data does not include pension distribution or investment in financial securities other than government bonds. We were not aware of reliable time series pension data at the first writing. Investment in equity securities and mutual funds did not become significant until recent years. However, the slower growth in currency in circulation and deposits in 1999 and 2000 is certainly related in part to the flow of money into the equity market in those years (89 billion yuan in 1999 and 102.4 billion yuan in 2000). The pension system in China has been largely a fragmented pay-as-you-go system operated by individual companies, though the government since the mid-1980s has initiated experiments to pool pension assets and put them under control of local or provincial governments. The pooling effort has brought more credibility to the system. One

Table 6.2
Pension Development in China: 1989–2000

	Number of Employees Contributing to Pension million	Number of Retirees Contributing to Pension million	Pension Contribution billion Yuan	Pension Distribution billion Yuan
1989	48.2	8.9	14.7	11.9
1990	52.0	9.7	17.9	14.9
1991	56.5	10.9	21.6	17.3
1992	77.7	16.8	36.6	32.2
1993	80.1	18.4	50.4	47.1
1994	84.9	20.8	70.7	66.1
1995	87.4	22.4	95.0	84.8
1996	87.6	23.6	117.2	103.2
1997	86.7	25.3	133.8	125.1
1998	84.8	27.2	145.9	151.2
1999	95.0	29.8	196.5	192.5
2000	104.5	31.7	227.8	211.5

can infer from the rapid growth in pension distribution in the 1990s (see table 6.2) that this would have had a negative impact on people's incentive to save for old age.[2]

3 Alternative Models of Saving Behavior

3.1 The Standard Keynesian Model and Its Rationale

In the standard Keynesian model, saving is supposed to depend entirely on current income. Furthermore, the saving-to-income ratio is expected to be an increasing function of income. That is, the national saving ratio would rise as per-capita income rises within a country or between countries. Accordingly, the saving function is typically approximated by a linear form:

$$S = s_0 + sY \tag{6.1}$$

implying (1′) $S/Y = s + s_0/Y,\ s_0 < 0$.

It is generally believed that this model can be used to explain the saving behavior of the relatively poor countries. The reasoning follows that people with low incomes may not be able to afford the sufficient level of saving when they are young and productive to support their consumption in old age, or at least not as much as people with higher incomes. The ability to make intertemporal transfers of resources constitutes the very foundation of the life-cycle hypothesis (LCH).

Even if one accepted this reasoning, the line, if any, that separates the rich from the poor could only be established empirically. In our view, the relevance of LCH for a country depends on the existence of a sufficiently large core of households that are able to carry over resources to provide for old age at a standard of living commensurate with that of preretirement. We hypothesize that China might well qualify, especially in view of two circumstances. The first is the absence of a national public pension system (social security) and scarcity of other pension institutions until the most recent decade.[3] The second, discussed below, is that, traditionally, the source of old-age support in past generations was children. This tradition was severely curtailed by the policy initiated in the 1970s to limit the number of children to one per family.

3.2 The Life-Cycle Hypothesis (LCH) Model and Its Implications for Steady Growth

The fundamental and novel implication of the LCH is that the national saving rate, S/Y, is unrelated to per-capita income but depends instead on the long-term rate of income growth. This result has been demonstrated in numerous earlier papers, including, in particular, Modigliani's Nobel lecture (Modigliani 1989). A short summary of the earlier demonstrations will be sufficient.

The model starts from the classic Fisherian postulate that individuals choose to maximize utility derived from their life resources by allocating them optimally between current and future consumption. Accordingly, it identifies life resources, instead of current income, as the budget constraint. This postulate, when combined with the assumptions: 1) stable preferences for the allocation of resources over a finite life are independent of the size of life income, and 2) a stable path of resources by age will give rise to a stable age pattern of the saving-to-income and wealth-to-income ratio. Now, suppose aggregate income grows in time at a constant percentage, say g. Consider first the case when the growth is due entirely to population growing at that rate, while per-capita income remains constant. Then, as time goes by, each age group, as well as aggregate consumption and income, all rise at the rate g, but the consumption-to-income, saving-income, and wealth-to-income ratios are constant. Thus, for any given g, the national wealth is proportional to income or $W = wY$, where w is a constant that is independent of income (though, possibly dependent on g). Since saying is the growth of wealth, we can infer:

$$S/Y = \Delta W/Y \equiv w\,\Delta Y/Y \equiv wg$$

where $g = \Delta Y/Y$, the income growth rate—that is, the saving ratio is independent of income. Instead, it is related to the income growth rate. If the growth rate is stable, say g, then S/Y should be a constant, wg. In particular, if income were stationary, so would wealth be, and saving would be zero, regardless of per-capita income. Similar conclusions hold when growth is the result of growing productivity (per-capita income). Furthermore, the relation between saving and growth does not seem to be affected much by the source of growth. That is, the value of w, for a given g, does not seem to be very different, whether g is due to a steady population growth, or productivity growth, or some mixture of the two.[4]

We can conclude that, as long as income is growing fairly steadily, the saving function implied by the LCH can be written as:

$$S/Y = s'_0 + s'g + e \tag{6.2}$$

where s'_0 should be close to zero, s' should be significantly positive and e is a random error term (iid).

An alternative way of arriving at the above result is to recognize that the assumptions stated earlier about the path of consumption and income imply that aggregate saving is a linear (homogenous) function of income and wealth. One can readily show that this equation implies that if income is rising at a constant rate, the saving wealth-income ratio will tend (asymptotically) to a constant, and therefore S/Y will tend to satisfy equation (6.1).

3.3 The LCH Consumption Function When the Economy Is Not in Steady Growth

3.3.1 Demographic Growth vs Demographic Structure

As was pointed out in Modigliani (1970), according to LCH, the saving rate increases with a steady population growth. But the relation is not a directly causal one. What truly affects the saving rate is demographic structure. In particular, the relation between the working and nonworking populations is the most important factor because the latter group tends to reduce national household saving, since it consumes without producing an income. The nonworking population includes the retired and those too young for regular employment. The latter will be referred to as the "minors" in this paper. Demographic structure is predictably related to population growth if, and only if, growth has been stable for a period long enough so that the structure has reached an equilibrium point for that growth.

When population growth has been varying over periods of time, the number of people in the different age cohorts grows at different rates. Then, the current growth rate will have no systematic correlation with the demographic structure and the saving ratio. These considerations are clearly relevant in China's case due to the wide swings in birth rates and age-specific mortality rates during the past half century. In these circumstances, Modigliani (1970) argued that population growth should be replaced by direct measures of demographic structure such as those mentioned above. He showed, on the basis of a large cross-section of countries, that both the ratio of retired population (age 65 and over) and of prework-ing population to the working-age population (20 to 65) had a strong and highly significant negative effect on the saving ratio. For China, there are a number of considerations, supported by some preliminary explorations, that have led us to conclude that the crucial demographic variable is the relation between the employed population and the number of minors, denoted hereafter as *E/M*. The typical age at which a person is classified as an adult is different from country to country. It tends to rise with the degree of development. For China in its early development stage, we concluded that people could be classified as minors up to the age of fifteen.

We wish to stress that the variable, *E/M*, has special relevance for China during the period we are studying because of its relation to a culture-related variable, the birth rate, which was greatly influenced by government policies. In the Chinese cultural tradition, the younger generation is supposed to take care of the elder members of the family, while the elders will bequeath the house and other assets to their children. In other words, an economic unit is the extended family rather than the nuclear family. Under such a system, a child is an effective substitute for life-cycle saving. Consequently, when strict birth-control measures came into effect in the 1970s, the accumulation of life-cycle (tangible) assets gained in importance as a substitute for children. One may also argue that even if the Chinese birth-control policy did not occur, the secular trends of 1) more nuclear families; 2) migration of families from their ancestral homes; and 3) less loyalty to elders would have had the effect of reducing the role of children as a substitute for saving, anyway.

Unfortunately, official information on minors is only available for a few selected years (1953, 1964, 1982, 1990, and 2000). However, as expected, it correlates well with the number of births recorded over the previous fifteen years. Accordingly, we have used this data series, available annually, to interpolate between the years for which we have official information.[5] The birth rate is shown in figure 6.2. *E/M* and *S/Y* are shown in figure 6.3. *E/M* reflects the effectiveness of the birth-control policy and economic expansion. It is relatively flat until the mid-70s, and then increases appreciably and steadily for about fifteen years until the end of the 1980s. It is apparent that *S/Y* follows the trend of *E/M*. The closeness of the two curves, confirmed by a simple correlation of 0.95, suggests that the close fit reflects not only the usual impact of *E/M* on the saving ratio (through the "less mouths to feed" effects) but also, in particular, the impact of the birth-control policy on the incentive to accumulate wealth for old age. The close correlation of the two series weakens after 1994 as *S/Y* starts to decline, while *E/M* continues to rise.

Figure 6.2
China's birth rate per thousand persons: 1953–1999

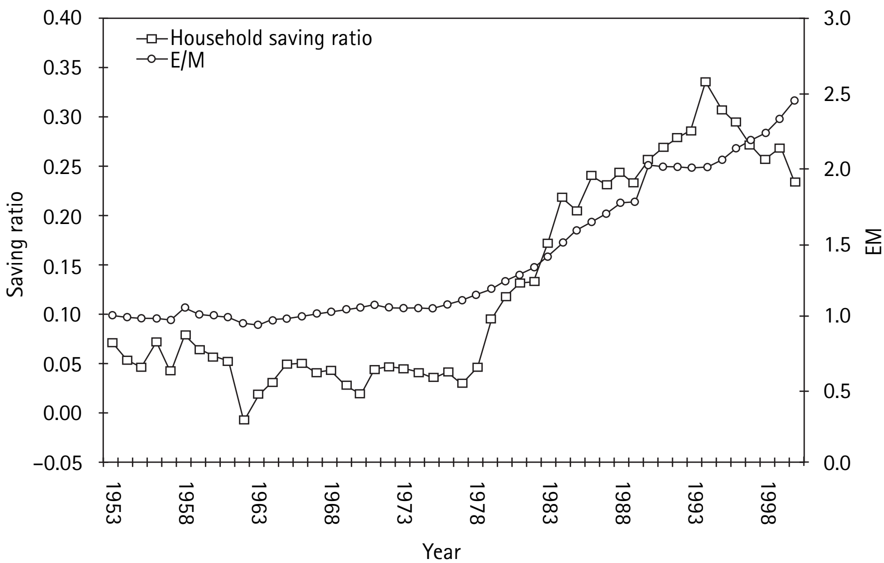

Figure 6.3
Saving ratio and E/M

3.3.2 Modeling Permanent Variation in the Per-Capita Income Growth Rate

With respect to the productivity growth component, the basic hypothesis is stated in (6.2). But there is a problem in that (6.2) has been shown to be an implication of the LCH in the presence of a lasting and stable income growth trend. Now it is obvious that in China the growth rate has been far from stable. Until the mid-70s, it averaged around 2 percent. It then rose almost monotonically, reaching over 10 percent in the early 90s and falling again in the late 90s. One possible first approximation can be obtained by measuring the growth trend not by the current or recent growth (which would be quite erratic), but by the average annual growth over an extended period of past years. In principle the period should be long, subject to the availability of data. In the case of China the data are an important limitation, as our estimates of disposable income growth begin only in 1953 and end in 2000. We have 48 years of data. Using a really long period would result in losing a large portion of our sample. We have settled for a compromise by measuring the growth trend for every year and by using the average annual rate of growth over the previous fourteen years (year one through year fifteen). This procedure results in the loss of the first fifteen observations, leaving us with a sample of 34 observations (1966–2000). However, many of the results presented below have been obtained using the entire sample, and approximating the growth trend of the missing previous fourteen years by the average annual growth rate for all the years available, up to the given year. This procedure yields a sample of 43 observations (1957–2000). This approximation is "courageous," but the early years are important because they are quite different from the later ones, and we submit that the approach is unlikely to cause serious bias since, in the early years, the growth is relatively constant and quite small.

The saving ratio and the per-capita income growth trend are shown in figure 6.4. The per-capita real income curve tracks remarkably closely with the wide fluctuations of S/Y from its trough in the 60s to its peak in the mid-90s. At the end, the S/Y decelerates faster than the growth trend. This difference will be explained below. On the whole, the graph provides surprisingly strong support for the LCH hypothesis as an explanation of Chinese saving.

The fifteen-year average growth rate is the basic variable used to estimate the growth effect. We will provide some refinements of this measure, which takes into account the possibility of differential effect for the more recent growth rates.

3.4 The Role of Inflation

3.4.1 Impact of Inflation on Measurement of Income and Saving

The presence of price variation (inflation) over the time period poses a number of measurement problems. First, in the presence of significant inflation, one must

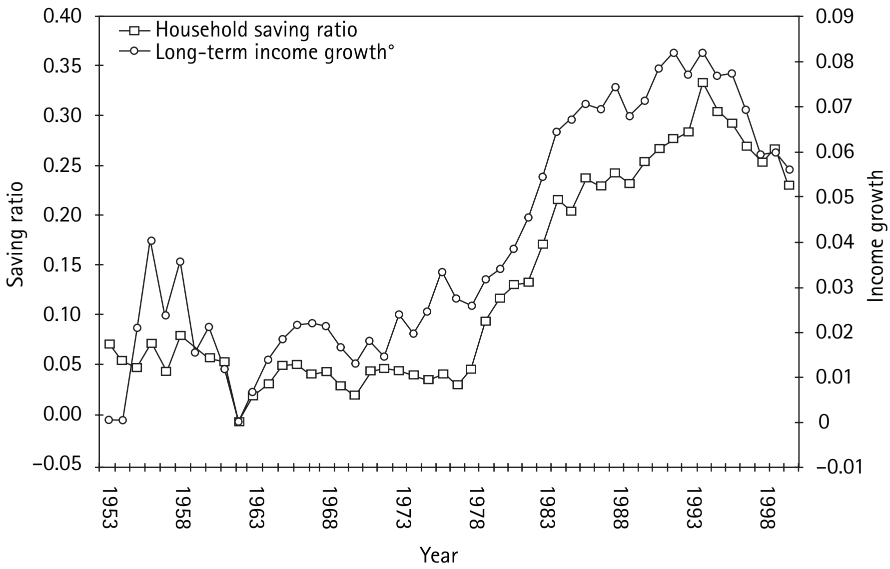

Figure 6.4
Saving ratio and long-term income growth

distinguish between current (nominal) and real values. The latter are conceptually measured in constant prices. In this study, we rely systematically on real values (i.e., measured in constant prices, estimated by deflating nominal values by an overall consumer price index, CPI1950 = 100.

Inflation poses another problem that is generally neglected. Basically the problem arises from the fact that saving is the difference between income and consumption. Consumption is well defined, but income, and the corresponding saving, may take on a number of different definitions. These different measures all coincide in the absence of inflation, but differ, possibly quite substantially, in the presence of significant inflation. Furthermore, these different measures respond differently to inflation. The alternative measures are described briefly below:

1. S(1), the national income accounts (NIA) definition of personal saving as the difference between net-of-tax personal income, Y, and consumption, C, or $S(1) = Y - C$.
2. S(2), the value of the net acquisition of personal assets during the period. This measure is conceptually identical to the first: i.e., S(2) = DA = Y − C = S(1). However, the statistical measure may differ because of error of measurement.

3. S(3), the personal sector contribution to the financing of national investment. In the absence of inflation this measure coincides with S(1) and S(2) above, and this is the basic reason why these measures are of special interest. But this coincidence ceases in the presence of inflation. With inflation, there are two distinct measures of the interest rate and two corresponding measures of income and saving. One is the conventional "nominal" rate, R, which measures the amount of money obtainable next period per unit of money delivered now. The other is the Fisherian "real rate," r, which measures the amount of "commodity baskets" next period per basket now. It can be derived from the current rate, by subtracting from it the rate of inflation, p, over the term of the loan contract, or $r = R - p$. In a perfectly rational market economy, with only short-term loans, inflation would have no effect on the real rate r (the classic neutrality of money), and therefore would raise the nominal rate by p, so that $R = (r + p)$. Actual experience suggests that R will in fact rise in response to inflation, but possibly less than one for one, at least initially, so that $R(p)$ can be written as $R = r + zp$, where z is between 0 and 1. That means that inflation causes conventionally measured income to rise by the increase in R, zp, times the amount of personal net wealth invested in nominally fixed assets, say A^*. However, it must be remembered that all nominal claims of the private sector on the private sector are offset by a corresponding debt, which results in increasing the wealth of the debtor. Thus, the private sector's net nominal wealth turns out to be essentially the interest-bearing portion of the government debt.

If we subtract from Y the inflation effect, zpA^*, we obtain an inflation-adjusted measure of income, $Y(3) = Y - zpA^*$. We suggest that, with inflation, the relevant measure of saving should be "adjusted" income minus consumption:

$$\mathrm{S}(3) = \mathrm{Y}(3) - \mathrm{C} = \mathrm{S} - \mathrm{zpA}$$

as this provides a better approximation than the conventional S(1) as a measure of the contribution of the personal sector to national capital formation. The reason is that, though the actual conventional saving is S or $S(3) + zpA$, the zpA component does not add to national saving because it is offset by an equal rise in the outlays (and deficit) of the government, reflecting a rise in the nominal interest paid by the government due to inflation.

4. S(4), the net increase in the purchasing power of the stock of assets owned. This differs from and exceeds S(2), or DA, the nominal value of net assets acquired during the period, insofar as the wealth carried over from the previous period contains any "nominal" assets, A. These assets will lose purchasing power between the beginning and end of the current period, through the erosion due to

inflation. The loss of purchasing power of assets outstanding at the beginning of period t can be written as $[p(t)\,A^*(t-1)]$, where $p(t)$ is inflation in period t. Hence to compute S(4) we subtract this quantity from the acquisition of assets, or:

$$S(4) = DA - p(t)A^*(t-1) = S(1) - pA^*(t-1) \tag{6.3}$$

Another useful way to understand the meaning of S(4) is to think of it as the result of subtracting consumption from NIA disposable income, properly corrected for the loss of purchasing power due to inflation. That corrected measure is $Y^*(t) = Y(t) - p(t)\,A^*(t-1)$. Subtracting C from Y^* yields $S^*(t) = S(t) - p(t)\,A^*(t-1)$.

Note that Y* and S* are the measures that would result if one computed the interest income from nominal assets using not the nominal rate, R, which is standard procedure, but the Fisherian "real rate" $(r - p)$. It is apparent that, in the presence of inflation, the conventional NIA measure of saving is an upward-biased measure of the increase in personal wealth.

3.4.2 The Impact of Inflation on Saving Behavior

Inflation can also affect saving behavior through a variety of channels, including deviations from rational behavior which result from difficulties of understanding its real implications. Let us consider first whether and how inflation might be expected to affect the conventional measure $S(1)/Y$, henceforth referred to as S/Y. It turns out that the answer to this question is more readily understood by looking at the ratio $C/Y = 1 - S/Y$. To understand the response of C/Y we must ask what will inflation's expected impact be on the numerator and the denominator. The denominator is affected, since, as we have argued above, the NIA measure of income Y is $Y(3) + zpA^*$. Here $Y(3)$ is supposedly broadly invariant with respect to inflation, z, is the impact of inflation $p(t)$ on the nominal interest $R = r + zp$, and could be anywhere between 0 (complete inflation illusion) and 1 (Fisher's law), in an economy with perfect markets, perfect foresight, and rational behavior. Hence, inflation has the effect of adding zpA^* to NIA income Y.

For consumption, the response should be the marginal propensity to consume, c, times the increase in income as perceived by the consumers, let us call it Y^{**}. This quantity will differ from Y because it may take into account, at least partially, the loss of purchasing power due to inflation. We may assume that the perceived loss is proportional to the actual loss, and can be expressed as $hp(t)A^*(t)$, where h should again fall between 0 (total inflation illusion by the personal sector) and 1 (no inflation illusion, so that the perceived loss coincides with the actual loss). So the effect of inflation on C, (dC/dp), can be expressed as $cA^*[z - h]$. With the help of these results, one can readily establish that the effect of inflation on the consumption ratio reduces to

$$d/dp(C/Y) = k\,A/Y \tag{6.4}$$

where $k = [(c - C/Y)z - ch] = -d/dp(S/Y)$ and k measures the effect of inflation on the consumption ratio ($-k$ the effect on the saving ratio) per point of inflation. Since A, Y, and p are all directly measured, (4) tells us that we can estimate k as the coefficient obtained by adding $p(A/Y)$ to the list of variables supposed to explain S/Y.

Our theory implies some definite bounds on the value of k. Since the marginal propensity to consume can be counted on to be smaller than the average, our theory implies that $k < 0$, i.e., that inflation must have a positive effect on the NIA saving ratio (except in the limiting case, when $z = h = 0$ or universal, 100 percent inflation illusion.) It can also be seen that (4) is a decreasing function of both z and h. Therefore $-k$ (the effect on S/Y) should not exceed C/Y, the average propensity to consume, that for China could be placed, on average, at around 0.8.

One can similarly establish that (see section 5.2.3)

$$d/dp[C/Y(3)] = k(3)\,A/Y \tag{6.5}$$

where $k(3) = c(z - h)$.

As indicated earlier, (6.5) yields what we regard as the most relevant measure of the effect of inflation on the supply of resources for capital formation—namely the negative of the effect of inflation on consumption. In the absence of inflation illusion, z and h are unity, and as expected, the effect is zero inflation has no real effects. In general the effect depends on the relative size of z and h. This is an empirical question and, most likely, it depends on the size and duration of the inflationary process, presumably approaching one and neutrality as the process continues.

Finally, one finds

$$d/dp[C/Y(4)] = k(4)\,A/Y \tag{6.6}$$

where $k(4) = k + C/Y > 0$, so that, as one might expect, inflation reduces saving defined as the net accumulation of real personal assets (because it fails to reduce consumption by as much as the loss of purchasing power of the initial nominal assets.).

4 Estimation

From the graphs presented earlier, it is apparent that the saving ratio, long-run growth, and the population are drifting together. This suggests that they can be co-integrated. To establish co-integration we must first check whether each series is integrated and contains a unit root. The results of the augmented Dickey-Fuller with one lag and a constant in the test equation are presented below:

	S/Y	ΔS/Y	g
ADF t-stat	–4.0***	–0.47	–0.55
	Δg	M/E	ΔE/M
ADF t-stat	–3.3*	2.75	–2.75*

***:1 percent, **:5 percent, *:10 percent MacKinnon critical values for rejection of hypothesis of a unit root.

The tests indicate that the three series contain a unit root. Given that each series has a unit root, we now test whether the series are co-integrated over the sample period and if so, what the co-integrating relationship is. To test co-integration we use two methods, the first being the Engle and Granger (1987) two-step method. The first stage consists of regressing the variables supposed to be co-integrated. The second stage consists of testing for a unit root in the residuals. Since the residuals are estimates of the disturbance term, the asymptotic distribution of the tests statistic differs from the one for ordinary series. (The correct critical values are obtained from Russell Davidson and James MacKinnon 1993, table 20.2).

Table 6.3 reports the results of the estimation (see also figure 6.5). The first row of each equation contains the estimated coefficients. The second row shows the corresponding t-values. The first two equations (I.1 and I.2) are the regressions on the saving ratio on the long-term growth rate and the E/M ratio,

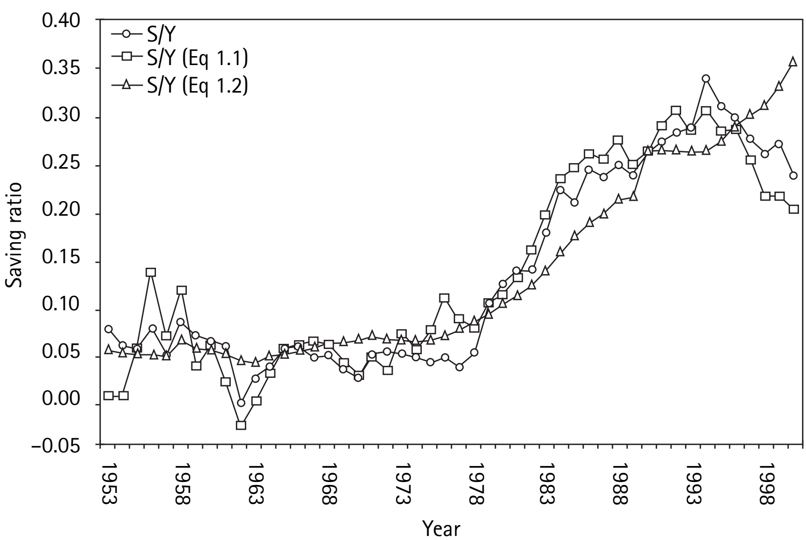

Figure 6.5
Saving ratio, long-term growth, and ratio of employment to minors

Table 6.3
Estimates of Coefficients of Saving Function

Coefficient	Variable	Coefficient	Variable
a_0	constant term	a_3	growth from previous year minus long-term growth
a_1	long-term growth (15 years)	a_4	inflation
a_2	E/M: dependency ratio	a_5	reciprocal of per-capita real income

Equation Number	Original #	Method	a_0	I. 1953–2000 (all years) a_1	a_2	a_3	a_4	a_5	ADF t-stat	N
I.1	1	OLS	–0.03	4.03					–2.87	1.00
$R^2 = 0.93$	t-value		–4.39	25.79						
I.2	2	OLS	–0.16		0.21				–1.71	1.00
$R^2 = 0.889692$	t-value		–9.96		19.26					
I.3	3	OLS	–0.10	2.57	0.09				–3.12	2.00
$R^2 = 0.972540$	t-value		–9.66	11.47	7.36					
I.4	4	OLS	–0.10	2.07	0.10	0.10	0.26		–2.83	2.00
$R^2 = 0.98$	t-value		–11.0	8.85	9.04	2.08	3.78			
I.5	5a	OLS	0.31					–0.25	–2.11	1.00
$R^2 = 0.85$	t-value		24.4					–15.9		
I.6	6	OLS	–0.12	2.16	0.11	0.10	0.26	0.01		
$R^2 = 0.98$	t-value		–3.50	7.85	7.33	2.15	3.69	0.65		

Table 6.3 (continued)

Equation Number	Original #	Method	a_0	a_1	a_2	a_3	a_4	a_5	ADF t-stat	Critical value
					II. 1953–1985					
II.1	10a		–0.01	3.18					–2.87	1.00
$R_2 = 0.85$	t-value	—	–2.1	12.8						
II.2	20	OLS	–0.26		0.30				–2.38	1.00
$R^2 = 0.845$	t-value		–10.3		13.1					
II.3	40	OLS	–0.13	1.52	0.14	0.14	0.74		–2.18	2.00
$R^2 = 0.92$	t-value		–3.23	3.50	3.04	1.95	1.79			
					III. 1978–2000					
Equation Number	**Original #**	**Method**	a_0	a_1	a_2	a_3	a_4	a_5	**ADF t-stat**	**Critical value**
III.1	10a		–0.03	4.00					–2.91	1.00
$R^2 = 0.88$	t-value	—	–4.00	25.00						
III.2	20	OLS	–0.06		0.16				–0.68	1.00
$R^2 = 0.71$	t-value		–1.43		7.26					
III.3	40	OLS	–0.10	2.52	0.09	0.13	0.18		–3.97	2.00
$R^2 = 0.97$	t-value		–6.22	8.80	7.90	2.23	2.81			

Critical values			N: number of I(1) series for which the null of non-cointegration is being tested.					
N = 1	1%	–3.577736295	N = 2	1%	–4.247013043	N = 3	1%	–4.619791871
	5%	–2.925594329		5%	–3.463602079		5%	–3.930802647
	10%	–2.600478072		10%	–3.134931947		10%	–3.586992439

respectively. The last two columns report, respectively, the ADF-t statistic of the regression residuals and the critical value. The first variable is the estimated growth trend, measure by the average rate of growth over the fifteen preceding years. The R_2 is 0.93, and t-value is 26, confirming the amazingly high explanatory power of this crucial LCH variable. The only qualm raised by this result is the coefficient of the growth variable; the value of 4 is a bit high, certainly higher than that obtained in earlier studies (see below) which typically runs around 2 or not much higher. But this anomaly will be shown to be, most likely, the "spurious" result of high correlation with the other major explanatory variable, population structure, E/M, whose coefficient is reported in equation I.2. This variable too fits the saving pattern remarkably well as demonstrated by an R_2 of nearly 0.9 and a t-value of 19. The regression coefficient, 0.21, is again on the high side compared with other studies but for the same reason.

The conjecture about the upward bias in the slope estimates of the previous two equations is confirmed by the results reported in equation I.3, where the two variables are introduced simultaneously. Both coefficients come down to reasonable levels while remaining highly significant.

In the remaining equations of part I we have tested a number of refinements to the basic model. First one can see from figure 6.1 that the growth of S/Y is not smooth but instead is interrupted by several "bumps" that represent short-term, transitory deviations from the long-term trend. According to LCH or the permanent income hypothesis, the propensity to save during such transients should be exceptionally high. We have introduced a new variable, the deviation of each year's growth from the long-term growth rate, as a rough but simple way of measuring the transient component. The coefficient of this variable reported in the a_3 (table 6.3) column has the expected positive sign and is moderately significant.

There remains to be tested the impact of inflation on the saving ratio. That impact, we recall, depends on the definition of income and saving. In the table the definition of the saving ratio is that adopted by the NIA or S/Y. The estimated coefficient is reported in the column a_4. Inflation is measured by the quantity pA^*/Y; therefore what is being estimated is the parameter k of equation 6.4, in section 3.4.2. From the analysis developed there we have concluded that according to our model the parameter $-k$, measuring the impact of inflation on S/Y, could be expected to be decidedly positive but appreciably less than 0.8. The estimate under a_4 is 0.26, which falls clearly within the stated limits and is highly significant with a t-value over 3.

Equations I.5 and I.6 present one last test focusing on a comparison of the relevance of the LCH versus the standard linear Keynesian model in accounting for the Chinese experience of the last fifty years. Equation I.5 gives the results of fitting the S/Y data to equation 1; according to which, S/Y is a linear function of

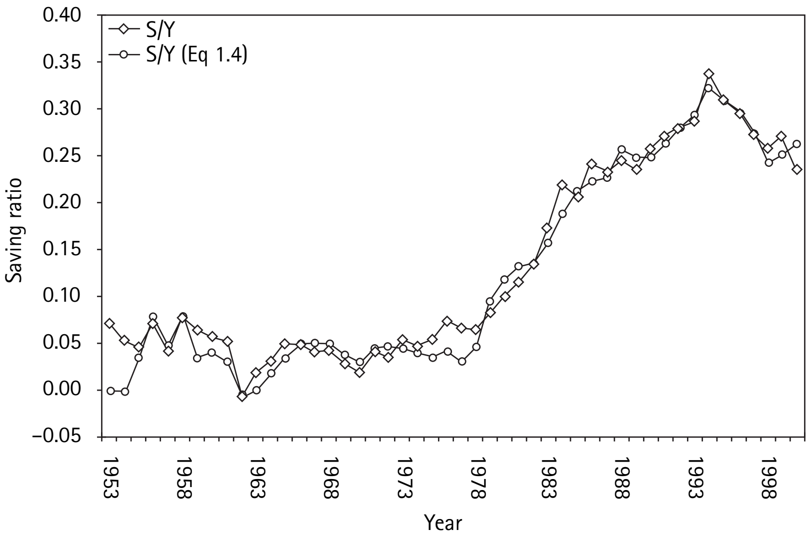

Figure 6.6
China's saving ratio 1953–2000: actual and computed values from equation I.4

the reciprocal of Y. Its coefficient, given in the column a_5, corresponds to the constant term of the linear equation. It is significantly negative, but this is not surprising since the accelerating pattern of income, which resulted in a rising saving rate, also resulted in a sharp rise in income. On the whole, S/Y and Y moved together. But the Keynesian equation is completely incapable of accounting for the initial stagnant period when S was flat though Y was rising moderately, or the sharp acceleration beginning in the late 1970s, or the decline after 1993. Equation I.6 consists of adding the Keynesian variable $1/Y$ to the (significant) variables of the LCH in equation I.4. The coefficient of this variable, reported in the column a_5, has become insignificant and has the wrong sign. We must conclude that there is no evidence whatsoever to support the conventional Keynesian Hypothesis.

The final hypothesis, represented by equation I.4, is seen to account closely for the highly unusual behavior of the Chinese saving ratio in recent decades. This can be inferred from the R_2 of 0.98. This value is strikingly high when one remembers that we are correlating dimensionless variables such as population structure, income growth rate, and inflation rate to the saving ratio. The closeness of fit can be judged from figure 6.6, which compares the actual pattern of the NIA saving ratio with the value calculated by using the fitted equation (I.4). It is apparent from figure 6.6 that the pattern of the saving ratio modeled by equation (I.4) follows very closely that of the actual saving ratio.

These results appear strongly supportive of the LCH and inconsistent with the traditional view, and this is confirmed by some tests of the reliability of the estimates, reported in the next two sections.

According to the conventional view, the *S*/*Y* ratio is supposed to be quite stable in time, at least for a given country. In the case of China, on the other hand, both actual and computed saving ratios are small—mostly below 5 percent—and relatively stable, from the early 1950s to 1978, a period characterized by modest growth rates (some negative)—except for 1958—very little inflation (some negative), and a low and stable E/M. Beginning in the late 70s with the economic reforms, the short-term and the long-term growth rates accelerate to a peak in 1994. During this same period, the enforcement of the one-child policy leads to a gradual reduction in the ratio of minors to employment, thereby, presumably reducing the consumption-to-income ratio. The net result is increasing the accumulation of marketable wealth. In addition, in the later years the saving-to-income ratio appears to be boosted by a couple of episodes of high inflation—one at the very end of the 1980s, the other even higher in the mid '90s. At the peak of the second inflation period, in 1994, our estimates suggest that it could have contributed as much as 6 percentage points to the peak of the saving ratio of almost 34 percent. After 1994, *S*/*Y* declines steadily as inflation is brought promptly to an end and growth decelerates steadily. These two variables outweigh the still expansionary effect of *E*/*M*.

Before endeavoring to draw some conclusions from our LCH Equation I.4, we report some tests of the reliability of our estimates.

5 Tests

5.1 Stability over Time

For this test, the entire sample is subdivided into two subsamples. The first consists (predominantly) of the years 1953–1985 (33 years). The second period consists of the years 1978–2000 (23 years). Several equations of part I were reestimated for each subsample and selected results are reported in parts II and III of the table. Equations II.1 and III.1 replicate the equation I.1 for each subperiod. The results confirm the crucial role of long-term growth in accounting for the behavior of the saving ratio in each subperiod. This is indicated by very high t-values and by the estimated coefficients, which are reasonably close. A somewhat similar result is found for the other LCH variable, *E/M*, equations II.2 and III.2. Equations II.3 and III.3 show the results of reestimating the full model of equation I.4 for each subperiod. It is seen that the estimated coefficients of the two basic variables, per-capita growth and the demographic ratio measured by *E/M*,

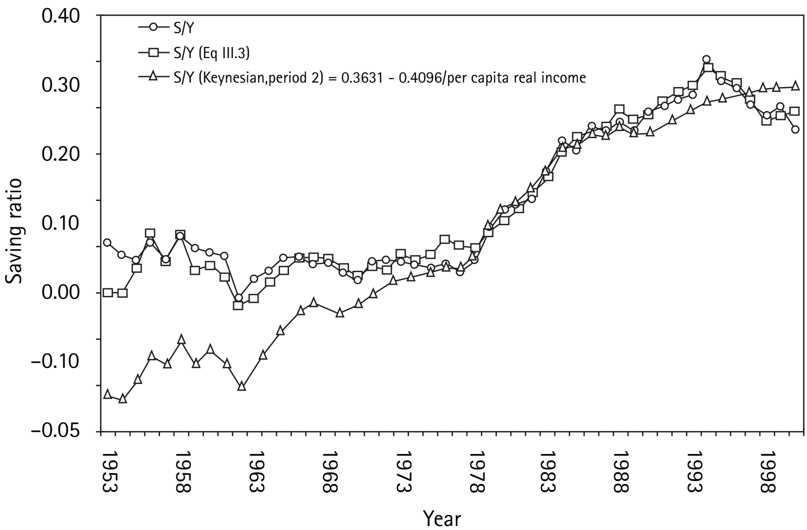

Figure 6.7
Saving ratio: actual and computed values from equation III.3

are all highly significant and remarkably stable; the variable measuring deviation from the growth trend is also significant and pretty stable. The remaining coefficient, that of inflation in column a_4, is less stable, being very high and barely significant for the first period.

But, there is a good reason for this event, namely that inflation in China appears to have been very sporadic. In a couple of cases it reached quite high levels, but those episodes occurred entirely in the second of our two periods (one in 1987–89; the second, more serious one in 1993–95). But from 1953 to 1985, prices were remarkably stable, at least as measured by the official consumer price index. Under these conditions we lack the information needed for a reliable estimate. And this lack of reliability is confirmed by the fact that the coefficient of 0.74 reported in equation III.3, col. a_4, is estimated with margin of error, as evidenced by a standard error of 0.4.

We have devised a test, however, to verify the validity of this explanation. If the coefficients of the first and second periods are basically the same up to statistical sampling error, then the second-period equation, which is "well estimated," should provide a good explanation for the behavior of *S*/*Y* in the first period. Our test then consists of using the parameter estimates for the second period to obtain "computed" values of *S*/*Y* for the first period. If our hypothesis is valid, we should find that these computed values fit well the first-period observed values. The results of this test are presented in figure 6.7. The curve marked with diamonds at the top of the graph is once again the observed saving rate. The curve that hugs it closely is the saving ratio computed from the second period equation, III.3, that is fitted to data from 1978 to 2000. It is apparent that these computed values fit the actual out-of-sample data from 1953 to 1978 remarkably well, with moderate underestimation in 1962 to 1964.

On the basis of this result it seems justified to conclude that the LCH model applies to the entire period with reasonably stable parameter estimates, despite the very notable changes in the underlying economy.

The third curve shown in figure 6.7 is an analogous test of stability of coefficients for the traditional Keynesian linear model. That is, we fit this model to the years 1978–2000, and find that this fit is bearably good though it misses the later decline. When we use the parameter estimates from the second period to compute the first period saving ratio, the result is a total failure; if saving were controlled by income the rise would have been much faster and there would have been a lot of "dis-saving" with the very low initial income. But the observed behavior is accounted for when Keynesian income is replaced by LCH variables, income growth, and population structure.

5.2 Comparison with Earlier Studies—A Survey

There are two major studies of which we are aware that have endeavored to test the LCH and estimate its parameters. First, Modigliani (1970) used a sample of 36 countries for which data were available from the UN Yearbook of National Account Statistics to test whether cross-country variations in the saving ratio could be accounted for by differences in growth rates and demographic structure, as hypothesized by LCH, instead of by differences in per-capita income according to the traditional view. This sample, which includes countries of quite varied backgrounds, is referred to hereafter as Cross I.

The second essay is Modigliani (1990) (hereafter Cross II), which used a sample of 21 OECD countries for a cross-country test of the forces accounting for the remarkable decline in the average saving ratio from 17 percent in the period 1960–70 to 10 percent in 1981–87, and it included inflation among the explanatory variables. The major conclusions from a comparison of the present study with the previous two are summarized as follows.

5.2.1 Long-Term Growth

This crucial LCH variable is found to play a major explanatory role in every case, with t-values running as high as eight (eleven in this study). As for the magnitude of the effect, the estimates for China, of around 2 percentage points increase in S/Y per 1-percent long-term growth, or a bit larger for the second period, agree quite well with those of Cross I—a little below two when growth is approximated by productivity, and somewhat lower (around 1.3) when growth is in per-capita income. Cross I, however, measured over a period appreciably shorter, mostly six to ten years, and we know from the Chinese data that shortening tends to attenuate the coefficient.

In Cross II the best measure of growth is found to be quite similar to ours (growth in the previous decade) and the estimates of the coefficient for the private saving ratio are clustered around 2. Cross II also reports the results from a sample of 81 developing countries assembled by the International Monetary Fund. The information for each country is rather limited in terms of the number of observations (1982–88) and quantity, but it includes estimates of the national saving ratio and average rate of growth of income over the six years. One finds that income growth plays a significant role for the sample as a whole and various subsamples. The t-values generally run between 4 and 6.5, and the coefficient estimates between 1.3 and 2.

We can conclude that there is substantial support and agreement among the studies about the central role of income growth and its quantitative impact. The Chinese results also provide a solid confirmation of the finding of Cross I that, contrary to the traditional wisdom, per-capita income is not a significant determinant of the saving ratio. Some evidence to the contrary in Cross I is shown to be the spurious result of cluster effects. However, there is one respect in which the results of Cross II seem to be at odds with those for China. Analysis of Cross I data confirmed the irrelevance of traditional Keynesian per-capita income for relatively high-income countries, but found some support for that variable for the poorer countries (t-value of 3). That result is not necessarily inconsistent with LCH, since it was initially advanced for developed countries, but is inconsistent with the results for China, which through most of the period was among the world's poorest countries. This discrepancy is one puzzle that remains to be solved.

5.2.2 Demographic Structure

For China the "dependency ratio" plays a major role, with t-value up to 9. The coefficient estimate is remarkably stable at 0.1. Both the minor-dependency ratio and the retiree-dependency ratio were also very significant in the studies cited above (see for example *R*/*W* and *M*/*W* in Modigliani 1970, table 2).

5.2.3 Inflation

There remains to consider the role of inflation that was estimated only for Cross II. There, the estimated effect was consistently negative though moderately significant (highest t-value of $4\frac{1}{2}$) with coefficients indicating a decline of between –0.5 and –0.7 percentage points in the saving ratio per 1 percentage point of inflation. This result appears again rather different from that for China, where the coefficient is consistently positive: for the period as a whole it comes to +0.25 with a t-value over 3, though it is not very stable in the subperiods. But the difference can again be easily reconciled by recognizing the differences in the measurement of the saving ratio with respect to both income and saving. For China the estimated coefficient measures the impact of inflation on the NIA measure of S/Y, which corresponds to k of equation (4). Cross II instead uses the inflation-adjusted concept, which subtracts from S and Y the loss in purchasing power of assets carried over. Therefore, the coefficient it estimates is $k(4)$ of equation (6). Now, it can be seen that $k(4)$ is more negative than k by the average propensity to consume C/Y. Therefore, for China, where C/Y is on the order of 0.8, the value of $k(4)$ can be placed around $0.25 - 0.8 = -0.55$, which is right in the range of the Cross II estimates.

Thus the present analysis confirms the finding of Cross II, to the effect that inflation has a strong negative impact on private wealth inflation-adjusted saving. This is especially true of China where nominal wealth is, in recent years, a large portion of private wealth (around ⅔ and somewhat larger than income). To illustrate, we recall that for China, there were a few years of high inflation, reaching a peak of 24 percent. With our estimated impact of 0.55 A^*/Y per point of inflation and a value of A^*/Y around one in recent years, this means a decline in adjusted saving (an impoverishment) close to 15 percentage points!

There are grounds, however, for questioning the interpretation advanced in Cross II to the effect that the large negative coefficient can be interpreted as evidence of serious inflation illusion. This results in a substantial decline in the contribution of personal saving to capital formation. It is true that with zero inflation illusion, and the Fisher law, the inflation-adjusted saving ratio is unaffected by inflation. This can be readily verified. But this conclusion does not justify attribution of a decline in the adjusted saving ratio to an increase in inflation illusion. Why? In the presence of large and/or persistent inflation, nominal interest rates rise, justifying higher consumption, without inflation illusion.

As suggested in section 3.4.1 and equation (6.5), the loss of contribution to capital formation through inflation is best approximated by the increase in $C/Y(3)$, or $k(3) = c(z - h)A/Y$. To estimate the value of this expression one needs to estimate z, which in turn would require information on prevailing interest rates paid to households during the inflationary episodes. But the official information available for China on nominal interest rates appears to be limited in time and quite spotty. It consists of a few points on the term structure of interest rates paid on deposits, at a few arbitrary dates (perhaps the dates at which interest rates are changed?). The information is presented in figure 6.8, in the form of time series of the rates, for each of three instruments on the term structure: three-month, one-

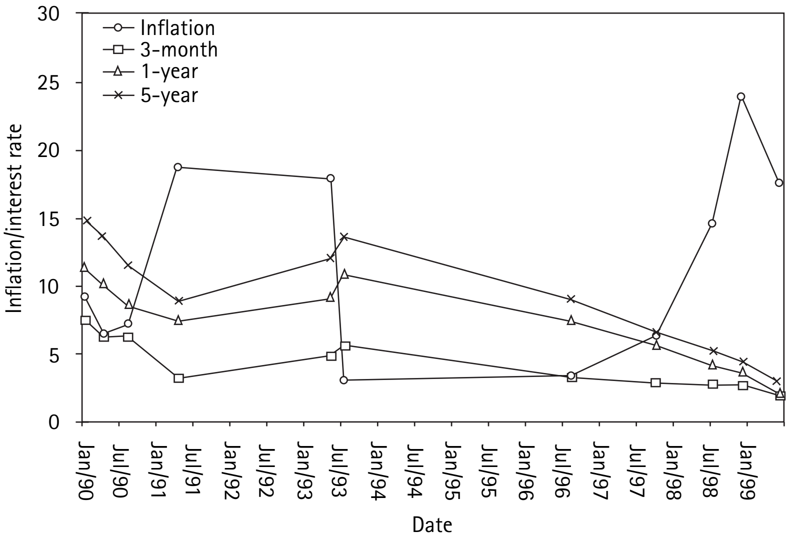

Figure 6.8
Inflation and interest rates

year, and five-year deposits plus the time series of the annual rate of inflation. It is apparent that interest rates do respond to inflation but with a lag and only partially. A more precise answer is especially difficult because of the lack of information for the interval from July 1993 to August 1996 which corresponds with the most dramatic explosion of inflation (24 percent in 1994) and ensuing reabsorption down to 7.5. Educated guesses about z suggests a negative value for $k(3)$ but of rather moderate proportion.

To summarize then:

1. According to the NIA measure of saving, inflation reduces C/Y (and increases S/Y) by some 25 basis points per 1-percent inflation, because it increases C less than Y. But this conclusion is highly misleading because the NIA measure is faulty in the presence of inflation.
2. Corrected for inflation, income does not rise and therefore consumption increases by $k(3)pA$, and saving by $(z - k(3)pA^*)$, presumably positive. However, government saving declines by zpA^*, so that national saving is reduced by the increase in private consumption, $k(3)pA^*$ whose sign we are unable to ascertain but is likely to be small.
3. Finally, inflation has a very large devastating effect on the value of the stock of accumulation of real assets, reducing by an estimated 55 basis points per point of inflation.

6 Conclusions: The Pervasive Role of Income Growth

Our conclusion can be most effectively summarized with the help of table 6.4. By the early 90s, the Chinese personal saving rate had reached a remarkable level of nearly 30 percent with a peak of over 33 percent (row 1). This occurred despite the fact that, even with the high growth rate, the per-capita income remained one of the lowest in the world. These saving rates are stunningly high in comparison with those of the United States, one of the world's richest nations. During those same years, the personal saving rate in the United States was 7.6 percent; and even the "private" saving rate, which is the sum of personal saving and corporate saving (profit retention), rises to only 10 percent (row 2). Since then the

Table 6.4
Normal Saving Ratio, Growth, and Population Structure for Selected Countries and Periods

I. Private Saving as Percentage of Private Income (Less Net Taxes)

	Country	Period	S/Y (%)	Y′ (%)	M/Pop (%)
1	China	1990–94	29.0	12.0	29
2	USA	1990–94	7.6 (10.0) +	2.3	29
3	China	1958–75	5.3	4.1	49
4	Iceland	1960–70	4.5	4.7	60
5	Japan	1971–80	24.3	9.5	35
6	Italy	1960–70	24.5	5.7	34
7	Average OECD*	1960–70	14.8	4.9	38

* —except Japan.
\+ —private saving rate: includes corporate saving.

II. National Saving as Percent of Net National Product

	Country	Period	NS/NY (%)	Y′ (%)	M/E (%)
1	China	1982–88	33.0	10.5	65
2	USA	1990–94	5.0	2.3	46
3	Japan	1971–80	25.0	9.5	35**
4	Korea	1982–88	31.9	8.2	64
5	Singapore	1982–88	42.5	4.9	67
6	Botswana	1982–88	35.3	10.9	47

** —M/Pop.

S/Y	
Y′	rate of growth of real GDP (lagged)
M/Pop	ratio of minors (under 20) to total population (%)
M/E	ratio of minors (under 15) to employment (%)
NS/NY	national saving to national income (%)

saving rate has slipped further with the personal down to 3 percent and the private rate down to 5 percent—though these low figures may reflect partly a transitory response to the boom in asset values.

When lay people are confronted with these figures, their usual reaction, after rubbing their eyes, is to attribute the huge gap to obvious differences in upbringing and education. This thinking reflects cultural-ethical values attributed to personal thriftiness and risk-taking in different cultures. But, if the reader has found the analysis of this paper persuasive, he should understand that that type of explanation is fundamentally baseless. And the simplest proof is found in this very paper, and summarized in row 3 of table 6.4, namely, that for a long stretch of very recent time, 1958–1975, the Chinese saving rate was quite low, around 5.3 percent or lower than in the United States.

What then is the explanation? The key to the puzzle, we suggest, is provided by the LCH and its implication that the major systematic determinant of the rate of private saving is to be found in the rate of growth of income and the demographic structure of the economy, while per-capita income, the traditional and commonsensical explanation counts little, if any. According to this model the extraordinary behavior of the Chinese saving ratio is the result of two nearly coincidental sharp turns in two key policies. The first is the movement initiated in the late 1970s toward a market-oriented economy, which, along with a number of very special characteristics of China's society and labor force made possible an explosive growth pattern such as was never seen before. With this development the growth rate jumps from a more or less stable rate of 4 percent (row 3) to a gradually rising (accelerating) pattern reaching some 12 percent but 25 years later.

The second turn regards demographic policies. Until the 1970s the Chinese government was not seriously concerned with population growth, and in fact for a while under Mao there was an endeavor to encourage births. But eventually the view prevailed that to improve the economic well-being of the Chinese population, it was essential to control population growth, and the new policy was announced and strictly enforced to limit the number of children per family (just one in the cities). As noted, this had a double profound effect on the saving ratio. The first was a drastic decline in the ratio of people under fifteen years to working population from 0.96 in the mid-70s to 0.41 at the turn of the century. The second was to undermine the traditional role of the family in providing old-age support to the parents by the children, thus encouraging provisions through individual accumulation. (Note that for purposes of comparison with other countries, the population structure, M/Pop is measured not by population below-fifteen/employed population, but by the ratio of the population below twenty/total population. Because of the larger denominator this ratio is around half as large as M/E.

According to our estimates (equation I.4 of table 6.3) each of these developments contributed equally more than ten basis points to the rise in the saving rate of some thirty basis points (from 3 percent to 33 percent) with the remainder largely accounted for by the spurious effect of inflation (five basis points).

As a further demonstration that the prodigious saving rate reached by the mid 90s does not reflect "ethnic" or cultural characteristics of China, we report in table 6.3 illustrative episodes for other countries and periods. First, in row 4, we show the case of Iceland in the 60s when the saving rate was about the same as that of China in the pre-reform years and the lowest of any OECD countries up to the end of the 80s. The growth rate is also low, like that of China, and the population structure is distinctively more unfavorable.

On the other hand, the last two rows of part I demonstrate that China is not the only portentous saver in the world. Japan in the 1970s (row 5) had an exceptionally high growth rate—though not quite matching that of China—and a quite favorable population structure that had a saving rate that nearly matched China's. One might be tempted to say that, after all, China and Japan share the heritage of the East. Unfortunately, such a simple explanation is refuted by row 6, which shows that in the 1960s, Italy saved even a larger fraction of its income than Japan. Yet, one cannot even use the excuse of the "Protestant ethic" since Italy is a solidly Catholic country and many think of it as the "paese della dolce vita!" The explanation can again be found in a growth rate not as high as the other two countries but still well above average (row 7) and a very favorable M/P that reflected the sharp decline in population growth, to which one can probably add the substantial loss of its tangible and intangible wealth during World War II. And a similar story can be repeated for France with a fairly high growth rate of 5 percent, saving 19 percent of its income, and of Portugal with a somewhat higher growth rate and a saving ratio of nearly 20 percent.

Row 7 shows the same information for the average of all the OECD countries for the decade of the 60s when the growth rate reached a peak of 4.9 percent and the saving rate also peaked for most countries with an average of 14.8 percent. Both are impressive by today's standards. By the 1980s, the average growth rate had declined to below 3 percent and the saving rate to 14.4 percent. The inflation-adjusted saving rate had declined even more, from 14.8 percent to 12.3 percent.

Part 2 of table 6.3 presents some further comparative statistics for the national saving rate—the sum of private and government saving expressed as a percent of national income. This measure reflects not only the behavior of the private sector but also the fiscal policy of the government, and to that extent, needs not respond closely to growth or demographic composition. In the case of China (row 1), as we have been before, it was a gigantic 33 percent—higher than the already mammoth personal saving rate. It reflected a fiscal policy aimed at supporting the rapid development of the country through high saving and investment.

On the other hand, for the United States (row 2) the national saving rate in the first half of the 90s was appreciably lower than the already low private saving rate, 5 percent versus 7.6 percent. This difference reflects the difficulty of reversing the policy of huge government deficits that characterized the Reagan presidency.

But the most impressive figures of part 2 are those in rows 4 and 5. The first shows that the Chinese record was nearly matched by that of Korea, another high-growth country bent on pushing saving. But row 5 reveals that both Korea and China were overshadowed by Singapore that had an incredible saving rate of 42.5 percent. But this huge accumulation did not result from growth, which was less than 5 percent. Instead, it was the result of a draconian policy that required all workers to make very large annual contributions to a pension fund (which can be used ahead of retirement for a variety of purposes other than current consumption).

The last row shows another bit of surprise provided by Botswana, a country for which the estimates are taken from Modigliani (1990, table 6.5), who in turn relied on information from the IMF. It appears that this country had an exceptionally high growth rate and a national saving rate even higher than China's.

In bringing this paper to a close, there is a strong temptation to try to use what we have learned to peek into the future of the Chinese saving ratio. All indications point to the likelihood that the ratio has already passed its crest. This presumes that China will manage to avoid the repetition of bouts of high inflation, as occurred especially in the early 1990s. The "dependency" ratio has already fallen dramatically to a level comparable with that of countries with little growth and there is not much room for further maintainable declines. As for the growth of income, it is difficult to imagine that China could improve on, or even maintain, the enormous current rate. It is instructive to note in this connection that Japan's growth rate after reaching 8 percent in the 1950s rose further to 9.5 percent in the next decade. It then declined gradually to less than 4 percent two decades later. In addition to these critical variables, one must recognize that private saving could be significantly impacted by a future public retirement system having universal or at least wide coverage.

We would like to express our heartfelt thanks to China for providing such a powerful experiment! There are considerations and limitations that underlie the "forecast" we have ventured to set forth above. But in the words of an old Italian saying, "He who shall live shall see!"

Appendix

The first of this series of papers was published in 1996. The data used were from 1953 to 1993. In 2002, we decided to update this paper to include data up to 2000.

For the 1996 study, most of the data were obtained from various annual issues of *China Statistical Yearbook*, published by the National Bureau of Statistics, People's Republic of China. During the course of the current study, we found that some of the data definition reported by the Chinese National Bureau of Statistics had changed in the interim. In 1999, the Chinese National Bureau of Statistics published *Comprehensive Statistical Data and Materials on 50 Years of New China* (referred to as *50 Years of New China* in this study), which contains data from 1952 to 1998. Some of the data in this publication are different from those found in the earlier editions of *China Statistical Yearbook*. The data reported in *50 Years of New China* are consistent with the data published in the 2000 and 2001 *China Statistical Yearbook*. We decided that it was best to use a consistent set of data. For the current study, most of the data are obtained from *50 Years of New China*. Additional data were obtained from the 2000 and 2001 *China Statistical Yearbook*. Some supplemental data were obtained from earlier edits of the *China Statistical Yearbook*.

The details of the data sources and some of the differences in the 1996 and the 2002 data sets are listed below:

Parameter	2002	Comments
Price index	Table A-20 Overall Price Indices of China, General Consumer Price Index, *50 Years of New China*, p. 20; *Statistical Yearbook of China* 2001, p. 281.	The 1994 issue of *Statistical Yearbook of China* includes overall retail price indices from 1978 to 1993. It also includes overall consumer price indices from 1985 to 1993. One would deduce that the complete series of consumer price indices was not available prior to 1985; the first study was conducted in 1994.
Personal consumption	Table A-6 gross domestic product by expenditure approach of China, household consumption, *50 years of New China*, p. 6, 2001 *China Statistical Yearbook*, p. 62.	The household consumption figures used in the 1996 study were slightly higher than those in the 2002 study.
Saving	Calculated: change in (currency + deposit) + bonds (new issues) + individual investment in fixed assets.	Currency in circulation: *50 Years of New China*, p. 65; *2001 China Statistical Yearbook*, p. 638. Total saving deposits: *50 Years of New China*, p. 25; *2001 China Statistical Yearbook*, p. 304. Bonds, domestic debt issued, payment for principal and interest of domestic debts, *2001 China Statistical Yearbook*, p. 249. Individual investments in fixed

Parameter	2002	Comments
		assets: 1953–79 from People's Bank of China; 1980–2000, *2001 China Statistical Yearbook*, p. 158.
Total population	Table A-1 Population of China: total population, *50 Years of New China*, p. 1; *2001 China Statistical Yearbook*, p. 91.	
Population profile	Interpolated from total population and population profile reported in: Table 4-4 *Basic Statistics on Nation Population Census* in 1953, 1964, 1982, 1999 and 2000; *2001 China Statistical Yearbook*, p. 93.	
Number of employed persons	Table A-2 Employment, staff and workers of China, total number of employed persons, *50 Years of New China*, p. 2; *2001 China Statistical Yearbook*, p. 108.	

Notes

1. World Development Indicators database, World Bank, July 2003.

2. As Modigliani and Sterling (1983) show, pension contribution does not automatically decrease household saving, in part due to early retirement induced by the existence of a pension program. Our casual observation indicates the latter effect was not apparent in recent Chinese history, which leads us to conclude the evolvement of the pension system in China had a negative impact on household savings.

3. The national pension system started to take shape in the 1990s. According to China's Ministry of Labor and Social Security, the system covered 104 million employees and 32 million retirees at the end of 2000. The contribution to the fund was 228 billion, compared to our personal savings estimate of 1322 billion. In addition, 62 million rural residents participated in the government-sponsored rural pension plan (see table 6.2).

4. Modigliani (1970) discussed in detail the identification problem, i.e. the interaction between saving, investment, and the growth rate.

5. Liao Min contributed to the development of this approach.

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II

UNEMPLOYMENT AND MONETARY POLICY IN THE EUROPEAN UNION DURING THE 1990S

7

AN ECONOMISTS' MANIFESTO ON UNEMPLOYMENT IN THE EUROPEAN UNION

Franco Modigliani, Jean-Paul Fitoussi, Beniamino Moro, Dennis Snower, Robert Solow, Alfred Steinherr, and Paolo Sylos Labini

1 Foreword

This manifesto challenges a pernicious orthodoxy that has gripped Europe's policy makers. It is that demand and supply side policies must have different aims, that a limited number of supply side policies are to be devoted to fighting unemployment, and that demand management (and particularly monetary policy) is to be devoted solely to fighting inflation. The prevailing orthodoxy also claims that the choice of policy instruments for combating unemployment is a political decision, in which each instrument is evaluated on a case-by-case basis.

In what follows, we outline various practical proposals aimed at a prompt reduction of unemployment. We are confident that if the advice is given proper attention by governments and monetary authorities, unemployment can be reduced significantly in a matter of a few years.

We will divide the proposed actions into those bearing on the revival of aggregate demand (demand policies) and those addressed to the reform of the labour and product markets and the system of benefits for the unemployed (supply policies). But we stress from the very beginning that we regard our proposals as strictly complementary with one another. Each proposal, applied in isolation, may produce little or even perverse effects, while the simultaneous application can be counted upon to yield the desired outcome. This holds in particular with respect to the relation between demand and supply policies. The underlying idea is that it is much easier to encourage people to look for jobs if there are jobs to be found and it is much easier to encourage firms to offer more jobs if there are more people willing to accept them.

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2 The Unemployment Problem

We share the view that at present time unemployment is the most serious and urgent problem facing the European Union (EU). Today (October 1998), the average rate of unemployment in these countries is 11 percent (19 million), with peaks of 15–20 percent, while in the 60s and early 70s it was almost universally well below 3 percent and nowhere over 5 percent. Such a huge rate of unemployment results in an immense waste of resources, through loss of output, that can be estimated at some 15 percent or more, and even larger loss of saving-investment potential. It is degrading and demeaning for the unemployed and with damaging long run consequences, especially for the young that represent, in most countries, the bulk of the unemployment. And it is also a source of dangerous social tensions.

We also share the view that the measures that have been proposed in numerous meetings of representatives of member governments at various levels, including the Amsterdam (June 1997) and Luxembourg (November 1997) meetings especially devoted to this problem, suggest that many of European leaders have not adequately confronted the nature of the problem. Consequently they have not succeeded in agreeing on politically feasible programs that have a chance of producing an appreciable decline in the current, high unemployment rate in the relevant near future.

3 False and Misleading Explanations for European Unemployment

The widespread acceptance of the timid program agreed so far seems to reflect in part the view, by now common in Europe, that unemployment is a calamity due to causes beyond the capacity of governments to manage, except possibly by increasing profits and generally increasing income inequalities. And this conclusion has led to a convergence of both the right and the left on the view that the scourge must be bravely endured for fear that any remedy might make matters worse.

Many possible causes have been advanced to account for the high and persistent rate of unemployment in the EU. They vary somewhat along the political spectrum. On the right, it has been argued that EU unemployment is primarily the outcome of i) the absence of the needed skills (there are jobs but the unemployed are not qualified to fill them), ii) the large share of long-term unemployed who lack motivation to seek jobs, and iii) the crushing burden of taxes. All these arguments contain grains of truth, but it is easy to be misled by them.

The first argument is supported by the observation that the rise in unemployment has fallen disproportionately on the less skilled and qualified segments of the labour force. But American experience over the past few decades suggests that when unskilled workers are not consigned to an "unemployment trap" through misguided welfare entitlements, then the demand for unskilled labour fluctuates with the availability of jobs. When available jobs shrink, workers with higher qualifications displace those less qualified, and when demand and job opportunities improve, the unemployment rate of the less qualified declines.

The share of long-term unemployment varies substantially among EU member states; but there is good reason to believe that a large share of long-term unemployed is more the effect than the cause of a high and persisting unemployment.

As for taxes, it is estimated that in 1997 total government levies amounted to about 43 percent of GDP in the EU versus 31.6 percent in the U.S. But these figures fail to distinguish between taxes that pay for government services and social security levies, that represent contributions toward pension benefits—i.e. saving—even if compulsory. If one leaves out social security contributions, the tax burden (direct plus indirect taxes) drops to 27 percent in Europe, versus some 23 percent in the U.S. The difference in the untaxed share of income—73 percent in Europe versus 77 percent in U.S.—is by no means dramatic and certainly cannot account for the fact that European unemployment is 8 percent higher than before the early 70s, while that of the U.S. is not higher. If unemployment were so sensitive to small differences in taxation, why is it that Germany—with a tax burden similar to that of the U.S. (23.3 percent)—has unemployment similar to the rest of the European countries (11 percent), while the U.K.—with a tax burden 6 percentage points larger than Germany (29.5 percent)—has much smaller unemployment (5.6 percent)?

It is of course true that in Europe social security levies take a much larger bite from income (16 percent of GDP versus 9 percent in U.S.). Of course the average rate of contribution for those workers actually covered by social security in many countries is much higher than 16 percent and the fiscal pressure is well over 40 percent for countries like Italy (44 percent) and France (46.8 percent), where the replacement rate (the rate of unemployment benefits to wages) is very high. These high levies, it is said, increase unemployment both by sapping the incentive to work and by raising the cost of labour to employers. But these assertions are fundamentally flawed. Firstly, the higher European levies do not, as is generally supposed, reflect the need to cover the higher costs of a more wasteful and intrusive government. They are instead the result of an explicit social choice of saving (in compulsory form) a larger portion of income in the working years in order to receive a larger pension in retirement (and to retire earlier), combined with the inefficiency of the pay-as-you-go public pension system. Secondly, social security levies generally have little influence on real labor costs in the long run because they are born primarily by labor, and not by profit earners, whether they are formally collected from the employee nominal compensation or from the employer.[1] A possible exception may arise for workers on a minimum wage, if that wage is fixed in terms of real take home pay. In that case higher social security levies cannot be shifted to the worker and will instead result in a higher real cost and price and thus higher unemployment.

As for the assertion that high social security levies reduce the incentive to find a job by reducing the difference between unemployment compensation and take-home pay, the conclusion is obviously valid only if the government pays the social insurance contributions of unemployed workers, or if pensions are independent of the workers' contributions, which is certainly not the general practice.

On the left side of the political spectrum, European unemployment has been portrayed as the outcome of iv) a crisis of capitalism, v) an excessively rapid rate of technological progress, and vi) competition with low-wage countries. All of these explanations are called into question by a very simple consideration: if they were valid, they should produce the same high rate of unemployment in all other developed countries. But in fact the sharp rise in unemployment since the 80s has no parallel among other advanced industrial countries. In fact, the unemployment rate of every OECD country nowadays is below the EU average and only two such countries have unemployment rates that are even close.

One further and very different line of thought that has supported tolerance toward the *status quo* is the argument that the demand and supply side factors above are an inevitable part of European political and social policy and reforms would be intrinsically undesirable. The restrictive demand side policies are commonly viewed as necessary prelude for the further economic and political integration of Europe; and the restrictive supply side measures are frequently seen as required to retain economic equality and social cohesion. It is held that governments must choose between two disagreeable options: a "flexible" labor market bedeviled by wide income disparities and an "inflexible" labor market crippled by unemployment. The "flexible" market, where people's wages reflect their productivity, is allegedly achieved by reducing job security, restricting unemployment benefits and welfare entitlements, eliminating minimum wages, bashing the unions, and opting out of the social chapter. The "inflexible" market, where people's earnings reflect politicians' judgements about fairness and social cohesion, is supposedly achieved by the opposite policies. This generates the conviction that the ultimate choice, then, is between inequality and unemployment. In this light, the high level of European unemployment is sometimes portrayed as a price that must be paid for the achievement of other important long-term objectives.

These presumptions are reflected in much of the European Union's policy approach to unemployment. The Luxembourg communiqué and those issued on earlier occasions restrict their purview to a very limited timid set of supply side policies and do not even mention the possible role of demand management policy, and monetary policy in particular, in affecting unemployment. Furthermore, they stress that unemployment is a problem that can and must be solved within each country, without explicit co-ordination of policies between EU countries even though, in joining the euro system, the member countries renounce the possibility of independent demand management policy, monetary or fiscal, and come into close competition on supply side policies.

By contrast, we believe that the bulk of European unemployment serves no useful purpose whatsoever. On the contrary, it is overtly harmful to the achievement of the objectives that have been used to rationalize the problem. Since work is the major avenue whereby people are able to claw their way out of poverty and overcome economic disadvantage, high levels of unemployment—particularly long-term unemployment—are deleterious to social cohesion and economic integration.

We call for rejecting the powerful pernicious myth that blinds policy makers to unemployment policies that could reduce unemployment without widening the gap between the rich and poor. Thus it is important to expose the myth and get on with the urgent business of fundamental policy reform. The trick is to recognize that much of the current employment policy is responsible for the disagreeable choice between unemployment and inequality.

It is thus extremely important to differentiate carefully between the genuinely promising policy proposals and those that are unpromising in the sense that they may reasonably be expected to turn out ineffective or counter-productive. Making this distinction is not easy because many of the policies that influence unemployment are highly complementary with one another. This means that potentially enlightened policy initiatives are often ineffective when implemented in isolation from one another. Employment-promoting supply side policies frequently enhance the effectiveness of employment-promoting demand side policies, and vice versa. Furthermore, counter-productive policies often emasculate the influence of enlightened policy measures. In the domain of unemployment policy, bad measures drive out the good, and good measures reinforce one another.

In the next section we review several of the major "conventional" policies which have been implemented or proposed and show that they belong to the "unpromising" set and are, in the end, an important source of European unemployment.

4 Misguided Policies as a Cause of High Unemployment

We hold the view that the European unemployment is, in important part, the result of policy errors. These errors involve both a mismanagement of aggregate demand (demand policies) and an unimaginative approach to the supply side of the economy. We are confident that these errors can be corrected promptly, putting an end to the unremitting longer-run growth of European unemployment.

4.1 Errors in Demand Management

The words aggregate demand policy have become familiar to economists ever since Keynes used it to provide an understanding of the Great Depression and the role played in that episode by central banks. Yet at present the concept has become taboo among many European central bankers and political leaders, even though there is plenty of evidence that, in recent years, it plays a significant role in accounting for the rising unemployment.

A first suggestive piece of evidence is provided by the observation made earlier that double digit unemployment is common only in Europe—or more specifically among the countries that are in (or are candidates for) the euro. In fact, the European countries not in the euro have substantially lower unemployment rates: in Norway the rate is 4 percent, in Switzerland 5.5 percent and in the UK 5.6 percent.

This observation has some powerful implications. It suggests that in order to gain insight into the constellation of causes responsible for EU unemployment, it is important to identify factors that are shared by most EU member states but are not in evidence in non-euro countries.

On the demand side one experience that the euro countries have shared in common in the last few years, and generally not shared with others, has been the very restrictive aggregate demand policies, both fiscal and monetary. They have been forced to pursue these policies as a result of their common endeavour to join the euro. The common fiscal policy was the result of the Maastricht parameters and it was very restrictive in the light of the huge unemployment and resulting depressed government revenues, and of the tight monetary policy. One by-product of this policy has been the slowdown of public sector infrastructure that is complementary with private sector investment. Similarly, monetary policy was made uniform by the fact that exchange rates were to be kept narrowly fixed while all restrictions on the free movement of capital were eliminated. Under these conditions, interest rates had to be the same for all the candidate countries and there was no room for the national banks to pursue an independent monetary policy. And the common monetary policy appears to have been also much too tight, especially in the light of the tightening of fiscal policy, resulting in a long period of excessively high real interest rates that have discouraged investment and swollen unemployment.

The relation between unemployment and the demand for labour provides further evidence. Since the beginning of the oil crisis, in 1973, the rate of growth of demand has fallen considerably below that of capacity output—the sum of productivity and labor force growth. In fact the growth of demand has been roughly the same as that of productivity. Thus demand could be satisfied without a significant increase in jobs and the growth in the labor force of around 2 percent went to swell the ranks of the unemployed. This process of jobs falling relatively to the labor force is confirmed by direct information on the available jobs, the sum of employment and vacancies. In most EU countries, the number of jobs offered in each year as a percentage of the country's labor force has tended to fall.[2]

We believe that one reason for the drastic European decline in the demand for labor relative to its available supply—and the resulting rise in unemployment—has been a decline in investment relative to full-capacity output. In this connection it is interesting to observe that the difference between the growth of unemployment since the early 70s in Europe (8.5 percent) versus the U.S. (0 percent) occurs mostly in two episodes since 1982. Up to that year joblessness had increased sharply on both continents as consequence of a restrictive monetary policy and resulting fall in investment, which was unavoidable to halt an inflationary spiral, ignited by the two oil crises. But after 1982 the shortfall and unemployment continued to rise in Europe till 1986, whereas in the U.S. both fell promptly and significantly. The second episode begins in 1992 and extends to the present. In both these episodes, the investment rose relative to full capacity in the U.S. but remained stagnant at peak levels in Europe.

4.2 Misguided Supply Policies

The measures to combat unemployment that have been suggested do not, in our opinion, reflect the best options available from the potential portfolio of feasible policy choices. One important source of the European unemployment problem are the misguided conventional policies that have been put into place to support the unemployed and protect the employed from job loss. The following provide three important examples.

Minimum Wage Legislation

Minimum wages are widespread in Europe and are potentially an important source of unemployment. The institute of the minimum wage (MW) is inspired by a lofty ideal, that any one who wishes to work should be able to secure a minimum decent living standard. The trouble is that the translation of this principle into practice typically takes a form which ignores basic economic laws and thus ends up creating great injustices and doing more harm than good. The form it takes consists in essence in forbidding firms to hire anybody for less than an imposed (fair) minimum wage or, equivalently, making it a "crime" for anyone to accept a job for less than that fair wage. Clearly this system will "work" for those that can in fact secure a job at the minimum wage. But if the number of people that would be willing to work at that wage or less exceeds the number of jobs that the system can offer at that wage, then it is obvious that the excess supply (if any) is condemned to unemployment, with all its negative economic and social implications. In practice these unemployed will largely consist of young people with no experience and little human capital.

It must be acknowledged that despite numerous studies attempting to measure the influence of minimum wage legislation on unemployment, to date there is little consensus about the precise nature of the impact. Although empirical studies have shown that relatively modest increases in the minimum wage may not raise unemployment, there is widespread agreement that large minimum wage hikes—wage increases sufficient to eliminate the major income inequalities between mainstream employees and workers marginally attached to the labour market—would have such an effect.

One further negative impact of minimum wages comes from their interaction with high social security levies. In so far as the minimum wage aims at assuring a minimum real take-home pay, higher Social Security levies cannot be shifted to the employee: an increase in compulsory saving will be borne by the employer and raise the cost of the employee. This is one of the important factors that make a minimum wage so high in Europe, discouraging the employment of less skilled labour.

Job Security Legislation

Some commentators have maintained that job security legislation helps reduce unemployment. The underlying argument is that such legislation reduces both firing (by making it more costly for employers to dismiss their employees) and hiring (by discouraging employers from taking on new recruits who may have to be dismissed in the future). But at given real wages, the firing costs generated by job security legislation discourage firing more than they discourage hiring, since firms that fire must pay the firing costs now, whereas firms that hire may have to pay the firing costs at some point in the future.

However, this argument rests on tenuous foundations. In the first place, even though firms may initially find it economical to employ more people than would be optimal in the absence of constraint, they eventually will find it advantageous to shrink their labor force, at least through attrition and aging and also rely more on overtime. Second, and far more serious, the slowdown in the flow of hiring greatly reduces the chance of outsiders, and particularly new entrants in the labor force, to find a job and is a major cause for the high, in some cases almost unbelievably high, incidence of unemployment among young people. This high rate is particularly striking in countries where no unemployment benefits are provided for people that have never held a job. In addition, a rise in firing costs cannot be expected to leave real wages unchanged. On the contrary, the greater the firing costs, the greater will be the market power of the incumbent employees (insiders) and thus the higher the wages these workers can achieve. Taking further into account the fact that the high cost of firing will add to labor costs both directly and through redundancy, we can expect a lower demand for labor, at least in an open economy. On the whole, job security legislation must be regarded as a major negative influence on unemployment, especially youth unemployment, even if it might have desirable effects in other directions. Consequently, the lower will be the demand for labor. For these various reasons, increasing job security is far more likely to eventually decrease jobs and employment rather than the reverse.

Work-Sharing and Early Retirement

There can be no objection to people reducing their work week or retiring early if they are prepared to accept the corresponding reduction in weekly pay, but we hold that there is no justification for the government to provide incentives for people to work shorter hours or retire earlier.

The logic underlying work-sharing and early retirement is of course elementary. If there is a fixed amount of paid work to be done in the economy, and if this work falls very unequally across the population—with a majority of people enjoying full time employment while a minority is saddled with long periods of unemployment—then considerations of equity and social cohesion make it reasonable to seek policies that share the burden of unemployment more democratically. In short, if the pain and impoverishment of unemployment are inevitable, it may the best to spread the misery as evenly as possible.

The problem with this approach is that the underlying premise is false. We do not believe that the European unemployment problem is unavoidable. The amount of work to be done in the economy is not fixed. When the economy is in recession, an increase in production and employment—in response, for instance, to a rise in export demand or on private investment—will lead to a rise in purchasing power and thereby generate a further increase in production and employment. In this sense, unemployment is not inevitable. Policy makers who see it as such are being unduly defeatist; they should spend more thought on bringing unemployment down rather than on spreading it more thinly.

We therefore agree that those measures are not appropriate as the central pillar of a policy strategy to reduce unemployment. In fact, they pose some dangers of becoming counter-productive in the sense they might reduce the total number of hours worked in the economy even if they succeeded in increasing the number of people working. It has proved very difficult to implement them without raising non-wage labour costs (particularly costs of hiring and training) and thereby discouraging firms from creating more jobs. Furthermore, by diminishing the number of people competing for jobs, these measures may indirectly put upward pressure on wages and thereby on prices. Governments or monetary authorities may then feel called upon to dampen inflation through contractionary fiscal and monetary policies, thereby generating further unemployment.

The push for shorter work week as a device to reduce unemployment by work sharing has taken recently a dangerous turn when some of its sponsors, in an effort to gain popular support for the measure, have proposed that the reduction from 40 to 35 hours should be accompanied by an unchanged weekly pay. We regard this version as little more than demagogy. It would compound the difficulties already encountered in reducing individual hours while maintaining the hours worked by the firm, by imposing a rise in hourly wages by 5/35 or nearly 15 percent. The effect could not but be disruptive. The increase in labor costs could hardly be expected to come out of profits but could be expected, instead, to be passed along in higher output prices. This would result in a weekly real wage equivalent to 35 hours and/or in successful demands for higher nominal wages, initiating a wage-price spiral. But with fixed exchange rates or a single currency, the rising prices would also reduce the share of the country's foreign and domestic markets and prove a new source of unemployment. This effect might be mitigated if all countries undertook the measure simultaneously, but the inflationary spiral would be reinforced.

5 Proposed Policies for a Timely Reduction of Unemployment

In what follows, we set forth a number of practical proposals aimed at a prompt reduction of unemployment. We are quite confident that if the advice is given proper attention by governments and monetary authorities, unemployment can be reduced by 4 or 5 percentage points in a matter of a few years and without compromising the recent gain in subduing inflation.

Our proposals cover both demand and supply side policies. We wish to emphasize that these policies are not to be assessed on a case-by-case basis. Our recommended policy package is not to be viewed as a portfolio of independent measures, from which policy makers can pick and choose. Rather, as noted at the beginning of the manifesto, we regard the policies as complementary to one another, with the demand side policies creating a need for the new jobs that the supply side policies make available.

The failure to exploit policy complementarities may be an important reason why so many of the partial, piecemeal labour market reforms implemented in EU member states have done little to reduce Europe's unemployment problem. In Spain, for example, a labor market reform has been introduced in 1984, whose main aim was to achieve a greater flexibility in labor contracts. Among other things, this reform introduced fixed-term labor contracts with low firing costs. As a result, fixed-term labor contracts have grown quickly and Spanish firms have used them to buffer fluctuations in demand by changing the number of fixed-term employees. But, at the same time, this policy reduced the risk of unemployment for workers with permanent contracts, which reinforced the bargaining strength of the insiders. Since wage bargaining agreements mainly reflect the interests of the latter, this reform has turned out to cause more rigidity rather than more flexibility of the wage rate. To mitigate this unwanted effect, Spain has recently reintroduced some restrictions on fixed term contracts and has reduced firing costs for all workers.

In France, several acts have been passed aimed at introducing a greater flexibility in the labor market and at preventing the negative effects of both minimum wages and the highest payroll taxes among OECD countries. Moreover, restrictions on part time work have been eased, and work-sharing has been encouraged. But nothing has been done in this country to reduce the stringency of job protection legislation and the bargaining power of insiders.

In Italy, a reform of the labor market was first passed in 1991, which allowed small- and medium-size firms to dismiss redundant workers, but only with the agreement of the unions, The so-called "mobilità lunga" (long mobility) was also introduced, which consists in the possibility to put the unwanted workers in the social security system (thus aggravating its operating costs) before giving these workers the right to definitely retire. A second reform has been recently introduced in 1997, which permits firms to hire workers temporarily from appropriate employment agencies.[3]

Also in Sweden, some reforms have been approved in order to increase labor market flexibility. In this country, unemployment benefits are of comparatively short duration, but the replacement ratios are high. Thus, jobless people can move from unemployment benefits to training programs and back, while generous welfare state entitlements encourage leisure relative to employment. In general, the welfare benefits in this country are so generous to render the condition of inactivity, especially for medium-aged people, more appealing than employment.

The United Kingdom and the Netherlands are the only two European countries that have witnessed appreciable reductions in unemployment from their labor market reforms over the last two decades. These successes may well be due to the broad-based nature of their reforms, enabling them to exploit significant policy complementarities. The U.K. for instance, introduced legislation restricting strikes and secondary picketing, decentralizing wage bargaining, liberalizing hiring and firing restrictions, reducing the duration of unemployment benefits and tightening the associated eligibility criteria. Moreover, minimum wages have been abolished (soon to be reintroduced by the current Labour government) and unemployment benefits have been reduced, and, at the same time, new procedures have been implemented in order to facilitate the search for a job for the unemployed people. These reforms, together with the decision to opt out from the European Monetary Union, at least in the initial stage, have spared this country the need to adopt restrictive aggregate demand policies and have greatly contributed to the fall in the U.K. unemployment rate from 10.5 percent in the 1993 (approximately equal to the EU average) to 5.6 percent in 1998. This result, moreover, has been achieved without substantially changing other welfare state entitlements, such as housing benefits, or by a thoroughgoing drive to improve education and training systems.

The experience of the U.K. and the Netherlands also highlights the dangers of leaving particular policy complementarities unexploited. In both countries the tightening of the unemployment benefit system was not matched by a correspondingly fundamental reform of the sickness and incapacity benefit systems. Consequently budgetary pressures have shifted from unemployment benefits to sickness and incapacity benefits. Since the latter have a longer duration than unemployment benefits, the shift created more serious conditions of dependency from publicly-provided income support than unemployment insurance. Thus in the Netherlands, which has one of the most generous disability benefits systems among the OECD countries, the percentage of persons directly involved in social benefits reaches 17 percent.

In sum, European countries have not, on the whole, sought to reduce unemployment by implementing a coherent strategy of fundamental reforms across a broad range of complementary policies. In the main, these countries have adopted a number of *ad hoc* measures that attempt marginal corrections to the most egregious distortions stemming from existing labor market policies or regulations. We argue that, since only marginal, piecemeal changes have been implemented, existing restrictive institutions and regulations that are complementary to each other continue to interact, blocking the effectiveness of the recent reforms and prolonging unemployment.

Accordingly, our recommended policy strategy is *a*) to implement a broad spectrum of supply side policy reforms that give employers a greater incentive to create jobs in response to increases in demand and give employees a greater incentive to accept these jobs, and *b*) to implement demand side policies that enable the European economies to raise their growth rates of capital formation and productivity, and to use the productive potential that has been released through the supply side reforms.

5.1 Aggregate Demand Policies

We believe that the demand side strategy for reducing EU unemployment should involve policies that stimulate a broad revival of investment activity, taking care not to ignite inflationary pressures or increase the size of the national debt relative to national assets. The process of stimulating investment is, to a very considerable extent, self-reinforcing, because of a well-established mechanism known as the accelerator effect. As investment rises, increasing employment and output, the initially existing excess productive capacity will become more fully utilized and there will soon be a need for additional capacity, which will require new investment.

It is generally agreed that labor and capital are often complementary in the production process, so that an increase in the capital stock usually leads to a rise in labour productivity. Provided that the economy is kept out of recession—so that the danger is avoided that firms employ as little labor as possible to meet a given, deficient product demand—and provided that there are sufficient supply incentives in place to encourage employers and potential employees to exploit profitable job opportunities, increases in labor productivity will generally lead to increases in labor demand and consequent reductions in unemployment.

The endeavour to expand the rate of investment need not, and should not, be limited to private investment. The constraints on public investment are currently felt with particular stringency because of the large public-sector debt existing in many European countries, and because of the consequent limitations on fiscal deficit imposed by the Maastricht parameters, together with the unfortunate circumstance that, in computing the deficit, all expenditures, whether on current account or for investment, are treated identically. Under these conditions, governments have frequently found it expedient to cut investments, even if highly desirable, rather than cut the budget for public employment (e.g., by reducing the number of employees). Given the prospective difficulties many EU member states face in satisfying the Maastricht criteria, this under-investment is likely to continue.

In order for an expansion of public investment to produce the same beneficial effects on unemployment as private investment, it is necessary that it should be financed neither by cutting other expenditure—except for transfer payments whenever it is possible—nor by raising taxes (which at present would be practically impossible anyway). This means that the additional investments must be financed, for most of the countries, in just the same way as private investments are typically financed, namely by raising the money in the capital markets in the form of debt or equity. Private-sector finance of public investment, along the lines currently being explored in some EU countries such as the U.K. needs to be expanded as well.

In this context, it would be important to introduce a distinction, long overdue, between the current and the capital account deficit, and to redefine the budget deficit, for the purpose of the Maastricht agreement and the later stability pact, as consisting of the current account deficit only. The Current Account Budget should include all current expenditures and receipts (expenditures that benefit those present and receipts collected from them) and it is appropriate to require that this budget be balanced, as this places the cost of current expenditure on the current beneficiaries.

The amount of public capital expenditures, on the other hand, should be primarily limited by the requirement that each project should have a return over its life at least as competitive as market returns (with proper adjustment for taxes). However the difference, if any, between the cash receipts and the annual cost of providing the services, including the interest cost, and the depreciation, would be charged to the Current Account as a current expense (if negative) or treated as a current income (if positive).

Of course, deficit financing of government expenditure, when there is no room for an expansion of employment, tends to crowd out private investment, and thus burden future generations by depriving them of the return on crowded out capital. But we hold that the program of government investment we advocate will not harm and may even improve the lot of future generations. In the first place, when there is an enormous reserve of unemployed resources, investment will increase income and thus saving, at least to the extent of financing the investment without any crowding out. In the second place, infrastructural investment increases the marginal product of both capital and labor in the private sector, which will have expansionary effects. Finally, we agree that debt financing of capital expenditures satisfying the above criteria, unlike that financing a current account deficit, would not be harmful to future generations, even if it displaced an equal amount of private investment, because its return would at least compensate for the return lost on private investment.

We propose to concentrate public investment on specific infrastructures capable of giving returns in the short run. To finance these investments we propose that the existing European Structural Funds should be more used than in the past. Such funds, already considerable (153 billions of ecus for the period 1994–99), should be enlarged and their regulations should be re-negotiated, especially with regard to the procedure to spend them, since in the future they could become the main financial instrument of the European strategy to cure unemployment and promote growth. Today these funds can be spent only if the state using them provides simultaneously an equal amount of funds. This regulation is wise, but an interval should be granted when there is a proper guarantee. Moreover, to avoid delays, a particular care should be devoted to the organization of the structures in charge of the projects.

In any case, some public support for job creation is perfectly justified. The European Structural Funds were created for the purpose. Although important, their grant nature leads to inefficient uses and quarrels between providers and recipients. Therefore, there is little hope for a substantial increase of these funds in the EU. We propose to augment the grant of the Structural Funds with loans at interest rates at or below market rates. Such loans could be provided by the European Investment Bank (EIB), which has the resources and the experience in project evaluation to ensure that these funds finance sound and job-creating investments. The EIB already received the mandate of the Amsterdam and Luxembourg meetings to attach top priority to job creation. We argue that this mandate should be scaled up. For such a programme, interest rates below market rates are important otherwise the potential to stimulate investment and job creation remains limited. In countries like Italy, in which it is particularly important to keep at a low level the public deficit, recently achieved after a long and costly effort, it would be advisable to partly finance the infrastructural investments by also using a share of the receipts obtained from the privatization of public enterprises.

However, the success of the operation requires a revision of the principle that has emerged from the meetings in Amsterdam and Luxembourg, namely that the solution of the unemployment problem is not a collective responsibility but a task to be tackled by each country on its own. This approach is mistaken and will make a prompt solution very unlike. It springs from the view that unemployment is mainly due only to the malfunctioning of the labor market. But while we all concur that labour relations greatly contribute to the problem, we also share the view that demand plays a major role, relying on the evidence presented above as well as other evidence and reasoning. But the agreements of the two recent summits actually hamper the exercise of demand policy, because, after assigning to the individual states the task of reducing unemployment, they deprive them of all the classical tools of demand management: i) monetary policy, because individual central banks have already little control over interest rates and will have, gradually, even less; ii) fiscal policy, because of the rigid constraint on fiscal deficit; and iii) exchange rate policy. And the European Union, besides being exonerated of any responsibility, has no tools either: the Bruxelles Commission has no resources to spend, and the European Central Bank (ECB) is to concern itself exclusively with price stability.

The solution we are advocating, by contrast, requires coordination of policies of EU member states. Indeed, if any country were to engage in a demand side expansion alone, then, as is well known, its effect on unemployment would be much smaller than under a coordinated policy approach, because much of its beneficial impact would be lost to it and would spill over to others, through higher imports. The resulting deterioration of the current account could be so severe as to make the expansion inadvisable. But when the expansion is simultaneous and symmetric, then the increased imports will be offset by an increase in exports resulting from the increased imports of the other countries, and this will both help the current account balance and restore the potency of the expansion of investment. In short, in a simultaneous expansion, countries would be helping each other.

The Role of the European Central Bank

In addition to the supply side measures illustrated in section 5.2 below, our proposed plan advocates a significant revival of aggregate demand and that revival in turn is expected to come from a strong and long lasting inversion of the persistently declining or stagnant trend of private investment activity. We expect that some of this inversion may come from the supply measures below; but most of it, especially in the early stages, must come from the long acknowledged, classical tool of investment control: monetary policy. But this policy is the prerogative of the central banks, which, hereafter, means essentially the new European Central Bank (ECB).

This has one very basic implication: if Europe really intends to achieve a rapid reduction in unemployment, it is necessary to give a broader and more constructive interpretation to the statutes that define the role of the ECB than that which is currently widely accepted. According to that interpretation, the Bank has but one target (one single front on which to do battle), namely preventing inflation. We urge a fundamental broadening of that interpretation—analogous to that of the U.S. Federal Reserve—to include, on an equal footing, another target: keeping unemployment under control. And we are confident that it can do so without renouncing or sacrificing its commitment against inflation.

There are three major considerations that support this view of the proper role of the ECB on the path of return to high employment. In the first place, making price stability the overriding target at this time is much like using all your military budget to fight the last war, an enemy that is no longer there. Inflation has been a most serious problem because of, and during, the two oil crises and their aftermath (including German reunification). But since 1991 inflation has been falling steadily for the group as a whole, and within each country, with hardly any exception. It is now around 2 percent, clearly a small number especially when taking into account the unquestionable upward bias of all inflation indices.

In summary, the perils of inflation as a result of a revival of investments are negligible at the present time. And that danger will be further reduced by applying several of the supply measures advocated below, which will increase the incentives to accept jobs as they become available. We submit therefore that, under present conditions, assigning the ECB the single task of fighting inflation should not be acceptable. It leaves it far too much unnecessary leeway, e.g. given that wages are rigid, it can satisfy that target by a prudential policy of raising interest rates *ad libitum*, reducing investment and raising unemployment further.

A second reason why the ECB should not make price stability its single, overriding focus is that, realistically, it has very limited control over the price level, at least in the short run. Indeed, its policy instruments—the money supply or interest rate policy—do not directly affect prices when there is slack in the labor market. Given large-scale unemployment, they can affect prices only indirectly by affecting the rate of economic activity, and hence the rate of unemployment (and utilization of capacity) and thereby the growth of wages and finally prices. But unemployment is not a very potent instrument to control inflation when there is already plenty of slack while it has a major impact on society's welfare.

The third and crucial reason for the central role of the ECB in the program of investment expansion is that, as control over monetary policy shifts from the states to the ECB, the latter becomes the only institution that has substantial power to influence investments. The other possible approach to stimulate investments could be through fiscal measures (subsidies, tax rebates, tax credits), but such measures cannot play a significant role at this time in view of the severe fiscal squeeze resulting from the Maastricht parameters.

One can think of various objections to this reinterpretation of the role and responsibilities of the ECB. One is that the Bank lacks the power to stimulate investment. This objection is especially popular among central bankers. But this argument is disingenuous. How can a central bank claim that it can control prices if it cannot control demand and how can it control demand if it has no control at all over investment? Another objection is that the euro needs to establish itself as a prestigious, credible currency in the world capital markets. To satisfy this need what is required is a policy that will be viewed as continuation of the tough policies of the Bundesbank, involving high interest rates that will attract capital and help to support high exchange rates, especially with respect to the dollar, seen as the major competing world currency. We believe that it would be a deadly mistake for the ECB to focus on a competitive struggle with the dollar, fought through the escalation of interest rates, and at the expense of an economic revival.

The high value of the dollar is the result of the strength of the American economy achieved through a policy of full employment pursued with "benign neglect" of the international "pecking order." The ECB must adopt the same attitude of independence aiming at fostering an economy as vigorous and prosperous as the American economy.

The awesome responsibility of the ECB for maintaining high growth in Europe have become even more serious with tragic events of the last few weeks in Asia, and Russia and the sharp set back in the equity markets. It is up to the ECB and to Europe with its still sound fundamentals to engage in a policy of supporting domestic investment and demand offsetting the expected decline in net exports.

5.2 Supply Side Measures

We do not believe that a widespread liberalization of the labor market in Europe, along the lines of that existing in the American or Japanese labor markets, is advisable, even if it were feasible. It is important to keep in mind that the present European welfare systems spring from different cultures and from different ways to interpret the solidarity and equality principles. However, we think that, in order to fight unemployment, it is necessary and feasible to introduce a substantially higher degree of flexibility in the European labor and product markets including, where necessary, a relaxation of job security legislation, a reduction in the coverage of collective bargaining agreements, and a reduction of barriers to entry of firms and of barriers to geographic mobility of labor. We believe that if such measures are combined with the reform strategy outlined below, both the efficiency and equity of European labour markets can be improved. The economic instruments now available in many European countries to pursue these efficiency and equity goals are insufficient. The portfolio of policy instruments needs to be expanded, along the lines suggested hereafter.

The labor market flexibility policies, unlike the macroeconomic management, cannot be uniformly adopted by all the European countries; on the contrary, they should be adapted to the different situations of each country and region. We begin by recalling that an important aspect of European unemployment is found in the regional differences that characterize this phenomenon. We believe that one important source of differential unemployment within countries is the uniformity of wages imposed by unions or custom in national negotiations, disregarding the glaring fact of important regional differences in productivity. We are in agreement that to remedy this problem requires recognizing the need for regional differentiation in labor cost per hour reflecting regional differences in productivity. But in order for these reforms to obtain a large social consensus, it is necessary that they be accompanied by measures that compensate for their negative effects on income distribution. In fact, it is evident that fundamental labor market reforms are very difficult to implement because usually, while they have readily identifiable distributional consequences for specific groups of people, it is not always easy to readily see their advantages for everybody. For this reason, it is likely that the most radical components of the reform packages will probably face strong opposition from the groups most affected by such reforms. Compensating the prospective losers is important to mitigate this difficulty. With respect to the realignment of labor in line with productivity one basic approach is not to put all the emphasis on reducing wages but on reducing cost to the firm through appropriate subsidies. Some suggestions for accomplishing this task efficiently are suggested below (see e.g. the paragraph on the benefit transfer program).

There is also evidence that the lower productivity and higher unemployment in some regions, like the South of France, Italy and Spain, reflects a paucity of entrepreneurs. We share the opinion that in these less developed regions, more active policies are needed in order to encourage new firms and help small- and medium-size firms, whose growth can be accelerated by some appropriate measures. For example, Italy has had some success with industrial districts. By industrial districts we mean here some horizontal aggregations of small- and medium-size firms, where each firm operates in an autonomous way from the others, but whose production is in fact coordinated with others in the district, with resulting external economies.

A successful example of regional development pushed on by the diffusion of industrial districts is the functional integration of small- and medium-size firms that occurred in many Italian regions like Toscana, Marche, Veneto and Emilia Romagna. But, for the regional development to further proceed by the implementation of this model, we need to introduce some reforms of the industrial districts, so as to reinforce them and make them more effective and dynamic. Such measures should aim primarily at creating advisory institutions in the field of bureaucratic, fiscal, financial and technological matters. As for the new technologies, it is fitting to emphasize the importance of the re-organization and the expansion of institutions for labor training and of the relations between firms and universities and other research institutions.

Another characteristic common to many underdeveloped regions is the rationing and high cost of credit, which affects new and small firms in general, reflecting both the cost of processing small loans, their risk, as well as the monopolistic power (and sometime the inefficiency) of the local banks. In these regions the spread between lending and borrowing rates has been huge and nearly prohibitive, discouraging small firms and new initiative. In some countries like Italy a great improvement in the availability and cost of loans has been obtained through the formation of cooperative of borrowers. The members of the cooperative, in exchange for the availability and lower cost of credit, must be willing to assume some personal responsibility to guarantee the repayment of the overall obligation assumed by the cooperative, something they are willing to do because of personal knowledge and trust of those who are admitted to the cooperative. In addition local governments (regional or subregional) typically have provided a rotating fund which also serves to increase the guarantee offered to the banks.

We advocate, where necessary, a broad supply side reform package that includes the following well-known elements:

—*Job creation policies* and product market reform to reduce barriers to employment creation.

Examples of such policies include tax reform or relaxation of regulations restricting the entry of firms, restrictions on land use, regulations limiting product market competition, as well as measures to avoid penalizing flexible time schedules and part time work. Measures to encourage part time leasing of workers may help not only to provide currently unemployed workers with a stepping stone into the world of work, but also help firms restructure their organization of production and work in accordance with the new advances in information technology and flexible manufacturing.

The tight regulation of the atypical working contracts now existing in many European countries deserves special attention in formulating labor market reforms.

—*Restructuring of the minimum wage institution.* We have indicated that the minimum wage tends to be harmful because it is a potential source of unemployment. Yet we cannot advocate the simplistic solution of abolishing that institution altogether because we share the view that its purpose to ensure a decent minimum standard of living to a full time worker is a worthy one. The trouble with the present structure is that a "decent" wage may prove to be higher than what would be required to induce employers to absorb the excess supply. If so, the only way to reconcile the "fairness" principle with high employment is to create a wedge between the remuneration received and the cost to the employer.

In reality such a wedge already exists in most economies, but in the direction opposite to the desired one: that is, the cost to the employer is typically substantially higher than what the worker actually receives in his pay-envelope. In many European countries this so called wedge may be close to 50 percent: the pay envelope may be not much more than half the cost to the employer.

This has led to simplistic solutions to the problem of lowering the cost without reducing the income—namely to abolish the wedge, misleadingly regarded as "fiscal levies" on the firm. But this solution ignores the fact that in reality the wedge is part of the remuneration of the worker, even if it is not in the form of cash. It consists in the first place of the income tax, which is part of workers' income, even if it is withheld from this income. The rest consists mostly of social security contributions, or compulsory saving (amounting in much of European around to one third part of income), of which some one third may be formally taken out of wage, while the other and major part is paid by the employer, but on behalf of the worker. On top of this the employer may pay some insurance for the worker. It is obvious that these amounts are part of the worker's remuneration, as they provide him with personal benefits, such as a pension or insurance against unfavorable events, even though they may be mandated. If the wedge, or part of it, were abolished, the worker would in effect receive a smaller remuneration. In addition, the social security system, being generally on a pay-as-you-go basis, would run into deficit. This could be avoided if the cost of the wedge were taken over by the government, as has been done some time on a limited scale. But this would be expensive and inconsistent at present with the need to satisfy the Maastricht criteria, unless more taxes were raised, hardly conceivable at present. What we seek then is a way to reduce the cost to the employer, while maintaining the take-home pay and minimizing the negative effects on the budget and the social security system.

We suggest that this might be achieved along the following lines.[4] Let the employer continue to pay the same take-home pay, and some fraction, say one third, of the social security contributions. Suppose, in addition, the government agreed to forego the income tax levied on this income, say around 20 percent. Then labor cost per hour would be reduced by some 40–45 percent.

Let us next stipulate that this special kind of treatment would be reserved to those unemployed that are least employable, namely those who have never held a job and the long-term unemployed in the depressed areas.[5] We submit that in this case there would be no significant loss to the government because the recipients would likely have remained unemployed and paid no taxes anyway. For the same reason, the social security would not lose revenue but more likely gain, because of the payment of one third of the contribution. Would then any one bear any loss from our proposal? At first sight the answer may seem: the worker who has been deprived of the two thirds of the employers contribution toward his pension benefits. But on the close examination this answer does not hold. In the first place, if he remained unemployed, the contribution to his pension would be zero, compared with one third under our proposal. It remains true that he will have lost two thirds of the contribution compared with what he would have secured had he found a job at standard wage (which he could not). But that only means that his current compulsory saving is below standard. All this should be perfectly acceptable in a life cycle perspective where we expect people to save little, if any, when they are young and relatively poor. We should expect then to make up for the low saving by saving more when they can afford it better. We propose to use this lead by incorporating into our scheme the right of any one accepting our "special" minimum wage (which is in any event not compulsory) to make up for say another third of this initially lost contribution. (The employer could be asked to match this contribution partly, as further inducement to a delayed contribution.)

The above is merely meant as an illustration of the appropriate strategy to deal with minimum wages: not abolishing them but finding appropriate acceptable ways of reducing the wedge.

—*An extension and generalization of fixed-term and part time jobs* could favor youngsters and women, whose possibility to work is often tied to these more flexible kinds of contracts. Usually, on the contrary, the existing laws favor only permanent or long-term labor contracts, which are much less flexible.

—*Reform of job security legislation policies* to reduce the ratio of firing costs or average wages.

We have criticized earlier the policies that have pushed job protection, to the point where firing of workers was a nearly impossible task. Although the situation has generally improved, there seems little question that the present institutions in many European countries still contribute to the unemployment problem by discouraging firms from hiring permanent employees even in the presence of a rising demand. We therefore share the view that the reforms needed to reduce unemployment must include substantial reforms of the job security provisions.

We do not believe that it would be possible or advisable to push reforms as far as the American system, where job security provisions are largely absent. But there must be a marked liberalization of the ability of firms to eliminate surplus labor, and some with respect to dismissal of individual workers for cause. This is particularly important in order to deal with youth unemployment, which is a serious problem in many European countries.

However we recommend that these reforms be postponed to a more suitable time. To carry them out now, when the demand is greatly depressed and there is plenty of unemployment, probably a good deal of redundant labor in many firms and few vacancies would have simply the effect of condemning many workers to join the rank of the unemployed, initially reducing instead of increasing employment. It would therefore meet with an understandably bitter opposition of union and workers who might well succeed in maintaining the *status quo*. In our view, therefore, the reforms should be postponed until, and made conditional upon, the realization of more favorable labor market conditions, which should hopefully not take very long if our program is pursued. But it would seem feasible and desirable to spell out promptly the conditions for proceeding with the reforms, e.g. when unemployment first reaches 7 percent. Furthermore, the content of these reforms should be agreed upon promptly. This two-stage approach should make the reforms far more palatable to labor while at the same time encouraging employers to assume more labour as the demand expands in the expectation that, if eventually the new employee assumed should prove redundant, they would be able to scale down their labor force.

—*Search promoting policies* to reduce labor market search costs, such as job counselling, information provision to unemployed workers and firms with vacancies. The U.K. experience with its Restart Programme and its counselling initiatives associated with the Welfare to Work policy indicates that such search promoting policies have an important role to play in enhancing the effectiveness of other employment creation measures, such as employment vouchers and training initiatives. These latter measures are likely to have a strong influence only if the currently unemployed workers are aware of them and help to make use of them as part of explicitly formulated strategies of gaining long-term employment in accordance with their idiosyncratic abilities.

—*Policies to stimulate worker mobility,* such as policies to increase the portability of housing subsidies, as well as the portability of health insurance and pensions between firms; and

—*Unemployment benefit reforms.* Unemployment benefit systems should be reformed in such a way as to give unemployed people appropriate incentives to seek work when jobs are available for them and to support them when such jobs are absent. Accordingly, the size of unemployment benefits could be made to depend on the ratio of vacancies to unemployment. The greater the number of vacancies relative to unemployed (in specified skill categories), the lower the unemployment benefits would be (within these categories). This proposal would promote efficiency, since it would give the unemployed a greater incentive to search, the greater is the firms' demand for their services. It would also help fulfil governments' equity objectives, since unemployed people are in greatest need of support when they are unable to find work.

This policy would generate a favourable complementarity between demand side and supply side policies. A government stimulus to aggregate demand (and thereby vacancies) could then be financed, partially or wholly, through the associated drop in unemployment benefit payments.

The political economy of unemployment benefit design could be influenced through the device of charging the cost of unemployment compensation to the public in the form of a separate tax. At present, the cost of unemployment, both its social cost and the cash cost of the benefit systems, is not well known, because, as has been frequently noted, it typically affects but a small fraction of the population, and the cash cost is not the very visible. This has the consequence that, on the one hand, the public does not put enough pressure on the government and the central bank to correct the situation, and on the other hand it makes voters frequently inclined to favor programs that grant excessive unemployment benefits through mechanisms that are economically wasteful. The above segregation would improve voters' information on this issue and permit them to make better informed policy choices.

Finally, we should like to stress that while the enforcement of specific micro-supply policies is primarily a matter for the individual member governments, all the member states share a common interest in the design of the unemployment policies and in making sure that these policies are forcefully pursued everywhere. This conclusion rests on the consideration that, because of the rising degree of factor mobility within the EU, as well as EU regulations concerning open market access and cross-border competition, the appropriate level of subsidiarity in unemployment policy making does not lie exclusively at the level of the EU member states. In addition, each member state has a very real and tangible interest in the reduction of unemployment in other countries as it contributes to reduce its own. Therefore the European Commission needs to take the lead in providing a legal and institutional framework within which necessary labor market reforms can take place and in making sure that the reforms are promptly carried through.

In addition, we believe that EU governments should also consider some more innovative supply side policy proposals that are designed to reform the incentives that employers and employees face. Currently EU governments spend massive sums of money on unemployment support, further education, and training. We believe that the question that should be asked is whether these funds could be redirected to create more incentives for employers to generate jobs and workers to become employed. The following are some illustrative policy measures that take this tack.

Conditional Negative Income Taxes

This measure may be seen as an alternative to supporting jobless people through unemployment benefits. The conditions attached to the proposed negative income tax would be analogous to those attached to current unemployment benefits. For instance if, under the current unemployment benefit system, people must provide evidence of serious job search in order to qualify for unemployment benefits, then they must also be required to provide such evidence under the proposed conditional negative income tax system; if unemployment benefits decline with unemployment duration under the current benefit system, then so too must the negative income taxes.

The Earned Income Tax Credit (EITC) in the U.S. belongs to this family of initiatives. The socially desirable relation between the magnitude of the negative income tax and the individual level of income has yet to be analyzed rigorously. The EITC is hump-shaped (so that the magnitude of the negative income tax rises with income at low income levels and then falls toward zero at higher income levels), whereas many of the proposed negative income tax schemes involve a strictly inverse relation between the size of the tax and the income level.

The broad argument in favor of a switch from unemployment benefits to negative income taxes is that this policy could meet the equity and efficiency objectives of current unemployment benefit systems more effectively than the unemployment benefit systems themselves. Although conditional negative income taxes would generate the same type of policy inefficiencies as unemployment benefits, the former would tend to do so to a lesser degree than the latter. For example, negative income taxes may be expected to discourage job search, but by less than unemployment benefits, for when a worker finds a job, he loses all his unemployment benefits, but only a fraction of his negative income taxes.

It is worth noting that a major criticism of the traditional negative income tax schemes—namely, that they make people's material well-being less dependent on employment and thereby discourage employment—obviously does not apply to conditional negative income taxes, since these taxes are conditional on the same things as current unemployment benefits.

Furthermore, conditional negative income taxes also tend to be more effective than unemployment benefits in overcoming labor market inefficiencies generated by credit constraints (e.g. people being unable to take enough time to find an appropriate job match or unable to acquire the appropriate amount of training on account of credit constraints), since the presence of these constraints is more closely associated with low incomes than with unemployment.

The Benefit Transfer Program (BTP)

The aim of the Benefit Transfer Program is simply to redirect the funds that the government currently spends on the unemployed—in the form of unemployment benefits, temporary layoff pay, redundancy subsidies, poverty allowances, and more—so as to give firms an incentive to employ these people. The BTP gives the long-term unemployed people the opportunity to redirect some of the benefits to which he is entitled to a voucher that can be turned over to a firm that will hire him.

The magnitude of the vouchers is to be set by the government, and depends on the magnitude of individuals' unemployment benefits (the higher the benefits, the higher the vouchers) and unemployment duration (the longer the unemployment spells, the higher the vouchers). The size of the vouchers is set so as to be financed from the unemployment benefits and other welfare entitlements foregone when people move from unemployment into jobs. Once a person is hired, the voucher gradually declines as the duration of employment proceeds. The vouchers could be given either to the prospective employers or employees. When unemployed people find jobs, they give up their unemployment benefits in exchange for the wage they earn.

The vouchers come in two varieties: “recruitment vouchers” and “training vouchers.” The former are granted solely on the condition that a previously long-term unemployed person is recruited; the latter are conditional on the employer being able to prove that the voucher is spent entirely on training the new recruit at nationally accredited training schemes. The recruitment and training vouchers both are related in the same way to the duration of a person’s previous unemployment and the duration of that person’s subsequent employment, but the level of the training voucher (other things equal) is higher than that of the recruitment voucher.

Since the BTP is voluntary, it extends the range of choices open to the unemployed and their potential employers. The unemployed will join only if it is to their advantage, i.e. if the wages they would be offered are higher than their unemployment benefits. At the same time, employers will join only if they find it profitable. Once again, many could well do so, since the vouchers could reduce their labor costs. In short, employees may wind up receiving substantially more than their unemployment support, and many employers may find themselves paying substantially less than the prevailing wages. The BTP has the unique capability of making most participants in the labor market better off: the unemployed who earns more, the employer who secures a lower cost, and the government that reduces its expenses incurred for the unemployed. This “free lunch” is possible since the BTP induces people who were previously unemployed to become productive, and the proceeds of the output they generate may be divided among the economic agents above.

The BTP has been implemented in various forms in the U.K., the Netherlands, and several other OECD countries. Empirical studies of the program indicate that there are three major obstacles to its effectiveness: i) displacement of current employees by the targeted groups of workers, ii) dead-weight (paying vouchers to unemployed people who would have found jobs anyway), and iii) substitution (the employment of the targeted group rather than unemployed workers outside the targeted group). The first obstacle can be mitigated by confining the vouchers to firms that increase their total employment relative to their industry average.[6]

The second and third obstacles can be reduced by targeting the employment vouchers at the long-term unemployed (since they have a relatively low probability of finding jobs anyway and are often imperfect substitutes for the short-term unemployed). However, these measures can only reduce, but never completely eliminate displacement, dead-weight and substitution. Nevertheless, evaluations of the program in the U.K. and the Netherlands have shown that, when the program is appropriately designed, it is able to create significant additional employment without putting upward pressure on wages. Moreover, even if the vouchers lead some firms to substitute their current employees for subsidized workers to retain the subsidized workers only so long as their vouchers last, the program will still succeed in substituting short-term for long-term unemployment. This would still lead to a fall in aggregate unemployment, since the short-term unemployed have higher chances of employment than the long-term unemployed.

Beyond that, the BTP is not inflationary, since it reduces firms' labor costs and since the long-term unemployed have no noticeable effect on wage inflation. If designed properly, it costs the government nothing, since the money for the employment vouchers would have been spent on unemployment support anyway.

By offering higher vouchers for training, the Program could become the basis for an effective national training initiative. Clearly, firms will spend the vouchers on training only if they intend to retain their recruits after the subsidies have run out. Thus the training for the unemployed would automatically come with the prospect of long-term employment. This is something that the existing government training schemes do not offer. Many existing schemes also run the risk of being ill-suited to people's diverse potential job opportunities, whereas under the BTP firms would naturally provide the training most appropriate to the available jobs. And whereas the existing training schemes are costly to run, the BTP is free.

Finally, the BTP could play a vital role in tackling regional unemployment problems. Regions of high unemployment would become areas containing a high proportion of workers with training vouchers, thereby providing an incentive for companies to move there and provide the appropriate training.

Auctioning Off Unemployment Benefits and Employment Vouchers

Existing unemployment benefit systems could be radically reformed to improve the incentives for job creation and job search without exacerbating disparities in incomes. Auctioning employment vouchers and auctioning unemployment benefits may be useful in this regard.

Regarding the former proposal, the government could auction employment vouchers to the firms. Firms would qualify for a number of vouchers equal to a) the number of previously unemployed people they intend to hire minus b) the number of employees they fire (or separate from).[7] Firms' entitlement would also be withdrawn if they used the vouchers to displace current employees. To make this provision credible, employees who believe they have been displaced would have the right of complaint, to be investigated by an independent body.

If the complaint is found to have been justified, the firm in question would be fined.[8]

To prevent the short-term unemployed driving the long-term unemployed out of the market, there would be separate options for workers belonging to broad groups with different unemployment durations. Another possibility is for the government to auction employment vouchers to unemployed people.

The aim of the supply side proposals here presented, then, is to transform unemployment benefits and other social security grants in incentives to firms to create more jobs and to workers to accept them.

6 Conclusion

In sum, we believe that the EU unemployment problem needs to be attacked on two fronts: through a broad spectrum of supply side policies and the demand management policy. The expansion of aggregate demand is necessary to increase both investment and employment. However, unless supply side measures are also taken, demand expansion can result in more inflation instead of more employment, because of the mismatch between the demand and supply of labor. What is important to stress is that both demand and supply side policies must be adopted together by all European countries, in order both to avoid beggar-my-neighbor problems, and, at the same time, to catch all the possible complementary effects of these policies.

Notes

1. Social security levies may be directly deducted from the take-home pay. But even if they are not, say, because they are levied on the employer, they will tend to be added to nominal labor cost and passed on into higher price—much like an *ad valorem* tax—thereby reducing the real take-home pay by the extent of the levy, at least to a first approximation.

2. For example in France in 1973 there were 101 jobs offered for every 100 persons in the labor force, but by 1993 the jobs available had shrunk to 89. As one would expect, unemployment moved inversely to job availability: as jobs kept shrinking well below the people seeking them, vacancies dwindled from 4 percent of the labor force in 1973 to a mere 1 percent in 1986; search time for an unemployed person grew longer and thus unemployment rose from 2.7 to 11.6 percent. There was only a short span of years, between 1986 and 1990, when demand rose temporarily somewhat faster than productivity, jobs increased from 90.7 to 92.7 percent, and unemployment promptly fell from 10.9 to 8.1 percent. A similar story may be told of other EU member states.

3. However, in a typical display of Italian partisan economic obtuseness, it has been suggested that the people to be rented out on a part-time basis should be hired by the agency on a permanent basis!

4. What follows is a broad description of the proposed approach, with the caveat that the details have to be elaborated taking into account the existing institutions.

5. Note that e.g. in Italy the first of this two groups alone represents about half of total unemployment.

6. The condition must be formulated relative to the industry average, for otherwise the scheme would be less effective in economic downturns—when the need for employment creation is greatest—than in upturns.

7. This difference may be adjusted for average changes in employment within that sector. Specifically, if sectoral employment is shrinking (expanding), then firms receive a number of vouchers greater (less) than the difference between the number of unemployed people hired and the number of employees fired. The reason for this adjustment is to avoid the possibility that the effectiveness of the voucher policy may diminish as the sector falls into a recession.

8. This anti-displacement provision has been successfully tried in Australia.

8

THE SHAMEFUL RATE OF UNEMPLOYMENT IN THE EMS: CAUSES AND CURES

Franco Modigliani

1 Introduction and Summary

Europe is presently suffering from an acute attack of the worst disease that can afflict a market economy—mass unemployment. In the 1960s unemployment among the members of European Community (EC) was prevailingly between 2 and 3 percent. But at the end of 1994, according to the figures from the latest *Bulletin of Labor Statistics* of the International Labor Officer, 1995-1, of the 11 countries now in the EC (excluding Luxembourg), only one had an unemployment rate of less than 7 percent (Portugal), 2 had a rate of between 7 and 10 percent (Holland and Greece), while two-thirds had rates near 10 percent and over, including the United Kingdom (10 percent), Italy (11.3 percent), France (11.5 percent), Belgium (14.1 percent), Ireland (14.6 percent), and Spain (20 percent). A tragedy and a shame!

Jobless rates of this magnitude had not been seen before, except possibly during the Great Depression, and even then, probably only in the United States. Macroeconomists like me thought that we had finally come to understand pretty well the nature of the market failure responsible for that calamity, and the cures needed to avoid a repetition, from the teaching of the economic giant John Maynard Keynes. And indeed, we have learned it to a considerable extent in the U.S. Since the end of the Second World War, unemployment has barely exceeded 9 percent in only two years and then by design; since 1982 it has been consistently below 8 percent, and mostly by substantial margins, and by this year it is a little higher than in the 1960s, actually down to physiological levels (below 6 percent). But in Europe, with unemployment since 1982 almost uniformly above 9 percent on the average, and mostly by a substantial margin, it would appear that at least the Central Bankers have forgotten the lesson that I have been

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teaching my students for nearly 50 years and which is punctually confirmed by the European experience, namely that exorbitant long-term interest rates produced by a serious shortage of real money supply, together with a fiscal policy much too tight for that monetary policy, imply an insufficient aggregate demand, and therefore insufficient numbers of jobs and vacancies, and finally mass unemployment.

My first task then will be to establish that the huge rise in unemployment primarily reflects demand causes—an insufficient aggregate demand reflecting extremely and mostly unnecessarily tight monetary (and even fiscal) policies; and that supply effects, including excessively high real wages, play a relatively minor role contrary to what is widely held in Europe and elsewhere. I will next endeavor to explain the reasons why Central Banks of so many EMS—and not only EMS—countries were led to simultaneously pursue such extremely tight policies. The basic explanation during this period is provided by the endeavor of central banks to maintain fixed exchange rates on the way to Maastricht. In the presence of high and increasing mobility of capital, to maintain stable exchange rates with but limited reserves can only be achieved by maintaining the same level of interest rates in all countries. Generally that meant enforcing the highest interest rate suited to any country which typically was that preferred by the Bundesbank. There were times during which that policy was also suitable for other EMS countries, as before and during the two oil crises, but there were other times when this was not true. In fact, since the second half of the 1980s, the interest rates imposed by the Bundesbank have been much too high to be consistent with full employment in (most of) the other members of the EMS. And the situation deteriorated further with the escalation of German interest rates following the annexation of East Germany. It pushed unemployment both in France and in the EMS as a whole to the highest levels of the postwar period.

These considerations lead me to face the last and most challenging question: whether it might be possible in the future to maintain fixed exchange rates and free movement of capital and still avoid the tragic consequences in terms of employment that we have witnessed, especially in recent years. It is acknowledged from the start that the problem of maintaining the real money supply at a level consistent with an appropriately low level of unemployment and a stable price level now appears more complex than we used to think, even in the context of an isolated economy. The complexity arises in part from recognizing the existence of "real," in contrast to Keynes' "nominal wage rigidity," and in part from the new top priority we have come to assign to price stability. Clearly, in order to control the *real* money supply, we now need to keep under control price and hence wage inflation.

This conclusion holds *a fortiori* if a group of countries intends to achieve not only price stability and full employment but also fixed parities and free movements of capital. It is argued that, in principle, all of these targets can be achieved simultaneously but that this would require far greater coordination of policies beyond the monetary and fiscal area, and especially with respect to wage policies, than seems to be presently contemplated.

2 Supply Versus Demand Explanations of the Rise in Unemployment

2.1 Supply Explanations

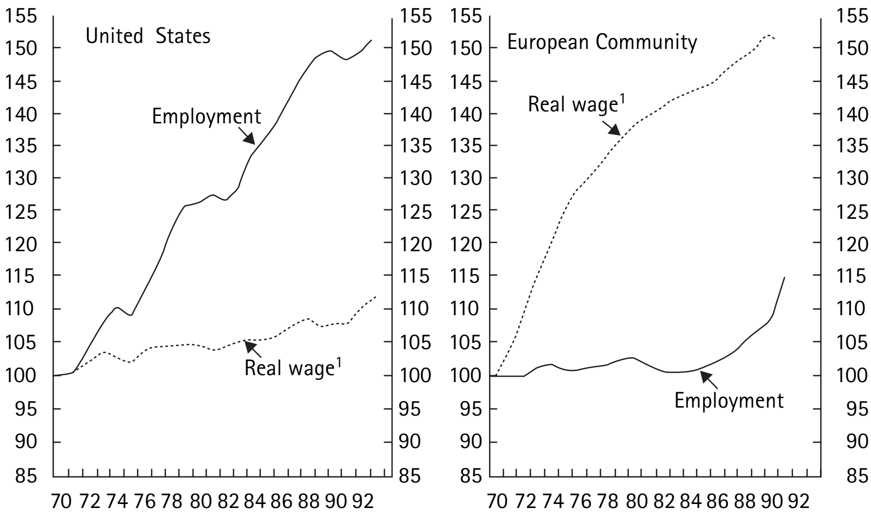
It is apparent from the summary that I attribute the major responsibility for European unemployment to demand factors. By contrast, in Europe, especially among politicians, it is now very popular to maintain that the main reason for the rise which has brought European unemployment above that of the U.S., Japan and other industrial countries, is to be found in "supply factors," that is, in the growth of forces or institutions which reduce employment relative to the labor force, for any given level of aggregate demand. The list of the supposed culprits has by now grown very large. A sample is provided below. However, there is one among them that has by now become so dominant and widely accepted as to require a separate and somewhat extensive treatment.

2.1.1 Exposing the Fallacy of a Crucial Role of the Real Wage

According to this explanation, the reason why unemployment grew in Europe but not in the U.S. is to be found in the very different behavior of real wages. The evidence for this conclusion is provided, *e.g.*, by figure 8.1, which by now is familiar to all, and by table 8.1, adapted with little change from table 1 in Drèze (1995, p. 38).

It is seen from the left panel of the graph that since 1970, U.S. wages grew very little, only 12 percent over the entire period or just over 1 percent per year (table 8.1, row 2). But that was accompanied by a very strong growth in employment of some 2 percent per year, cumulating to over 50 percent, and sufficient to keep up with labor force growth, preventing a rise in unemployment. But in Europe, where the growth of real wage was much larger, nearly 3 percent per year, or a total of nearly 50 percent, employment was almost stationary until the mid-80s, and over the entire period grew only 15 percent compared with 55 percent in the U.S. With a similar growth of the labor force, the effect had to be the large-scale rise in unemployment that we saw before.

The explanation for these observations that seems to be generally accepted in Europe, is that an increase of the real wage reduces employment through three



1. Total compensation per employee deflated by the GDP deflator.
Source: OECD *Economic Outlook* database.

Figure 8.1
Employment and real wages in the U.S. and the European community 1970–1993 (index numbers 1970 = 100)

mechanisms: 1) by encouraging the replacement of labor with capital; 2) by reducing international competitiveness; and 3) by reducing profitability and thus discouraging investment. Since strong unions in Europe succeeded in pushing the real wages much faster than in the U.S. (see table 8.1), employment grew much slower.

This explanation seems straightforward, though it leaves certain "coincidences" unexplained. In particular, why is it that in both regions the growth of the real wage is just about the same as that of productivity? And why it is that the growth of the real wage in the U.S. happens to be such as to lead to an increase in employment just matching that of the labor force and leaving unemployment stable?

I would like to advance a rather different explanation, which can readily account for the coincidences. My point of departure is the observation that while strong unions can set nominal wages, it is the employers that set the price, given the wage, within the constraint of a competitive environment (and possibly some further constrains imposed by the international economy, depending on the exchange regime), and it is this price that finally determines the real wage. In particular, the conventional explanation neglects the basic identity that the change in the real wage is equal to the change in productivity plus the change in the labor

Table 8.1
Growth of Output, Productivity, Employment, and Real Wage—Europe and U.S.A. 1960–1990
Annual Rates of Growth, %

	Europe	U.S.A.
GND	3.3	3.0
Real Wage	2.9	1.1
Employment	0.3	1.9
Productivity	3.0	1.2
Change in Labor Share	–0.1	–0.1

share of income. But the change in productivity clearly is not directly affected by labor nominal wage demands, while the change in the share of income is the reciprocal of the "mark-up" which is at the core of employers' price policy.

Now suppose that price behavior reflects a policy of a relatively stable mark-up on unit labor cost, or that for any other reason, the share of labor is relatively stable, as it has been in the U.S. and to a good extent also in Europe (whether because of a stable mark-up of for any other reason). Then the differential growth of the real wage in each region should (i) coincide, as it does, with the differential growth of productivity; and (ii) be the result of that differential rather than of differential union power. In addition, if the growth of the real wage coincides with the growth of productivity, then the three propositions of Drèze are no longer valid and the differential growth of the real wage can no longer be used to explain the different growth of employment.

But is it not possible that in the face of a strong wage push, firms might find it impossible to set a price that will maintain the mark-up? In principle this possibility exists and circumstances in which it may actually materialize are discussed below in section 5.1. Two possible scenarios are developed there. The first is the case of fixed exchange rates, where raising the price results in a loss of foreign and domestic markets. Here firms may choose to sacrifice some mark-up while still losing some sales, output and employment. An analogous situation could arise in the presence of a rigidly non-accommodating monetary policy. However, these situations are characterized by the fact that the rise of the real wage above that of productivity is compensated by a decline in the mark-up and hence by a rise in the labor share of output at the expense of the non-wage share. It is therefore important to recognize that between the early 1970s, when unemployment was at a minimum, and the second half of the 1980s, when unemployment reached its peak, the share of labor declined appreciably and with no significant exception or, the increase in real wages was smaller than that of productivity. Thus the increased unemployment cannot be attributed to the excessive growth of real wages. The exceptions are countries like Italy and Spain, where the labor share increased but unemployment grew more than elsewhere.

What then explains the growth in employment in the two regions? The basic answer is *aggregate demand* within the bounds imposed by the available labor force. In particular for the U.S., I suggest that the observed growth was the result of a conscious and successful management of aggregate demand. It resulted in a growth of aggregate demand consistent with that of potential output, which is the sum of labor force and productivity growth. As a result, the growth of employment, which equals the growth of output minus the growth of productivity, coincided with the growth of the labor force. In the case of Europe on the other hand, the growth of output, around 3 percent, was well below the growth of potential output, amounting to some 5 percent (2 percent for labor force and 3 percent for productivity). So unemployment kept growing throughout the period and may have just peaked.

Thus to understand European unemployment, we are finally led back to the question: was the failure of output to keep up with the labor force and productivity the result of an insufficient growth in aggregate demand (the demand hypothesis) or was it due to a failure of employment to expand adequately, even though the demand was there (the supply factors hypothesis)?

2.1.2 Other Supply Factors

The supporters of the supply hypothesis have pointed their fingers to a very large number of additional culprits. (See, for instance, Alogoskoufis et al. (1995). Following is a list of those that appear to have been more broadly accepted, together with short comments:

1) One cause that is almost unquestionably accepted is the relatively generous European welfare provision for the unemployed. These benefits, when combined with the high payroll taxes and similar taxes paid by the employers and employees to cover social insurance programs, have greatly reduced the spread between take-home pay and unemployment benefits, and thus sapped the incentive to actively seek a new job. But the CEPR report of April 15 casts serious doubts on the relevance of this institutional difference by pointing out that the European provisions were already more generous in the 50s and 60s when unemployment in Europe was well below that in the U.S. (a phenomenon that appeared readily understandable at the time), while during the 1970s, when European unemployment became conspicuously larger than in the U.S., the increase in benefits was not very large. In fact of late, the benefits have been cut substantially but unemployment, after a modest temporary abatement, is now back to peak levels.

Next we have two supposed supply effects where everybody agrees with the symptoms but there is wide disagreement about the nature of causation.

2) The first is the role of long-term unemployment. There certainly is much more long-term unemployment in Europe than in the U.S. and the supply-siders say that this makes a substantial contribution to the lower employment to labor force ratio, because the long-term unemployed are the least desirable candidates for jobs and because they reduce their search activity. But American experience suggests that the causation is the other way around, i.e., that high and prolonged unemployment generates a growing number of long-term unemployed. When a responsible aggregate demand policy shrinks unemployment, the long-term rate shrinks even more, until it goes back to physiological levels.

3) The other argument is that of "mismatch": there are enough jobs to employ the current labor force, but an important proportion of the labor force does not possess the qualifications needed to fill the jobs. The evidence to support this contention is that the rise in unemployment was accompanied by a disproportionate rise in the unemployment rate of the less skilled and educated. This phenomenon occurred and received a lot of attention in the U.S. in 1961, when overall unemployment reached a postwar peak of 6.7 percent, from 4.1 percent in 1956. In a relatively low-skill group like Blacks, the rate of unemployment reached 12.4 percent from 8 percent in 1956. These statistics were widely interpreted as evidence of mismatch, arising from the new age of computers. But this interpretation was given up when, with the expansion of demand from the Vietnam War, black unemployment fell back to 4.5 percent. It was replaced by a quite different explanation, namely that when unemployment rises, the loss of jobs is exacerbated among the least qualified because they get bumped off by the newly unemployed, with higher qualifications (and hence over-qualified). Thus, when unemployment is high, it is especially high among the less qualified. But when demand rises again, the over-qualified move back to the positions for which they were qualified and release the lower jobs back to the less qualified.

In other words, the observed higher incidence of unemployment among the least qualified is the effect rather than the cause of cyclically low employment.

4) Another very popular explanation is that developing countries are stealing European jobs. This argument has long been known to be nonsense, reflecting economic illiteracy. If foreigners are willing to sell us their goods at less than we can produce them, that must improve our lot by improving our terms of trade. Increased foreign competitiveness in foreign markets could deteriorate our terms of trade but should not have much of an effect on employment.

5) Many other supposed supply factors are reviewed in Alogoskoufis (1995). But in the end, there is only one whose negative effects on unemployment is not in question, and that is the effect of minimum wages. To illustrate, suppose that the larger size and growth of employment in the U.S. was due to industries with low productivity and hence low real wages and relatively low growth, and that

in Europe such employment was cut off as a result of minimum wages and was instead reflected in unemployment. That could account for the higher growth in U.S. employment and also for the smaller growth in real wages and productivity. So qualitatively it provides an explanation of the facts different from mine, but also different from the conventional view, in that the differential behavior is not the result of strong unions but instead of the economic effects of minimum wages. But it is questionable whether the minimum wage provision could account for much of the differential behavior of employment and unemployment since the 1970s, for, as pointed out in the *CEPR* reports, it is neither a new institution nor one specific to Europe (though it has been quantitatively more important there.)

Be this as it may, it would be hopeless to try to measure how much, if any, each of the supposed supply effects contributed to the rise in unemployment. I propose instead to rely on a rather different construct known as the Beveridge curve. I will demonstrate how this device can be used to measure the total number of job openings in the economy and to estimate to what extent variations in unemployment can be traced back to changes in jobs (demand effects) and to other causes (supply effects).

3 Tests of the Role of Demand Based on Beveridge Curve Tests

3.1 The Beveridge Curve

The Beveridge curve can be derived from two basic hypotheses. The first is that all unemployment is basically frictional—that is, due to the fact that when people become unemployed, some time is required before they can find another vacant and suitable job. Given a steady flow of people into unemployment, F (measured say, as people per month), then if the average time it takes for an unemployed to find a job is T months, the average amount of unemployment, say U, will amount to T-months flow, $U = FT$. By expressing U and F as a percentage of the labor force, this can be rewritten as $u = fT$.

The second hypothesis is that T depends on the existing vacancies rate, v. If there are but few, it will take a long search to find one. We may thus hypothesize that T is roughly inversely proportional to v, say $T = c/v$ where c is some constant. But that implies $u = fc/v$ or $uv = fc$, which is a standard way of formulating the Beveridge curve.

Figure 8.2 shows a hypothetical Beveridge curve for a given country and point of time. It is a "rectangular hyperbola" in the (U, V) space and its position depends on the two parameters f and c that characterize that economy. If, and only if, these parameters change in time, the curve will shift. The actual value of U and V at

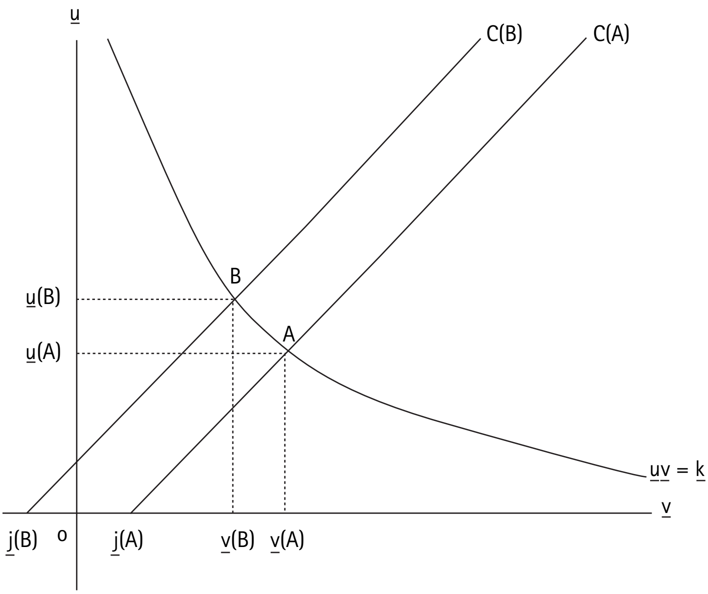


Figure 8.2
The Beveridge curve

date t, U_t, V_t is given by the intersection (simultaneous solution) of the Beveridge curve at that date and the labor market constraint of budget equation, tying U to V. Denoting by J the total number of job openings in the economy, whether filled of not, and by LF, the total labor force, or sum of employment E plus unemployment, that constraint takes the form: $V = J - E = J - (LF - U)$ or expressing both sides as a percentage of LF, $v = (j - 100) + u$, where v and u are the unemployment and vacancy rate, respectively and $j = 100J/LF$, the number of jobs expressed as a percentage of the labor force.

This constraint can be represented in figure 8.2 as a *linear* relation between v and u with slope one and an intercept $(j - 100)$ such as the line BB. The coordinates of the point of intersection A with the Beveridge curve expressed as a relation between v and u, u_a and v_a represent the current unemployment and vacancy rate. (Note that J can be directly observed, being the sum of employment and vacancies.)

3.2 The Response of the Unemployment and Vacancy Rate to a Change in Jobs

Now suppose there is a decrease Δ_j in the number of job openings. If LF is constant, the constraint BB will exhibit a downward parallel shift by $100\Delta_j/LF$, determining a new point of intersection B, with coordinates u_b and v_b (see figure 8.2). Clearly $u_a - u_b$ measures the increase in the rate of unemployment due to the loss of jobs, on the assumption of an unchanged Beveridge curve. Thus, if we know the availability of jobs at different dates as measured by *j*, we can compute the level of unemployment corresponding to *j* and a given Beveridge curve.

3.2.1 Computation of the Unemployment Rate Implied by Availability of Jobs—Comparison with Actual Unemployment

Table 8.2, column (3) reports for Germany the rate of unemployment calculated from job availability *j* and a reference Beveridge curve, which for the table is that of 1960 ($uv = 2.7$). It can be seen that the figures of column (3) track very closely the actual jobless rate *u* reported in column (1): out of the 23 years in which *u* exceeds 1 percent, there are only three in which the computed unemployment figure of column (3) differs from that of column (1) by more than 10 percent. (When *u* is less than 1 percent, the percentage difference is not informative.) Thus in Germany (and Europe, see below) the level and movement of the jobless rate is largely explained by a single factor: the demand for labor, as measured by the availability of jobs *j* of column (2), and its effect on vacancies. In particular, the huge rise in the jobless rate, from 1 percent in 1973 to 9 percent in 1986, is accounted for by a similar decline in column (3), reflecting the massive decline in jobs. Between 1986 and 1991 there is a modest pick up in jobs, promptly accompanied by a fall in *u*. But already by 1993, *u* is back to near peak levels, as vacancies have dried out.

Table 8.2, column (2) reports similar calculations for the U.K. The results are very similar, except that the figures of column (3) tracks those of column (1) even more closely than in the case of Germany. For France, reported in table 8.2, column (3) the tracking is a little less close.

We must at this point give some consideration to the difference between actual *u* of column (1), and u^*, the computed *u* of column (3). Both figures reflect the actual job availability in the given year. However, the actual figures of column (1) result from the intersection of the *current* labor market constraint and the *current* Beveridge curve, while those of column (3) are based on the *reference* Beveridge curve of the base year. Thus the difference of between 0 and 10 percent between the two columns reflects the effect of the shift of the Beveridge curve. If it is positive, it implies an *outward shift* of the Beveridge curve since an outward shift generally increases *u*. By failing to take into account a shift, column (3) underestimates *u*. And a prevailing outward shift is consistent with the hypothesis that there are some supply effects of the type claimed in the European literature. Each of these effects (except for the minimum wage discussed below) implies a longer duration of *u*, and hence a larger *u*, for given vacancies. However,

Table 8.2
Unemployment and Job Availability

1. Germany				2. United Kingdom				3. France			
Date	*u*	*j*	*u**	Date	*u*	*j*	*u**	Date	*u*	*j*	*u**
1960	1.23	101.0	1.25	1965	1.42	99.70	1.42	1970	2.45	99.17	2.45
1965	0.66	102.3	0.88	1970	2.55	98.26	2.40	1973	2.70	101.35	1.43
1970	0.69	103.1	0.73	1973	2.10	99.09	1.80	1974	2.83	100.45	1.78
1973	1.27	102.5	0.84	1974	2.15	99.01	1.85	1975	4.03	97.82	3.36
1974	2.65	98.7	2.45	1975	3.35	97.26	3.2	1976	4.40	97.67	3.47
1975	4.73	96.3	4.34	1976	4.50	95.98	4.38	1977	4.95	96.82	4.14
1976	4.61	95.5	5.05	1977	4.77	95.84	4.51	1978	5.15	96.35	4.53
1977	5.54	96.7	4.00	1978	4.65	96.17	4.21	1979	5.85	95.69	5.09
1978	4.35	97.5	3.34	1979	4.27	96.66	3.76	1980	6.25	95.32	5.41
1979	3.80	97.5	3.34	1980	5.42	95.10	5.21	1981	7.38	93.88	6.71
1980	3.88	97.5	3.34	1981	8.47	91.90	8.29	1982	8.10	93.44	7.12
1981	5.56	95.5	5.23	1982	9.85	90.62	9.55	1983	8.30	93.19	7.35
1982	7.69	92.7	7.67	1983	10.67	89.89	10.26	1984	9.68	91.22	9.21
1983	9.21	91.7	8.62	1984	11.07	89.54	10.61	1985	10.20	90.72	9.69
1984	9.12	91.7	8.62	1985	11.27	89.38	10.77	1986	10.33	90.67	9.74
1985	9.28	91.2	9.11	1986	11.35	89.40	10.75	1987	10.50	90.62	9.79
1986	8.93	91.7	8.62	1987	10.00	90.91	9.26	1988	10.00	91.32	9.12
1987	8.89	91.8	8.53	1988	8.15	92.80	7.41	1989	9.42	92.15	8.33
1988	8.74	92.0	8.34	1989	6.27	94.54	5.74	1990	8.92	92.71	7.80
1989	7.88	93.3	7.09	1990	5.85	94.79	5.50	1991	9.43	91.88	8.58
1990	7.15	94.0	6.44	1991	8.10	92.34	7.86	1992	10.33	90.82	9.59
1991	6.30	95.0	5.51	1992	9.87	90.59	9.58	1993	11.60	89.37	10.99
1992	6.67	95.0	5.51	1993	10.35	90.16	10.00				
1993	8.28	92.6	7.76								

u = Unemployment rate; *j* = Ratio of available jobs to labor force; *u** = Unemployment computed from the initial year Beveridge curve (1960) and job availability of column 2.

the results in tables 8.2, columns 1–3 are consistent with our claim that supply effects, though not absent, are secondary in comparison with the effects of the huge decline in jobs.

One might be tempted to blame the decline in jobs on an increasing shortage of aggregate demand. However, one must acknowledge the possibility that other causes might have contributed, and in particular, minimum wages. It is certainly conceivable that this constraint could have reduced the number of jobs offered because the price required to make it profitable to produce certain outputs was so high as to substantially reduce the demand for that output and hence the demand for labor to produce it. It is hard to estimate what proportion of the dramatic decline in jobs could be reasonably attributed to the effect of the minimum wage. But there were no dramatic changes in the institution of the minimum wage since the mid-1970s, although their unfavorable effect on labor demand could have become somewhat more pronounced because of competition of lower wage countries. But since the rise of unemployment was widespread also in industries where the minimum wage effect was not important, one can safely presume that the enormous decline in jobs beginning there reflects a growing insufficiency of aggregate demand; in other words, it is of the type described and analyzed by Keynes in the *General Theory*.

4 Keynesian Unemployment

What distinguishes this type of unemployment or excess supply of labor from others is that its final cause is a shortage in the real money supply, that the shortage is not self-correcting, but can be cured through an expansion of the real money supply (provided that is feasible).

4.1 The Classical versus the Keynesian Monetary Mechanism

The underlying view of the monetary mechanism, *i.e.* of the mechanism responsible for ensuring the reaching of equilibrium in both the money and commodity (labor) markets, is radically different from and inconsistent with that envisaged by the traditional (classical) model.

That view postulates that prices are flexible and that if there is an excess demand in a market, prices will rise promptly: the rise will in general reduce demand and/or increase the supply, reducing the excess demand, until an equilibrium is reached in which demand equals supply and prices stop adjusting; similarly so for an excess supply.

In the specific case of the money market, we can deal with nominal or real quantities. In particular, the real demand is proportional to "full employment output" and the real supply is the institutionally given nominal supply of money divided by the price level.

An excess demand for money should cause a rise in its price in terms of commodities, and hence a decline in the price level, which is its reciprocal. Similarly, the excess supply of labor should reduce wages and prices; thus both market disequilibria should reduce the price level. With the (reasonable) Keynesian assumption that prices are closely tied to wages, most of the adjustment should come through wage flexibility.

The fall in the price level contributes directly to the clearing of the money market by increasing the real money supply and presumably the demand for commodities, until the supply matches the demand and equilibrium is re-established in both markets. Thus the essence of the traditional view is that, thanks to wageprice flexibility, the real money supply is endogenous and behaves in stabilizing fashion, preventing lasting disequilibria.

Keynes's monetary mechanism is very different. First, he ingeniously redefines equilibrium in the labor market as a situation in which wages have ceased to adjust (at least at a reasonable rate), whether or not the other classical condition is satisfied, namely the equality of demand and supply. Second, he realistically rejects the hypothesis that nominal wages are flexible enough to ensure a prompt equilibrium in the money market through the real money supply, but shows that nevertheless the money market may be cleared through a number of novel mechanisms. The first is "liquidity preference," *i.e.* the fact that the demand for money per unit of output (or the velocity of circulation) depends on interest rates, (or more precisely on the spread between the rate on money and that on less liquid substitutes—the cost of carrying money). When money demand exceeds the supply, what is out of balance is the composition of the portfolio that has too little cash relative to other assets. The response is to try to acquire cash not by "selling more (or buying less) goods" but by selling close substitute securities. This *per se* helps to re-establish equilibrium by increasing the yield on money substitutes, thus reducing the real demand for money. But in addition, it sets into motion another equilibrating mechanism: the rise in interest rates working through the investment function reduces real investment which, through the multiplier, also reduces consumption and income. Thus, interestingly enough, unemployment is an important part of the mechanism ensuring the clearing of the money market. It works by reducing the excess demand for money through a fall in the *demand* for money brought about by a decline in output and employment. If, in addition, prices and nominal wages are flexible, they will fall, increasing the real money supply, thus supporting the other two demand-reducing mechanisms which are not instantaneous. But it might take a long time. In addition, a return toward full employment could result from various shocks, including expansionary fiscal polices.

The final outcome of the process is an "equilibrium" in which the money market—as well as other markets with flexible prices—clear, but the labor market may end up with an excess supply because there is not enough demand for output to absorb the full-employment labor supply. But it will still be in Keynesian equilibrium, for with wage rigidity, the excess supply will not cause wages to adjust further. (In the case of an excess demand for labor, nominal wage rigidity could only be transitory and eventually result in higher wages and prices, which would reduce demand along the lines of the traditional model.)

The essence of the Keynesian Revolution is that the money market is cleared—*i.e.* the demand for money is brought into equality with the Central Bank determined money supply—not through an imaginary high flexibility of wages, but through movements in unemployment and interest rates.

4.2 Nominal Wage Rigidity versus Real Money Supply

It has been argued that the market failure in this case reflects the rigidity of wages and prices that do not respond to the excess supply. But when unemployment is due to nominal wage rigidity, it is just as correct to conclude that it reflects a shortage of real money supply. For if the equilibrium could be re-established by an x percent decline in nominal wages, then to the extent that nominal wages are truly rigid up to a neighborhood of full employment, equilibrium could equally be re-established by an x percent expansion of the real money supply, whether by a fall in prices of a rise in the nominal money supply, or and combination of the two.

4.3 Nominal Wage Rigidity, A Source for Concern or an Opportunity for Improved Stabilization?

The hypothesis that nominal wages were rigid, or very sluggish, was widely accepted in the early postwar period. It was initially a source of concern about economic stability. For clearly, the real money supply required for full employment could be expected to vary over time as a result of parameter shifts and other shocks. But with rigid prices, there was no mechanism to ensure that the real money supply would tend to move toward the appropriate level; hence the danger of significant discrepancies and corresponding movements in unemployment.

However, there soon was a more fundamental shift to optimism as it was realized that, if wages are prevailingly rigid, the *real* money supply could be readily controlled by the simple operation of changing the *nominal* money stock to accommodate demand. Hence the exhilarating feeling, lasting through the 1960s and early 1970s, that the problem of unemployment had largely been brought under control, and that there was an important task for the macroeconomist in advising on the proper course of the nominal money supply—or on the level of interest rate consistent with full employment—letting money demand determine money supply.

4.4 The New Pessimism: Real versus Nominal Wage Rigidity

What then happened in the 1970s and thereafter to kill this optimism and replace it with deep pessimism about being able to control excess unemployment and perhaps be able again to eliminate it? The answer, in my view, is the growing realization that what is rigid is not the *nominal* but rather the *real* wage. While Keynesian (nominal) wage rigidity implies that labor's (minimum) wage target is set in nominal terms, real wage rigidity means that the target is set relative to the price of commodities bought (the cost of living).

When prices are relatively stable, as is the rule among western countries, the two hypotheses have similar implications and it is hard to discriminate among them. But when prices vary widely, the test becomes simple, since the two hypotheses have opposite implications. A rigid real wage implies a very flexible nominal wage, chasing after prices, whereas nominal wage rigidity implies great real wage flexibility. The evidence from inflationary episodes of any duration clearly supports the hypothesis of rigid real wage targets.

4.4.1 Problems with Targeting Real Money under Conditions of Real Wage Rigidity

Unfortunately, when labor uses its power to control nominal wages, over which it generally has considerable control in the EMS, to reach a real target, the monetary authority may become unable to adjust the real money supply to accommodate the full employment needs, and it is even doubtful that it should.

To illustrate, suppose that labor, in an endeavor to enforce a higher *real* wage target, successfully presses for increases in nominal wages larger than productivity growth. The push is based on the expectation that the rise in nominal wage would result in a similar rise in the real wage. This expectation is a fallacy which is widely believed, and which the Italian labor movement has formalized in a famous slogan: "the wage is an independent variable" (meaning that it effects no variables other than real wages and profits). And quite possible employers have frequently been prone to acquiesce to the push in wages on the equally fallacious belief that since they control prices, they could transfer the higher wages onto higher prices at no loss to themselves without concern that the higher prices could trigger higher wages (the mark-up is the independent variable!).

In reality, when nominal wages rise faster than productivity in pursuit of a high real wage target, unit labor cost rises, and because prices are largely based on a mark-up on unit labor costs, the initial response to the rise in unit labor cost is to transfer it through higher prices. What actually happens thereafter depends critically on the response of the monetary authority.

4.4.2 The Basic Dilemma: To Accommodate or Not

The fundamental choice confronting the Central Bank is *to accommodate* or *not to accommodate*.

Non-accommodation means not changing the money supply and the exchange rate (keeping both on the initial path). It is not a very appealing policy, because it can be expected to have a series of costly consequences such as an insufficient real money supply, a rise in unemployment and a fall in net exports (see 5.1 below). This last effect implies that non-accommodation may not be pursued indefinitely, as it is likely to result in an unsustainable drain of reserves and of accessible foreign credit, terminating in a balance of payment crisis. This in turn would necessitate further measures restricting demand and employment or force abandonment of fixed exchanges and of non-accommodation.

But accommodation can also be very risky. It means trying to maintain an unchanged real money supply by letting the nominal supply rise in proportion to prices. This policy, together with a floating exchange or an appropriate devaluation of the nominal exchange rate, would make it possible for the producers, who largely control prices, to transfer the whole wage rise into higher prices, with no real effect on output or employment. While this would protect employment, it would do so by totally undoing the gain in the real wage that labor had sought and initially obtained. In the light of their new *real* target, they can be counted upon to renew their push for nominal wage increase, which, if accommodated, will lead to another round of price increases and so on, opening the way to the traditional wage-price spiral.

In other words, to the extent that wage increases higher than the rise in productivity lead to rising prices, a policy of accommodation of that rise may not produce just a one time price increase, but risks to open the way to a steady inflationary spiral. And this conclusion will be reinforced with escalator clauses aimed at preventing higher prices from eroding the real wage.

Because of the high cost of non-accommodation, there might be a strong temptation to start on an accommodation path maintaining the real money supply, considering also that, in the beginning, the cost may be only a modest rise in inflation. But the problem is that, failing a lucky break, the solution risks to prove unstable, as the rise in prices could eventually turn into an inflationary spiral.

It would therefore seem unadvisable for a central bank to engage in a policy of accommodation unless it has sound information that the accommodation would not turn into an inflationary spiral that the central bank could not tolerate, especially in today's climate when central banks are assigned the major responsibility for price stability. In that case, accommodation would eventually have to be given up. Then the real money supply would become inadequate and unemployment would rise. In that case, in the end, the problem of unemployment would not have been eliminated but only postponed and possibly made worse by the momentum that inflation might have picked up. However, even in the absence of accommodation, unemployment may not grow forever. Its rise may eventually persuade workers to stop pushing and settle for less than the intended, while at the same time the loss in sales may persuade employers to be more accommodating in terms of make-up. Thus eventually the inflation will slow down, reducing the unemployment needed to resist it, and with some luck, both may even come to an end.

As we have shown, during the last twenty years there have been frequent situations when the real money supply has proved insufficient to support an output close to full employment. In the next section, we propose to consider a number of such episodes and ask what caused the shortage of real money and whether it could have been avoided through more effective policies. In the concluding section, we propose to use what we have learned to examine the possibility of designing more effective policies for the future.

4.5 Some Case Studies

4.5.1 The Italian (and Spanish) Case

Italy offers a very clear example of the consequences of a powerful union movement capable of a strong monopolistic control over labor supply and the accommodation dilemma this poses for the monetary authority.

The relevant episode begins at the end of 1969, when wages were pushed up some 30 percent in a few months and terminates in 1992, when the major unions agreed to join a compact with employers and government aimed at bringing inflation to an end.

Throughout this period, the unions were able to push wages faster than productivity. As pointed out before, the initial response tends to be a rise in prices to preserve the mark-up and with full accommodation, this is possible without affecting output or employment. But the situation is not stable and tends to degenerate into an inflationary spiral, which can only be truncated by an end to accommodation.

Let us next consider more closely the cost of non-accommodation. As indicated, it implies an insufficient real money supply generating a rise in unemployment and a deterioration in the balance of trade. There are several considerations pointing to that conclusion. In the absence of accommodation, there is a trade-off between the extent of transfer and loss of markets. A larger transfer will cause a rise in the price of domestic output *versus* the presumably unchanged price of foreign output. This will reduce the competitiveness of domestic goods both in the foreign and domestic markets, causing (as a rule) a decline of exports and rise in imports. Thus the non-accommodated rise in wages by increasing wages relative to both domestic and foreign prices, will in the end result in an increase in the real wage but at the cost of reduced output and therefore higher unemployment as domestic labor and its output get progressively *priced out* of the world market.

A further decline in demand may come from the fixed money supply, from the rise in interest rates, reducing investment and from the magnification of these declines through multiplier effects.

In Italy, as wages kept rising faster than productivity, the monetary authority was continuously confronted with the unsavory choice between accommodation and inflation on the one hand and non-accommodation and rising unemployment on the other.

So despite the enormous increase in productivity and per capita income (roughly tripling between 1960 and the present day), the country led a very troubled life, with the monetary authority understandably wavering between accommodation and non-accommodation (see Modigliani and Padoa-Schioppa, 1978). During periods of accommodation there was high inflation, with the price level rising nearly 20 times between 1960 and 1990. Periods of non-accommodation were characterized by balance of payment problems and rising unemployment from 3 percent to 13 percent in 1989.

Of course, to be able to push up wages in the face of gigantic (for the time) and rising unemployment required a tight monopolistic control to prevent the competition of the unemployed and to reassure those that still had a job but were no longer so safe. This was accomplished with a variety of devices, including extreme forms of employment securities for those who still had jobs, (which contributed to discouraging firms from new hires) and the famous "Cassa Integrazione" which provided large unemployment benefits for those who had been insiders, even though there were hardly any benefits for the unlucky ones who had been frozen out.

The deteriorating picture of inflation and unemployment only came to an end after the repudiation of "accommodation" which became credible with Italy's entrance into the EMS. But the real relief from the anguish of continuing inflation and rising unemployment came in July 1992 when after an amazingly fast maturation, the unions joined a tripartite compact which programmed a path for inflation and nominal wages consistent with it. That pact has since successfully withstood renewals and, at least so far, even a devaluation due to political instability and not to excessive wage settlements, which cut into the purchasing power of workers.

The history of Spanish inflation and unemployment is very similar, almost a caricature of Italy, typically with some lags; pushing by labor was even more extreme and was accompanied by more damaging measures, especially in the area of industrial relations and job protection. It is not surprising that unemployment in Spain in recent years has reached nearly twice the gigantic Italian level.

4.5.2 The Great Oil Crises

Another important example of real wage targets and the accommodation dilemma is provided by the oil crises. One must first begin with the consideration that the real target wage which is relevant is the one in terms of domestic output prices (the real wage in terms of domestic output). What actually happened is that workers endeavored to push wages in order to maintain a stable purchasing power in terms of consumables. But since those prices, which included energy, rose more than the prices of domestic goods, wages were meant to rise relative to domestic goods. As the rise was not compensated by higher productivity, it would have meant a shift in the income distribution to compensate labor for the Sheik tax. But there was no reason why employers should have accepted such a redistribution. Since they controlled prices, the rise in unit labor cost was largely passed on to higher prices of domestic output. Accordingly, the rational solution would have required maintaining unchanged the nominal wage, thus preventing inflation. Real wages would of course have declined immediately by an amount reflecting the Sheik tax. But that loss could not be avoided if workers successfully pushed for higher nominal wages to maintain a stable purchasing power in terms of commodities, as actually happened. The superimposition of domestic inflation on imported inflation initially led to a price-wage spiral but with an initial phase of acceleration. Between 1972 and 1975, the average rate of price inflation for the 12 older members of the EC rose from $6\frac{1}{2}$ to 14 percent and that of wage inflation from $11\frac{1}{2}$ to 19 percent.

These developments confronted the authority with the usual cruel dilemma. However, during that episode there was less and less willingness to accommodate, as price stability came to be more generally accepted as the first priority, with the responsibility for enforcing it falling on the central bank.

So the inflationary spiral was accompanied by rapidly rising interest rates decreasing investment and output, and rising unemployment. These developments, together with a slowdown of imported relative to domestic inflation, eventually resulted in a gradual decline of inflation back to $10\frac{1}{2}$ percent, in 1978, but at the cost of a substantial rise in unemployment from 2.7 to 5.1 percent. So the period 1973–1978 was characterized by the combination of high unemployment and high inflation that has since come to be known as "stagflation"—it rises from the fact that the policy of non-accommodation designed to reduce inflation through unemployment is successful in raising unemployment, but not very effective in reducing inflation. Thus the traditionally accepted causation from low unemployment to inflation through excess demand is replaced by a new one, from inflation (possibly exogenous) to unemployment, through restrictive (and inefficient) central bank policy.

By 1979–1980, the spiral is rekindled by the second oil crisis and inflation returns to the peak level of 1975. But now a staunch non-accommodating policy raises interest rates quickly back to 11 percent. By 1983 wage inflation was down to $7\frac{1}{2}$ percent and price inflation to $8\frac{1}{2}$ percent, but unemployment had risen to 10 percent.

The analysis of the last two sections suggests that the high unemployment of countries like Spain or Italy, or the world-wide unemployment during the oil crisis could be readily understood in the light of a need, which had gradually emerged, for a non-accommodating central bank policy in order to cut short an inflationary spiral. However, it does not help to explain the high unemployment from 1983 to the end of the decade, which remains a puzzle.

4.5.3 The Puzzle of the 1980s

The puzzle arises from a set of seemingly inconsistent circumstances: i) during this period, unemployment reached a very high postwar peak and remained close to it until the very end of the decade; ii) the members of the EMS pursued very tight monetary policies which were, presumably, largely responsible for the high unemployment; iii) these policies were pursued despite the fact that inflation declined steadily and became very moderate, at least by the middle of the decade. These "facts" are illustrated in table 8.3 which shows for Germany, France and the European Community the path of inflation (column (2)), and nominal and "real" interest rates (column (3)) during the 1980s. At the top of each column, we report for reference the mean value of the variables in the period 1961–1970. The path of unemployment is given in table 8.2.1 for Germany, 8.2.2 for United Kingdom and 8.2.3 for France.

For Germany, unemployment is very high in 1982 and 1983, but inflation is also relatively high by German standards, and this could explain the very tight policy in those years. That policy raises unemployment to a peak, but it also brings inflation down in 1984 to a very modest 2 percent, a level that is pretty much maintained until 1988; and yet the real rate remains quite high.[1] Over the five years 1984 to 1988, the nominal rate averaged 6.8 percent, or the same as in the 1960s. But with inflation appreciably lower, the implied real rate was 4.6 percent instead of 3 percent, which presumably kept unemployment at its high level.

This policy is hard to understand unless the Central Bank had become so imbued with its role and responsibility as the guardian of price stability that it

Table 8.3
The Puzzle of the 1980s

	Germany			France				EMS		
	GDP Deflator	Long-term interest rates		GDP Deflator	Long-term interest rates			GDP Deflator	Long-term interest rates	
Year	% Change (1)	Nominal (2)	Real (3)	% Change (1)	Nominal (2)	Real (3)	ER* (4)	% Change (1)	Nominal (2)	Real (3)
’61–’70	3.8	6.8	2.9	4.4	6.5	2.0		4.4	6.7	2.2
1982	4.4	9.0	4.4	11.7	15.7	3.6		11.5	14.3	2.5
1983	3.2	7.9	4.6	9.7	13.6	3.6	3.0	8.5	12.6	3.9
1984	2.1	7.8	5.6	7.5	12.5	4.7	3.1	6.9	11.8	5.6
1985	2.1	6.9	4.8	5.8	10.9	4.8	3.0	6.0	10.9	4.6
1986	3.2	5.9	2.6	5.2	8.4	3.0	3.2	5.5	9.2	5.3
1987	1.9	5.8	3.9	3.0	9.4	6.2	3.4	4.1	9.4	5.1
1988	1.5	6.1	4.6	2.8	9.0	6.1	3.4	4.4	9.3	4.7
1989	2.4	7.0	4.5	3.0	8.8	5.6	3.4	4.9	9.8	4.7

* Exchange rate: Francs/Mark.

could not tolerate even a modest, non-escalating inflation as slow as 2 percent (and subject to a substantial measurement error) and refused to accommodate it, at the cost of high unemployment. This policy might be justified if the high unemployment had the effect of moderating the cost push inflation resulting from wage demands in excess of productivity. However, the evidence suggests that this costly mechanism is at best highly inefficient; despite the high unemployment, the inflation never descended appreciably below the 2 percent level first reached in 1984. Could there be some mechanism less economically costly and socially destructive to achieve acceptable price stability? This is the issue discussed in the last section of this paper.

Table 8.3 providers similar data for France and the other EMS countries. As can be seen from column (3), since 1984 French long-term real interest rates fluctuated between 3.0 and just over 6 percent, a level much higher than in the 1960s, even though the availability of jobs was at the lowest level and unemployment was consistently at the distressing level of around 9 percent.

Up to 1986, this policy could be justified by the goal of resisting a rate of inflation which, though steadily decreasing, was still quite high. But by 1987, wage increases in France and in the rest of the community abated substantially, and in 1987, inflation came down in France to a fairly modest level of 3 percent, which was maintained throughout the rest of the 1980s. Yet monetary policy in France was not loosened, but rather became even more restrictive. One possible explanation could be that the Central Bank was resisting that 3 percent as the Bundesbank appeared to resist 2 percent. But a more credible and fundamental explanation is that by then the members of the EMS had decided to move toward a stable exchange rate, with a progressively smaller range of fluctuations around parity, on the way to a unified currency.

In the case of France, this policy became operative at least by 1986, as can be seen from column (4) for France, showing that the exchange rate with the Mark (in French Francs per Mark) is rigidly fixed since then (the so-called Franc Fort policy). But the only way to maintain a stable exchange rate when capital is highly mobile, as has been more and more the case in recent years, and the Central Bank has limited reserves, is to pursue a monetary policy resulting in French rates matching (being as attractive to investors as) interest on Mark-denominated instruments. It is this requirement that accounts for the continuing high French rate, even higher than the high German one because of less "credibility."

This obligation of toeing Bundesbank policy had been a positive feature of the EMS during the late 1970s and part of the 1980s for countries that suffered from chronic wage and price inflation, as did many countries at that time. Joining the rigorous monetary and exchange policy of Germany, inspired by and consistent with price stability and high employment, forced countries to accept its straitjacket, giving them an opportunity to resist the inflationary trend, with the threat of unemployment or devaluations. But after the mid-eighties, the continuing tight German policy became clearly unsuited for France and other EMS countries. The very high interest rates caused investments to decline. In addition, they caused an appreciation of the Mark relative to the non-EMS countries, which in turn resulted in an equal appreciation of the other EMS currencies reducing net exports (foreign investment). The fall in domestic and foreign investment due to the high interest rate required holding down the real money supply and aggregate demand, causing a rise in Keynesian type unemployment. Elementary Keynesian Economics!

We are thus led to the conclusion that much if not all of the huge unemployment, at least during the second half of the 1980s, can be accounted for by an insufficient aggregate demand, due to a combination of a mercilessly tight German monetary policy and the endeavor of the members of the EMS to maintain a stable exchange rate with the Mark, while fostering freedom of capital movement. This combination forced them to adopt the tight German policy.

4.6 The Case of the EMS in the 1990s

4.6.1 Up to August 1993

What happened during the 1990s is basically a repetition of what happened in the 1980s, only on a larger and more distressing scale. The reason is the unification of West and East Germany, which resulted in a large increase in German demand. Having decided not to make primary recourse to higher taxes, the prevention of excessive demand pressure required and increase in interest rates. And indeed, between the first half of 1989 and mid-1990, German interest rates escalated from below 7 to over 9 percent, remaining at that level through 1992. As the high interest rates began to be effective in restraining demand, one might have expected some abatement in the second half of 1992. There actually was none, as the Central Bank had to contend with a serious wage push in 1991 and 1992. An appreciable decline finally came in 1993, but the level remained substantially higher than in the second half of the 1980s.

Remembering that German rates in that period had already been appreciably too high for the other EMS countries, it is not surprising that the further escalation of the 1990s was disastrous. When the escalation came, if not earlier, France, with an unemployment rate already up to 9 percent and inflation stable at 3 percent, should have freed itself from the shackles of German monetary policy, abandoning the fixed parity and pursuing a policy of rapid expansion of the real money supply and deep cut of interest rates, and letting the exchange rate depreciate. The same can be said for most other EMS countries. The lower interest rates would have expanded domestic investment while the depreciation would have increased net exports, especially to Germany, which in turn would have helped Germany by increasing the resources available for East Germany and by holding down inflation.

There is some evidence that the Bundesbank actually favored this approach, to relieve both French unemployment and the pressure on the Bank to ease their monetary policy faster than they (rightly or wrongly) deemed best for Germany. But France's view was that devaluation must be avoided regardless of domestic costs because: i) it meant abandoning the EMS and European unity; ii) it might result in loss of credibility—that is, that it might result in French interest rates having to command a positive spread over German rates, thus reducing or even eliminating the possibility of bringing French rates below the German ones; and probably by far most important, iii) it would have meant a loss of face, inconsistent with French pride. It therefore held on to the "Franc Fort" policy.

Given that decision, there was no way to avoid huge unemployment, except by adopting a fiscal policy of government deficit as large as that of Germany and even higher, to compensate for the lack of the exogenous stimulus; but such an alternative was not even entertained.

4.6.2 August 1993 and Thereafter

With this dismal picture in mind, by August 1993 a group of MIT economists, including myself, reached the conclusion published in the *Financial Times*, that the time had come for France to put unemployment above other considerations and cut interest rates. As this view spread in the market, speculation (led by French, *not* Anglo-Saxon, interest) forced abandonment of parity.

The scenario we foresaw was 1) a sharp cut of the interest rate, 2) followed by appreciable devaluation—most likely through a period of floating—and possibly a joint one by the one-German EMS. What actually happened was quite different. The EMS was saved by broadening the band to 15 percent, which would have made it possible for the non-German EMS to cut interest rates without abandoning the EMS. But the amazing and unexpected thing was that, despite the wider band, exchange rates in the following days, weeks and months were kept in the old narrow band by a policy led by France of holding on blindly and at any cost to the "Franc Fort" policy and hence to the Bundesbank high interest rates. This policy proved to be suicidal. It cost France an increase in unemployment from a very high 8.9 percent in 1990 to 11.6 percent in 1993 and 12.6 percent in 1994. The cost was similar for the European Union as a whole. It transformed the EMS and the march toward Maastricht into an instrument of torture.

The reason why participation in the EMS has been so disastrously costly can be readily understood. It results fundamentally from the decision to maintain rigidly fixed exchanges, at first as a member obligation, and after August 1993, by voluntary consensus of all the central banks remaining in the system. Now it has long been well-known that a payment system based on fixed exchanges suffers form a fundamental and unavoidable shortcoming, namely that it lacks a mechanism for the effective transfer of resources from a country with excess full employment saving at the common interest rate to those that have insufficient saving. As we have seen, the problem could be handled with a flexible *real exchange rate* by pursuing an easier monetary policy, letting domestic rates decline and the real exchange depreciate. The lower interest rates, by increasing domestic investments, reduce the excess to be transferred, and the real depreciation allows for the transfer of the rest by increasing net exports. But with a fixed *nominal exchange*, the real exchange rate can only change through relative changes in the nominal level of wages and resulting prices. Thus a prompt adjustment of the real exchange rate toward the desired level cannot be achieved unless wages (and prices) are very flexible, a condition that was clearly not satisfied in the EMS countries. If the real exchange rate is rigid, then the country with excess full employment saving will end up with an insufficient aggregate demand and unemployment, as in the case of France and the rest of the EMS, illustrated above. Thus the EMS system was one of fixed *real* exchange rates, in which countries with excess saving ended up with insufficient aggregate demand and Keynesian unemployment. Given the high interest policy imposed by the Bundesbank, and the unwillingness to accept huge Government deficits, that meant that practically all member countries were forced to accept an intolerably high jobless rate, with the possible exception of Germany. Germany, as the leader, had the option to avoid the unemployment, but it frequently failed to use that option. Note that the excess full employment national saving could have been "sterilized" through a fiscal policy of government deficits, but much larger than the substantial deficits that already existed. Such a policy was never given serious consideration and in fact was sternly opposed by Germany and its supporters abroad. And in any event, a sterilization policy is a wasteful one (see Modigliani, 1972).

Why should we continue the present potentially destructive path which generates such devastating effects in terms of waste of resources and human suffering over one of quickly reducing interest rates and reabsorbing the excess unemployment? Some reject that solution on the economically ignorant argument that lower interest rates will have no effect on aggregate demand (other than the demand for Marks), an argument rejected by the evidence, including the recent experience of the U.S. Others, especially among Central Bankers, have objected on the grounds that as long as the Bundesbank rejects that approach, it would require some exchange rates adjustments; and to reintroduce some needed flexibility in the exchange rates would do irreparable damage to the ideal of Maastricht and the march toward that target.

In contrast, in the view of many observers outside governments and central banks (who seem to have a natural addiction for, and fascination with, imitating Germany), current policy is actually most likely to produce disaffection with the idea of unity, at least of the monetary kind, by giving tangible evidence of how disastrously costly such a policy might prove. And more damage will come as we wait for a "spontaneous recovery."

I suggest therefore that it is time to consider some fresh approaches, beginning with an emergency package to promptly reduce the disruptive, exorbitant rate of unemployment, to be followed by some basic reforms of the system designed to put an end to the "excruciating dilemma" between maintaining employment through accommodation and risking an inflationary spiral, between floating exchanges and fixed exchange rates with widespread unemployment.

5 What are the Remedies?

5.1 The Emergency Package

The first step is for the EMS to acknowledge that the unemployment problem has reached crisis proportions with little prospects of a quantum improvement, at least through the next year. This calls for prompt, emergency measures, aiming at a step reduction in unemployment toward historical levels, while maintaining price stability. The measures should be of two types.

i) EMS member countries (with the possible exceptions of Germany and Holland) should agree on a program of rapid expansion of the real money supply and sharp cuts in short and long-term interest rates, aimed at reviving demand and reducing unemployment until it reaches some target level. To the extent that the rise in demand turns deficits into surpluses, or reduces them drastically, there would be an opportunity for some fiscal expansion, especially in the form of public investments, to support the monetary policy (see Drèze and Malinvaud, 1994).

ii) Step i requires an expansion of the *real* money supply. But the monetary authority can only control the *nominal* one. It is essential therefore to ward against the danger that the expansion of nominal money and employment will be accompanied by a new push for higher nominal wages, threatening a resumption of inflation, interfering with the expansion of the *real* money supply and calling for non-accommodation, and higher interest rates. The only way to eliminate that danger is to secure the participation of labor, employers and government into an agreement that will ensure the stability of prices. First and foremost, workers must accept a freeze on wages (within limits consistent with the unexpired portion of past contracts) or at the least to keep any increase within the limit of productivity. Similarly, employers should agree to freeze mark-ups, though such an agreement might be less easy to enforce, and the government should commit itself to a freeze on taxes or transfer programs. The acceptance of this commitment in turn should make it possible to maintain a stable price level and ensure a maintenance of the purchasing power of wages. In fact, labor should gain in real terms, through rising productivity. The freeze should last until unemployment has been reduced to an "acceptable" target level, agreed upon in advance.

Clearly the hardest part is to secure agreement of labor and the organization representing it, but then acceptance would be part of a package which includes the protection of the real wage and a large targeted expansion of jobs—and is limited in time until that target is reached. Refusing their collaboration would bluntly place the responsibility for the continuing blight of unemployment on them.

The path of interest rates should be coordinated by participating countries in order to make it possible for them to maintain fixed parities within the group, floating jointly with respect to Germany and third countries. The joint float could be orchestrated by France (possible with the technical help of the now developing European Central Bank), and this new role could provide France with a consolation for the loss of prestige coming from the devaluation *vis-à-vis* the Mark.

There are many other policies that have been suggested as antidote against some supply effects, which might contribute to reduce unemployment, such as taming minimum wage provisions and reducing levies on low wages; however since such measures usually work slowly relative to the expected short duration of the emergency period, they do not deserve much emphasis.

6 Beyond the Emergency Period

6.1 Some General Principles

What is to happen when the emergency period is over, unemployment is back to reasonable levels, and the wage freeze terminates?

The answer I would like to suggest rests on a basic consideration. Given today's prevailing view that price stability has precedence over all other targets, there are large externalities in private wage contracts. In addition to the private benefits and costs, they have important social consequences as they affect the wage level directly and indirectly by way of their impact on other wage contracts through imitation and pecking order; and the wage rate changes in turn play a major role in price movements. In particular, when wage increases exceed the productivity trend, this will tend to produce some inflation and non-accommodating response by the monetary authority responsible for price stability, foreshadowing a decline in output and employment. Hence, society has a direct stake in preventing inflationary wage settlements.

This suggests that the path of inflation, wages, and related items should be the subject of regular tripartite consultation involving the "social parties," government (and the monetary authority), labor and employers plus the unemployed. The object of the consultation would be to reach agreement on a consistent path of nominal wages and prices, acceptable to labor, employers, the government and one that the Central Bank is prepared to accommodate. This is essentially the procedure that has been followed successfully in Italy since July 1992. It was largely inspired by the dedicated work of a brilliant young labor economist and trade union adviser, Ezio Tarantelli (see Tarentelli, 1988, pp. 556–561; 1995), who was tragically murdered in March 1985 by the infamous Red Brigades for his efforts to promote an understanding between labor and management. Wages and prices have closely followed the agree upon declining path and labor relations have been uncharacteristically peaceful. That agreement has recently withstood the stress of a sizable devaluation, not due as in the past to excessive wage demands, but rather to a loss of confidence in the country due to the squabbling between the political parties about the appropriate financial program, even though the devaluation has cost the workers some loss of purchasing power. At the moment this is creating some concern with respect to the forthcoming renewal, which hopefully can be resolved with a recouping of the exchange rate and some goodwill.

The basis for such a negotiation rests on an identity and a few other relations. The identity states that the percentage change in the price level or the rate inflation—say at an annual rate—is equal to the percentage change in the nominal wage rate (wage inflation) minus the percentage change in the real wage rate (wage inflation minus price inflation). This means that the negotiations can be broken down into two separate components—one dealing with purely nominal issues, namely price inflation, and the other with real issues, namely the real wage.

6.2 Programming Inflation

Concerning the first component, we note that once the issue of the real wage has been settled (along lines discussed below), one can achieve any desired inflation target by simply setting the wage target at the targeted rate of inflation minus the agreed change in the real wage.

Labor should be willing to accept any targeted range of rate of change in the *nominal* wage, since it would have no effect on the *real* wage which has been set separately, but only has an effect on inflation. In other words, labor has nothing to gain but only to lose from a higher nominal wage increase, since it will be reflected entirely in higher prices, higher interest rates and a likely loss of purchasing power of their financial assets.

From these considerations, one might conclude that is a strong case to program zero, or near zero, inflation from the very beginning. Actually, there are a number of considerations in favor of a gradual descent, at least if we start from non-negligible inflation. First, for reasons of equity, the programmed path of wages must take into account the unexpired portion of existing contracts; and second, one may wish to avoid substantial quick declines in nominal interest rates and the resulting large transfers to lenders from borrowers who entered into the borrowing contract at high interest rates on the expectation of a sluggish inflation rate and nominal interest rate.

6.3 Programming the Real Wage, the Case of a Closed Economy

With respect to the real wage, let us conveniently start with the case of a closed economy. We begin by recalling the basic identity:

$$p = w - rw \tag{8.1}$$

i.e. price inflation (p) = wage increase (w) – increase in real wage (rw). We next recall that under the presumption of a relatively stable mark-up, (or stable sharing), the increase in the real wage is equal to the increase in productivity

$$rw = y. \tag{8.2}$$

We can then conclude that one can achieve any desired rate of inflation, say $\hat{p}$ by programming a nominal wage increase equal to $\hat{p}$ + productivity growth

$$\hat{w} = \hat{p} + y. \tag{8.3}$$

This conclusion can be readily verified by substituting for w in (1) the $\hat{w}$ given by (8.3), and from (8.2), substituting y for rw, obtaining $p = \hat{p}$.

More generally, to achieve consistent programming of wages and prices, p must equal $w - y$, or w must equal $p + y$.

6.4 Programming Real and Nominal Wages to Achieve Price Stability, Fixed Exchange Rates and Full Employment in the EMS

We now proceed to consider a set of open economies, a subset of which, like the members of EMS, share the desire to achieve those goals. We suggest that it is possible to reach them simultaneously, but that this will require much greater coordination of policies than the present one, which is largely limited to the area of interest rates and possibly fiscal policy. In particular, it would have to be extended to the coordination of wage policies.

We may again conveniently begin our analysis by considering the case where the member countries happen to be satisfied with their current degree of external competitiveness and therefore have no desire to change the real exchange rate. In this case, all the participants must agree to have the same path of price inflation. They only need to agree on what that rate should be, which should not pose a serious problem. Suppose it has been agreed that for the current period the inflation should be p^*; then they all should program, for the current period, a rate of inflation p^* for the price index of domestic value added or of the GNP deflator rate. Referring back to the results of section 6.2 and 6.3, it is seen that any given country, say country j, will have to program a rate of increase of wages, w, equal to the sum of p^* and the productivity growth specific to that country in that period, say y_{jt} or $w_{jt} = p^* + y_{jt}$.

This approach, and a reasonably stable mark-up price policy, will ensure that the real wage in each country will tend to grow at the rate of productivity growth in that country. To the extent that productivity grows at different rates in different countries, real wages will also end up growing at different rates; but if, as seems likely, there is a trend for productivity to catch up to that of the most productive, the same will be true of real wages.

We can next consider the more complex and realistic case where some countries initially suffer unemployment because of their inability to transfer their potential surplus of saving at full employment. The remaining countries should suffer some degree of shortage of saving which might manifest itself in some inflationary pressure or more likely in very high interest rates, as in the German case.

The indicated solution in this case would be a devaluation of the real exchange rate of the surplus countries, giving rise to a transfer of resources through the resulting increase in net exports.

If inflation is the same in all countries, then the real exchange rates will be constant; this was program suggested in the last section where the countries were satisfied with existing real exchange rates. But if they wish to bring about a change in the real exchange rate, then programmed inflation should be different in different countries.

Specifically, if it is agreed that country a should increase its competitiveness relative to country b at the ratio of x percent per year, then, since we assume the nominal rate of exchange is intended to stay constant, the result can be achieved by programming in country a a rate of inflation x percent smaller than that programmed for country b. But the chosen differentials between each member of the group of, say n countries must be consistent, in the sense that if the desired differential between a and b is x_1 and that between b and c is x_2, then the differential between a and c must be $x_1 + x_2$. The desired result can be obtained most efficiently by first agreeing on an average inflation rate for the group, say p^* (however, it should now be a weighted average, weighted by GDP weights). Then for any given country, say country j, inflation is programmed at $p^* + a_j$ and the a_j, $j = 1 \ldots n$ can be obtained by simultaneously solving n equations consisting of the definition of p^* above, and $n - 1$ equations of the form $a_1 - a_j = x_j$, and x_j is the desired differential between a reference country 1 and country j.

From the last section we know that to ensure for country j the rate of inflation $p_j = p^* + a_j$, we need to program a wage increase:

$$w_j = p_j + y_j = p_t^* + a_{jt} + y_{jt}. \tag{8.4}$$

Thus we will end up with some differences within the group, not only in the rate of inflation, but also in the rate of growth of nominal wages, arising from two sources: differences in productivity and differences in the need to correct the real exchange rate in order to increase competitiveness, net exports and employment. Note that in the equation above, p must be interpreted as a measure of the change in the price index of domestically produced commodities or more precisely, of domestic value added, which is the GDP deflator. Thus $(w - p)$ is a measure of the change in real wages, but measured in terms of purchasing power over domestically produced commodities, which in the stable mark-up model is always equal to productivity growth. On the other hand, the change in the price of commodities purchased, say P, rather than produced within the economy, can be approximated by a weighted average of the rate of change of the GDP index and of the price of imported commodities, say p^f, weighted by the share of the two types of commodities in the domestic basket, say D and $1 - D$ or:

$$P_j = D_j p_j + (1 - D_j) p_j^f. \tag{8.5}$$

Then if we conveniently approximate the change in the price of imported commodities, p_j^f, by p^*, the average rate of inflation for the group, one can establish the following result: the change in the real wage measured in terms of purchasing power over commodities purchased is:

$$w_j - P_j = w_j - p_j + (1 - D)\left(p_j^* - p_j\right), \tag{8.6}$$

but $w_j - p_j$ is productivity growth y_j and $p^* - p_j = a_j$, the differential rate of inflation or increase in competitiveness. Thus, the increase in the real wage programmed for country j turns out to be:

$$w_j - P_j = y_j + (1 - D_j) a_j, \tag{8.7}$$

that is, it is equal to its productivity growth, plus the programmed relative appreciation, multiplied by the import share.

This brings out the fact that for a country (or more precisely its labor) there is a trade-off between the growth of the real wage and the rate at which it chooses to gain competitiveness and reduce unemployment; and on this point one can expect with great probability, a conflict between the employed, on the one hand, who presumably will be primarily interested in obtaining the highest possible increase in real wages through revaluing or minimizing any devaluation of the real exchange rate, and the unemployed workers and the net exporting sector on the other hand, that stand to gain from devaluation. For this reason, I suggest that, especially when unemployment is high and has been long-lasting, the unemployed should be invited to participate in the decisions, at least at the level of setting the desirable real rate of depreciation.

There can be little doubt that the approach suggested would increase employment in the devaluing countries, in particular through increased net exports. But will it not increase unemployment in the countries asked to accept a relative revaluation, e.g. by reducing net exports? The answer is definitely not, provided the monetary authorities make sure that the interest rates common to all countries decline properly by an appropriate expansion of the real money supply. If that is done, the resources released in Germany, plus any resources that might have been "idle" before, should be reabsorbed in Germany by a large expansion of demand. First, there should be a hefty increase in consumption spurred by the increase in real income resulting form the gains in terms of trade and by the expansion of employment. In addition, there should be a strong expansion of investment induced by the fall in interest rates and the rise in income, particularly in the case of inventories. At the same time, as pointed out earlier, the expansion of income in other countries, arising from the larger investments in net exports, fixed capital and inventories, will reduce the amount of transfer or net exports needed to return to full employment.

It seems quite likely that by the end of the adjustments described above, many economies would have undergone significant changes in the structure of production and employment. Such changes, while desirable from the point of view of efficiency, should be allowed to occur, but gradually. For this reason among other things, the adjustment of the real exchange rate should be programmed to occur over an appropriate span of time.

But the central banks should understand that a very substantial expansion of the real money supply will be required to finance the expansion in the volume of transactions plus a substantial decline in interest rates. It is above all the Bundesbank which needs to understand and accept the policy needed to keep the EMS together. Perhaps the best hope is through some kind of central bank not dominated by Germany.

7 Conclusion

There is no denying that the scheme of cooperation outlined in section 6.4 is highly ambitious and probably equally complex. I would suggest however that the greatest difficulty would be encountered in the first round. Thereafter, all that should be required are marginal adjustments.

But in my view, unless one can secure the kind of coordinated, responsible behavior suggested here, it will not be possible to satisfy in the EMS the reasonable, and in principle, perfectly consistent and achievable ambitions of exchange stability, price stability and low unemployment.

Appendix: An Alternative Decomposition of the Effect of the Availability of Jobs and Shifts in the Beveridge Curve on Unemployment

The issues involved can be best understood by reference to figure 8.3. Suppose that initially the labor market conditions can be described by point A generated by the intersection of the initial constraint C(A) with the base Beveridge curve *k*

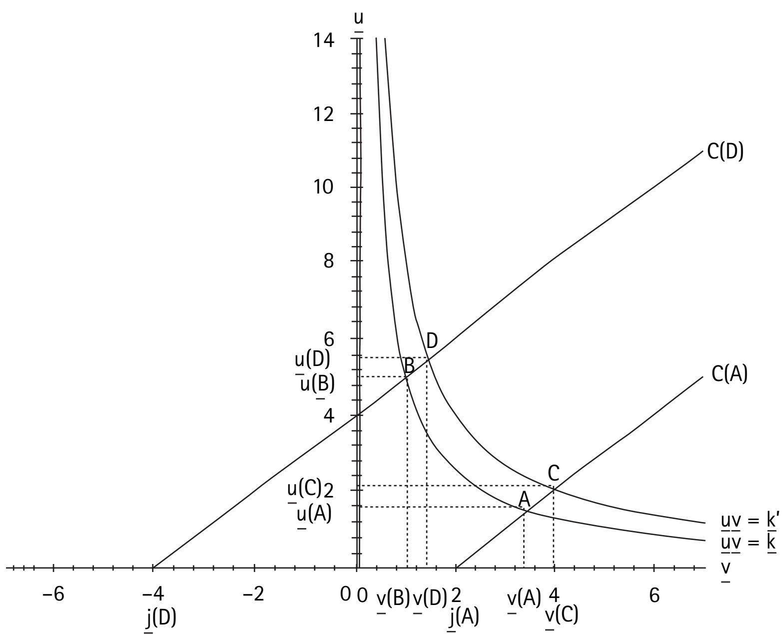


Figure 8.3

and resulting in unemployment u(A) and vacancies v(A). Suppose that at some later point in time market conditions have shifted to a new point D, given by the intersection of the new constraint C(D) and a different Beveridge curve k' with unemployment increased to u(D). In this section of table 8.4 and figure 8.3, we have decomposed u(D) into two components: the first is u(B), at the intersection of the initial Beveridge curve with the new constraint, which is shifted to the left because of the large decrease in the number of jobs as a percent of the labor force (in the graph the decline is from 102% down to 96 percent); the second component is u(D) – u(B) and measures the effect of the shift in the Beveridge curve from k to k',with the constraint kept fixed at the new level, C(D). This shift adds to unemployment as must happen if and only if, the curve shifts in an outward direction. As noted in the text (p. 246), in table 8.2 this second component is given by the difference between u (which corresponds to u(D) in the figure) and u^* (which corresponds to u(B)).

For present purposes it is more relevant to concentrate on the increase in unemployment. Which in our figure is represented by u(D) – u(A). Following the approach in the text, this increase can be decomposed into two components: i) u(B) – u(A) which measures the pure effect of the decline in jobs and vacancies reflected in the upward shift of the constraint from C(A) to C(D) with a fixed Beveridge curve, namely the initial one (*i.e.* k); and ii) u(D) – u(B) which, as before, measures the effect of the shift of the Beveridge curve from k to k', with the constraint fixed at C(D). But it can be readily seen from the figure that there is an alternative possible decomposition, by moving form A to C and from C to D. In this decomposition u(C) – u(A) provides an alternative measure of the effect of the shift of the Beveridge curve, but with the constraint kept unchanged at the *initial* position. Then u(D) – u(C) measures the pure effect of the shrinkage of jobs, but with the Beveridge curve fixed at the new level, k'. An interesting question is whether either of the alternative and equally consistent ways of measuring the jobs and shift effect will systematically yield a higher estimate of, say, the jobs effect. It is seen that, in the specific case illustrated in figure 8.3, the results demonstrate that the Beveridge shift effects u(D) – u(B) and u(C) – u(A) are relatively small and not very different from each other, though the alternative calculation seems to produce a slightly larger result (and hence a smaller measure of the job effect). But in reality one cannot give a definite answer to the question, as can be verified from the year by year calculation for Germany reported in table 8.4.

In table 8.4, column (1) reports the change in unemployment (positive if there is an increase, negative if there is a decrease) relative to the benchmark year, which is 1960 for Germany, and corresponds to point A in the figure. The second column reports the job effect on the basis of the original approach, *i.e.* u(B) –

Table 8.4
Alternative Computation of Jobs and Beveridge Shifts Effects Germany 1960–1993

Date	Change in u over 1960 u(D) − u(A)	Job Effect Original u(B) − u(A)	Job Effect Alternative u(D) − u(C)	Beveridge Shift Effect Original u(D) − u(B)	Beveridge Shift Effect Alternative u(C) − u(A)
1960	0	0	0	0	0
1965	–0.57	–0.35	–0.32	–0.22	–0.25
1970	–0.54	–0.50	–0.49	–0.04	–0.05
1974	1.42	1.22	1.18	0.21	0.24
1975	3.50	3.11	2.97	0.39	0.53
1976	3.38	3.82	2.87	–0.44	0.51
1977	4.31	2.77	3.61	1.54	0.70
1978	3.12	2.11	2.62	1.01	0.50
1979	2.57	2.11	1.99	0.46	0.58
1980	2.65	2.11	2.06	0.54	0.59
1981	4.33	4.01	3.81	0.33	0.53
1982	6.46	6.44	6.30	0.03	0.16
1983	7.98	7.40	7.95	0.59	0.03
1984	7.89	7.40	7.74	0.50	0.15
1985	8.05	7.88	7.68	0.17	0.37
1986	7.70	7.40	7.03	0.31	0.68
1987	7.66	7.30	6.88	0.36	0.78
1988	7.51	7.11	6.65	0.40	0.86
1989	6.65	5.87	5.54	0.79	1.11
1990	5.92	5.21	4.68	0.72	1.24
1991	5.07	4.28	3.96	0.79	1.11
1992	5.44	4.28	4.32	1.16	1.12
1993	7.05	6.53	6.05	0.52	1.00

u(A). These figures correspond to u^* of column (3) in table 8.2.1 minus the initial unemployment (1.23 percent).

It is seen here that the job effect accounts for a very large fraction of the rise in u. Out of the twenty years in which unemployment exceeds 1 percent there are only two (1997 and 1978) in which proportion is not close to 80 percent or higher, and only four more (1989–1992) in which that proportion does not approach 90 percent.

The next column shows the results of calculating the jobs effect using the alternative approach. In many years the results are very similar—within 0.2 percent. In the years with larger differences there are four in which the alternative calculation yields a larger job effect (e.g., 1977–1978). However, in the remaining 40 percent of the years, the alternative gives a smaller estimate. But on the whole one can conclude that the exercise in this Appendix does not fundamentally change the conclusion in the text that the job effect explains the bulk of the change in unemployment.

Note

1. With the only exception of 1993, which seems due to the exceptionally low German rate, which in turn may well reflect an underestimation of the rise in inflation.

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9

WHY THE EMS DESERVES AN EARLY BURIAL

Olivier Blanchard, Rudiger Dornbusch, Stanley Fischer, Franco Modigliani, Paul Samuelson, and Robert Solow

The current turmoil in the currency markets and the broader crisis of rising unemployment in Europe is the most striking evidence yet that European monetary policies and exchange rate arrangements are profoundly counerproductive.

By far the most important factor esponsible for this debacle is the Bundesbank's policy of high interest rates, combined with certain features of the European Monetary system which have forced other member countries to follow suit: these include the mobility of capital across member countries and the fact that central banks have interpreted the rules of the exchange rate mechanism to require not only the maintenance of fixed exchange rates but also the avoidance of party adjustments.

The result of high interest rates across Europe has been that unemployment has risen to record levels. Governments have done little but seek a variety of excuses for the loss of jobs. It looks as if the 1930s are being re-enacted. Then, it was felt to be imperative to hang on to gold at any price: today the feeling is to hang on to the D-Mark.

It is not useful to spend too much time allocating blame—on a Bundesbank that fights inflation of "only" 4 percent, on EMS member countries such as France, which opts for a hard currency at the price of an entirely unwarranted recession, or on a country like Spain, which believes that being in Europe means being a member of the ERM.

In our view, the essential issue is for the EMS countries to shift priorities, putting unemployment at the top of the list and recognising that much labour can be reabsorbed through reflationary policies, beginning with a sharp cut in interest rates.

Of course, countries would have to accept the depressing implications that such a move would have on their exchange rate with the D-Mark, without committing their reserves in an attempt to support the current parity. This means they must

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be prepared to see their exchange rates fall below their lower limits, and hence face ejection from the ERM.

Ideally it might be hoped that France would take the initiative by slashing interest rates and letting the franc float. Other European countries, including Belgium and Spain, would follow. At this point two possible scenarios are conceivable: the first is that, faced with the prospect of a sharp loss in competitiveness, Germany would limit the damage by cutting interest rates. Within just a few weeks the longawaited cut in European interest rates would have materialised.

Second, the Bundesbank would refuse to cut rates beyond another token amount, and then Germany would be left alone in the ERM (thereby floating too). This would mean the end of the EMS as presently constituted. But this should not be a cause for regret.

To be sure, there was a time when the ERM, with fixed parities and Bundesbank leadership, performed a very useful function. But in their quest for moderate inflation, countries adopted an unwavering D-Mark peg, which is now becoming a problem.

There is no reason to equate yielding on existing ERM parities with sacrificing the entire anti-inflationary effort of the 1980s. France has no inflation problem—unlike Germany. France can, therefore, afford to do without the recession which the Bundesbank feels it must impose on Germany. Spain has a phenomenal unemployment problem which should be the immediate focus of attention, not a currency peg that is killing growth and prosperity.

Some might bemoan the ensuing incapacitation of the EMS as a rude shock to the dream of European unity. But such a concern is unwarranted.

First, the central element of the European Community is the set of common market institutions, and these should and would remain in place even if the EMS were temporarily impaired. Second, plans for European monetary union are already in tatters. Thus, leaving the present system would only acknowledge a truth that many in Europe refuse to face. Forsaking the present arrangements is the only way to move quickly towards a new European monetary system in which a crisis such as the current one cannot be repeated.

10

NO REASON TO MOURN

Olivier Blanchard, Rudiger Dornbusch, Stanley Fischer, Paul Krugman, Franco Modigliani, Paul Samuelson, and Robert Solow

In the straitjacket of the exchange rate mechanism, the European economy in 1993 was lost to recession and 1994 was unlikely to see recovery. European unemployment, already at 22 million, was expected to rise in 1994 to an all-time high. The demise of the narrow ERM now opens the door to far better performance in most economies.

It is wrong to believe that something precious was lost last weekend; on the contrary, the liberation of currencies previously trapped in the ERM offers a significant opportunity to recapture the buoyant spirit that animated Europe in the run-up to 1992.

The decision to loosen exchange margins was inevitable; central banks could postpone, within limits and at escalating cost, the time of crisis, but not the ultimate occurrence. The markets understood the basic dilemma: the Bundesbank had made clear its unwillingness to cut interest rates to preserve the existing exchange rates. Whatever the rhetoric, Denmark, Spain, Belgium and ultimately France lacked the reserves and the resolve to sustain exchange rates at the price of visibly and rapidly rising unemployment. Uncertainty about the timing and extent of German interest rate cuts and the urgent need for relief in the distressed partner countries opened up a credibility gap. Such a situation is a standing invitation for speculators who understand which way rates must move.

Somethimes currency speculation may deserve the bad name it has; by prematurely hardening exchange rates, the central bankers and finance ministers of Europe gave speculators the proverbial one-way bet. Even so, in this case the speculators were the best friend of the unemployed, and—even though we will not hear that admission—of the monetary officials who had assumed unsustainable commitments.

There has undoubtedly been some loss of face for policy-makers who proclaimed that they would never devalue, but it would be wrong to dwell on that,

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Rather than look back and dream of punishing speculators, officials need now to exploit the newfound freedom to fight unemployment, of course paying due respect to inflation risks.

The decision to maintain the format of the ERM—exchange rate margins, but 15 percent—is sound and pragmatic. The European Monetary System was a good convergence device for quite a while, but it hardened prematurely with the insistence that further realignments would destroy the accumulated gain in credibility. With limited margins and no realignments, the room for divergent interest rate developments vanished, just at the time when high German inflation made far more flexibility highly desirable. The wide margins adopted in the present form can accommodate big divergences in interest rates without the prospect of creating yet another crisis—at least in the near future.

What strategies should countries pursue to use enlarged scope for interest rates and currency movements? There is no common and simple answer for each of the countries gaining freedom of manoeuvre.

All must be concerned to avoid a recurrence of inflation, a task easier for some than for others. But they also must give urgent priority to expansion, because that is the only way to bring down unemployment. Low interest rates are the fastest affordable way, given actual or imagined constraints to fiscal action, to get there. Finally, they all must look beyond recovery to give more emphasis to the supply side; more room for incentives, more flexibility, less status quo. But beyond these general common targets, the differences in constraints and opportunities deserve spelling out.

France enjoys a privileged position for action. With moderate inflation, it can go hard for growth and will succeed. France should cut interest rates rapidly to reach a level of 4 to 5 percent in just a few months. There is no reason to hold off. In fact, given the long lags of monetary policy in stimulating recovery—particularly when unaided by fiscal stimulus, as the U.S. demonstrates so clearly—there is no place for complacency. Even with immediate action, it will take at least until the beginning of 1994 to see results in terms of growth.

In the case of Belgium the need for moderate interest rates is even more imperative. The extraordinarily high debt ratio—perhaps the highest in the world—makes the country hypersensitive to even the appearance of unsustainable strategies. The country has a good reputation now, but it can lose it in no time if interest rates stay high.

Spain faces far more serious contraints. Inflation is not moderate and the instinctive response to a weakening of the currency is a resurgence of inflation. Of course, keeping the tight money lid on does not solve the problem. Lower interest rates are essential; growth is paramount; the status quo of pervasive cor-

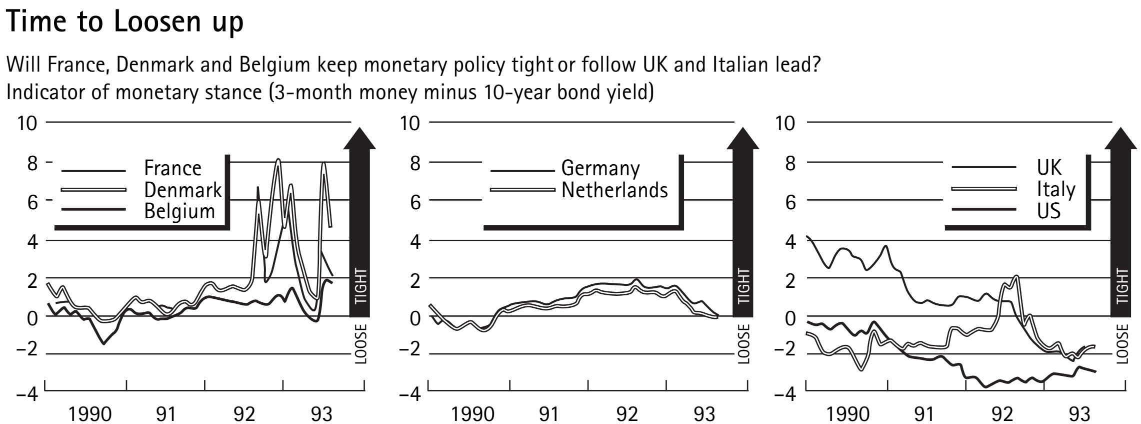

Figure 10.1
Time to loosen up
Will France, Denmark, and Belgium keep monetary policy tight or follow the UK and Italy's lead?
Indicator of monetary stance (3-month money minus 10-year bond yield)
Source: Datastream, FT calculations

poratism, lack of competition and mounting unemployment needs a co-operative frontal attack.

That there is another way is demonstrated by Switzerland, Italy, and the U.K. Switzerland has interest rates of less than 5 percent, far below Germany's. The U.K. when it was pushed out of the ERM last autumn opted for growth and is well on the way, without signs of strain or loss of financial stability. Italy's demise at the hands of speculators became the foundation for growth and for far-reaching domestic reform. Italy demonstrates that unions can be far-sighted and willing to co-operate in a growth strategy that does not translate into inflation.

Interest rate cuts cannot be accomplished without *some* depreciation of currencies. Only with the expectation of an appreciation relative to the D-Mark can a currency have lower interest rates than Germany. The practical question then is how much the French franc, say, must decline to support moderate interest rates. Our view is that the necessary depreciation is very limited, perhaps 5 to 7 percent. After all, France is just moving ahead of German rate cuts by six to twelve months or so, and that hardly warrants big swings. Much the same argument applies to Belgium and Denmark. Thus the extent of depreciation need not be large and stabilising speculation can be counted on to limit the fall.

There is, of course, a strong argument for limiting unnecessary volatility and uncertainty by broadly and informally co-ordinating the strategy among the floaters. For the most part they should be able to cut interest rates in line with one another, and that will limit excess volatility. Where they part company will depend on their attitude towards unemployment, their performance on inflation, and their success in reducing rates without overly large depreciation.

If interest rate targeting takes advantage of the new room for letting exchange rates move and growth resume, there is also the question of when to tighten the margins and return to the EMU project.

The immediate priority is flexibility and that precludes formal commitments to unsustainable exchange rate targets. There is no reason, however, to rule out pragmatic trading ranges around newly found levels of the exchange rates, once interest rates have been cut. Thus we do not expect extreme volatility, just because the margins are wide. Ultimately, 18 months or two years from now, Europeans can reexamine whether the preconditions for stable rates or even monetary union are in place, how to remedy shortcomings, how to assure better co-ordination, and how to proceed.

Whether ultimately there is a common money or not, a common Europe has already shown its worth in the establishment of a market where goods and services flow freely. The good name of Europe will be all the better if further integration yields prosperity and not mass unemployment.

III

MISCELLANA: INFLATION, RISK, LEGAL INSTITUTIONS, AND UNEMPLOYMENT

11

LONG-TERM FINANCING IN AN INFLATIONARY ENVIRONMENT

Franco Modigliani

There is a widespread view among economists that inflation is neutral, i.e. it has only nominal but not real effects on the economy. This conclusion supposedly derives from the well-known proposition that the real economy should be unaffected by the level of nominal prices, it should perform in exactly the same way with a price level twice as high or one third smaller. But if the price level does not matter, then inflation, which is the change from one level to another, should also have no effects on the real economy. But this conclusion is in fact quite wrong, at least for existing economies, as is evidenced by recent experiences with inflation.

To be sure, one can imagine an economy in which all contracts are indexed instantaneously (and have forever been indexed). But the reason why inflation would have no effect in this economy is that, in effect, that economy has abandoned the use of money as the numeraire and adopted a different numeraire, namely the "commodity basket," in terms of which there is no inflation (though even in this economy there would be some real effects through the demand for money).

But in reality no system has ever been 100 percent indexed and for good reasons, including inertia, inflation illusion, and the difficulty and cost of indexing, especially past contracts. In the absence of total indexation, inflation has disruptive effects on the real economy increasing with the rate of inflation. These effects can be traced to five basic factors: (1) continuing reliance on nominal institutions, both public and private; (2) effects of unanticipated inflation on preexisting nominal contracts, e.g. redistribution from lender to borrowers; (3) uncertainty of future prices; (4) inflation illusion or misunderstanding of the real implications of nominal contracts; and finally (5) the one effect that is generally recognized, the Keynesian money economizing syndrome.

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I would like to focus on the problems created by inflation in connection with various types of lending contracts, especially long-term amortized contracts such as in home financing.

Consider first the case of debt instruments of the pure balloon type (to repay with accrued interest at the end of the loan): these could be short-term, one-period instruments or zero coupon bonds. This instrument happens to be totally unaffected by the rate of inflation, as long as it is fully anticipated and there is no money illusion. The nominal interest rate increases, and hence also the terminal balloon payment, for a given amount borrowed, but this extra payment is fully made up by the loss of purchasing power of the terminal payment; so in real terms, leaders and borrowers pay and receive at each point the same real amount independent of inflation, even though they contract in nominal terms at the nominal rate. The outcome is what some seem to have in mind when talking about neutrality. But suppose instead the instrument is again a balloon type but paying periodic interest. In this case, the periodic payment will go up to the extent that interest rates rise because of inflation. As a result there will be a higher initial real payment stream. However, the payment will gradually decline in real terms as the price level rises. And when the terminal balloon payment comes due it will be lower in purchasing power than the amount received with the difference paid out of the overpayments of the early years. Thus inflation is seen to change radically the nature of the contract, requiring in particular, an earlier repayment (front loading) and with the difference being large if inflation is large. One might object that the original real contractual path could be retrieved by substituting the initial nominal contract with an alternative one, or by an indexed contract. Unfortunately, indexed contracts are generally unacceptable to lenders and possibly borrowers. As for alternative nominal contracts, what would be required is for the lender to lend additional nominal amounts in the early phase to permit the borrowers to pay the extra interest; but this negative amortization is hardly what "conservative bankers" want to engage in, given their nature and their inability or unwillingness to distinguish between a nominal and a real payment.

The situation gets to be much more unwieldy in the case of fully amortized, level payment fixed rate instruments, such as the traditional mortgage, which has been the standard way of financing housing for many decades. In the absence of inflation the traditional mortgage possesses some very desirable characteristics: it enables the debtor to repay his loan—including accrued interest—fully within a preestablished time with payments constant in time in real terms. But the same instrument in the presence of inflation loses all these nice properties (except the fixed term). The source of the problem is that the payment is constant in *nominal term*; hence, it must decline in *real terms*, as the price level rises at the rate of inflation. Thus the repayment schedule must be tilted forward with an overpayment in the early years and underpayment toward maturity. The initial overpayment can be quite large even with moderate inflation. To illustrate, with a real rate of 5 percent and a rate of inflation of 10 percent, the nominal rate trebles from 5 percent to 15 percent, and for a long-term mortgage (say 30 years) the initial payment nearly trebles in real terms. Of course, these high payments are made up by much lower payments toward the end of the contract, but this is no compensation, at least for buyers who have cash constraints and can look forward to more income as they get older. Many such people will be frozen out of buying a house or at least one commensurate with what they can afford in terms of life income.

Another problem is that high (and variable) inflation leads to high and uncertain interest rates. This is especially serious when long-term mortgages are financed from short-run deposits as has been the case for the United States. The mismatch of maturities implies high risk of losses and probability of bankruptcy which is precisely what happened to the Saving and Loan industry in the U.S.—they lent very long and financed it with very short liabilities in a period of rising interest rates.

To avoid this mismatch problem one must lengthen the duration of the liabilities or shorten the maturity of the assets. Lengthening the liabilities has little chance, at least in the U.S., as the S&L industry's role has been that of offering a liquid short-term asset. As for shortening assets, nothing can be done directly since a mortgage, by nature, requires a long maturity. However, it is possible to shorten the relevant measure of duration of the mortgage, by replacing the traditional fixed long rate with a rate floating with a short-period rate (say a one year), reset at the termination of each period (one year). From the point of view of matching maturities this instrument has the same maturity as the short rate on which it is floating. Each time the interest rate is reset the periodic payment is recalculated using the new rate, the debt balance carried over, and the remaining years to maturity.

This approach offers an effective but partial solution to the maturity mismatch problem. It has been utilized fairly extensively in the U.S. under the name of Adjustable Rate Mortgage (ARM), and in other countries with middling rates of inflation, like the U.K. However, it suffers from two serious drawbacks. The first and most serious is that the approach does not remedy or improve the major problem of the tilting of the real repayment schedule. That problem depends on the early (real) payments rising with inflation because inflation raises nominal interest rates. With the ARM, the interest on which payments are computed is a nominal market rate (though a short-term one) and will therefore also rise with inflation, tilting the payment schedule just as much as a conventional mortgage. Thus the ARM, though it may be of considerable help in solving the lenders mismatch problem does not provide significant relief from the tilt effect on cash constrained borrowers.

The endeavor to come to grips with this problem has resulted in a number of different approaches. One approach, called the Graduated Payment Mortgage (GPM), rests on a mathematical property of the mortgage formula: if one replaces the level payments with one rising at some rate, say g, and at the same time reduces the interest rate from r to $r - g$, the resulting stream of payment has the same present value as the original stream. Now we know that the ideal mortgage calls for constant *real* payments. Suppose that, over the life of the instrument, the inflation could be taken as fairly constant, at the rate g. Then, if the nominal payment could be made to rise at the rate of inflation g, the real payment would be constant. It follows that one could reproduce the optimal repayment path by offering a graduated payment mortgage with nominal rate $r - g$ and payments rising over the contract at the constant growth rate g. The trouble with this instrument is that it can achieve a constant real payment only if inflation is known and constant, while the future rate of inflation is generally neither constant nor known. Of course we can take a guess, but given the typical great uncertainty over a period of decades, we are likely to end up creating great uncertainty that the repayment path, instead of being level in real terms, might end up uneven and badly tilted either way. It is thus not surprising that GPM, and variants thereof, have not enjoyed and lasting success.

Another possible approach is represented by the so called "Price Level Adjusted Mortgage" (PLAM). In this instrument, which is based on "real" considerations, the first payment is computed using the *real* long-term rate and the agreed upon maturity, while the payments for later years are obtained by multiplying the first payment by an inflation adjustment factor equal to the cost of living for the given year relative to the first year.

This instrument has a number of desirable properties and has been used fairly extensively in the postwar period in highly inflationary environments (Finland, South America, Israel). Its main desirable feature is that it solves effectively the tilt problem. In fact, since the first payment is based on the *real* rate, it will be largely independent of the rate of inflation and further payments, since they increase with the cost of living, are constant in real terms at the initial level. Its main shortcoming is that lender and borrower must agree to receive and pay a fixed *real* rate set initially for the duration of the contract. The borrower, and the lender, especially if he finances himself by short-term borrowing, might prefer using financial indexation a variable short-term rate, as in the ARM, if a way could be found of eliminating the tilt problem.

Since the 1970s, several alternative designs have been proposed which satisfy the two requirements above: they allow the parties to opt for a short-term floating rate—a financial indexation—while eliminating or minimizing the tilt problem.

The so-called "French Mortgage" eliminates the tilt by scheduling the periodic payment exactly as in the PLAM, i.e. computing the initial payment using the fixed long-term real rate and the chosen maturity and indexing later payment to the price underscore. The balance outstanding at the end of a payment period is obtained by adding to the opening balance the amount of interest due for the period and subtracting the amount actually paid in the period. The interest due, in turn, is obtained by applying to the initial balance, the *floating short-term rate* for the period. When the balance due reaches zero the mortgage is fully paid. How long will this take? It can be shown that if the difference between the floating nominal and the fixed real rate coincides with the rate of inflation (which is implied by the so-called "Fisher Law"), then the maturity will coincide with the one embedded in the first payment. Unfortunately this condition may fail to hold, and in this case, the time required before the debt is finally paid may be shorter or longer. In other words, the instrument has a floating maturity. This is the major drawback of the instrument. It should be noted that, though two distinct rates are involved in the specification, the floating short rate is the one that is effectively paid by the borrower while the fixed rate only controls the duration of the instrument.

An interesting variant of the French Mortgage is the so-called "Mexican Mortgage." It is the same as the French Mortgage except that the periodic repayments are indexed on a wage index rather than on the cost of living. It aims to avoid the danger that, in the course of inflation, wages may fail to keep up with prices, causing the periodic payment to grow faster than labor income. Under the Mexican scheme the periodic payment cannot rise any faster than the wage index. Of course if a decline of the real wage does occur, then it will take longer for the debtor to fully repay his debt.

Another type of mortgage, labeled the Inflation Proof Mortgage (IPM), was developed at MIT in the mid 70s. Like the French and Mexican, it gives the lenders and borrowers the opportunity to replace the fixed with a floating rate but is designed so to insure a maturity fixed in advance, as in the traditional mortgage. To this end the real repayment schedule is not fixed in advance at a constant level but is instead recalculated at each payment date by reamortizing the remaining balance over the remaining life of the contract. The balance at any point is computed as in the French Mortgage. Note that the amortization over the balance of the contract insures that the last payment, due at the fixed maturity, will complete the payment of the entire debt. Aside from this feature, what makes the IPM attractive is that the periodic payment will tend to stay pretty constant in real terms provided Fisher's Law holds (i.e. the short-term floating rate exceeds the long-term real rate by the rate of inflation). In this case, the nominal payments called for under the IPM as well as under the French Mortgage will rise at the rate of inflation, i.e. will be constant in real terms.

To the extent that the Fisher Law fails to hold, real payments with an IPM are not entirely constant while under the French Mortgage they will remain perfectly constant but the maturity becomes uncertain. This serves to illustrate the trade-off implied for borrows and lenders in the choice between the alternative instruments. However, the IPM has one additional valuable feature: by choosing a real long rate above or below the expected real rate one can produce negative or positive tilting of any desired extent, slowing or speeding amortization. This may be a desirable feature for some borrowers or lenders.

We have reviewed the several instruments which are available to avoid the severe problems which arise with the traditional mortgage in the presence of significant inflation, and in particular the tilt problem. Given the variety of instruments available, it seems truly amazing that a large number of countries, including the U.S., has done nothing whatever to encourage the development of tilt-proof instruments. It is particularly surprising, in that the cost of the inflation-induced tilt must be rated as quite high when one includes the effect on the construction industry and employment, the derived effects on aggregate demand and output, and the effects on the welfare of those who have been forced to forego or delay the acquisition of a house or to acquire facilities below the level that would have been appropriate in the absence of inflation. At the same it is hard to see that adoption of one or more of the above proposals, especially on an optional basis, would have involved significant costs or dangers, other than adjusting to novelty. I would like to conclude, therefore, with the suggestion that this whole episode is a glaring illustration of the devastating effects of inflation, brought about by inflation illusion and inertia, and with the hope that the future will both see less need for the devices proposed here and more readiness to accept them.

12

RISK-ADJUSTED PERFORMANCE: HOW TO MEASURE IT AND WHY

Franco Modigliani and Leah Modigliani

Investors and financial analysts have long been interested in measuring the performance of portfolio managers. Initially performance was evaluated by comparing the total returns of a managed portfolio with those of an unmanaged portfolio chosen at random (the dartboard portfolio). Later the concept of efficiency was introduced, and managers were benchmarked against the unmanaged "market" or a capitalization-weighted portfolio consisting of the entire market. Recently, the benchmarks have been revised to reflect more closely the investments relevant to the portfolio manager under evaluation.

While the benchmarks may have improved, the industry continues to focus almost exclusively on total return. Yet total return is an incomplete measure of the performance of a portfolio because it ignores risk. It is well known that investors can increase expected returns simply by accepting a greater level of risk, or uncertainty in the range of possible outcomes, implying a greater chance of loss. Uncertainty, and therefore risk, can be measured by "dispersion." That is, for any given level of expected return, the greater the dispersion in possible outcomes, the riskier the investment.

The explanation commonly offered for this trade-off between risk and return is that investors do not like risk, and therefore require compensation for uncertainty in the form of a "risk premium." In fact, this explanation does not provide an operational basis for assessing risk, because appropriate premium levels would vary across individuals and with the composition of a person's holdings. Instead we base our analysis on the simple observation that, regardless of *individual* preferences, the *market* offers investors the opportunity to trade off risk (dispersion) for expected return.

In this context, when one fund has a higher total return than another, we are compelled to ask whether this performance could have been achieved simply by

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gaining exposure to a higher level of risk. In other words, we want to know whether a fund produces a return over and above the return that could have been achieved simply by exploiting the opportunity that the market offers any investor—to obtain higher expected returns in exchange for greater uncertainty.

The answer to the question of how to evaluate performance requires an approach that takes risk differentials into account. In March 1995, the Securities and Exchange Commission (SEC) solicited comment on how to improve mutual fund risk disclosure. It received an overwhelming 3,700 comments and letters in response. While there was general agreement that more information on risk is desirable, there was little consensus on a suitable quantitative measure of either risk or risk-adjusted performance.

Currently, the most common measure of risk-adjusted return is the "Sharpe ratio," which converts total returns to excess returns by subtracting the risk-free rate, and then divides that result by a common measure of dispersion, the standard deviation or sigma, to get a measure of "reward per unit of risk" (see Sharpe [1966]). Other methods devised by Jensen [1968] (Jensen's alpha) and Treynor [1966] (the Treynor ratio) adjust excess returns for the capital asset pricing model's beta.

While experts may find these measures helpful in comparing funds, the resulting figures are difficult to interpret, at least for the average investor not intimately familiar with regression analysis and the modern theory of finance. This inaccessibility at least partially explains the investment community's lack of consensus as to an appropriate measure of risk and the SEC's decision not to adopt a quantitative measure of risk or risk-adjusted performance, at least for the time being.

An Alternative Measure of Risk-Adjusted Performance (RAP)

We propose an alternative measure of risk-adjusted performance (RAP) that is grounded in modern finance theory and yet easy for the average investor to understand. Following conventional methods, we propose measuring the performance of any managed portfolio against that of a relevant unmanaged "market" portfolio. Unlike prevailing methods, however, RAP makes the comparison in performance only after properly adjusting the portfolio return for risk.

The basic idea underlying RAP is to use the market opportunity cost of risk, or trade-off between risk and return, to adjust all portfolios to the level of risk in the unmanaged market benchmark (e.g., the S&P 500), thereby *matching* a portfolio's risk to that of the market, and then measuring the returns of this risk-matched portfolio.[1] Like the original return, the risk-adjusted performance of any portfolio i, RAP(i), is expressed in basis points—which investors are familiar with and understand—and can be compared with the risk-adjusted performance of any other portfolio j, RAP(j). The difference is again expressed in basis points.

In particular, RAP(i) can be compared with the average return of the market over the same period of time (call it r_M). The difference tells us by how much, in basis points, portfolio i outperformed the market (if the difference is positive), or underperformed the market (if the difference is negative), on a risk-adjusted basis. Because the benchmark market portfolio is in principle a viable alternative investment to any portfolio i, the difference performance ($RAP(i) - r_M$) can be regarded as a clear and easily computable standard to assess whether portfolio managers are worth their keep.

Definitions and Notation

Before proceeding to operational formulas, we summarize the notation to be used as:

r_f	short-term risk-free interest rate;
r_i	average return of portfolio i;
$r(i)$	average return of risk-equivalent (or matched) portfolio, or the risk-adjusted return of portfolio i;
e_i	average excess return of portfolio i ($e_i = r_i - r_f$);
$e(i)$	average excess return of risk-equivalent portfolio i ($e(i) = r(i) - r_f$);
σ_i	standard deviation of r_i and e_i;[2]
$\sigma(i)$	standard deviation of $r(i)$ and $e(i)$;
S_i	the Sharpe ratio = e_i/σ_i;
r_M	average return of the market portfolio;
e_M	average excess return of the market portfolio ($e_M = r_M - r_f$); and
σ_M	standard deviation of r_M and e_M.

RAP: A Measure of Risk-Adjusted Performance of Any Portfolio

Using this notation, we can readily show how risk-matching can be operationally accomplished. To this end, we rely on a well-know proposition in the field of finance. Given any portfolio i, with total return r_i and dispersion σ_i, it is possible to construct a new version of that portfolio having *any* desired level of dispersion (risk). This can be accomplished by a financial operation called levering or unlevering the original portfolio. By unlevering a portfolio we mean selling a portion of that portfolio and using the proceeds to buy riskless securities (such as short-term government securities).

Intuitively, we know that this operation will reduce the risk, but also decrease the expected return of the portfolio (provided the original portfolio had a positive excess return). Indeed, if we sell, say, d_i% of the portfolio and use the proceeds to purchase riskless securities, this will reduce the dispersion (sigma) of the returns of the portfolio by d_i% (because d_i% of the returns will have been made constant). It also reduces the excess return of the portfolio by the same d_i%.

Similarly, by levering a portfolio we mean increasing the investment in the portfolio through borrowing. Intuitively, we know that this will increase the risk, and also increase the expected return of the portfolio (again, assuming a positive excess return on the original portfolio). If an additional amount, say, d_i%, is invested in the portfolio, and this investment is financed by borrowing, then both the sigma and the excess return of the portfolio will increase by d_i%.

The risk-adjusted return of portfolio i, or RAP(i), is the return of portfolio i, levered by an amount d_i (d_i positive or negative), where d_i is defined as the leverage required to make portfolio i risk-equivalent to the market, i.e., to make its sigma, $\sigma(i)$, match that of the market. The value of d_i can be inferred from this definition:

$$\sigma(i) = (1 + d_i)\sigma_i = \sigma_M \tag{12.1}$$

which implies:

$$d_i = \sigma_M/\sigma_i - 1 \tag{12.2}$$

Taking into account the interest on d_i, which is the amount borrowed (if d_i is positive) or lent (if d_i is negative), we find:[3]

$$\text{RAP}(i) = r(i) = (1 + d_i)r_i - d_i r_f \tag{12.3}$$

Substituting equation (12.2) into equation (12.3), we can rewrite RAP as:

$$\text{RAP}(i) = (\sigma_M/\sigma_i)r_i - [(\sigma_M/\sigma_i) - 1]r_f = (\sigma_M/\sigma_i)(r_i - r_f) + r_f \tag{12.4}$$

Using the definition of e_i, we can also rewrite RAP as:

$$\text{RAP}(i) = (\sigma_M/\sigma_i)e_i + r_f = e(i) + r_f \tag{12.5}$$

where

$$e(i) = (\sigma_M/\sigma_i)e_i \tag{12.6}$$

Thus RAP(i) can be computed from total returns, using equation (12.4), or from excess returns using equation (12.5).

We see from equation (12.5) that e(i) = RAP(i) − r_f. That is, RAP(i) and e(i) differ only by r_f, a constant in the sense that it is the same for all portfolios. This suggests an alternative, slightly simpler measure of risk-adjusted performance, based exclusively on excess returns:

$$\text{RAPA}(i) = e(i) = (\sigma_M/\sigma_i)e_i \tag{12.7}$$

RAP and RAPA are essentially interchangeable in that they rank portfolios identically, but the reference value for RAPA is the excess return of the market, e_M, frequently referred to as the equity risk premium.

Sigma as a Measure of Risk

Our analysis relies on the fact that the market as a whole is willing to pay a price—give up expected returns—to avoid variability. A portfolio consisting only of short-term riskless assets faces no dispersion in returns—it bears essentially no risk. If one is prepared to bear risk—say, for instance, the risk associated with holding the market—one can increase the expected return by the market mean excess return, which on average has been substantial. This excess return is the premium one can earn for bearing the market risk. Its existence offers anyone the opportunity to trade off an increment of excess returns for an equal percentage increment of sigma.

The observation that the market rewards greater uncertainty with higher expected returns implies that, on average, investors are risk-averse. That is, investors tend to behave as though incremental gains are less and less valuable (or produce decreasing incremental satisfation). If this is the case, then an investment returning \$10,000 with certainty—i.e., without dispersion—is preferable to one that has the same expected value but is dispersed, such as an investment that is equally likely to return \$5,000 or \$15,000. It takes more than the prospect of an extra \$5,000 to offset the equal probability of falling \$5,000 short. More generally, dispersion has to be compensated by higher expected return because upside dispersion is accompanied by more costly downside risk.

For this reason, the view that dispersion is a valid measure of risk is consistent with the assertion that what really matters to investors is "downside risk." While here we have employed the traditional notion of sigma as a measure of risk, in fact, the RAP methodology is applicable to many alternative definitions of risk including several measures of downside risk.

RAP as a Tool for Optimal Portfolio Selection

We have said that any portfolio with dispersion σ_i can be transformed into another portfolio with different risk, say, σ_p, by levering it through a choice of $d = \sigma_p/\sigma_i$

– 1 = $d_i(p)$. This will, at the same time, change the original excess return, e_i to $e_i(p) = (1 + d_i)e_i = (\sigma_p/\sigma_i)e_i$. Thus, an investor wishing to achieve a certain specific risk level can obtain that level with any portfolio j. The excess return for that risk will be:

$$e_j(p) = (\sigma_p/\sigma_j)e_j = \sigma_p(e_j/\sigma_j) \tag{12.8}$$

It follows that to achieve the *maximum* return for any desired level of risk, an investor should lever the portfolio that attains the highest e_j/σ_j. From equation (12.5), remembering that σ_M and r_M are market values that are the same for all portfolios for the period under consideration, it is evident that the portfolio with the highest e_j/σ_j is precisely the portfolio with the highest value for RAP or RAPA. Therefore, independently of individual preferences, RAP unambiguously identifies the "best" portfolio, i.e., the portfolio with highest return for *any* given level of risk.

The best performance is therefore attained by separating the decision regarding which portfolio to hold, from the decision as to how much risk to bear. Investors achieve the highest risk-adjusted returns by choosing the "best" portfolio—that with the highest risk-adjusted performance (RAP)—and then levering or unlevering that portfolio to the desired level of risk. (Note that investors with risk measures other than sigma will want to verify the applicability of the RAP approach, which we discuss later.)

Finally, we note that leverage is a key tool in achieving optimal investment performance because it can easily be used to tailor the risk of a portfolio. If investors or their managers fail to consider using leverage, it may constrain them from taking advantage of a large and highly relevant subset of efficient portfolios. A levered low-risk portfolio may produce superior results with less risk than an unlevered portfolio (including the market portfolio). Regulations that limit leverage, classifying it as risky per se, may severely limit the ability of investment managers to deliver the best possible performance to investors.

RAP versus Other Measures of Risk-Adjusted Performance

Sharpe Ratio

Rewriting (12.7) as:

$$\text{RAPA}(i) = \sigma_M(e_j/\sigma_i) \tag{12.9}$$

and remembering that e_i/σ_i is precisely the Sharpe ratio—call it S_i—we can see that ranking by RAPA or RAP coincides with ranking by S. The portfolio that is best by the RAP criteria is also best by the Sharpe measure (and conversely).

Nonetheless, it should be recognized that RAP(i) and S_i provide very different measures of risk-adjusted performance. RAP gives an answer in basis points that is readily understandable by non-experts, while Sharpe produces a ratio that is difficult for the average investor to interpret.

Jensen–Treynor Measures

The Jensen and Treynor formulation measures risk-adjusted performance by comparing the excess returns of portfolio i with those of a market portfolio that is matched to the risk of portfolio i. The risk-matching is accomplished by multiplying e_M by the ratio of β_i to the β of the market (which is the same as β_i since the β of the market is one).

We see two problems with this approach. First, using the ratio of the betas is fundamentally different from using the ratios of the sigmas. To the extent that the returns of portfolio i are less than perfectly correlated with the returns of the market benchmark, β_i may differ appreciably from σ_i/σ_M. In short, RAP reflects the proposition that in considering which portfolio to select for investing a major portion of wealth, what matters in judging its performance is *total* risk, not just systematic risk.

Moreover, these measures account for risk by adjusting the market portfolio to match the risk of portfolio i, instead of adjusting portfolio i to match the risk of the market, as in RAP. While these approaches are closely related, the Jensen–Treynor measure is less useful in that it may produce misleading rankings. The portfolio with the highest risk-adjusted return by the criteria of Jensen's alpha or the Treynor ratio will not necessarily be the portfolio capable of achieving the highest return for any level of risk. This is in contrast to RAP and S, which unambiguously identify the optimal portfolio for any desired risk level.

Graphical Representation of RAP

Figure 12.1 illustrates RAP in graphic form. Measuring standard deviation on the x-axis and return on the y-axis, any portfolio i can be represented as a point P_i, with coordinates (σ_i, r_i). The point P_M represents the market portfolio, with standard deviation σ_M and total return r_M. Similarly, P_0 represents a portfolio of riskless fixed-income securities with sigma σ_0 of zero and return r_f. Drawing a straight line from point P_0 through any portfolio point P_i gives us the "leverage opportunity line," or l_i, for that portfolio.

For any given level of sigma, the vertical distance between l_i and the x-axis represents the total return of portfolio i (similarly, the distance between l_i and the horizontal line through r_f represents the corresponding excess return). Levering

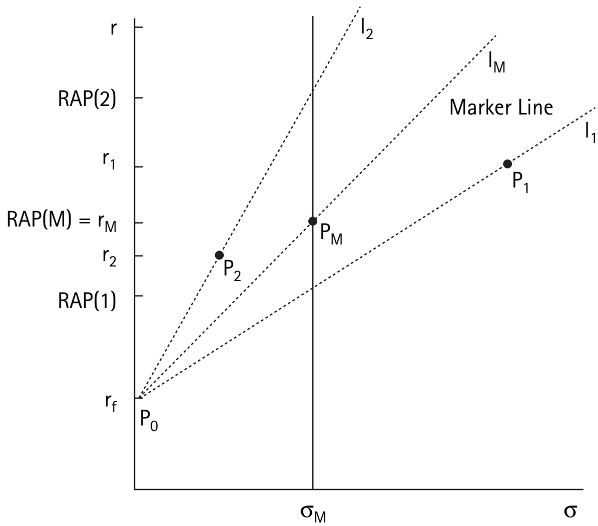

Figure 12.1
RAP: Total and risk-adjusted return
Legend:
y-axis: r = return; x-axis: σ = standard deviation = risk
P_0 = a portfolio of risk-free assets with the risk-free rate of return, r_f;
P_i = portfolio i with total return r_i, risk σ_i, and risk-adjusted performance RAP(i);
where:
P_M = the market portfolio;
P_1 = portfolio 1;
P_2 = portfolio 2; and
RAP(i) = the risk-adjusted performance of portfolio i

or unlevering a portfolio moves it up or down along its leverage opportunity line to alternative levels of standard deviation and total and excess return. Levering a portfolio by d_i% increases the standard deviation as well as the excess return of that portfolio by the same d_i%.

The points on a portfolio's leverage opportunity line l_i represent portfolio i at various levels of d_i. Any point on l_i that is to the right of the initial portfolio P_i constitutes borrowing and levering the portfolio (with a positive value for d_i). Any point on l_i that to the left of the original portfolio P_i represents lending and unlevering the portfolio (with a negative value for d_i). The market leverage line l_M (or "market line") is a straight line from the risk-free rate to point P_M, representing the market trade-off between risk and return. Notice that the slope of any portfolio's leverage opportunity line is that portfolio's Sharpe ratio $S_i = (r_i - r_f)/\sigma_i$.

Thus, from the graph it is easy to identify the portfolio with the highest return per unit of risk as that with the steepest slope or greatest Sharpe ratio (P_2 in figure 12.1). Moreover, the graph can be used to compute RAP(i) and demonstrate its relation to the market return r_M. RAP is found by moving along the portfolio's leverage opportunity line, l_i, until reaching the perpendicular line through σ_M, and then reading the corresponding return on the return-axis. At that point, the difference between RAP(i) and r_M represents the over- or underperformance of portfolio P_i in basis points.

From figure 12.1 we see that while portfolio 1, represented by point P_1, has a higher total return, r_1, than the market, r_M, its risk-adjusted performance, RAP(1), is lower than that of the market. This indicates the relative to the market benchmark, portfolio 1 does not achieve a return sufficiently above the market to compensate investors for its extra risk. Similarly, while portfolio 2, represented by point P_2, has a lower total return, r_2, than the market, r_M, its risk-adjusted performance, RAP(2), is higher than that of the market. This indicates that investors are more than adequately compensated for the risk that they take with portfolio 2 compared with the market benchmark.

Also, portfolio 2, with the highest RAP, achieves the highest level of return at not only the market level of risk, σ_M, but also at every level of risk. An investor levering or unlevering portfolio 2 would move along its leverage line, l_2, thereby achieving the highest return for any given level of sigma corresponding to a desired level of risk.

Application of the RAP Measure

In table 12.1 we have applied RAP to a selection of portfolios to illustrate the difference between total and risk-adjusted return and to demonstrate the usefulness of the RAP measure. The exhibit shows that some of the most glamorous funds turn out to be far less attractive on a risk-adjusted basis.

Consider, for example, the T. Rowe Price New Horizon fund. Over the last ten years, the New Horizon fund produced an average annual return of 16 percent, significantly outperforming the 14.1 percent return of the S&P 500. It was substantially more volatile than the market, however.

An evaluation of risk-adjusted performance requires that portfolios be measured on a risk-equivalent basis. In order to "dampen" the New Horizon fund's volatility to match it to that of that of the S&P 500, it would have been necessary to liquidate 36 percent of the fund and invest the proceeds in risk-free assets (see last column of table 12.1). The yield of this restructured portfolio is precisely RAP and, as reported in column (5), is but 12.2 percent, well below the market's 14.1 percent. By the same token, investors who preferred the greater level of volatility associated with the New Horizon fund could have made higher returns by levering the S&P 500 (22 percent versus 16 percent).

Table 12.1
RAP: Analysis of Selected Mutual Funds

Mutual Funds (in order of total return)	Average Quarterly Total Return (at annual rate)	Quarterly Standard Deviation	Sharpe (quarterly)	Risk-Adjusted Rank	Quarterly Risk-Adjusted Return (RAP) (at annual rate)	Leverage Factor
S&P 500	14.1	7.2	0.30		14.1	1.0
AIM Constellation	19.7	12.3	0.29	3	13.9	0.6
20th Century Vista Investors	16.7	14.0	0.20	7	11.3	0.5
T. Rowe Price New Horizon	16.0	11.3	0.23	6	12.2	0.6
Fidelity Magellan	15.4	8.6	0.29	4	13.8	0.8
Vanguard Windsor	13.0	7.5	0.25	5	12.7	1.0
Fidelity Puritan	12.0	4.7	0.34	2	15.4	1.5
Income Fund of America	11.3	4.0	0.36	1	15.9	1.8
T-Bill	5.5					

On the other hand, funds with average total return below the market may have significantly higher risk-adjusted returns, causing them to outperform the market. The Income Fund of America, for instance, has a total return of only 11.3 percent for the ten years ending in mid-1996 compared with a much higher average return of 14.1 percent for the S&P 500 over the same time period. On a risk-adjusted basis, however, the Income Fund of America returned 15.9 percent, significantly more than the S&P 500.

In fact, of the seven funds listed in table 12.1, the Income Fund of America has the *lowest* total return but the *highest* risk-adjusted return, illustrating the importance of adjusting for risk. It is the "best" fund in table 12.1, in that, levered to any desired level, it would have produced the highest return for *any* level of risk.

Qualifications to the RAP Approach

While we believe that RAP is a broadly applicable and practical measure of risk-adjusted performance, we recognize that there are legitimate qualifications to our approach. We have found five worth considering, and outline them below. (A more thorough treatment will have to wait for a companion article.)

Historical versus Future Performance

Our approach measures historical performance, which, as any investment prospectus will tell you, is not necessarily indicative of future performance. A number of studies have attempted to evaluate the predictive value of historical performance, with mixed results.

We conclude that while historical results may not be a perfect indicator of future performance, they remain useful and interesting information. Before deciding how to invest going forward, investors are likely to want to know how various fund managers performed in the past, and whether they were adequately compensated for the risks to which they were exposed.

Alternative Measures of Risk

In keeping with modern portfolio theory, we have chosen standard deviation as our measure of risk, and return as our measure of reward, deriving our equations accordingly. A result of this choice is that RAP ranks portfolios the same way the Sharpe ratio does (probably the current most popular measure of risk-adjusted return). It may well be that investors feel more comfortable with other measures of risk such as "downside risk," however.

We therefore note that the RAP equation is a valid measure of risk-adjusted performance that will identify the "best" portfolio for any measure of risk such that an increase in leverage increases the risk and return of a portfolio in the same proportion. Measures of downside risk that meet this standard include semi-variance, or average deviation, below the mean (assuming that the distribution of returns is roughly symmetrical around the mean).

Moreover, the fundamental RAP approach of risk-matching is applicable to any measure of risk, provided that it satisfies two more general conditions: 1) leverage changes the risk and reward of portfolios in the same direction, and 2) leverage does not change the ranking of portfolios at any level of risk. With such an alternative measure of risk, the RAP approach would still call for levering or unlevering a portfolio until its risk matches the risk of the market, and then measuring the return of that risk-equivalent portfolio. Again, this approach will identify the portfolio with the highest return for any level of risk.

We identify "probability of loss" as an interesting measure of downside risk that meets these criteria. If returns are approximately normally distributed, then the probability of a negative return (loss) can be inferred from the ratio of the mean to the standard deviation.

Arithmetic versus Geometric Returns

Compliance with the Association for Investment Management and Research (AIMR) performance presentation standards requires the use of geometric rather than arithmetic average returns when reporting investment results. This has become the standard for the investment industry.

The arithmetic mean is always higher than the geometric mean, and the difference between the two grows larger, the greater the variance in returns. The two averages measure different quantities. The geometric mean measures the return of an investment that grows in each period at precisely the rate of the return of the portfolio (before personal taxes). The arithmetic mean measures the return of an investment that is held constant at the initial level. Both geometric and arithmetic measures can be considered equally meaningful (and some would argue that both should be reported).

Deviating from the standards for computing total returns, but in keeping with the common practice for calculating risk-adjusted performance, and the Sharpe ratio in particular, we use arithmetic returns in our calculations for this article. Analyzing the dispersion in returns using geometric figures involves significantly more complex calculations. The standard deviation associated with the geometric mean return is approximately the standard deviation of the log of $(1 + r)$. More important, the effect of leverage on geometric returns is less clear-cut than that on arithmetic returns and cannot be represented simply.

Nonetheless, we checked our results using geometric returns, and for the selection of funds analyzed, the results are very similar. Given the industry standard for geometric returns, we plan to devote further study to this issue.

Recombining Portfolios for Superior Performance—Protection Against "Idiosyncratic" Risk

While RAP unambiguously identifies the "best" single portfolio for any set of portfolios with the same benchmark, it does not consider new combinations of those portfolios that could be optimal. It is conceivable that an even better portfolio could be constructed by combining the set of existing portfolios.

For instance, in theory, a new portfolio could be created consisting of the combination of the second- and third-ranked portfolios to produce a superior risk-adjusted return and ranking above the number one-ranked portfolio. The lower the correlation between any two portfolios, the more likely it is that combining them would produce a superior risk-adjusted return.

One special case of this condition is the market benchmark combined with any other portfolio. For the selection of funds we analyze in table 12.1, we looked at whether any portfolio could be recombined in any proportion with the S&P 500 to produce a higher risk-adjusted return than the number one-ranked portfolio. We found that the funds tend to be highly correlated with the market, and therefore recombining them with the S&P 500 does not produce superior performance.

Nonetheless, as with any other portfolio selection method, portfolio optimization techniques should be employed to evaluate the recombination of assets for superior performance. (Note that performance should always be measured on a risk-adjusted basis as with RAP.)

Similarly, when selecting a fund that is to be combined with other investments in a portfolio, investors should consider the correlation between asset returns. Given the composition of an investor's total portfolio of wealth (including, for instance, earning capacity or real estate) and the resulting exposure to specific idiosyncratic risk, an investor may wish to consider diversifying into specialized funds (such as an energy portfolio). A fund that is best by the RAP criterion is not necessarily the optimal addition to an existing portfolio. Portfolio optimization techniques should also be employed to help identify the best addition to an existing portfolio as the addition that leads to the best risk-adjusted performance (RAP) for the portfolio as a whole.

Tracking Error and the Information Ratio

One performance measure that has been gaining in popularity is the so-called information ratio, defined as the portfolio's average excess return above the market benchmark, divided by the standard deviation of the difference between the portfolio return and the market return. The ratio is essentially a measure of the probability that the performance of a portfolio will fall below that of its benchmark (assuming that the distribution of the difference in returns is approximately normal). The information ratio is a useful concept, although its primary relevance is for money managers (portfolio managers or pension fund managers) who are likely to be judged by their "tracking ability," i.e., performance relative to the market.

Our preliminary analysis has led us to conclude that this measure can be quite misleading. The information ratio is similar in definition to the Sharpe ratio as it includes the standard deviation in its denominator. Yet, it is the standard deviation of the tracking error, and does not take into account the overall risk (dispersion in possible outcomes, or probability of loss) of the portfolio under evaluation. Therefore the information ratio in *not* a risk-adjusted measure of performance.

For purposes of illustration, we looked at the selection of funds in table 12.1, and found that the two portfolios with the highest RAP (Puritan and Income Fund of America) would have been rejected by the information ratio criterion, because these two funds have total returns less than the market benchmark, and therefore produce a negative information ratio. Yet it can be shown that when properly levered, these funds produce an information ratio that is higher than that of any other fund in the sample.

Use of the information ratio as the primary factor for portfolio selection decisions is a questionable approach. The information ratio should be used in conjunction with a risk-adjusted performance measure such as RAP.

Conclusion

We have presented a measure of risk-adjusted performance that is applicable to any portfolio. RAP is grounded in the theory of modern finance, and at the same time is intuitively clear and easily computed from readily available information. RAP provides a framework for analyzing risk and return. It uses the market opportunity cost of risk (in terms of return) and the financial operation of leverage (borrowing and lending) to alter the risk of portfolio returns easily and precisely.

The key to RAP is that it adjusts every portfolio to the level of risk in its unmanaged benchmark, and then measures the performance of this risk-equivalent portfolio. With all portfolios on the same scale, RAP allows us to compare apples to apples and, with proper qualifications, draw conclusions about the relative performance of portfolios and their managers. RAP also allows us to identify the "best" portfolio, the portfolio that has the highest return for any level of risk. This is in contrast to the conventional method of evaluating the performance of portfolios using total return, which can be highly misleading.

RAP provides a simple answer to the question, "do returns adequately compensate us for the risk that we bear?" Ranking portfolios by RAP yields the same results as ranking portfolios by the Sharpe ratio, except that RAP yields a score expressed in basis points, which is a much easier measure for the average investor to understand.

Finally, the recognition that the risk of a portfolio can be readily altered provides the basis for an important corollary of the RAP approach: In the pursuit of superior performance, investors should separate decisions as to which portfolio to hold from decisions as to how much risk to bear. The portfolio to hold is the portfolio with the highest RAP, because it will yield the highest return for any level of risk. Risk can then be tailored to individual preferences through leverage. We note that this renders leverage a key tool for risk management in the pursuit of optimal investment performance.

Appendix: Relative Measures of Risk-Adjusted Performance

One may prefer a measure of risk-adjusted performance that is relative to the performance of the unmanaged market. An obvious measure is therefore RAP relative to the total return of the unmanaged market portfolio:

$$\text{RAP}^*(\text{i}) = 100(\text{r}(\text{i})/\text{r}_\text{M}) \tag{12.A-1}$$

which shows by what percent the risk-adjusted performance of portfolio i exceeds that of the market.

Similarly, the same measure expressed in excess return is RAPA relative to the excess return of the market:

$$\text{RAPA}^*(\text{i}) = 100(\text{e}(\text{i})/\text{e}_\text{M}) \tag{12.A-2}$$

Both indexes will equal 100 for a portfolio doing just as well as the market and exceed (or fall short) of 100 for superior (or inferior) performance. We note that the two indexes will tend to produce similar, but not necessarily identical rankings of portfolios; in fact, only RAPA* will rank portfolios in the same optimal way as RAP. We therefore conclude that RAPA* is the best relative measure of risk-adjusted performance. We also note that equation (12.A-2) can be expressed as S_i/S_M, where S_M is the Sharpe ratio for the market (which can also be thought of as the market price for a unit of risk).

Finally, one may wish to consider a variant of (12.A-2) that measures the difference between risk-adjusted return and the return on the market as a percent of the market return:

$$\text{RARP} = 100(e(i)/e_M - 1) \qquad (12.A\text{-}3)$$

Notes

1. An alternative to the S&P 500 could be any unmanaged market portfolio (e.g., a global index fund) that represents the universe of investments relevant to the portfolio manager (or a close approximation).

2. The variance of r_i, $V(r_i)$, is not necessarily the same as the variance of e_i, $V(e_i) = V(r_i - r_f) = V(r_i) + V(r_f) - 2Cov(r_i, r_f)$ unless r_f is treated as a constant, making $V(e_i) = V(r_i)$. We have found $V(e_i)$ to be very close to $V(r_i)$ and have treated r_f as constant throughout this article. Note that one way to make r_f constant is to set it equal to the rate (yield) on an n-year fixed-income security at the beginning of the n years being evaluated (this is the expectation of the short-term rates for the future n years).

3. Note that here we have assumed that the borrowing and lending rates are the same. Given that investors can borrow against the equity portion of their portfolio, we do not expect the difference in borrowing and lending rates to be great. Nonetheless, changing this assumption is a relatively minor adjustment, which involves using the appropriate rate based on whether d_i is positive or negative.

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13

THE RULES OF THE GAME AND THE DEVELOPMENT OF SECURITY MARKETS

Franco Modigliani and Enrico Perotti

Introduction

Rationales for economic legislation

An important contribution of economic theory has been the elaboration of a normative theory of economic policy. While positive economic theory explores the implications of individually rational behavior, normative economic analysis concerns desirable allocations from a social point of view, with the purpose of defining principles to guide economic policy.[1]

A well-recognized purpose of economic legislation is to limit market failure, or at least to attenuate its consequences. Macroeconomic policy and microeconomic regulation are instruments to correct circumstances in which uncoordinated private agents fail to achieve efficient outcomes through market transactions. Examples are the containment of monopoly power and the regulation of externalities; the residual risk function of certain institutions, such as the lender of last resort; or finally, macroeconomic stabilization policy in the face of monetary disturbances or excessive wage rigidity.

Equally important, the legal rule absolves also the primary purpose of facilitating private arrangements by offering an impartial enforcement mechanism. This ensures that individuals can bind themselves contractually to efficient actions, so that all potential gains from trade are realized.

The former function has been dubbed the Hobbes Principle, referring to Hobbes' mistrust of the capacity of individuals to behave cooperatively in the absence of coercion; and the latter the Coase Principle, concerning the opposite belief that private bargaining, in the absence of transaction costs, can achieve efficient allocations (Cootner and Ulen, 1988).[2]

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The economic analysis of the law does not address issues of equity or income distribution, which certainly represent a crucial purpose of the government action. It is well known that considerations of equity and efficiency tend to conflict; rules established to generate incentives to efficient behavior often give rise to the ex post temptation of altering the promised allocation of benefits in order to favor a broader constituency (e.g., patent laws). This important aspect of economic legislation has not been explored extensively. It concerns the reliability of the legal rules. i.e., the degree to which the government is committed to the present set of laws, as well as the extent of strict and neutral enforcement of these laws.

In the first part of this paper we briefly summarize some important contributions to the normative analysis of government intervention in economic development, without pretence of doing full justice to either the literature or to individual contributions. In the second part we present our conjecture concerning both the extent and the reliability of actual regulation and its impact on the development of security markets.

I Theories of Government Intervention and Financial Development

An early explanation of the link between economic growth and development of financial markets was contributed by the economic historian Gerschenkron (1963). His view interprets the structure of financial intermediation in a nation to be a reflection of the timing of development. Thus early-developed nations, such as England, grew slowly, by promoting risky commercial ventures by merchants and technological innovations by entrepreneurs; security markets ensured supply of funds with a broad redistribution of risk. Late developers, on the other hand, needed to accumulate capital investment rapidly in order to catch up; therefore the development of banking was indispensable to rapidly expand credit to industry. Moreover, direct or indirect government control of banks was utilized to channel funds into large-scale investment.

This view, although certainly helpful, does not quite explain the persistence of different financial systems beyond the initial stage of development, nor does it fully explain the different levels of success among nations in promoting economic growth. However, it does highlight the crucial role played by government intervention in economic growth, and its impact on the dominant form of financing investment.

Legislation represents an important form of government intervention.[3] The economic policy of an individual country may be predominantly influenced by one of the two ideal functions of the law discussed in the introduction, of either substitution for or support of market transactions.[4] Economic policy in developing countries has traditionally been biased toward a direct involvement of government in the economy. This historic tendency toward an active role of the State has been justified in the context of an industrializing nation, where rapid capital accumulation to incorporate known technological knowledge may best be achieved ex imperio. Moreover, whenever there are large externalities in investment, so that the private benefit does not coincide with the social benefit, there is a strong rationale for direct government intervention.

This view is predominant in recent developments in growth theory, sparked by Romer's contribution (1986). This literature suggests that growth is self-perpetuating due to increasing returns to knowledge (interpreted as accumulated human capital). Knowledge is produced under traditional decreasing returns to scale, and gives rise to short-term competitive advantage; however, unlike most goods, its benefits are easily captured by other agents in the economy, generating aggregate increases in productivity which further increase the incentive to innovate. (For a survey of this literature, see Solow, 1990.)

This literature maintains that it can account for the radically different rates of growth achieved by LDC nations, a fact that contrasts with the neoclassical prediction of economic convergence. Eckaus (1989) has criticized this claim. He questions the implicit assumption that the beneficial spillover effect of investment stops at national boundaries. If language is not an unsurmountable barrier, why should the spillover not spread to neighboring countries, thus equalizing growth rates?[5]

The presence of an externality suggests that a government could promote long-term growth by taking appropriate action. In some models it could do so by supporting the accumulation of knowledge, and in others by providing an initial impulse on the demand side (Murphy et al., 1989).

However, the historical evidence does not fully endorse the view that government intervention ultimately delivers the good of faster growth. The generalized failure of centrally planned economies has brought forward the view that decentralized decision-making is ultimately more successful at producing economic development.

An important expression of this view has come from the early "financial deepening" literature on financial deregulation. This school of thought, developed in the late sixties by Stanford economists McKinnon (1972) and Shaw (1973), maintained that government regulation of financial and money markets, although justifiable in principle, damages the capacity of an LDC economy to harness its full growth potential. Their neoclassical analysis led to an emphasis on the great economic benefits that may derive from financial liberalization. In particular, they highlighted the small incentives given to promote individual savings which were caused by the LDC governments' inclination to force savings into financing capital investment (and government budget deficits) at subsidized rates. They also insisted that excessive regulation and restrictions were the causes of the limited extension of financial markets, in particular security markets.[6]

The "financial deepening" school has succeeded in attracting interest in the advantages of generating increased savings in LDC countries; their recipe for policy was very successful in certain circumstances where it was applied in a controlled fashion (such as Korea). However, most historical attempts to broaden financial deregulation aimed at promoting domestic savings and investment have not been successful; see Diaz (1985) or Dornbusch and Reynoso (1989). The development of domestic financial markets in countries which liberalized their financial legislation was very short-lived, and compounded the LDC debt problem when it allowed a vast flight of private capital before and during the crisis. It would therefore appear that something besides freedom to transact is necessary to support the success of domestic capital markets in channelling funds to investment.

Another important variable is believed to be the effect of cultural values over economic organization. To suggest that growth requires a favorable institutional framework is to restate a frequently repeated claim which has failed to become explicit. Our ambition is to add some content to this notion by focusing on some economic consequences of legislation.[7]

We attempt a characterization of an underdeveloped institutional framework and offer a partial analysis of its economic consequences, in particular for the emergence of a developed security market.

There are important questions to be answered. What will explain the different size and liquidity of security markets in different economies? What accounts for the more limited size of these markets in developing nations, and for the dominant role of credit institutions in their capital markets? Why is there such a propensity for noncontractual, long-term relations in certain systems? Even more fundamental, what is the role of security markets in economic development?

II Government Intervention and the Character of Legal Rules

Legal rules are a common good: their use by an agent does not limit the access of another. They are indispensable to economic exchange: individuals need them to write and enforce private arrangements, in order to be able to bind themselves credibly to future actions. The existence of an explicit, neutral code of law which upholds private arrangements permits enforcement of commitments to contractual performance and allows desirable coordination of individual actions.

Normative welfare economics postulates that this function is offered by an enlightened, selfless ruler, programmed to produce the maximum economic welfare.[8] However, actual governments are not selfless, preprogrammed entities. When enforcement is an at least partially excludable good, the provider of enforcement can extract some surplus from transacting agents.[9] That is, a ruler with some degree of self-interest will not supply a "neutral" legal rule unless they (or their constituencies) can capture some benefits. Thus they may pursue an "opportunistic" rule, altering laws or the degree of their enforcement to capture or redistribute rents.[10]

It is generally recognized that sovereign governments have vast power in determining the "rules of the game" through legislation; and that this power cannot be credibly renounced (although it may be limited by constitutional law or international treaties). Rulers are subject to the temptation to use this rule-making capability to favor their preferred constituencies (or in the case of entrenched regimes, themselves). Thus even when governments may ex ante prefer not to take such actions, they are not able to commit not to do so.

The Historical Experience of Government Intervention

The history of economic policy in many countries, but particularly in LDC nations, has seen governments maintain broad discretionary powers on the allocation of resources. This has taken the form of large bureaucracies endowed with detailed functions of supervision, a large presence of government-owned firms, and more generally of institutions in many sectors (such as transport, telecommunications, utilities, credit) which are governed by politically appointed management. This structure is complemented by a pervasive tendency to regulate in detail, or to retain discretion to dictate, permissible forms of economic activity.

There are indeed solid examples of successful promotion of growth by government intervention. However, in many countries the historical experience of government intervention cannot be described as enlightened, benevolent "dirigisme." A most damaging character of the form of the practiced intervention has been its erratic and discretionary nature, the result of a continuous process of readjustment of political favors to different entrenched interests, which ultimately benefits only the mediating political class. When the judicial branch of power lacks strength in the face of a dominant executive branch, the written legal rule has no credibility since it does not hold against the wishes of the rulers.

This condition, however, is in no way specific to autocratic regimes. An executive which is in itself weak and overruled by political factions conduces to a state of confused and ambiguous legal rule, with dispersed, when not conflicting, competencies and vast administrative discretion. A perverse incentive is then created for everyone to enter into collusive arrangements with political factions in order to avoid being left without some indispensable form of patronage. This generates and perpetuates a form of dependence of the private sector on administrative or political assent.

The absent public good, the law, is substituted by privileged access to public officials, a private commodity because it permits appropriation of its benefits to the exclusion of others. This form of limited access of many economic entities from full use of the law produces economic losses.

Such a process of persistent, arbitrary interference ultimately damages the credibility of economic policy, wears out the thread of public confidence in the legal process, and destroys the necessary sphere of contractual autonomy of the private sector. At the same time a vast amount of resources is wasted in the purchase of influence required to facilitate, or more simply make possible, economic transactions.

It is important to realize that the greatest damage to economic efficiency is not caused by the wasteful investment in political influence. This might appear as an additional cost to economic activity, the analog to loss of velocity due to friction. Instead, the lack of credible enforcement has distortionary effects due to the degeneration in private transacting. Private parties cannot credibly commit to contractual compliance in the absence of neutral enforcement. Agents will tend to develop noncontractual enforcement mechanisms, which bind them to collaborate through threats of retaliation or ostracism. There is indeed ample evidence of such networks of mutual collaboration, from the "old boys network" and private clubs to "guanxi" to crossholding networks[11] to family connections (see Greif 1988; Milgrom et al. 1989). Such private enforcement mechanisms are considerably less efficient than a common legislation, since agents unconnected through a scheme are unable to transact. Thus the loss is larger than the benefits enjoyed by some agents, since there is loss of valuable exchange opportunities. Finally, lack of explicit contracting is reflected in a general loss of information concerning opportunities to transact, since the terms of these transactions are not observable and the price at which exchange takes place is not conveyed.[12]

III The Development of Security Markets

A major consequence of an ambiguous legal framework is the stifling of the development of security markets.[13]

Financial transactions are by nature particularly sensitive to the legal framework in which they take place. Securities are nothing more than transferrable legal claims representing liabilities of economic entities. The contractual content of one of these certificates is obviously quite limited (compared with, say, a lengthy bank loan document). Their value, therefore, depends crucially on the enforcement of the associated rights stipulated by economic legislation on their issuance, circulation, and participation in income and control. Lack of adequate protection for small investors, particularly in the matter of fiduciary responsibility and accountability of corporate directors, is reflected in an unequal distribution of gains between small and large shareholders (in particular when transfers of control over assets occur in private transactions outside the Stock Exchange). This in turn results in scarce participation by small investors, leading to depressed demand for securities and ultimately to thin trading and poor liquidity. These market conditions, aggravated by unsuppressed insider trading, increase required yields, and therefore depress prices.

Poor liquidity in turn feeds back to further discourage transacting on the security markets. Participation of a large number of investors in the market is not important only for liquidity; it is also essential to support the search for and circulation of information which generate meaningful prices, and to justify the development of physical and human investment necessary for sophisticated market transacting.

The absence of an impartial and effective legal framework has strong consequences in the chosen form of transacting. In the first place, funding of investment tends to be provided in the form of unconditional claims, i.e. claims demanding an unconditional payment of a fixed amount at a stipulated date. In other words, debt tends to dominate. Creditors often require, moreover, that their claim be strictly secured with current assets (as opposed to being supported on prospective returns).

Second, large organizations, or institutional relationships, tend to prevail over markets as a form for organizing transactions. One reason is that governments active in allocating credit wish to centralize the collection of savings, and are likely to favor banks through fiscal and administrative means. There are, however, other reasons. Large banks offer an acceptable, if undiversified, form of investment of savings: their liabilities are unconditional claims, very simple to enforce; they have huge investments in fixed capital (the branch network) as well as in reputational capital, so are not likely to disappear overnight; and they are often owned, or sponsored, by the government, so they can guarantee at least the nominal value of their liabilities.

On the asset side, their financing of investment relies on long-term relations where performance is ensured by noncontractual enforcement, i.e. the threat of refusing credit. Since a long-term lender is presumably better informed on the firm prospects, other funding sources will be unwilling to second-guess the bank and will be in general more expensive.

Finally, transacting tends to take place predominantly among individuals or groups linked by virtue of some private enforcement device. Agents in the private sector need to establish long-term relationships, be closely related by kin, or develop complex bonding schemes, such as cross-holdings, to ensure compliance.

The absence of public capital markets is in particular an obstacle to the growth of small entrepreneurial firms. Their expansion is constrained to be financed with internal funds (which in general will not correspond to the investment opportunities available) or with expensive bank intermediation. The disadvantages of bank credit lie in its lack of risk sharing, the vast bargaining power of local institutions with information advantage over other lenders, and especially in its incapacity to support rapid expansion based on future prospects as opposed to current net worth. It is unsuitable for emerging industries where widely held risk capital is essential to redistribute the idiosyncratic risk of new ventures, and for sectors where the costs of financial distress are considerable.

Credibility of Macroeconomic Policy

The reputation of a financial center as a reliable locus for transacting is therefore crucial to its success. Its prerequisites include a reliable political environment in which a consistent economic policy ensures macroeconomic stability and reduces damaging uncertainty, such as on future freedom of movement of capitals. At the microeconomic level, a first obvious requirement is the absence of discrimination among the participants, including market outsiders, small investors, and foreign institutions.

Government credibility concerns macroeconomic as well as microeconomic policy. A sound monetary and fiscal policy is necessary to avoid discrediting the value of the currency and ensuring its constant convertibility. Ready acceptance of government-issued money is also a public good. It also requires some restraint of government influence on the credit market, which can be accomplished by ensuring central bank independence and independence of the banking system.[14]

The decision to transfer control of credit market institutions to private ownership and management is a particularly crucial test of the political will to ensure an independent future for the financial market. Privatization of these institutions, as well as of productive firms which have no reason to belong to the public sector, is a first obvious step to commit to a credible policy of restraint by political interests in the control of credit. This decision offers the possibility to reduce budget deficit, multiply the capitalization and liquidity of the domestic market, and promote widespread share ownership. However, to avoid increasing the concentration of control that creates market power and discourages small investors, privatization must be combined with a strong regulatory code that supports competition, and with limitations on the acquisition of dominant stakes by large private groups.

Our interpretation has clear implications for the debate about financial liberalization in developing countries. It highlights the danger of a rapid loosening of restrictions on financial transactions in an inadequately structured, or politically ambiguous, legal framework. In the absence of a credible macroeconomic and regulatory policy, capital flight toward countries with more established legal protection and macroeconomic credibility is the natural result.

In general, the establishment of credibility by a legal system requires time. The reason is that a sovereign government can prove its intent to abstain from discretionary interference only by maintaining a rigorous policy and building reputation capital. In effect, once such capital has been established, its economic benefits may well restrain even an opportunistic government from excessive interference.

Perhaps this notion can be utilized to interpret the pattern of development of the financial superstructure, impressively documented by Goldsmith (1969, 1985). If the credibility of the legal infrastructure, and thus favorable conditions for arm's-length transacting, develop slowly, acceptance by private agents of financial instruments which facilitate exchange will be gradual and will be highly correlated with the general increase in economic specialization and productivity. However, as their credibility improves, financial assets will grow at a rate faster than that of the economy; and their composition over time will tend to shift from collateralized or self-enforced credit (agricultural and merchant loans), to direct or indirect nominal claims on the government, to claims issued by independent economic units in deficit and subscribed by units in surplus through enforceable, transferrable claims.

This may explain why very few developing countries become major financial centers. Credibility is established mostly on consistency of policy record. Interestingly, the only exceptions, Hong Kong and Singapore, are small city-states with a tradition of common law and a tremendous dependence on foreign investment in their capital markets. In their case, a rigorous policy is credible because too much would be lost by opportunistic action.

IV Conclusions

We have linked the development of security markets to the emergence of a credible enforcement rule designed to protect minority interests as well as small and foreign investors.

We summarize our definition of an inadequate legal rule as follows. Legal inadequacy is the consequence of a regime of intervention which is at the same time excessive and insufficient: excessive because executive and administrative discretion is too pervasive and ends up fostering corruption and opportunism rather than supervision; and insufficient because the legal rule is intentionally left lacking clarity, completeness, or even more simply, enforcement. We argue that the regulatory structure is often intentionally maintained in its obtrusive form in order to perpetuate the ubiquitous involvement of political factions in the government of the economy.[15] In the context of an ambiguous legal rule, a large proportion of economic decisions must be negotiated or mediated by political intervention.

Government policy is crucial along two important lines: in restraining fraudulent or destabilizing behavior, and in maintaining favorable and reliable conditions for private transactions. It is usually believed that the former purpose requires regulation while the latter suggests deregulation. In contrast to this view, we have argued that legislation aimed at facilitating private transacting must offer a contained level of regulation together with juridical certainty and impartiality: some clear and reliable "rules of the game." Although regulation is a constraint on private transacting, ultimately an inadequate or underenforced system of regulation is counterproductive, since it results in uncontrolled "laissez faire" where there is some benefit at the expense of others. The credibility of the transacting mechanism therefore collapses, with a general loss of gains from trade.

The Potential for the Development of Asian Security Markets

The arguments laid out in this article of some relevance in evaluating the challenge facing the emerging stockmarkets of Southeast Asia. These economies have made great strides on the road to full economic development. The growth of their economies, and notably that of our host country, Thailand, has been truly astonishing, not only in comparison with the sluggish developed economies but even in comparison with the growth rates achieved by Japan in the 60s. In the process, their security markets have also made great leaps forward. Recent data gathered by the International Finance Corporation indicates that their Asia Index, which tracks stocks in Thailand together with Malaysia, South Korea, and Taiwan, increased 9½ times between 1984 and 1989, in comparison with a mere 2½ for the S&P index. Similarly, the increase in the market capitalization of the seven emerging Asian markets has been an impressive 8½ times. This group of countries shows, moreover, an increase in listings of 40 percent, although for Thailand the figure is as high as 75 percent.

These developments reflect in part, in our view, the fact that these countries have broadly met the conditions conducive to the development of capital and equity markets. On the one hand, the credibility of Asian governments has been nurtured by a responsive fiscal policy, and at least so far by rigorous monetary policy, cautious deregulation and privatization, liberalization of foreign exchange, and freedom of capital movements; and in general by a progressive retreat from direct government intervention. All this has been accompanied by strengthened regulation aimed at increasing the attractiveness of the capital markets to the small investors. Some of these rules were introduced after the crash of 1979. These include measures against insider trading, and certification requirement for quarterly and annual reports. Yet a good deal more seems desirable and possible. For instance, there are gaps in the authority's ability to enforce the rules on insider trading. Verification of accounting statements by the establishment of a SEC-type commission to supervise and police trade practices has been in process for some time but it is not yet complete. Similarly, the establishment of rating agencies remains to become operational.

It is important to stress that new regulation should not necessarily mimic exactly the extensive regulation of security markets, as in the most developed Western markets. Asian economies might well be dominated by first-generation entrepreneurs, keen to retain control and unwilling to quote their companies if very strict and costly requirements and constraints are imposed on their operating procedures. Certainly it is necessary that minimum requirements be set up to ensure that minority investors receive adequate protection. However, if greater latitude of operating procedures for controlling shareholders is allowed, this may simply be reflected in higher required rates of return demanded by passive investors for corporate securities. There has been in fact some afterthought on the received notion that full security market development and the rise of the public company amount to the best possible form of financing. A radical separation of ownership and control may generate serious agency costs; and the reliance on the market for corporate control to contain the problem might be too disruptive to be desirable. It is possible that systems where concentrated ownership dominates, possibly at the cost of reduced protection of minority rights, may offer, on the balance, a better performance. The fundamental feature to respect is the solidity of legislation that defines investors' rights and their form of enforcement.

In conclusion, the successful Asian economies are now facing crucial choices about the types of financial markets they wish to encourage in order to provide maximum support to economic development. The dominance of bank financing and the overwhelming influence of government in the allocation of financing flows are being challenged. This challenge should prove beneficial at their stage of development. Investing firms must face realistic prices for capital, and funds must find their optimal allocation among industries. At the same time, unbridled liberalization brings along serious risks in the presence of underdeveloped legal infrastructure. Insufficient ground rules, inadequate monitoring capacity, or policy credibility can cause extensive damage through episodes of fraud, speculative excesses, or capital flight leading to loss of credibility. Most Asian economies come to this important passage already commanding strong institutional and government credibility. We hope that the considerations set forth here will prove relevant in their deliberation and may contribute to their success.

Notes

1. In this exploration it has increasingly joined ranks with the law and economics literature, which aims both to explain and improve on existing economic legislation. For an introduction to the economics and law approach, see Cootner and Ulen (1988).

2. Obviously, the two principles are partly complementary and partly in conflict. Some circumstances require mandatory behavior; others are more conducive to coordination of individual actions (provided there is a way for credible commitment by all parties). In general, all economic legislation contains some balance of these considerations.

3. In principle, in a democratic society legislation is determined by the legislative branch, while government represents the executive branch, with the task to implement the law. In practice, even in a democracy the political majority controlling the legislation also controls the executive, so the two notions are less distinguishable. What we refer to as government should be understood to be the entity in a position to impose rules.

4. One could conceive in this abstract context of two extreme and opposite examples of policy orientation, a communist regime and a pure “laissez faire” capitalistic system. The former substitutes state authority on the allocation of resources over market transactions, claiming that the private decision-making system fails to deliver equity; the latter system is built around the notion that the government should stay away from intervening in the economy, since the market process in itself guarantees an efficient allocation of resources. However, the characterization of these systems as respectively over-regulated and under-regulated is very misleading. All historical examples of communist rule, as in all situations when the government does not face periodic verification of popular assent, have degenerated in an arbitrary exercise of power that benefits the “political elite,” whose ability to expropriate its subjects has been ampler than ever because of the delegitimization of individual rights (in this sense, it rather resembles feudal law). On the other hand, a laissez faire system actually relies enormously on a strong legal enforcement of certain claims, specifically ownership rights, guaranteeing to property very strong capacity of exclusion as well as protection from claims of public interest.

5. On the other hand, our argument suggests that boundaries are relevant because different legal systems rule across them, engendering more or less favorable climates for private contracting, and ultimately investment and therefore growth.

6. To quote McKinnon (1972), “Whether authorities decided to nourish and expand the real stock of money . . . or allowed it to remain shrunken and heavily taxed has critically affected the relation between savings and income and the efficiency of investment in a number of countries.”

7. North and Thomas (1976) raise the point that “The factors we have listed (innovation, economies of scale, education, capital accumulation, etc.) are not causes of growth; they are growth. . . . Growth will simply not occur unless the existing economic organization is efficient.” Elsewhere, North states “The creation of a state is an essential precondition for economic growth. . . . Model of the state should be an explicit part of any analysis of secular changes”(1979).

8. Moreover, the monopoly of the legal use of force by one institution (the State) is believed to be justified in order to provide order at a low cost through specialization in enforcement.

9. Keohane (1984) has developed a theory of “hegemonic stability,” arguing that a provider of enforcement, in order to be willing to supply stability and policing to economic trade, must be able to capture some benefit from its role.

10. Note that these intentions may be well-meant from an ethical point of view; after all, economic efficiency is defined in terms of Pareto-optimality, which eludes the issue of equity.

11. See Berglof and Perotti (1989) on the Japanese keiretsu.

12. Another detectable consequence of the lack of legal certainty and neutrality is the tendency towards concentration of economic power, since larger firms are more able to bargain directly with the power centers while small-scale private companies are forced to escape outside the shadow of the law to survive, giving rise to a thriving but uncontrolled underground economy.

13. A classic example can be found in the comparison between the development of financial markets and the evolution of law in France, a centralized and often autocratic country, and England, where the independent power of the common law courts restrained the monarchy. The inconsistency of public policy and the arbitrary treatment of individual claims and responsibilities during financial crises observed in France contrast sharply with the more rigorous enforcement of contracts and apportioning of losses in England (see Kindleberger, 1985, on the Mississippi and the South Sea bubbles). The founding of the Bank of France came no less than 106 years after the founding of the Bank of England. As a consequence, in France commercial banking and the diffusion of financial instruments, including banknotes, lagged behind England's for more than a century, possibly contributing to lagging economic performance (Kindleberger, 1984).

14. Ultimately, credibility derives from adherence to principles of fairness and balance. Democratic countries with viable opposition parties benefit from having a feedback mechanism punishing governments which are perceived to have trespassed certain boundaries.

15. Moreover, maintaining the status quo might be in the interest of private coalitions which have developed preferential access to enforcement power, or which may have elaborated private enforcement mechanisms which other private agents may prefer to use over unreliable public courts. Privately generated enforcement is probably preferable to none, but it falls short of a public good because of its exclusion features. It certainly could represent an explanation for the more extreme distribution of income in developing countries.

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14

EMERGING ISSUES IN THE WORLD ECONOMY

Franco Modigliani

The topic is emerging trends, emerging features of the world economy, and I will relate the past to the future and indicate how we have moved to this point and what we can expect to follow.

We are now experiencing a period, unfortunately, of worldwide economic problems. The most serious of these affects the developed countries, in particular the OECD countries that have experienced low growth in recent years. You will find that within this period the U.S., Japan, and Germany each has reported a quarter or two in which output has actually declined in fairly unusual circumstances. One may ask whether this means we are approaching an era of no growth or even negative growth.

To answer this you will notice one feature that is unique to the situation. It has occurred many times in the past that the U.S., Germany, Japan, France, and many other countries have experienced simultaneous contractions, a condition which can be attributed to a spreading of infection. That is, if one country gets sick, this tends to spread to others, essentially through international trade. When one country has problems, a shock that pushes down income, that country will then import less; they get into trouble, into a contraction, and essentially the other world economies then follow.

What is special about the current situation is that world economies are all moving out together but for completely different reasons. There is absolutely no evidence that the contagion is of any consequence. Each country has developed its own peculiar reason. Take the case of NSTs. The reason for their decline is perhaps not entirely clear, but seems to be a factor of consumers not spending; they suddenly decide to go into retrenchment particularly for durable goods. They simply cut down, and this trend seems to be the expectation for the future.

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People are worried about the economic situation. They see people who are unemployed, a condition they have not seen for awhile, and they begin to be concerned that this may happen to them. The apprehension leads them to spend less, and that certainly has been an important phenomenon.

We have seen that the recovery phenomenon began in 1992. It appeared as though the economy was on its way, except that recovery began with a fairly severe contraction following which recovery petered out, essentially died, and almost became even negative. Economists love to call this situation a "double dip", where you go down, then you go up, and you go down again. Of course you can have triple dips but they are seldom seen.

We got down to almost zero growth, then we picked up again, and now the economy is moving. There remain many negative indications, so that one cannot be absolutely sure that we are through, but, and I think this is the prevailing view, we are growing something like 3 percent, not a good growth but almost adequate. This is the case of the U.S., where consumption is reinforced as a result of speculative phenomena.

We have had, in the 1990s, an overbuilding of office space. This speculative activity caused damage to banks. Many banks reached the verge of collapse, which in turn has made the present contractions more serious because the banks are reluctant to lend to business but are willing to lend on regular long-term projects. All these conditions are self corrective—I don't think that the expectations can continue to worsen. Once the situation stabilizes, you can only go up.

Now from the U.S., look at the case of Germany. Their case is of course all related to the reconstruction of East Germany that led to very large, very high expenditure, and a large deficit. Germany, as you know, thinks that there is nothing worse than inflation—not even debt is as bad as that. The banks began to tighten, creating a situation of very high interest rates that discouraged investment. In Germany there has never been significant fear of going down for any length of time, but they did experience a marked slowdown which is still underway.

The case of Japan is, again, totally unrelated to the situation in Germany. Fundamentally, what is wrong with Japan is the stock market. The stock market was a bubble, an unsustainable situation. Some people claim it can be accounted for, but those who say so can do this only by taking into account the overpriced land. The sharp decline in the market, and the uncertainty as to how far this may go, has resulted in a number of consequences. It has discouraged investment, in part because of pensioners and in part because of loss of capital. It has to sell more stocks in order to get more money, which has discouraged investment.

The people of Japan are now responding to the foreign market by consuming less and by rebuilding the capital they have lost. High savings would serve this purpose. To some extent this contraction could have been offset by fiscal policies through higher, more expansionist policies to observe the slack temporarily. But the Japanese government has been unwilling to do this, so Japan has experienced quarters of negative growth. It is now, however, heading for a period of low growth for next year, presumably around 3 percent.

We can see that this particular picture, which at the beginning sounded threatening, does not portend anything much for the future. I think it very unlikely that we will return to a situation like this, where diverse causes come simultaneously into play. We can look forward to growth, but unfortunately the resumption of growth does not seem to be very strong.

If we look at the situation, we find that we have accomplished gigantic things from 1950 to date. There has never been a period in the world where so many people have had their income rise so sharply over such a long period of time. There has been an enormous spread of well-being, accompanied by the enormous improvement that higher income brings, more expenditure in health and education, and in addition, increased trade. The boom in trade is of great mutual benefit to all, not the least of which it has produced a climate favorable to peace. No conflict, because trade requires peace. So here is essentially a situation where everything is going in the right direction and the rates of growth have, in principle, been notably large.

There was, however, a natural break in this situation, which occurred in the 70s and which corresponds to a combination of two terrible shocks: the oil crises. These two events occurred within a matter of a couple of years. It is unfortunate that since that time we have never gone back to the kind of growth that the whole world enjoyed. Even in the early periods everything was growing fast. The U.S. was growing close to 4 percent, Japan was growing 10 to 12 percent, and of course at that time there was no new phenomenon such as Singapore or Thailand with their enormous rates of growth.

Japan clearly experienced the fastest growth of this period. The situation changed suddenly in the early 70s and never regained these rates of growth. In the developed countries up to the 60s the growth still averaged about 5 percent for all countries, and in the following decade it fell to 3.5 percent. In the next decade it came down to 2.5 percent. This pattern represents a dramatic reduction in the dynamics of the system, and a question one should like to understand is, what is the reason for this decline? Why did we stop growing this fast?

There are a couple of reasons, one being the fact that technological progress is appreciably lower, particularly for the industrial sector. And of course technological progress is an important component of total growth.

Another thing that has been suggested, and that is well worth considering, is the fact that the oil crisis was a very disruptive mechanism for a decade or so, for the whole of the 70s and early 80s. You can explain a lot by the effect of disruption of oil. The very high price of oil affected not only the ability to produce but also the cost of production, and in addition these prices gave rise to inflation, very high inflation. This in turn led the government to take measures to reduce inflation, which led to diminished activity and decreased employment. When you have inflation you get into a momentum, a situation in which the inflationary condition feeds itself. Higher prices lead to higher wages, which in turn lead to still higher prices and higher wages, and the system is fed into itself. To stop this trend you must prevent the growing as fast as inflation. When this happens, the economy becomes short of money and interest. You get unemployment, which in turn reduces inflation, and eventually you kill it, which in the case of the U.S. was pretty severe. This was stabilized in 1979–80, a period in which monetarists played a critical role.

What was done in the U.S. was essentially to maintain the money supply, fixed on the grounds of monetarism. Then after inflation was broken, and the monetarists wanted to continue with the same policy, we turned around and permitted rapid rise. Mr. Volker was never a monetarist, but used monetarism as an excuse to promote his policies. So, clearly, through the 70s one reason for the decline was the drop in productivity, which accounted for the continued problem in the 80s.

A hypothesis worth considering has to do with the change of exchange regimes, from fixed to floating. The abandonment of fixed exchanges found many people willing to go along with the change, with the expectation that they would get a system which would work much better. At no cost you get many benefits in particular that each country would have more freedom to do what it wanted. What people did not understand was that fixed exchanges imposed a certain discipline as to what kind of policy you can have. When you have various exchanges you do not have to agree, but then there is nothing to keep the exchange fixed. There is nothing to prevent the exchanges from moving even more widely.

So in principle we have moved into a period of high variation in exchange rates. Of course this period owes a great deal, I am afraid, to the deeds and misdeeds of Mr. Reagan, who singlehandedly decided that he would cut taxes and spend more on armaments simply because it was a “good thing to do.” People do not like to pay taxes so why not cut taxes, and this of course created a gigantic deficit, of which you are aware, of up to 4 to 5 percent of income. It created instability. When you do this, then of course the amount of domestic ceiling available for capital formation is reduced because it is absorbed by the government deficit, so you have less ceiling. Interest rates rise, and you attract foreign capital. Foreign capital is attracted by way of the movement of the exchange rates. Essentially it is evaluation: it takes a higher exchange rate to import the capital needed to satisfy the lesser capital that is available from domestic sources. So, we had a period of pretty sharp change in rate of exchange, and this was against the philosophy of those who favored the exchanges.

Now, can one expect that some of the weakening of growth and trade really can be brought back to the instability of exchange rates? This raises the issue of what the future holds, how we should handle this, and what the hopes are for stabilizing the exchange rates. No one wants to return to fixed exchange rates, having experienced the freedom that comes from variable rates, despite all consequences.

A fixed exchange would have to rely on the key currency. It would be fixed with respect to one currency and that currency would be the U.S. dollar. But Mr. Lee has shown you that you should not rely on the U.S. dollar because the U.S. can do erratic things, so in the long run you have to set a complicated system of multiple reserve currencies or else essentially this is not a feasible thing.

Some believe that you can have a look at the future by seeing what has been done in the past in Europe. The countries of the Common Market have essentially decided that they were tired of moving up, down, and across. Remember that in principle you have several countries. If the dollar fluctuates then the relation between currencies will follow, because the rise in the price of the dollar will differ from that of imported and exported goods.

And so what have they done? First they have formed a union, a custom union which is coming to fruition any time now. It is in the process. Some questions exist as to exactly when and where, but the union is aiming for a common currency and they have established certain conditions to be fulfilled for countries to join—conditions which are far from being met at the moment. One of these conditions for entering the union is that the debt must not be higher than 60 percent of GNP. But in Italy it is 110 percent whereas Belgium is very close to meeting the standard. So it is difficult to go from 110 percent to 60 percent, no matter what you do, unless you do not pay the debt, which I don't advise. But the important point is that, even with the move over the current cut, there have been many years of fairly stable exchange rates. This provides a pathway to stability. Of course you still have some instability with respect to a currency which is still floating vis-a-vis that of the rest of the world. But you would have a strong stability, acquired in this way.

This implied stability suggests that perhaps in the future there may be a tendency to move towards more such groupings, because European grouping is itself subject to much growth. Many countries, for example, Sweden and Switzerland, have joined the system to get going again. In principle Eastern Europe might also join. It's quite a possibility, quite a range, so you have here a big block.

The United States, as you know, is negotiating a similar treaty with Mexico, so that there would be an American Common Market which could then possibly be extended to other countries. If you did this you would emerge probably with a new world, essentially a tripolar world, a world of three poles: the U.S. and Company, Europe, and Japan, with a situation which is quite normal. You have essentially three equal groups.

From the point of view of exchange stabilization, this situation has possibilities. First, if you have large groups, the exchange rate, even though reasonable for individual countries, is much more stable. Essentially the whole change in exchange produces a very large drop because it comes from a large group of countries, so these three groups were floating something. The situation probably would not be too bad assuming the U.S. has learned from the past and is not going to repeat. I think you know for certain that the U.S. would not repeat past errors, because the system requires the maintenance of certain fiscal policy standards—Japan has a pretty good tradition so just left alone might not be so bad.

Second, there is reasonable hope of some coordination. In other words there is hope that, being just three blocs, they might come to agreement on certain policies and measures. There are various forms that these things can take. One possibility is target zones, a situation where each country has a zoned target within which it intends to stay, but within that zone the exchange is free—so you do not intervene. This system is promising and has been widely discussed.

Now I come to a different story, that is, the question of the past and future role of the U.S. and Japan—and some other Asian countries you may recognize—but most importantly, I would like to discuss these two countries. I point out here that the U.S. has lost much of its position. It once had well above half the world trade, up to 70 percent immediately following the war. But after the war this figure declined gradually to well below 50 percent. Where as the U.S. has lost much ground, Japan has grown accordingly. The question is, why has this happened. I take the view held by the Japanese and many people around the world, that the American economy lost its vigor, whatever that means, and it has a huge deficit in the current account. It was unable to meet competition. In addition, it was the culprit of many bad things; for example it had a very low ceiling interest rate compared to other countries. In Japan productivity is high, people are working. People are not making financial deals. Savings is high over the population, as high as 30 percent in some years, whereas Americans have been spendthrifts. The Japanese have been virtuous, putting large sums aside for the future. And they have a wonderful government, no corruption, everything perfect. The problem is, there is very little difference between the U.S. and Japan.

Let me take this one-by-one. International trade: Why did we develop this enormous deficit? Simply, this was an elementary and necessary consequence of the government deficit. If you have a government deficit you will attract capital. You have less capital left, and if you attract capital you must have imports in excess of exports. How much in excess depends on elasticity, on the response to the demand for investing, the U.S. elasticity differential, and the level of the elasticity of domestic investment with respect to interest rates. We simple import a lot more than we export, because there is no way the poor manufacturers could do otherwise—the exchange rate moved against them and they could no longer sell. They had to import the capital. The exchange rate had to move against them.

By the way, it was reinforced by the regulation that it has a tremendously high exchange rate against the dollar, in part because of the U.S. deficit. As the dollar has returned pretty much to normal levels, our tremendous deficit has disappeared. In fact during one quarter last year the balance of trade showed surplus, but this did not last. We still had a deficit, and not negligible, about $20 billion. From $160 billion it came down to $20 billion.

Of course, in my view the exchange rate is still too high. Why is it too high? Because the deficit remains too high. If we take on less deficit in order to be able to reverse the process with smaller deficit, we must lower interest rates, in which case we will be pushing out capital instead of attracting it. This phenomenon does not imply that there is anything inferior or incapable or less vigorous. So from this point of view the situation has now changed. There is nothing new in our savings over the last few years. The reason Japan saved more is simply that saving essentially depends on growth. The more you grow the more you save—with the exception of Singapore. No matter how fast you grow, you save 40 percent. But everywhere else this law applies. There is plenty of evidence by which Japan's high saving rates are largely explained.

Investment has declined appreciably. The net investment has about halved, which presumably has something to do with growth. The question of the extent to which investment affects growth is delicate. I have done work recently and am convinced that there is evidence that growth does depend on investment. In the case of Japan the very high savings rate helps. In my theory, the relation between growth and saving has nothing to do with individual decisions. It is not the case that when there is rapid growth people decide to invest and to save more. It is, instead, a mechanical effect. The faster you grow, the more you save. This has nothing to do with upbringing or any such feature. In the case of Japan, one important feature is that of bequest.

There was one further topic I planned to discuss, an aspect of what might happen in the financial market—the whole question of globalization. But in deference to punctuality I will close now.

15

THE KEYNESIAN GOSPEL ACCORDING TO MODIGLIANI

Franco Modigliani

Introduction

My purpose in this essay is to try to explain Keynes's path-breaking contribution to the understanding and cure of mass unemployment, the most serious shortcoming of a market economy. My ambition is to do so in a way that is understandable to a reader with little economics background, provided he is prepared to do a little hard work. I hope to achieve this result by using a novel approach that stresses the communality between pre-Keynesian models—"the classics"—and that of Keynes. They basically share the formulation of the demand and supply of money; the only real difference is in modeling the labor market—the response of nominal wages to unemployment. The classical model turns out to be a special case of the Keynesian general model, in which wages are assumed (counterfactually) to decline promptly in the face of unemployment. But the difference between rigidity and perfect flexibility turns out to make an enormous difference for the "monetary mechanism"—the mechanism that ensures the clearing of the money market—and the understanding of unemployment.

As I will try to demonstrate in the following pages, Keynes's contribution can be summarized as follows: before his work, mass unemployment (in developed countries) was considered a random and transitory aberration of the system. Like catching a cold—sometimes it's light, sometimes it's a serious problem, but there is no certain remedy, though if you are patient, "it will go away." Instead Keynes in the *General Theory* develops a radically different interpretation of unemployment.

1. Offering a systematic explanation of this illness, proving that it is not a random accident, but a physiological response to certain disturbances and in particular, an insufficient *real* money supply. This explanation, I hold, applies to Europe's

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mammoth unemployment (though Europe's opinion leaders refuse to understand it, blaming other fictitious causes).
2. Teaching how this illness could be cured (a teaching that EMS countries still refuse to learn).
3. Proving that, if one fails to understand and apply the appropriate cures, this illness could last for a long time (as EMS countries are learning).

My purpose here is to develop a synthetic model of Keynes's construction designed to lay out its essence in readily understandable form. This is not my first endeavor to do so. But the present version differs from my previous papers, published throughout my career, starting from my first on "Liquidity Preference." The difference springs in part from the fact that the new presentation is meant to be understandable by a non-technical audience, but in part it reflects my recent realization, that it is possible to use a model different from the prevailing one, which stresses the communality between Keynes and the classical theory. In fact, I will argue that the classical model is but a special case of Keynes's General Theory. It applies only to an economy in which wages (and prices) are highly flexible downward in response to "excess supply," and financial markets are unimportant. It should be obvious that this special case is of very little relevance to the present-day developed economies, and so are the analytical and policy conclusions that follow from it.

It is my ambition that anyone who accepts the validity of the classical theory of the demand for money—the foundation for the so-called quantity theory—will, at the end of this essay, fully understand the scope and nature of Keynesian unemployment, and come to admire with me the greatness and originality of his contribution. A reader impatient to know what this paper is all about is invited to glance at the first paragraph under "Concluding Remarks."

I The Concept of Equilibrium and the Money Market

I.1 The Classical Model of Market Equilibrium and the Role of Price Flexibility

Economists model the whole economic system as composed by a series of markets in each of which the quantity exchanged and its price tend to an equilibrium or sustainable value. If the market is competitive (i.e. there are no legal or economic impediments to the entry and exit of buyers and sellers), the market mechanism generating the equilibrium can de described by the well-known "law" of demand and supply. The demand is described by a schedule indicating the quantity that would be bought at different prices. It can be represented graphically by a curve like DD in figure 15.1, in which the quantity (**q**) is measured

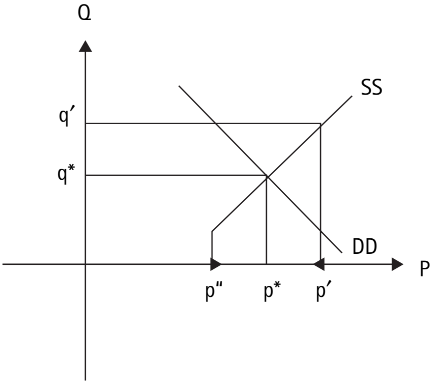

Figure 15.1
Supply and Demand in competitive markets

along the vertical axes and price, p, along the horizontal axes. It normally falls from left to right since we except the quantity bought to be smaller the higher its price. SS represents the supply curve (function)—a schedule of the quantity offered at different prices—that typically rises as price increases. The basic classical proposition is that in each (competitive) market the price and quantity tend to a stable equilibrium value given by the coordinates of the point of intersection of the demand and supply curves, labeled q* and p* in figure 15.1, where demand equal supply (the "market clears").

This conclusion rests on a fundamental assumption or postulate, which may be labeled the "postulate of price flexibility" (The Postulate). It states that, if at any given market price, the supply exceeds the demand, as at p′ in the figure, then the unsatisfied suppliers, unable to sell at least part of what they intended, will endeavor to sell their unsold supply by bidding the price down toward the equilibrium value, p*, as shown by the arrow pointing toward the left on the horizontal axis. This movement will persist as long as **p′** will exceed **p*** and **q′** is above **q***. A similar mechanism operates when the price is below p*, like p″ in figure 15.1, and the quantity demanded exceeds that supplied: the unsatisfied would-be buyers will bid the price up. Therefore, provided the postulate of price flexibility holds, equilibrium can be described as a situation where the price and quantity comes to rest, or where demand equals supply (because only there can price and quantity come to rest).

I.2 Money Market Equilibrium and the Monetary Mechanism

In an economy that relies on money (rather than barter) to carry out transactions, (as most economies do today), one of the relevant markets—and an exceedingly important one at that—is the so-called "Money Market." Money is defined as the collection of instruments (one or more) that in a monetary economy are used as "means of payments" for the purchase or sale of goods and services, that is, to execute "cash" transactions. (In the following presentation, in order to simplify the presentation and focus on the critical issue, we will, as a rule, focus on the behavior of this market in an economy closed to international trade.)

In the Money Market, as in every other market, there is a supply and a demand curve. Because the essential function of money is to carry out transactions, it has long been recognized—indeed for centuries—that the demand for money depends on the value of transactions. And this view is accepted by Keynes, as well as the classics. In current presentations, the value of transactions is, conveniently, assumed proportional to the value of Gross National Product expressed in terms of money, at an "annual rate" (Y). Thus, the "classical" demand for money can be expressed as:

$$\mathbf{M^d = kY}$$

Here **k** is a constant that measures the stock of money held, or demanded, by the economy relative to domestic income.[1] The value of business transacted per year, which we have approximated with Y, can be expressed as the product of the quantity of units sold per year times the price per unit, or

$$\mathbf{Y = PX}$$

where **P** is a price index (e.g. the price of a representative basket of goods traded) and X is an index of the quantity (of baskets) traded. The demand for money can therefore be written as

$$\mathbf{M^d = kXP} \tag{15.1}$$

Equation (15.1) can also be rewritten in a form, suggested by Irving Fisher, to wit:

$$\mathbf{M^d = PX/V} \tag{15.1a}$$

where **V = 1/k** is again a constant that Fisher labeled the "Velocity of Circulation," because it is a measure of the value of the transactions executed, on average, during a year, by each unit of currency.

The supply of money, $\mathbf{M^s}$ is usually considered exogenous from the market system. In particular, in a modern (closed) economy the stock of money is determined by the Central Bank, or $M^s = \bar{M}^s$.

This money supply, together with the demand equation, gives the equilibrium condition in the money market

$$kPX = M^s \tag{15.2}$$

In what follows, we will refer to the "*monetary mechanism*" as the mechanism that insures the clearing of the money market, i.e. that equation (15.2) is satisfied.

We will show that the mechanism proposed by the classics is unacceptable for it relies on the counterfactual assumption that wages are *highly flexible downward*, i.e. that they decline promptly in the presence of excess supply of labor or (non-frictional) unemployment. Keynes instead elaborates a mechanism that is valid regardless of the degree of rigidity of wages. It is the formulation of this mechanism that constitutes his crowning achievement.

II The Classical Monetary Theory

II.1 The Determinants of the Price Level and the Quantity Theory of Money

The essence of the classical monetary theory is very simple: money and the money market has one single role to play: to determine the price level, while having no effect on any other real variable. It rests entirely on the (counterfactual) Postulate of wage-price flexibility: in the presence of excess supply, wages-prices decline promptly toward the equilibrium level reducing the excess supply until it has disappeared. Since this assumption insures that every market (including the labor market) always clears, aggregate output X, must be fixed, pegged at the "full employment output," say $\bar{X}$, which at any time can be taken as a given. If one substitutes $\bar{X}$ for X in (15.2), one obtains the classical demand equation

$$M^d = k\bar{X}P$$

Since at any point of time, $\bar{X}$ and k are given "real" variables, independent of M, reflecting the size of population, the state of the arts (productivity), payment habits of M, they can be taken as constant, so that the demand for money turns out to depend on a single variable P, and to be proportional to it. This demand equation is shown in figure 15.2, as MM. It is a straight line through the origin of slope $k\bar{X}$. The supply can again be taken as exogenous, M^s, and represented by the horizontal line at that height. Equilibrium P, $\bar{P}$ is found at the intersection of demand and supply. Its value is given by the solution for P of the equilibrium condition $M^d = k\bar{X}P = M^s$ or

$$\bar{P} = \mathbf{M}^s / \mathbf{k}\bar{X} = \mathbf{M}^s\, \mathbf{V} / \bar{X} \tag{15.3}$$

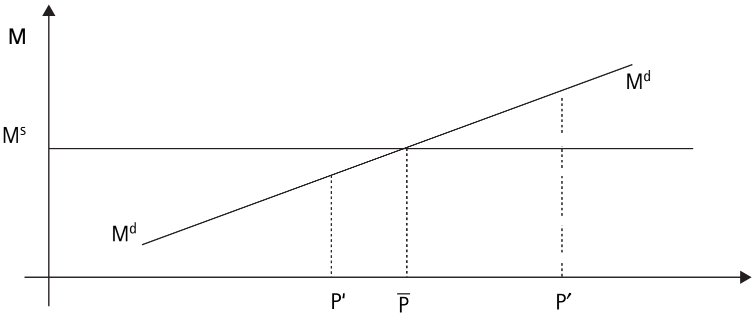

Figure 15.2
Classical Money demand (fixed income)

With $V/\overline{X}$ independent of M^s, *M^s determines P and nothing else* (Money is neutral with respect to all nonmonetary or "real" variables in the economy). Furthermore, at any point of time, P is proportional to M^s. This is the essence of the classical *Quantity of Money Theory of the price level.*

II.2 The Classical Monetary Mechanism

But what supposedly insures the neutrality of money? The answer is simple: *it is The Postulate*. It insures the continuous clearing of all markets, including the money market and the labor market (except for those occasional, unexplainable "colds" referred to earlier).

This can be demonstrated with the help of figure 15.2. Suppose that at a given point of time the price level happened to be P′ which exceeds the equilibrium price $\overline{P}$; this would result in an excess demand for money, a-b. In the classical model, the excess demand for money will cause a prompt decline in the price level P—the Postulate. The explanation provided in the classical literature is essentially the following: when people find that to transact their business they need more cash than they actually have, they endeavor to increase their endowment of money by promptly offering to sell commodities at reduced prices. In essence, in the classical model, an excess demand for money is equivalent to an excess supply of commodities; and according to the Postulate, the excess supply results in a fall in the price of commodities, P, toward $\overline{P}$. This in turn reduces the nominal demand for money, and hence excess demand, till it disappears at P = $\overline{P}$. A similar argument holds if P were too low, causing an excess supply of money.

(An alternative more usual formulation is that an excess demand for money leads to an increase, in its *price in terms of commodities*. But the price of money in terms of commodities—commodity baskets per unit of money—is simply the reciprocal of the general price index—money per basket. It is usually referred to as the *purchasing power of money*. Thus an excess demand for money must result in a rise in its price, or purchasing power, of money which means a fall in the price level P, toward the equilibrium level P*.)

There is an interesting alternative formalization of the classical model, which has been suggested more recently. Let's rewrite the equilibrium conditions (15.3) in the form:

$$\mathbf{k}\hat{X} = \mathbf{M}^s/\mathbf{P} \tag{15.3a}$$

The left side of the equation is the nominal demand divided by the price index, and is usually referred to as "real" demand for money, that is the quantity of money demanded expressed in terms of commodity baskets. The right hand side of (15.3a) is a measure of the "real supply" of money, which is the existing stock, expressed in terms of number of baskets. In this formalization of the classical model, the real demand for money is fixed, being proportional to full employment income. On the other hand, though the *nominal* supply is fixed at M, the *real* money supply is variable, moving inversely to the price level. If at a given time the real demand were to exceed the supply, the price of money in terms of commodities—or its purchasing power—would rise, that is, the price level P would fall. This increases the real money supply, reducing the gap until it disappears at the equilibrium price **P***.

To summarize, according to the classical model aggregate supply or output is fixed at full employment $\bar{X}$, and aggregate demand is sure to match that supply promptly (except perhaps for occasional random shocks) through the prompt adjustment of wages and prices in response to an excess supply of commodities (excess demand for money) assured by the Postulate.

The fundamental, irremediable shortcoming of the classical model consists, as Keynes points out, in the fact that the classical postulate is manifestly counterfactual (see below). As a result, that model is unable to provide a systematic explanation of unemployment or to give any guidance on how to control it.

III Keynes: A Generalization of the Quantity Theory to an Economy with Rigid Prices: The Theory of Under-Employment Equilibrium

III.1 The Rigidity of Wages and Prices

The General Theory starts by rejecting as an unrealistic fairy tale the classical postulate that wages and prices are sufficiently flexible in both directions so that the demand for money quickly adjusts to any given supply. At least in this century, after the enormous decline of the role of agriculture, and the appearance of trade unions, with their power to exclude unemployed people from the bargaining table, the flexibility of nominal wages on the down side does not exist, if it ever did, and the same of course applies to prices which are very closely tied to wages. The reasons why such a downward rigidity prevails, and why it is not grossly inconsistent with rational behavior has been analyzed and explained in many papers [e.g., Dreze 1999, Akerloff 2002], which it is not necessary to summarize here, especially since the classical postulate is patently refuted by empirical evidence. In fact, it was precisely the failure of wages to decline in England, despite the great and persistent unemployment rate prevailing in the 20s and 30s and until World War II, that inspired Keynes's *General Theory*. The rigidity has certainly not diminished since then. This conclusion is strikingly supported by the attached table 15.1 (table 29 taken from the European Economy 2001, No. 72), which reports the annual rate of change of "nominal compensation per employee" for each of the 15 members of the European Union, the U.S., and Japan, a total of 17×42 or 714 observations. Of these, only 3 were declines (Japan, 1998–2000), and this, in a period where unemployment rates reached distressingly high level, as high as over 20 percent, as can be seen from table 15.2, unemployment rate (table 3 from the same source), and the same is largely true for prices. The experience in the USA in the post-war period is just a confirmation of that in Europe.

The problem that Keynes confronted in the General Theory, is that of analyzing the behavior of an economy in which wages are "downward rigid," i.e. will not fall or at best very slowly, in the presence of excess supply of labor (unemployment above the minimal frictional level).

III.2 A Generalization of the Notion of Market Equilibrium

Keynes's first step in the reconstruction is to recognize that the phenomenon of price/wage rigidity in the face of excess supply requires a major redefinition of the notion of "market equilibrium." In the classical model there were two alternative ways of characterizing market equilibrium: 1) when demand equals supply; and 2) when price has reached a stable level (at least in the short run). But when prices (wages) do not fall, despite the presence of an excess supply, the two definitions are not equivalent. Keynes chooses the second definition of equilibrium, which is applicable regardless of whether prices are rigid or flexible: *a market reaches equilibrium at a point where quantity and price stop adjusting, independently of whether at that point there exists an excess demand or supply*. And the choice is a very appropriate and general one, since it applies equally to the classical model of equilibrium with price flexibility or the Keynesian model of price rigidity: that is, equilibrium can always be operationally inferred from the (at least local) stationariness of price (and for Keynes, only from that).

Table 15.1
Nominal compensation per employee; total economy

(*national currency; annual percentage change*)

	B	DK	D([1])	EL	E	F	IRL	I	L	NL
1961	3.3	12.9	10.2	4.6	12.9	10.6	8.3	8.2	2.9	7.4
1962	7.4	11.1	9.1	6.6	15.2	11.6	8.5	13.4	4.8	6.8
1963	8.1	4.6	6.1	7.7	21.1	11.4	5.2	19.7	8.0	9.3
1964	9.9	10.7	8.2	13.3	13.7	9.2	13.7	11.6	13.3	16.5
1965	9.6	13.8	9.5	12.2	15.6	6.5	5.3	8.2	4.2	11.7
1966	8.8	10.2	7.6	12.6	18.1	6.0	8.5	8.0	5.0	11.1
1967	7.5	11.0	3.3	9.5	14.7	7.0	8.0	8.3	2.8	9.3
1968	6.4	10.1	6.7	9.3	8.8	11.9	10.6	7.6	5.9	8.6
1969	8.5	11.4	9.5	9.6	11.8	10.9	13.9	7.6	5.6	13.2
1970	9.3	11.3	16.0	8.8	9.4	10.4	16.8	15.3	15.1	12.6
1961–70	7.9	10.7	8.6	9.4	14.1	9.5	9.8	10.7	6.7	10.6
1971	12.2	12.1	11.4	8.0	13.6	11.3	14.8	13.7	7.8	13.9
1972	14.2	8.7	9.6	12.6	17.7	10.1	15.8	10.6	9.7	12.9
1973	13.5	13.7	11.9	17.2	18.3	12.4	18.8	17.5	11.4	15.6
1974	18.0	18.9	11.4	19.3	21.3	17.8	18.0	22.9	22.9	15.8
1975	16.5	14.4	7.0	20.3	22.5	18.7	28.9	21.0	12.4	13.6
1976	15.8	12.1	7.7	23.2	23.4	14.8	19.6	21.1	11.1	11.0
1977	9.1	10.1	6.6	22.0	26.8	12.2	14.9	20.6	9.9	8.5
1978	7.2	9.8	5.5	23.1	24.8	12.4	15.5	16.4	5.9	7.0
1979	5.8	10.1	5.8	22.1	19.0	12.8	18.9	19.0	6.7	5.6
1980	10.6	10.7	6.8	15.7	17.3	14.4	21.1	21.7	9.2	5.4
1971–80	12.2	12.0	8.3	18.3	20.4	13.6	18.6	18.5	10.6	10.9
1981	6.4	9.8	4.8	21.3	15.6	14.0	18.1	22.5	8.3	3.4
1982	7.0	12.3	4.2	27.5	13.8	14.3	14.2	16.1	6.9	5.9
1983	5.9	8.8	3.6	21.6	13.9	10.2	12.8	15.8	6.9	3.1
1984	6.9	6.2	3.4	20.8	10.2	7.4	10.7	11.7	7.1	0.3
1985	5.1	5.4	2.9	21.0	9.6	6.9	9.2	10.0	4.3	1.3
1986	3.9	5.0	3.6	12.0	9.7	4.4	5.1	7.5	5.7	2.1
1987	2.0	8.5	3.2	11.3	7.3	3.3	5.1	7.9	4.1	1.4
1988	2.2	5.6	3.0	20.1	7.6	4.4	7.0	8.2	3.4	0.9
1989	3.9	4.2	2.9	23.2	7.5	4.1	6.5	8.6	7.7	0.7
1990	7.6	4.1	4.7	17.9	10.2	5.1	4.2	10.4	5.2	3.2
1981–90	5.1	7.0	3.6	19.6	10.5	7.3	9.2	11.8	6.0	2.2
1991	7.5	3.9	5.9	15.5	10.3	4.1	4.3	8.8	6.5	4.5
1992	5.8	4.2	10.5	11.8	11.3	4.4	7.0	5.8	5.3	4.7
1993	3.7	2.3	4.1	9.8	7.4	3.0	6.4	4.6	5.4	3.3
1994	4.0	3.5	3.0	10.9	3.7	2.1	2.5	3.0	4.0	2.8
1995	2.4	3.5	3.6	12.9	3.6	2.6	2.0	4.2	2.3	1.9

Table 15.1 (continued)

(*national currency; annual percentage change*)

	B	DK	D([1])	EL	E	F	IRL	I	L	NL
1996	1.6	3.3	1.3	8.8	4.5	2.7	3.5	6.1	2.3	1.4
1997	2.9	3.5	0.3	13.6	2.1	2.3	3.6	4.0	3.1	2.2
1998	2.0	3.8	1.1	6.0	2.8	2.3	4.4	–1.5	0.9	2.8
1999	2.3	4.2	1.1	4.8	2.8	2.0	5.1	2.4	3.1	3.0
2000	3.2	3.9	1.2	5.0	4.0	1.6	7.9	2.9	5.1	4.1
1991–2000	3.5	3.6	3.2	9.8	5.2	2.7	4.6	4.0	3.8	3.1
2001	3.1	3.6	1.7	5.4	4.1	2.7	9.8	3.0	4.6	4.5
2002	3.0	3.7	2.5	5.7	2.9	3.1	8.8	2.9	3.7	4.5

([1]) 1961–91: D_90.

(*national currency; annual percentage change*)

	A	P	FIN	S	UK	EUR-II([1])	EUR-12([2])	EU-15([3])	US	JP
1961	9.3	5.8	7.9	8.1	6.5	9.6	9.5	8.7	3.2	13.2
1962	7.6	4.8	9.2	9.9	4.5	10.7	10.6	9.0	4.4	14.1
1963	7.9	8.1	10.8	9.4	4.9	11.7	11.6	9.7	4.0	13.2
1964	9.3	8.3	15.0	9.9	6.9	10.4	10.5	9.6	5.1	13.1
1965	9.1	11.0	9.6	8.6	6.7	9.2	9.2	8.7	3.7	11.9
1966	9.3	9.9	8.1	8.9	6.4	8.6	8.6	8.2	5.0	11.2
1967	9.5	13.7	9.7	9.2	6.2	7.2	7.2	7.1	4.3	12.1
1968	7.3	3.6	10.9	6.6	7.7	8.5	8.5	8.3	7.4	13.7
1969	8.3	10.0	7.4	6.9	7.1	9.9	9.9	9.2	7.3	15.8
1970	8.0	22.6	9.4	7.9	12.9	13.1	13.1	12.9	7.6	16.7
1961–70	8.6	9.7	9.8	8.5	7.0	9.9	9.9	9.1	5.2	13.5
1971	12.6	11.5	15.2	9.0	11.3	12.3	12.2	11.9	7.3	14.6
1972	11.0	15.8	14.6	8.5	13.0	11.3	11.3	11.5	7.3	14.2
1973	13.2	17.7	18.1	6.9	13.1	14.3	14.4	13.9	6.9	21.0
1974	13.9	35.1	24.0	12.9	18.7	17.4	17.5	17.6	8.2	25.7
1975	12.7	34.6	28.3	16.9	31.2	16.1	16.1	19.0	9.1	16.2
1976	9.2	24.5	16.3	17.9	14.8	14.9	15.0	15.0	8.3	11.1
1977	8.5	24.2	8.9	12.2	10.6	13.6	13.7	13.0	7.7	10.1
1978	9.0	18.8	6.1	10.9	13.3	11.9	12.0	12.2	7.7	7.5
1979	5.8	19.9	11.4	8.6	15.2	12.0	12.1	12.5	8.8	6.0
1980	6.6	25.7	13.1	10.9	19.8	13.2	13.2	14.3	10.2	6.5
1971–80	10.2	22.6	15.4	11.4	16.0	13.7	13.7	14.1	8.1	13.1
1981	8.1	21.0	13.5	9.2	14.1	12.1	12.2	12.6	9.5	6.4
1982	6.3	21.5	9.6	6.2	8.4	10.7	11.0	10.5	7.7	3.8
1983	4.7	21.8	10.0	7.9	8.7	9.3	9.4	9.3	5.4	2.2
1984	5.1	21.2	10.4	8.2	5.9	7.3	7.5	7.2	5.1	3.9

Table 15.1 (continued)

(*national currency; annual percentage change*)

	A	P	FIN	S	UK	EUR-II([1])	EUR-12([2])	EU-15([3])	US	JP
1985	5.3	22.5	10.3	7.5	7.6	6.6	6.8	6.9	4.6	2.9
1986	5.5	21.6	7.3	8.7	8.0	5.5	5.7	6.2	4.1	3.2
1987	4.0	14.4	7.7	7.0	7.4	4.7	4.8	5.4	4.2	3.3
1988	3.8	13.1	8.9	7.5	8.3	5.0	5.2	5.8	4.8	3.8
1989	4.6	15.2	10.2	11.3	9.3	5.1	5.3	6.2	3.2	4.8
1990	5.2	19.2	9.3	11.3	9.0	6.8	7.0	7.5	5.2	5.5
1981–90	5.3	19.1	9.7	8.5	8.7	7.3	7.5	7.7	5.4	4.0
1991	6.2	18.1	6.4	6.8	9.0	6.7	6.8	7.2	4.6	4.6
1992	5.8	16.3	2.2	3.9	5.3	7.7	7.8	7.2	5.3	1.3
1993	5.3	6.0	0.9	4.4	4.4	4.3	4.3	4.4	2.8	0.8
1994	3.8	5.6	3.1	4.8	3.4	3.0	3.1	3.2	2.4	1.8
1995	5.0	7.2	3.9	2.8	2.6	3.5	3.6	3.4	1.8	1.3
1996	1.5	4.9	2.7	6.8	3.7	2.9	3.0	3.2	2.5	1.1
1997	1.3	3.7	1.7	3.8	4.4	2.1	2.2	2.7	3.1	1.0
1998	3.4	3.7	4.1	3.3	4.9	1.4	1.5	2.2	4.4	–0.6
1999	2.9	4.2	2.7	1.3	5.2	2.0	2.1	2.6	4.0	–0.9
2000	2.1	5.6	4.0	7.0	4.1	2.4	2.5	2.9	4.8	0.7
1991–2000	3.7	7.4	3.2	4.5	4.7	3.6	3.7	3.9	3.6	1.1
2001	2.7	5.8	4.0	3.9	4.2	2.9	3.0	3.2	4.7	–1.2
2002	2.0	4.2	3.5	4.0	4.4	3.0	3.0	3.3	4.7	–0.1

([1]) PPS weighted; EU-15 excluding DK, EL, S, UK; 1961–91: including D_90.
([2]) PPS weighted; EU-15 excluding DK, S. UK; 1961–91: including D_90.
([3]) PPS weighted; 1961–91 including D_90.

III.3 The Implications of Wage-Price Rigidity

What are the implications of a rigid wage for the behavior of the price level? On this issue Keynes, somewhat surprisingly, by and large, goes along with the classics in assuming that producers behave as if they worked in a perfectly competitive environment, adjusting employment to the point where the marginal product of labor equals the real wage. This means that price equals marginal labor cost, i.e. marginal labor input times the wage. Marginal labor input in turn is supposed to be a non-decreasing function of output. Thus for any given output, the price P is proportional to the wage. But the ratio P/W would be an increasing (non decreasing) function of aggregate output. However now a day it is widely recognized that very few markets are perfectly competitive. For non-competitive markets a better approximation—embodied, e.g., in the so-called (short-run) fixed "*mark up*" model—is to assume that the price is roughly proportional to "unit

Table 15.2
Unemployment Rate; Total. Member States: Definition Eurostat

(*percentage of civilian labour force*)

	B	DK	D([1])	EL	E	F	IRL	I	L	NL
1960	2.3	1.3	1.0	5.6	2.4	1.4	5.8	5.7	0.0	0.7
1961	1.9	1.1	0.7	5.5	2.4	1.3	5.3	5.1	0.0	0.5
1962	1.7	1.2	0.6	4.8	2.0	1.6	5.4	3.6	0.0	0.5
1963	1.5	1.6	0.6	4.8	2.6	1.5	5.0	5.0	0.0	0.6
1964	1.4	1.2	0.5	5.0	2.2	1.6	5.1	5.4	0.0	0.8
1965	1.6	0.9	0.4	4.8	2.6	1.5	5.0	5.0	0.0	0.6
1966	1.7	1.1	0.5	5.0	2.2	1.6	5.1	5.4	0.0	0.8
1967	2.4	1.0	1.4	5.4	3.0	2.1	5.5	5.0	0.0	1.7
1968	2.8	1.0	1.0	5.6	3.0	2.6	5.8	5.3	0.0	1.5
1969	2.2	0.9	0.6	5.2	2.5	2.3	5.5	5.3	0.0	1.1
1970	1.8	0.6	0.5	4.2	2.6	2.4	6.3	6.1	0.0	1.0
1961–70	1.9	1.1	0.7	5.0	2.5	1.8	5.4	4.8	0.0	0.9
1971	1.7	0.9	0.6	3.1	3.4	2.7	6.0	5.1	0.0	1.3
1972	2.2	0.8	0.8	2.1	2.9	2.8	6.7	6.0	0.0	2.3
1973	2.2	0.7	0.8	2.0	2.6	2.7	6.2	5.9	0.0	2.4
1974	2.3	2.8	1.8	2.1	3.1	2.8	5.8	5.0	0.0	2.9
1975	4.2	3.9	3.3	2.3	4.5	4.0	7.9	5.5	0.0	5.5
1976	5.5	5.1	3.3	1.9	4.9	4.4	9.8	6.2	0.0	5.8
1977	6.3	5.9	3.2	1.7	5.3	4.9	9.7	6.7	0.0	5.6
1978	6.8	6.7	3.1	1.8	7.1	5.1	9.0	6.7	1.2	5.6
1979	7.0	4.8	2.7	1.9	8.8	5.8	7.8	7.2	2.4	5.7
1980	7.4	5.2	2.7	2.7	11.6	6.2	8.0	7.1	2.4	6.4
1971–80	4.6	3.7	2.2	2.2	5.4	4.1	7.7	6.1	0.6	4.4
1981	9.5	8.3	3.9	4.0	14.4	7.3	10.8	7.4	2.4	8.9
1982	11.2	8.9	5.6	5.8	16.3	8.0	12.5	8.0	2.4	11.9
1983	11.0	9.0	6.9	7.1	17.5	8.1	13.9	7.5	3.5	9.7
1984	11.1	8.5	7.1	7.2	20.2	9.7	15.5	8.0	3.1	9.3
1985	10.4	7.2	7.2	7.0	21.6	10.2	16.8	1.3	2.9	8.3
1986	10.3	5.4	6.6	6.6	21.2	10.3	16.8	9.0	2.6	8.3
1987	10.0	5.4	6.4	6.7	20.6	10.5	16.6	9.8	2.5	8.1
1988	9.0	6.1	6.3	6.8	19.5	9.9	16.2	9.8	2.0	7.6
1989	7.5	7.3	5.6	6.7	17.2	9.4	14.7	9.8	1.8	6.9
1990	6.7	7.7	4.8	6.4	16.2	9.0	13.4	9.0	1.7	6.2
1981–90	9.7	7.4	6.0	6.4	18.5	9.2	14.7	8.7	2.5	8.5
1991	6.6	8.4	4.2	7.0	16.4	9.5	14.7	8.6	1.7	5.8
1991	6.6	8.4	5.6	7.0	16.4	9.5	14.7	8.6	1.7	5.8
1992	7.2	9.2	6.6	7.9	18.4	10.4	15.4	8.8	2.1	5.6
1993	8.8	10.2	7.9	8.6	22.7	11.7	15.6	10.2	2.6	6.5

Table 15.2 (continued)

(*percentage of civilian labor force*)

	B	DK	D([1])	EL	E	F	IRL	I	L	NL
1994	10.0	8.2	8.4	8.9	24.1	12.3	14.3	11.1	3.2	7.1
1995	9.9	7.2	8.2	9.2	22.1	11.7	12.3	11.6	2.9	6.9
1996	9.7	6.8	8.9	9.6	22.2	12.4	11.7	11.7	3.0	6.3
1997	9.4	5.6	9.9	9.8	20.8	12.3	9.9	11.7	2.7	5.2
1998	9.5	5.2	9.3	10.9	18.8	11.8	7.5	11.8	2.7	4.0
1999	8.8	5.2	8.6	11.7	15.9	11.2	5.6	11.3	2.3	3.3
2000	7.0	4.7	8.1	11.0	14.1	9.5	4.2	10.5	2.2	2.8
1991–2000	8.7	7.1	8.1	9.5	19.6	11.3	11.1	10.7	2.5	5.4
2001	6.5	4.6	7.8	10.5	12.8	8.5	3.8	9.8	2.0	2.6
2002	6.1	4.5	7.1	9.9	11.9	7.8	3.5	9.3	1.8	2.4

([1]) 1960–91: D_90.

(*percentage of civilian labor force*)

	A	P	FIN	S	UK	EUR-II([1])	EUR-12([2])	EU-15([3])	US	JP
1960	2.5	1.7	1.8	1.7	1.4	2.5	2.6	2.3	5.5	1.7
1961	1.9	2.0	1.5	1.5	1.2	2.2	2.3	2.1	6.7	1.4
1962	1.9	2.3	1.6	1.5	1.7	1.9	2.0	1.9	5.5	1.3
1963	2.1	2.4	1.8	1.7	2.1	1.8	1.9	2.0	5.7	1.3
1964	2.0	2.5	1.8	1.6	1.4	1.9	2.0	1.9	5.2	1.1
1965	1.9	2.5	1.7	1.2	1.2	2.1	2.2	2.0	4.5	1.2
1966	1.8	2.5	1.8	1.6	1.1	2.2	2.3	2.0	3.8	1.3
1967	1.9	2.5	3.5	2.1	2.0	2.7	2.8	2.6	3.8	1.3
1968	2.0	2.6	4.8	2.2	2.1	2.8	2.9	2.7	3.6	1.2
1969	2.1	2.6	3.4	1.9	2.0	2.5	2.6	2.4	3.5	1.1
1970	1.4	2.6	2.3	1.5	2.2	2.4	2.4	2.3	4.9	1.1
1961–70	1.9	2.5	2.4	1.7	1.7	2.3	2.3	2.2	4.7	1.2
1971	1.3	2.5	2.7	2.5	2.7	2.6	2.6	2.6	5.9	1.2
1972	1.2	2.5	3.1	2.7	3.1	2.9	2.8	2.8	5.6	1.4
1973	1.1	2.6	2.8	2.5	2.2	2.8	2.7	2.6	4.9	1.3
1974	1.3	1.7	2.1	2.0	2.0	2.9	2.9	2.7	5.6	1.4
1975	1.0	4.4	2.7	1.6	3.2	4.2	4.2	3.9	8.5	1.9
1976	1.8	6.2	3.9	1.6	4.8	4.7	4.6	4.6	7.7	2.0
1977	1.6	7.3	5.9	1.8	5.1	5.0	4.9	4.9	7.1	2.0
1978	2.1	7.9	7.3	2.2	5.0	5.4	5.3	5.1	6.1	2.2
1979	2.1	7.9	6.0	2.1	4.6	5.7	5.6	5.3	5.8	2.1
1980	1.9	7.6	4.7	2.0	5.6	6.1	6.0	5.8	7.1	2.0
1971–80	1.6	5.1	4.1	2.1	3.8	4.2	4.2	4.0	6.4	1.8
1981	2.5	7.3	4.9	2.6	8.9	7.3	7.2	7.4	7.6	2.2

Table 15.2 (continued)

(*percentage of civilian labor force*)	A	P	FIN	S	UK	EUR-II([1])	EUR-12([2])	EU-15([3])	US	JP
1982	3.5	7.2	5.4	3.3	10.3	8.6	8.5	8.7	9.7	2.4
1983	4.1	8.2	5.5	3.7	11.1	8.9	8.9	9.1	9.6	2.6
1984	3.8	8.9	5.2	3.3	11.1	9.8	9.7	9.7	7.5	2.7
1985	3.6	9.2	5.0	2.9	11.5	10.1	10.0	10.0	7.2	2.6
1986	3.1	8.8	5.2	2.7	11.5	10.0	9.9	9.9	7.0	2.8
1987	3.8	7.3	4.8	2.2	10.6	10.0	9.9	9.7	6.2	2.8
1988	3.6	5.9	4.2	1.8	8.7	9.6	9.6	9.1	5.5	2.5
1989	3.1	5.2	3.1	1.6	7.3	8.9	8.8	8.3	5.3	2.3
1990	3.2	4.8	3.2	1.7	7.0	8.2	8.1	7.7	5.6	2.1
1981–90	3.4	7.3	4.7	2.6	9.8	9.1	9.0	5.0	7.1	2.5
1991	3.4	4.2	6.6	3.1	8.8	8.1	8.1	8.1	6.8	2.1
1991	3.4	4.2	6.6	3.1	8.8	8.2	8.2	8.2	6.8	2.1
1992	3.4	4.3	11.7	5.6	10.0	9.2	9.1	9.2	7.5	2.2
1993	4.0	5.7	16.3	19.1	10.5	10.8	10.8	10.7	6.9	2.5
1994	3.8	6.9	16.6	9.4	9.6	11.6	11.5	11.1	6.1	2.9
1995	3.9	7.3	15.4	8.8	8.7	11.3	11.2	10.7	5.6	3.1
1996	4.3	7.3	14.6	9.6	8.2	11.5	11.5	10.8	5.4	3.4
1997	4.4	6.8	12.7	9.9	7.0	11.5	11.5	10.6	4.9	3.4
1998	4.5	5.2	11.4	8.3	6.3	10.9	10.9	9.9	4.5	4.1
1999	4.0	4.5	10.2	7.2	6.1	9.9	10.0	9.2	4.2	4.7
2000	3.7	4.2	9.8	5.9	5.6	9.0	9.1	8.3	4.0	4.7
1991–2000	3.9	5.6	12.5	7.7	8.1	10.4	10.4	9.9	5.6	3.3
2001	3.4	4.6	9.1	5.2	5.3	8.4	8.5	7.7	4.6	4.7
2002	3.2	5.1	8.4	5.0	5.1	7.8	7.9	2.2	4.7	4.8

([1]) EU-15 excluding DK, EL, S, UK; 1960–91: including D_90.
([2]) EU-15 excluding DK, S, UK; 1960–91: including D_90.
([3]) 1960–91: including D_90.

labor cost" i.e., to the e wage (adjusted for productivity trend). Thus we take the mark up, at any point of time, as fixed—i.e., independent of output, (though it will tend to increase in time). This approximation implies that, if at any time the wage is fixed at Wo, the price itself is fixed as some corresponding multiple P_0.

We find it convenient to rely on this approximation which does not materially affect the substance of the argument, but greatly simplifies the exposition, for when combined with the assumption of a fixed wage Wo, it enables us to treat price P in equation (15.3) as a constant, P_0. The demand for money then becomes

$$\mathbf{M^d = kP_0X} \tag{15.4}$$

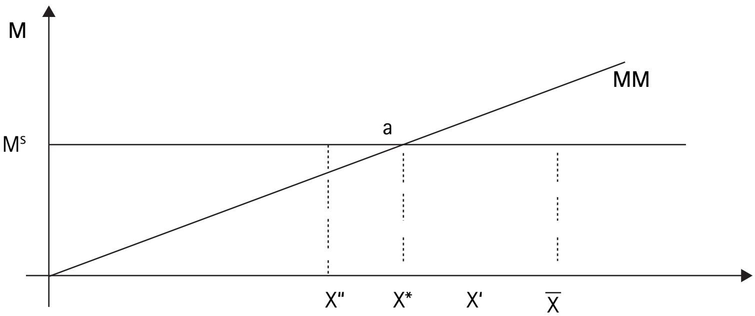

Figure 15.3
Classical Money demand (fixed Price level)

It is evident from (15.4) that, assuming for the moment that **k** is a constant, (in line with the classical model), at a given time the quantity demanded depends *only on the level of Real Income*, *X*. The demand function is reported in figure 15.3 as the straight line MM with slope of $\mathbf{kP_0}$. On the vertical axis we have marked with $\bar{X}$ the output corresponding to full employment of the labor force. The money supply can again be considered as exogenous, and therefore in figure 15.3 is represented as an horizontal line at the height $\mathbf{M^s}$. The equilibrium of the Money market is given by the condition that the demand for money equal the supply, or zero excess demand

$$\mathbf{KP_0X - M^s = 0}$$

But since KP_0, and M^s are exogenously given at any point of time, it is apparent that the only variable that affects the excess demand is output X. Thus the money market is cleared not by the classical price level, but by output. The market clearing output is given by

$$\mathbf{X^* = (M^s/P_0)V} \tag{15.5}$$

Graphically the money market reaches a Keynesian equilibrium at a point (a) where the demand for money equals the supply or existing stock. In the graph the *equilibrium income, X* is below the full employment income, X″*; nevertheless (a) represents an equilibrium because, in the money market, the demand for and the supply of money are equal, while in the labor market there is unemployment or excess supply, but because of wage rigidity, the wage and hence P is stationary. Thus **X*** can be characterized as a *Keynesian under-employment equilibrium*.

But to establish that X* is indeed an *"equilibrium," stable, and unique* it is not enough to show that, at that level of output, the demand and supply of money are equal: one must also show that, no other value of **X**, would be sustainable. Keynes's masterly contribution consists in demonstrating that when X differs from X*, and there is therefore an excess demand (or excess supply) of money, a mechanism is triggered *m* that pushes **X** toward **X***, reducing the excess demand (or supply) until it is eliminated, when X equals X*.

In order to carry out this demonstration, he had to "invent" and elaborate a series of concepts and interactions that were either new or neglected by the pre-Keynesian theory, and in particular by the classical theory. In the process, he gave rise to a new branch of economics, known as *Macroeconomics*, including a sub-branch, *international macroeconomics and Finance*.

IV The Keynesian Mechanism

The foundations of the Keynesian construction are provided by four major building blocks: Liquidity Preference, The Investment Function, The Consumption Function, and The Investment Multiplier.

IV.1 Liquidity Preference

Liquidity Preference is generally identified with the proposition that the demand for money depends also on the interest rate, and is frequently regarded as an interesting minor refinement to the theory of money demand. In reality it brings to light a very fundamental misconception in the classical monetary theory. For that theory—explicitly or implicitly—assumes that money is used exclusively to buy commodities, and hence its price is its purchasing power in terms of commodities. But, it fails to recognize—that, besides the market for commodities, there is another market of overwhelming importance in a developed economy: the market for assets, especially intangibles. Thus, it neglects the fact that money is also extensively used (and growingly so) to acquire (and liquidate) *financial assets*, which represent claims to future cash returns. These may be characterized as instruments whose attractiveness depends on the nature of their return. In particular, reducing current cash holdings by a buck, one can acquire a buck's worth of, say, (short-term) "bonds" yielding a return measured by the interest rate. This transaction can be thought of as the exchange of a buck now for an instrument promising (1 + r) bucks next period. Similarly, one can acquire a spot buck at the cost of 1 + r bucks next period by selling a bond from one's portfolio or by *borrowing at the rate r* from the market or an intermediary. Thus (1 + r) is the quantity of money that has to be paid in the next period, per unit of money delivered in the current period, or the price of "money" (now) in terms of money next period. The classics refer to 1/P as the purchasing power or price of money in terms of commodities: 1 + r can then be referred to as the purchasing power of money over future money, or also as the *opportunity cost* of (holding) money. A rise in the opportunity cost will induce agents to economize on cash holdings, reducing the amount of cash demanded per unit of transaction, or k of equation (15.4). Thus, Keynes generalizes the classical demand for money to:

$$\mathbf{M^d = k(r)P_0X} \tag{15.4a}$$

Here k(r) is a decreasing function r, while the velocity V(r), increases as r rises.

There are of course many instruments representing claims to future money, each with its own return or interest rate, just as there are many purchasing powers in terms of individual commodities, but we usually summarize the purchasing power of money in terms of a broad price index. Similarly we generally use, as the representative interest rate in terms of which we measure the opportunity cost of money, the return on short-term risk-less debt instruments, (or the short-term return expected from holding longer term instruments—which should be equivalent, to a first approximation, except for uncertainty).

Liquidity Preference, understood as the recognition of the role of financial assets in the demand for money, far from being an elegant side issue, has a crucial role to play in the transmission mechanism from money to the rate of employment.

In order to bring out this role, let us suppose that $\mathbf{X}$ exceeds $\bar{X}$, as $\mathbf{X'}$ in figure 15.3, e.g., because the money supply has been reduced. At X′ there is an excess demand for money, that is the public holds less money in their portfolio of assets than they actually need or would like to have, at the current interest rate. According to the classical model, the public endeavors to increase its cash balances by offering discounts and even reducing net purchases at the expense of consumption, thus putting downward pressure on prices. The price decline reduces the nominal demand to match the supply (or increases the real money supply to match the increased demand). Keynes instead suggests a much more credible and realistic response: if the private sector happens to posses less money in its portfolio than the desired quantity, at the current interest rate, its response will be to sell some of the liquid assets that it holds, such as short-term bonds, or to borrow by selling newly created debt instruments (to banks or to the market). This very consideration explains why, typically, the private sector—families and enterprises keep stocks of liquid assets—short-term risk-free asset—and credit lines, that permit satisfying unexpected needs for ready cash, without having recourse to distress liquidation of goods, or to reduce consumption. It should be obvious to anyone familiar with the life-cycle model that being short of cash calls for an adjustment of the portfolio, and not for abandoning the smoothing of consumption.

For the purpose of our synthetic construction, the essential implication of Liquidity Preference is: What happens when there is an excess demand for money and the public tries to sell assets, or demands more credit from banks, in order to acquire money today in exchange for money tomorrow? Evidently the interest rate r, or the price of spot money, will increase. The first impact that the excess demand will have is probably on short term interest rates on risk-less loans like Treasury bills which are highly liquid, being traded in near-perfect markets. But, just as commodity prices tend to move together so will return on different instruments. Thus, the increase of the short term rate will spill over into longer term rates, though with some difference reflecting maturity and expectations of future returns (term structure effects).

We arrive therefore at the first novelty of the Keynesian mechanism: in the classical model an excess demand for money tends to reduce prices pushing the demand for money down toward the available supply; but according to Keynes this effect is at best slow and uncertain, while its sure effect will be to increase promptly the price of money in terms of future money, i.e. the interest rate. This increase, in the opportunity cost of holding money, in turn, will have two major effects: first, it will tend to reduce the demand for money, for a given value of transactions, thereby alleviating the excess demand itself, and second, it will affect investment outlays as is described by the second fundamental relation.

IV.2 The Investment Function

The essential characteristic of this function is that when the cost of capital increases, investment declines. For example, when the interest rate on mortgages decreases, the annual carrying cost of a given house decreases. Therefore, more and more people desire homes (or better homes), and construction activity or investment in houses—one of the most important components of the national gross investment—increases. The cost of capital is not synonymous with the interest rate; nonetheless one can use the interest rate as an indicator of the cost of capital even though the relation between the two may be subject to shift over time (see below).

The sensitivity of investment to the interest rate has been long debated. Keynes himself tended to be “interest skeptic,” and today in Europe, it is fashionable to claim that the cost of money has scarce influence on investment. The excuse for not adopting more expansive monetary policies aimed at lowering interest rates is that they would have little effect on unemployment, but would have the undesirable effect of increasing inflation. This view must be regarded as “pre-Keynesian nonsense” or plain irrationality. Those that understand economics and rational behavior know that, money might conceivably affect the price level but only by increasing demand and employment. A direct effect is credible only to the extent that opinion leaders and the public have been brainwashed into the irrational belief that an increase in the money supply will cause prices to rise, independently of any effect on the real economy. In this case, of course, firms might respond to an increased money supply by immediately raising prices and workers (or trade unions) by insisting on immediate wage increases; regardless of the status of commodity and labor markets. But in this case the opinion leaders (especially in the banking sector) should take great pains to explain to the public the basis and evidence in support of their irrational beliefs.

In fact, the empirical evidence, at least in the USA, strongly supports the conclusion that the cost of capital, which depends on the interest rate and on fiscal parameters, is a fundamental variable that explains investment. This result is consistent with econometric analysis in many countries and is confirmed by the recent vigorous and strong economic growth in the USA, which was started with low interest rates from 1992 to 1996. On the other hand, Europe's monetary policy orchestrated first by the Bundesbank and then by its successor, the European Central Bank, has been dictated by the obsession with inflation and "credibility" instead of focusing on the encouragement of investments. Investment has been very sluggish and lagged way behind those in the US, and so has economic development, in line with Keynesian predictions. Some evidence to this effect is presented below (Modigliani and Ceprini, 2000).

The essential implication of the investment function is that investment (I), is a decreasing function of the cost of capital, which we identify with its major component namely the interest rate, (r) or:

$$I = I(r).$$

IV.3 The Saving (and Consumption) Function

This function describes the relationship between Income (X) Consumption (C) and Saving ($S = X - C$). The determinants of Saving received a good deal of attention in the General Theory, but important further developments have occurred since. According to a broadly accepted basic view, first formulated in the Life-Cycle Hypothesis (Modigliani, 1954, 1956) and in the Permanent Income Hypothesis (Friedman, 1957), consumption is a relatively stable fraction of *permanent* income (life resources), mostly influenced by its long-run growth, but is not very sensitive to *transitory* fluctuations in earnings; this means that the marginal propensity to consume (c) with respect to short-run, cyclical variation of income, tends to be relatively small, and therefore the corresponding propensity to save ($s = 1 - c$) tends to be large. This behavior has the effect of stabilizing the economy against short-term shocks, an effect that is markedly increased by the presence of income taxation (see below).

IV.4 The Saving—Investment Identity. The Multiplier

At the heart of the Keynesian contribution there is the "famous" and "powerful" proposition that saving is "equal" to investment. It is famous because nobody had paid any attention to it and in the beginning people had a hard time understanding or accepting it. Yet it follows directly from the very simple consideration that one can distinguish two basic components of demand, consumption (C), which is responsive to current income, and in investment (I), which is not directly related to current income but mainly responds to interest rates (and the expectation of future demand growth). Total aggregate demand is then the sum C + I, and in equilibrium, this must coincide with aggregate output, X; thus equilibrium requires

$$X = C + I. \tag{15.5}$$

Subtracting C from both sides of the above equilibrium condition we find

$$I = X - C = S \tag{15.5a}$$

These two Keynesian equations have some powerful implications. The first tells us that aggregate output (and employment) is determined *not by the economy's capacity* to produce at full employment, $\bar{X}$, as claimed by the classical model, but by *aggregate demand*. The second tells us that aggregate demand (and hence output) is determined by investment (within the limit of capacity and the effective labor force). Indeed S (or X − C) depends on X, and for every I there will be a unique level of output that will result in a rate of saving matching I, that is, in a level of output matching demand (clearing the commodity market). It will be presently shown that, with wage rigidity, there is no mechanism to insure that aggregate investment will be such as to result in aggregate demand equal to $\bar{X}$.

In particular, a change in investment will give rise to a change in income such as to produce a matching increase in saving. The ratio of the rise in income to the rise in investment is the famous Keynesian "investment multiplier." It is equal to the reciprocal of the so-called "marginal propensity to save" (mps)—the increase in saving generated by a dollar increase in income. Thus if the mps is say 1/4, (an additional dollar of income raises saving by 25 cents) the multiplier is 4. A unit increase in investment will result in an increase in income of 4, for this will increase saving by $1/4 \times 4 = 1$.

The "multiplication" comes from the fact that when investment increases by one the increase in saving by one is accompanied by an increase in consumption by three units, and therefore total income rises by four units. Another way to explain the multiplier is that a new investment is a direct source of employment, and the newly employed will spend on consumption and therefore will increase employment in those industries that produce the goods they want to consume, and so on. (It is necessary to clarify that this identity is valid only in a closed economy, without a government with tax power, public expenditure, or public saving. But the identity is easily generalized—see below.)

V The Under-Employment Equilibrium—Synthesis

V.1 The Four Quadrants Analysis

Our remaining task is to show how the interaction of the four basic blocks produces a series of results which are typically Keynesian and anti-classic, including the crucial result, namely the demonstration of the existence of stable underemployment equilibria, such as that illustrated by X* in figure 15.3. I propose to develop my argument in graphical form relying on a construction known as the four quadrant analysis. It is a bit cumbersome but in my view very helpful for a solid understanding of the argument. The four quadrants are needed to accommodate the four variables of the model, Income X, Investment I, Saving S, and the Interest Rate r.

In principle, all of these variables are measured in real terms, but because of the maintained assumption that prices are rigid (in the sense that they do not respond to the remaining variables), real variables are indistinguishable from nominal. (This holds also for the interest rate, since fix-prices excludes expectations of inflation or deflation.) The four variables are measured along the four semi-axes that extend outward from the origin, as shown in the graph. Thus X is measured along the vertical axis extending upward from the origin; r is measured along to the horizontal axis, extending right-ward from the origin; and similarly I and S are measured along the remaining two semi-axes. As a result we end up with four quadrants that on the graph are labeled, counterclockwise, I to IV.

V.2 Income Depends on Investment

We begin our analysis by utilizing quadrant III, IV, and I to establish that output is determined not by the classical "capacity to produce" but by the rate of investment I (or equivalently, by aggregate demand C + I). Quadrant III gives a graphical representation of the last block (section IV.4) to the effect that in equilibrium, investment must equal saving. This relation is readily represented by means of the straight line through the origin of slope one. It says that to any rate of investment' say I* in the figure, there corresponds a saving rate of equal amount. S* on the S axis of quadrant III.

We next move to quadrant IV, in which we graph the saving function of block 3, which describes the relation between S and income X, measured on the corresponding axis of quadrant IV. Here we must make a distinction between the short run (cyclical) and the long run saving function. As suggested earlier, in the long run saving tends to be proportional to income, with a proportionality rising with the growth trend. This long run relation is represented in Quadrant III by the line S′S′, through the origin, with slope, say s, determined by the long run growth trend: s is also the long run average and marginal propensity to save. Next, along the X-axis mark a height $\bar{X}$, which is equal to the trend value for the current period. Then to $\bar{X}$ will correspond a point $\bar{S}$ on the long run function S′S′; but note that $\bar{S}$ is also a point on the short run saving function in the sense that is the saving rate that would prevail if current income happened to be $\bar{X}$.

We can now readily establish that the short run saving function must be given by a curve like SX in the graph, going through $\bar{S}$ but with an appreciably greater slope. Indeed, consider an income lower than $\bar{X}$ like X* in the figure. Clearly; because X* − $\bar{X}$ represents a negative transient deviation from trend, we can expect that a relative small portion of that decline will result in a decline in consumption, while most of it will be absorbed by a decline in saving. In short, SX will tend to markedly steeper than S′S′ because its slope is the "marginal propensity to save out of transitory income," which can be quite high relative to the long-run propensity which is the slope of S′S′. Note that if income is well below trend, saving may fall to zero, or even become negative, but this can only be a transient phenomenon.

From the saving function, we can see that to the saving rate S* there corresponds, on the X axis, a level of output X*, the unique level that results in the saving S*. Thus I* determines the level of output X* that clears the commodity and capital market, in the sense that demand, C + I equals output or income. Clearly a change in I would require a change in S and hence in X, by the "multiplier" times the change in I, and the (short run) multiplier is the reciprocal of the slope of the saving function (the marginal propensity to save).

The fact that the short-run saving function is steep (high propensity) has an important stabilizing effect on the economy. This can be readily understood by remembering that the multiplier is the reciprocal of the marginal propensity to save; Failure to distinguish between the average and marginal long and short run propensity to save lead to greatly exaggerate the potential instability of income in response to cyclical variations in investment: with a consumption income ratio averaging around say 90 percent the multiplier is 10; but a short run cyclical decline in income may reduce consumption a lot less, perhaps by 1/2 which implying a short-run multiplier of only 2. This stabilizing effect is markedly increased by the presence of income taxation (see below).

In figure 15.4, we report some evidence in support the Keynesian proposition that investment largely determines aggregate output and employment. It relates

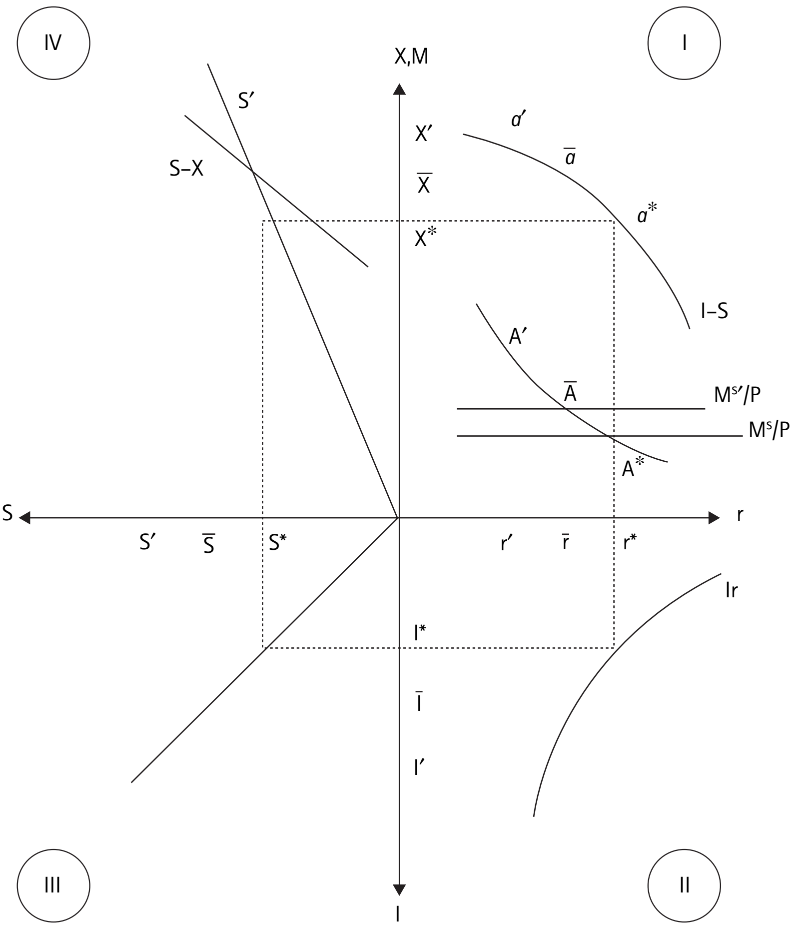

Figure 15.4
The Keynesian mechanism (a gospel)

to the Euro 15 countries over the period 1973 to 2001, and was presented at a hearing of the European Parliament in Brussels on April 4th, 2000. The Community provides a very useful case study because of the wide variations in both unemployment and investment rates during that period.

V.3 Investment and Output Depend on The Cost of Capital

As can be seen from the investment function I-I of Quadrant II, investment is a decreasing function of the cost of capital proxied by the interest rate, relying on the convenient approximation that the risk premium is either a constant or bears a stable relation to r, so that I can still be represented as a function r. It must be understood however that the I-r curve can be shifted around by changes in time in a variety of variables, including the risk premium (and taxation). Furthermore, the risk premium also accounts for one important feature of I-I: although I rises as r falls, there is a limit to how far it can be expected to rise. This is evidenced in Quadrant IV by the finite value of investment I corresponding to a zero interest rate. For even if the market r can be bid down to zero (which is most unlikely), the required return on investment will still be positive, equaling the risk premium, and one should expect that the volume of investment returning at least the risk premium should be very finite.

When we combine the result that X is determined by I and I by r, we can conclude that for every r there is an aggregate output X that clears the capital market and the commodity market. This critical relation is represented in Quadrant I. by the curve labeled IS. To locate this curve choose any value of r on the r axis, e.g., r*. Now we know that to r* corresponds the output X*. Accordingly in Quadrant I, we plot a point with coordinates (r*, X*), marked a* in the figure; clearly this point is a point on IS. The rest of IS can be traced by choosing successive value of r and, for each, plotting a point at the height, measured on the X axis equal to the X corresponding to that r. To summarize then, IS shows for every r, the corresponding commodity market clearing output.

We may note in passing that in the classic approach the primary determinant of saving was supposed to be a supply of saving function, increasing with the rate of return, on the ground that a higher return to capital makes accumulation more desirable. But this simplistic view has been abandoned, and the effect of the return on capital on saving has become a highly controversial issue. The reason is the realization that a higher return makes lenders richer, encouraging consumption, and in the Life Cycle, households are typically lenders throughout their life. Thus, while the sign remains controversial, it is generally acknowledged that the quantitative effect is unlikely to be large. In any event, one can readily show that allowing in the saving function for some small effect of r would not change our basic conclusions.

V.4 The Underemployment Equilibrium

The next question then is what determines r? In the Keynesian system, it is the demand and supply of money, not the "classical" demand of funds for investment and supply of funds through saving. The Keynesian money demand equation is given by equation (15.4a), which restated in real form becomes:

$$M^d / P_0 = k(r)X$$

It depends on two components, k and aggregate output, X. Now k, the economy's demand for money per dollar of transaction, is a function of the cost of money r; it can be shown graphically in the lower portion of Quadrant IV. It is typically a proper fraction and it decreases with r because, as money gets more expensive, both individual and firms endeavor to reduce their cash requirements. We next note that the second component, X, is also a unique function r, given by the IS curve. Thus, the demand for money can be expressed as a function of the single variable, r:

$$M^d/P_0 = k(r)X(r) = M^d/P(r) \tag{15.5}$$

This function is graphed in Quadrant IV, with the label $M^d(r)$; it is the product of the two curves, k(r) and the IS curve. It is lower than IS because k is generally less than unity, and it is flatter because the k curve has a negative slope.

To close the system, we need the (real) money supply M^s/P_0, which is shown in the figure by the horizontal line, so labeled. The Keynesian equilibrium is at the point A*, at the intersection of the money demand and supply. It results in an equilibrium r, equal to r*, shown on the r axis, and the equilibrium output X*, is at a* on IS, at the intersection of IS and the perpendicular through r*. It is an equilibrium because all markets are in equilibrium: the commodity market because X* is on IS, and the money market because A* clears the money market. To be sure, the labor market does not clear; it exhibits instead an excess supply, since the equilibrium output X* is seen to fall short of full employment $\bar{X}$, but is in Keynesian equilibrium since wage rigidity assures the stability of price and quantity in that market. And X* is stable and unique in the sense that, given the parameters underlying the figure, any value of X except X* would give rise to excess demand and supplies in non-rigid price markets which would push the system back toward X*. As an illustration, an output larger than X*, e.g., the full employment output, would not be maintainable. Indeed, from the IS in Quadrant IV, we can see that $\bar{X}$ would require a value of $r = \bar{r} < r^*$, and that the combination would result in a money demand shown by point A on the money demand curve, corresponding to r. But at that point, the demand is well above the money supply, M^s/P_0: there is an excess demand for money which bids up interest rates reducing I and X toward X*. And this process would continue as long as X is above X*, resulting in an excess demand for money. In short, X* is the only stable equilibrium despite the presence of unemployment, since there is no mechanism to push X from X* toward $\bar{X}$. One conceivable mechanism would be some tendency for wages to decline, at least when unemployment is large and persistent. One cannot rule this out, but recent experience confirms that this is not an effective mechanism to stabilize employment. It is for this reason that we can refer to X* as a stable Keynesian underemployment equilibrium.

A return to full employment can be achieved only through appropriate policies and, in particular, through an appropriate increase in the money supply, which creates an excess supply of money lowering interest rates toward the full employment rate $\bar{r}$ and full employment output, corresponding to point a.

VI Clarifications and Amplifications

VI.1 Wage Rigidity is Asymmetric: Downward Only

The rigidity (or stickiness) of prices and wages postulated by Keynes is asymmetric—downward only. In the presence of unemployment and unutilized capacity—demand is lower than supply—wages and prices do not fall, (and in fact are likely to continue to rise with productivity); but in any event they don't fall quick enough in order to reestablish equilibrium. They might decline to reestablish it in what economists call the "long run," but as Keynes has wisely warned, "in the long run we are all dead." A gradual adjustment is not enough to avoid significant and prolonged unemployment.

The situation is different if demand exceeds productive capacity. In figure 15.5, let us suppose that the Money Supply is allowed to increase above the full employment level $M^{s\prime}$ so that it intercepts the Money Demand curve at a point

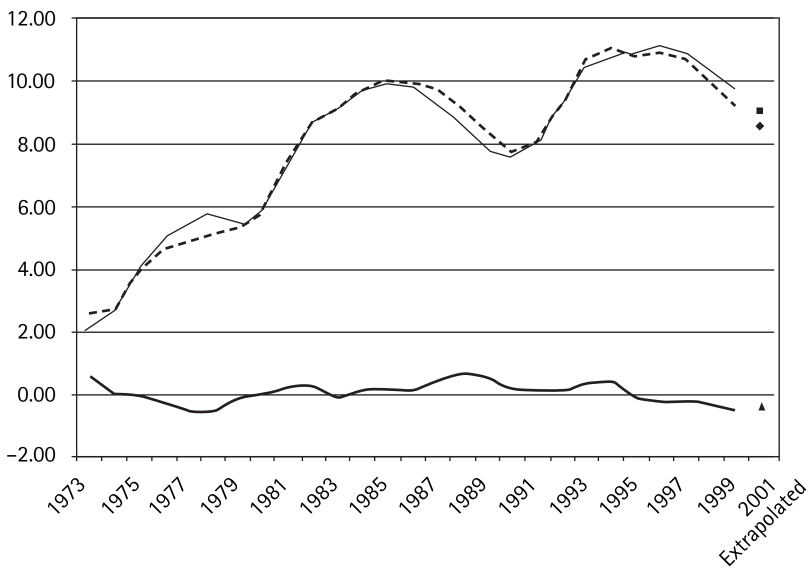

Figure 15.5
Unemployment vs. investment shortfall
Legend: grey broken: unemployment; grey solid: predicted; black solid: residuals Investment and unemployment in the Euro countries—1973–2001

above and to the right of b. To that point corresponds an aggregate demand X″ larger than full employment capacity output, $\bar{X}$. Demand exceeds capacity, firms bid up wages in an effort to attract more workers and will be able to pass on the higher wages onto higher prices. There will be inflationary pressures. If the quantity of money remains fixed, M^s/P will fall as P rises, and the **MM** curve will shift down (and r rise) until the equilibrium position is reached at b, where salaries will stop increasing. There will be price increases, but the increase will occur only once ("una tantum"). The risk is that the central bank tries to limit the increase of the interest rate needed to stop inflation, by increasing M as fast as P. In this case, inflation will continue and (possibly) eventually accelerate—the so-called (semi) vertical Phillips curve. (Semi-vertical because verticality (acceleration) occurs only for $X > \bar{X}$; but for $X < \bar{X}$ the Phillips curve is flat: the increase in wages is not appreciably affected by unemployment within an appreciable range.)

An interesting consequence of the asymmetry is the observed tendency, for at least a century, of the price level to follow a rising trend. Once it grows because of an excess demand or other reasons (wars) it never goes back.

VI.2 Keynes Unemployment—Wage Rigidity or Insufficient Money Supply?

Keynes's interpretation (as well as Hicks, 1937) has emphasized the fact that unemployment derives from an insufficient real supply of money. But what is the cause of the insufficiency? The classical point of view, supported by Keynes's critics, believes that it derives from a market failure, caused by the rigidity of wages. If wages-prices obeyed the classical flexibility postulate, then the unemployment in existence at point *a* would trigger a prompt decline in wages, and hence in prices, increasing the real money supply above M^s until it reached $\bar{M}^s$ and full employment. Thus, the postulate, if it were close to true, might ensure the continuous maintenance of the real money supply appropriate for full employment.

What is wrong with this criticism is that it is of no practical relevance for the purpose of preventing substantial and enduring unemployment, and for two reasons. In the first place, there are serious doubts that declining wages in response to unemployment could actually restore full employment (Keynes—chapter 19, 1936; Tobin, 1975). The reason for this conclusion is well known: a fall in nominal wages does not reduce real wages, which are the result of competitive mark-ups. Therefore, it could not increase consumption, since real income would be unchanged. The deflation would increase the real money supply and thereby lower nominal r; but not necessarily the real rate, which is equal to r + deflation. In the second place, there is no known device (perhaps short of totalitarian dictatorship), to force wages to behave according to The Postulate. Finally the experience of the last 50 years, cited earlier, makes it clear that there is no hope that labor will, on its own, behave according to The Postulate.

But, fortunately, Keynes has an operational and practical answer to offer: don't try to pursue the hopeless path of pushing wages down, try the easy way by pulling the nominal Money Supply up to the right level. As long as the nominal wage W is effectively rigid, unemployment is curable by increasing M. Since money is controlled by the central bank, unemployment is caused by an improper monetary policy. In my opinion this conclusion, although based on a "bare bones" model, contains the key explanation for the inexcusably high level of unemployment that Europe is suffering today.

VI.3 Too Little Money or Too High an Interest Rate?

Up to this point, we have stressed the fact that, with rigid nominal wages, employment can be controlled by monetary policy. But in reality, the Central Bank has an alternative tool for exercising its control over economic activity, namely the interest rate—though on condition of giving up control of the other tool, M. It is clear from figure 15.5 that if the bank were in a position to enforce a given value of r, like r* in Quadrant I, this would result in investment I*, hence income Y*, and point a* on the IS curve of Quadrant I.

But how can the central bank enforce r? Essentially by standing ready to buy or sell some specified kind of assets (e.g., Treasury Bills or other short-term loans) at a specified yield r*: that yield than becomes the "market rate." It can make that offer operational since it has "money" to buy whatever may be offered by the public at that rate, in virtue of the fact that it can pay with its IOU, which is money (or the source of money) for the economy. Similarly, if the public wants to buy, it will tender the bank's IOU that will be cancelled, reducing the money supply. Thus the Bank can make r* the market rate. But it is clear that by paying or receiving Central Bank cash in exchange for the asset, the public will cause a change in the nominal money supply. Thus the Bank is handing over to the public the decision of what quantity of M it wants to hold. That quantity will, of course, be the money demand corresponding to the point a* generated by the chosen r*. The money demanded and supplied will be an amount M* such that

$$M^*/P_0 = X^*(r^*)k(r^*)$$

Graphically, it is the money supply corresponding to the point A* on the money demand curve that results in a LM curve that goes through A* in Quadrant IV.

There is only one interest rate, $\bar{r}$ consistent with $\bar{X}$; it is the interest rate that the bank should choose, if it decides to use r as the instrument for policy. There have been long debates on whether it is more effective to fix short-run targets for the interest rate or the quantity of money, or to use some rules, which combine the two, taking into account the uncertainty of the real outcomes. In fact, this issue has played a central role in the famous "Monetarist controversy" of the 1960s and 70s with the Keynesian favoring an active interest policy and the monetarists a money supply policy, but of a very special passive type, namely keeping money growing at a forever constant rate. This position is an echo of the classical view that there is enough long-run flexibility in wages so that, no matter what the nominal money supply, the real supply will get to the full employment level. If so, interest policy has no role to play; there is no need for it, and, if the bank holds r at a level different from r, the result must be either unemployment (if too high) or inflation (if too low).

Central banks' behavior has reflected the monetarists controversy, tending at first to follow the Keynesians, then for a while shifting toward monetarism, but lately harking back to interest targets.

VI.4 The Liquidity Trap

In addition to demonstrating how wage rigidity can cause unemployment, Keynes can boast another result of great theoretical value, whatever its empirical relevance: a market economy using money as a medium of exchange may be unable to avoid involuntary unemployment.

This conclusion is readily established with the help of figure 15.4. It rests on one basic proposition: while, in general, an expansion of the money supply will reduce the price of money now in terms of money later—that is r—there is a limit to the extent to which r can be bid down. Without engaging in a lengthy argument as to what that limit might be, we can assert with confidence that it cannot be less than zero. Indeed, in a world in which money is essentially costless to store, who would ever want to exchange money now for less money tomorrow?

Back to figure 15.4: suppose that, at zero r, the rate of investment, I(o), is *smaller than the full employment rate of saving*. As noted earlier, this is perfectly possible, especially recalling the existence of a risk premium; this means that IS cuts the X axis at an X smaller than $\bar{X}$ or that to induce a rate of investment as large as full employment saving it would be necessary to pay a premium to investors in physical capital! Then output X must be smaller than full employment output, $\bar{X}$! No monetary or feasible interest policy can get the system back to full employment, since as a rule it cannot create a negative real interest fate, nor for that matter, could classical perfect wage flexibility. This is the Keynesian liquidity trap. Its empirical relevance is questionable, but some hold that it entrapped the U.S. economy at the bottom of the Great Depression and the Japanese economy of late.

VI.5 The Role of Fiscal Policy

So far in our analysis we have ignored the role of public expenditures and taxes, because we have not allowed for the existence of a public sector. Once we take into account the government, private saving is no longer equal to private investment, but to investment plus public deficit. This generalization can be readily accommodated by our figure 15.4, by simply replacing the 45o line of Quadrant III, with another, which is above it by the deficit. If the deficit is a constant independent of any other variable (a case convenient for exposition, though not very realistic) the new curve will be parallel to the old one. Consequently, to any given r and corresponding I there will correspond a higher saving and hence income. But this means that the deficit will have the effect of shifting up the IS curve by an amount equal to the deficit times the multiplier.

But what happens to X is finally dependent on the response of the monetary authority. In the elementary "text book" version, the Central Bank (CB) is (implicitly) assumed not to let the deficit affect the interest rate; then IS rises by the deficit times the multiplier; but since this raises the demand for money, the CB must be prepared to expand the money supply, to match so that the intersection of the new money demand and supply curves correspond to $\bar{X}$ (and the initial r) (all this is not shown in the graph which is already over clogged, but is easily constructed!). At the other extreme, suppose CB holds M^s unchanged, then X rises by a fraction of the deficit times the multiplier because the increased demand for money, with an unchanged supply, raises the interest rate, crowding out some I and offsetting a portion of the deficit, (the deficit could also displace some I as a substitute).

This result is important in that it implies that fiscal policy offers *an alternative tool to control employment.* To illustrate, suppose the economy finds itself in the underemployment equilibrium, *a**. Then, by creating an appropriate deficit through increased expenditure (or lower taxes), and with an accommodating monetary policy can be shifted until X* corresponds with $\bar{X}$. This opportunity could, obviously, be also exploited to counter a Liquidity Trap.

The Keynesian recognition of the possible employment effects of Fiscal Policy has given origin to a widely held view that the essence of Keynes is the advocacy of fiscal deficits. While this association might have had some basis in the early years of the General Theory in the immediate aftermath of the Great Depression, it is totally false at present. In particular my view, which I believe would be broadly acceptable to those who understand the fundamental Keynesian message, is that employment stabilization should be primarily the responsibility of monetary policies, except for automatic fiscal anti-cyclical stabilizers, while the main effect of non-cyclical budget deficits should be recognized as that of redistributing resources between generations.

VI.6 Nominal Versus Real Wage Rigidity

In the analysis presented so far, we have supposed a sticky nominal wage that does not decrease promptly in the presence of unemployment. This behavior was seen initially as a cause for pessimism. For one could not count on wage flexibility to provide an automatic offset to many potential internal and external shocks. But in the 50s and 60s, that perceived rigidity became a source of optimism, since it was believed that unemployment was principally of a Keynesian type (that is, caused by an insufficient real money supply) easy to control through the money supply (equivalent to real supply). The monetary authority (it was believed) held in their hands the instruments needed to insure employment stability, though there was a danger of an inflationary bias if the concern with unemployment dominated the concern with inflation. And this story is consistent with history from the 50s until the oil crisis—low unemployment but some inflation.

What happened in the 70s and 80s to cause the rising unemployment, reaching dramatic levels in the 90s (11.5 percent for the Euro countries), and to replacing that optimism with a wave of dark Euro pessimism on the feasibility of using economic policy to keep unemployment within acceptable limits? And how can these developments be reconciled with the Keynesian revolution and its promise to cure forever the problem of mass unemployment?

I have labored for many years searching for an answer to this intriguing question. But for lack of space, I must limit myself to summarize my findings by itemizing five major causes for the unfavorable developments.

Very high on the list is: i) the transition (probably temporary) from nominal wage rigidity to aggressive real wage targets, beginning in the late 60s but greatly (though temporarily) reinforced, between the mid-70s and mid 80s, as a result of: ii) the two great oil crises. During those crises, labor, confronted with the sharp rise in the price of energy, endeavored to maintain a real wage target by securing large increases in nominal compensations. But since productivity had not increased, there was a surge in unit labor costs, which was passed on into higher prices. That called for higher wages to maintain purchasing power which led to more price increases and so on, igniting a typical wage-price spinal.

Now consider the predicament of the bank: keeping the money supply constant would help tame the spiral but at the cost of a reduction of the real money supply leading to rising unemployment. But trying to maintain the real money supply by increasing the nominal one would have fueled the spiral, probably without regaining control of the real supply, now at the mercy of an upward flexible nominal wage level. This dilemma, between accommodating inflation or risking high unemployment, arises whenever there is endeavor to pursue a real wage target inconsistent with the mark-up target of firms, an event seen occasionally even before the oil crisis (see e.g., Italy and Spain in the 70s and 80s). In these circumstances, and in the absence of a political solution, the bank has no alternative but to check the growth of nominal money and accept the consequence: the so-called stagflation.

Other major causes of deterioration are: iii) some damaging labor market institutions (e.g., overprotection of the insiders), iv) mammoth Social Security contributions interacting with minimum wage legislation; and, last but not least, v) the many large errors in monetary policy, particularly in Europe since the mid-80s. They include the overly tight policies by the Bundesbank first and then ECB, which have kept targeting levels of output and employment greatly below full employment. This over-tightness, and its inconsistency with a very hardheaded, pro-cyclical fiscal policy, reflects a failure to understand Keynes teaching. And this failure is also reflected in the assigning to the ECB one single responsibility that of checking inflation, but no meaningful responsibility for unemployment, which is creeping down at a snail's pace.

VI.7 The Open Economy

An important amplification that has been omitted is that of an open economy. Unfortunately, that adds more complications then can be handled here. These are the complications (exchange rates, international flows of commodities and capital) have given rise to the new field of international macroeconomics and finance. But it is worth noting that the issue of price-wage rigidity, both nominal and real, still plays a major role even there, interacting with the two major exchange regimes, fixed and floating.

Concluding Remarks

I would like to bring this survey to an end with two brief remarks:

The content of this whole paper can be summarized in two short sentences. The classics (and European consensus) believe that demand (almost) always, adjusts spontaneously to the (feasible) full employment supply. Thanks to the flexibility of wages, activism can only end in inflation. Keynesians know that wages are stubbornly rigid downward and demonstrate that, under these conditions, supply adjusts to demand, which, therefore needs steering toward high employment by appropriate non-inflationary monetary (and as a last resort, fiscal) policies.

Let me end on a note of optimism about the prospects for mass unemployment: I have faith that the outlook, especially in Europe, is less grim than the recent past and the present. I believe that the actors are (slowly) learning from past mistakes, how to avoid them in the future, perhaps in part by a better understanding of Keynes's message.

Note

1. k reflects characteristics of the economy such as payment habits and technology, and can therefore be different for different countries or at different times (but does not depend on Y or M).

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IV

INTERVIEW

16

AN INTERVIEW WITH FRANCO MODIGLIANI

Interviewed by William A. Barnett and Robert Solow, November 5–6, 1999

Franco Modigliani's contributions in economics and finance have transformed both fields. Although many other major contributions in those fields have come and gone, Modigliani's contributions seem to grow in importance with time. His famous 1944 article on liquidity preference has not only remained required reading for generations of Keynesian economists but has become part of the vocabulary of all economists. The implications of the life-cycle hypothesis of consumption and saving provided the primary motivation for the incorporation of finite lifetime models into macroeconomics and had a seminal role in the growth in macroeconomics of the overlapping generations approach to modeling of Allais, Samuelson, and Diamond. Modigliani and Miller's work on the cost of capital transformed corporate finance and deeply influenced subsequent research on investment, capital asset pricing, and recent research on derivatives. Modigliani received the Nobel Memorial Prize for Economics in 1985.

In macroeconomic policy, Modigliani has remained influential on two continents. In the United States, he played a central role in the creation of a the Federal Reserve System's large-scale quarterly macroeconometric model, and he frequently participated in the semiannual meetings of academic consultants to the Board of Governors of the Federal Reserve System in Washington, D.C. His visibility in European policy matters is most evident in Italy, where nearly everyone seems to know him as a celebrity, from his frequent appearances in the media. In the rest of Europe, his visibility has been enhanced by his publication, with a group of distinguished European and American economists, of "An Economists' Manifesto on Unemployment in the European Union," which was signed by a number of famous economists and endorsed by several others.

This interview was conducted in two parts on different dates in two different locations, and later unified. The initial interview was conducted by Robert Solow

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at Modigliani's vacation home in Martha's Vineyard. Following the transcription of the tape from that interview, the rest of the interview was conducted by William Barnett in Modigliani's apartment on the top floor of a high-rise building overlooking the Charles River near Harvard University in Cambridge, Massachusetts. Those concluding parts of the interview in Cambridge continued for the two days of November 5 and 6, 1999 with breaks for lunch and for the excellent espresso coffee prepared by Modigliani in an elaborate machine that would be owned only by someone who takes fine coffee seriously.

Although the impact that Modigliani has had on the economics and finance professions is clear to all members of those professions, only his students can understand the inspiration that he has provided to them. However, that may have been adequately reflected by Robert Shiller at Yale University in correspondence regarding this interview, when he referred to Modigliani as "my hero."

Barnett: In your discussion below with Solow, you mentioned that you were not learning much as a student in Italy and you moved to the United States. Would you tell us more about when it was that you left Italy, and why you did so?

Modigliani: After the Ethiopian war and the fascist intervention in the Spanish Civil War, I began to develop a strong antifascist sentiment and the intent to leave Italy, but the final step was the close alliance of Mussolini with Hitler, which resulted in anti-Semitic laws, which made it impossible to live in Italy in a dignified way. At that time I had already met my future wife, Serena, and we were engaged. Her father had long been antifascist and preparing to leave Italy. When those laws passed, we immediately packed and left Italy for France. We spent 1939 in France, where we made arrangements to leave for the United States. We left in August 1939 for the United States on the very day of the famous pact between Hitler and Stalin, which led to what was the later attack by Germany on Russia. I came to the United States with no prior arrangements with a university. I wanted very much to study economics, and I received a scholarship from the New School for Social Research, thanks in part to the fact that the school had many prominent intellectual antifascists, and one of them, the renowned antifascist refugee, Max Ascoli, helped me to get the scholarship.

Barnett: Franco, I understand that after you had left Italy you returned to Italy to defend your dissertation. Can you tell us whether there were any risks or dangers associated with your return to defend your dissertation?

Modigliani: Yes, it is true that when we left from Rome to Paris, I had finished all of my examinations to get my degree, but I had not yet defended my thesis. In July of 1939, before leaving Paris for the United States, I wanted to have all my records complete, and I decided to go back to Rome to defend my thesis. That operation was not without dangers, because by that time I could have been arrested. I had kept my contacts with antifascist groups in Paris, so there was the possibility of being harassed or being jailed. Fortunately nothing happened. My father-in-law was very worried, and we had made arrangements for him to warn us of any impending perils by a code. The code was all about Uncle Ben. If he was not feeling well, we should be ready to go. If he was dead, we should leave instantly. We never needed to use that code, but I felt relieved when I was able to complete my thesis, and then late in August we left for the United States.

Barnett: The famous painter and sculptor, Amedeo Modigliani, was born in Livorno, Italy, in 1884 and died in Paris in 1920. Was he related to your family?

Modigliani: There is no known relation.

Solow: Franco, the first thing I want to talk about is your 1944 *Econometrica* paper, "Liquidity Preference and the Theory of Interest and Money." When you were writing it, you were 25 years old?

Modigliani: Yes, about that. I hadn't studied very much in Italy of any use. There was no useful teaching of economics. What was taught there was something about the corporate state. So all I picked up was at the New School of Social Research in New York with the guidance of Jacob Marschak.

Solow: When was that?

Modigliani: That was 1939 through 1941–1942.

Solow: So your main guide was Jascha Marschak.

Modigliani: Jascha Marschak was my mentor. We studied Keynes and the *General Theory* in classes with Marschak. I attended two different seminars, but in addition received a lot of advice and support from him. He suggested readings and persuaded me of the importance of mathematical tools, acquired by studying some calculus and understanding thoroughly the great book of the day by R.G.D. Allen, *Mathematical Analysis for Economists*, and studying some serious statistics (attending Abraham Wald lectures at Columbia); and last but not least he sponsored my participation in a wonderful informal seminar, which included besides Marschak people like Tjalling Koopmans and Oskar Lange. But unfortunately, to my great sorrow, Marschak in 1942 left New York for Chicago. He was replaced by another notable mind, Abba Lerner. I had a lot of discussions with him about Keynes. At that time, Abba Lerner was pushing so-called functional finance.

Solow: Yes, the famous "steering wheel."

Modigliani: Functional finance led me to the 1944 article. In functional finance, only fiscal policy could have an impact on aggregate demand. Therefore, it was an economy that belonged to what I later called the Keynesian case. I tried to argue with Lerner and to have him understand that Keynes did not say that. That was the origin of the 1944 article, trying to put Keynes in perspective.

Solow: Now, with Marschak or Lerner, had you read any of the earlier mathematical models of Keynesian economics, such as Hicks's, of course, or Oscar Lange's articles?
Modigliani: Well, I was familiar with the literature, and of course it had hit me, as is visible in my articles. Hicks's article on Keynes and the classics was a great article, and it was the starting point of my article, except that in Hicks the rigidity of wages was just taken as a datum, and no consideration was given to alternatives. It was just the one system, and fixed forever.
Solow: What he later called "a fix-price model."
Modigliani: Fixed price so that he could deal in nominal terms as though they were real. Money supply is both nominal and real.
Barnett: Prior to your arrival at MIT, you were at a number of American universities, including the New School for Social Research, the New Jersey College for Women (now Douglas College), Bard College of Columbia University, the University of Chicago, the University of Illinois, Carnegie Institute of Technology (now Carnegie Mellon University), and Northwestern University. Prior to MIT, what were the most productive periods for you in the United States?
Modigliani: The most productive period was unquestionably the eight years or so (1952 to 1960) spent at Carnegie Tech, with an exceptionally stimulating group of faculty and students, led by two brilliant personalities, the dean, G. L. Bach, and Herbert Simon, working on the exciting task of redesigning the curriculum of modern business schools, and writing exciting papers, some of which were to be cited many years later in the Nobel award: the papers on the life-cycle hypothesis and the Modigliani and Miller papers.
Barnett: I've heard that while you were teaching at the New School, you had an offer from the Economics Department at Harvard University, which at that time was by far the best economics department in the United States. But to the surprise of the faculty, you turned down the offer. Why did you do that?
Modigliani: Because the head of the department, Professor Burbank, whom I later found out had a reputation of being xenophobic and anti-Semitic, worked very hard and successfully to persuade me to turn down the offer, which the faculty had instructed him to make me. He explained that I could not possibly hold up against the competition of bright young people like Alexander, Duesenberry, and Goodwin. "Be satisfied with being a big fish in a small pond." Actually it did not take me too long to he persuaded. Then, after my meeting with Burbank, I had scheduled a lunch with Schumpeter, Haberler, and Leontief, who had expected to congratulate me on joining them. But they literally gave me hell for letting Burbank push me over. Nevertheless, in reality I have never regretted my decision. Harvard's pay at that time was pretty miserable, and my career progressed much faster than it would have, if I had accepted the offer.

Barnett: I understand that the great football player Red Grange (the Galloping Ghost) had something to do with your decision to leave the University of Illinois. What happened at the University of Illinois that caused you to leave?
Modigliani: In short it was the "Bowen Wars," as the episode came to be known in the profession. The president of the university brought in a new wonderful dean, Howard H. Bowen, to head the College of Commerce, which included the Department of Economics. But the old and incompetent faculty could not stand the fact that Bowen brought in some first-rate people like Leo Hurwicz, Margaret Reid, and Dorothy Brady. The old faculty was able to force Bowen out, as part of the witch hunt that was going on under the leadership of the infamous Senator Joseph McCarthy. The leader of the McCarthyite wing of the elected trustees was the famous Red Grange. I then quit in disgust with a blast that in the local press is still remembered: "There is finally peace in the College of Commerce, but it is the peace of death." My departure was greeted with joy by the old staff, proportional to their incompetence. But 40 years later, the university saw fit to give me an honorary degree!
Solow: Well, how do you look at the 1944 paper now? Would you change it drastically if you were rewriting it?
Modigliani: Yes! Not really in content, but in presentation. That is what I have been doing in my autobiography. I am revising that paper completely and starting from an approach which I think is much more useful. I am starting from the notion that both the classics and Keynes take their departure from the classical demand for money model, which is one of the oldest and best-established paradigms in economics. The demand for money is proportional to the value of transactions, which at any point can be approximated as proportional to nominal income (real income multiplied by the price level). The nominal money supply is exogenous. Therefore, the money market must reach an equilibrium through changes *in nominal income.* Nominal income is the variable that clears the money market.

Where then is the difference between classical and Keynesian economics? Simple: The classics assumed that wages were highly flexible and output fixed by full employment (clearing of the labor market). Thus the quantity of money had *no effect on output but merely determined the price level, which was proportional to the nominal money supply* (the *quantity theory of money*). On the other hand, Keynes relied on the *realistic assumption* that wages are rigid (downward). That is, they do not promptly decline in response to an excess supply of labor. Workers do not slash their nominal wage demands, and firms do not slash their wage offers, when unemployment exceeds the frictional level. What, then, clears the money market? Again, it is a decline in nominal income. But since prices are basically fixed, the decline must occur in real income and particularly in employment. When there is insufficient nominal money supply to satisfy the full employment demand for money, the market is cleared through a decline in output and employment. As Keynes said, the fundamental issue is that prices are not flexible.

Solow: Not instantly flexible.

Modigliani: That's right. They may very slowly respond, but a very slow adjustment of the real money supply can't produce the expansion of the real money supply needed to produce a rapid reestablishment of equilibrium. What, then, reestablishes equilibrium? Since wages and prices are fixed, the decline in nominal income can only occur through a decline in real income and employment. There will be a unique level of real income that clears the money market, making the money demand equal to the money supply.

Solow: No mention of the interest rate?

Modigliani: The interest rate comes next, as a link in the equilibrating mechanism. In fact, Keynes' unique achievement consisted not only in showing that unemployment is the variable that clears the money market; he also elaborated the mechanism by which an excess demand for money causes a decline of output and thus in the demand for money, until the demand matches the given nominal money supply. In the process of developing this mechanism, unknown to the classics, he created a new branch of economics: *macroeconomics.*

Macroeconomics, or the mechanisms through which money supply determines output (employment), stands on four basic pillars, with which, by now, most economists are familiar: (1) liquidity preference, (2) the investment function, (3) the consumption or saving function, and (4) the equality of saving and investment (properly generalized for the role of government and the rest of the world).

Liquidity preference is not just the fact that the demand for money depends on the interest rate; it brings to light the profound error of classical monetary theory in assuming that the price of money is its purchasing power over commodities (baskets per dollar) and that, therefore, a shortage of money must result in a prompt rise in its purchasing power (a fall in the price level). In reality, of course, money has many prices, one in terms of every commodity or instrument for which it can be exchanged. Among these instruments, by far the most important one is *"money in the future," and its price is money tomorrow per unit of money today*, which is simply $(1 + r)$, where r is the relevant interest rate.

Furthermore, experience shows that financial markets are very responsive to market conditions: Interest rates (especially in the short run) are highly flexible. So, if money demand is short of supply, the prompt reaction is not to liquidate the warehouse or skimp on dinner, forcing down commodity prices, but a liquidation in the portfolio of claims to future money (or a rise in borrowing spot for future money), leading to a rise in the terms of trade between money today and tomorrow—that is, a rise in interest rates. And this starts the chain leading to lower output through a fall in investment, a fall in saving, and thus in income and employment. It is this fall, together with the rise in interest rates, that reduces the demand for money till it matches the supply.

Solow: Yes, so the interest rate is a key price.

Modigliani: Actually it's one plus the interest rate. If you are short of money, and the system does not have enough money, the first thing it attempts is to get more spot money by either liquidating assets or by borrowing, which is borrowing money today against money tomorrow. Interest rates rise, reducing investment, and then comes the great equation: investment equals savings—an identity that is so far from the classical view that in the beginning they would not even believe it.

Solow: Right.

Modigliani: And income then adjusts so that the demand for money is finally equated to its supply. This will result in both a higher interest rate and a low r income. The two together will serve to equate the money demanded with the given supplied. And how much must interest rates rise or income decline? That depends upon the parameters of the system (demand elasticities).

Solow: But exactly! And Keynes's fundamental contribution then was to say that it's not the interest rate and the price level, but interest and real output.

Modigliani: Yes, precisely. I think this is the way to look at it. It is the output that adjusts demand and supply.

Solow: What you just described is maybe a different way of telling a story and saying what's important, but it's not fundamentally different from what's in the 1944 paper or in the ISLM apparatus.

Modigliani: Absolutely. But I suggest that to think of unemployment not as a transitory disease, but as a variable that clears the money market, is a useful and significant innovation. Unemployment is an equilibrating mechanism. It seems like a dysfunction, since we think that full employment is what an economy should produce. But unemployment is a systematic feature of an economy relying on money to carry out transactions. To avoid unemployment, it takes continuous care by either setting the right money supply or fixing the right interest rate. There is no other way to get full employment. There is nothing automatic about it.

Barnett: Some work on monetary policy has emphasized the possibility that the monetary transmission mechanism works through a credit channel. The implication of this research is that monetary policy may affect consumer and business spending, because it affects the quantity of credit available to agents, rather than the interest rate. This work is often motivated by the observation that the

interest elasticity of spending is too low to explain the large impact monetary policy appears to have on real economic activity. Do you think there is an important credit channel for monetary policy?
Modigliani: My attitude toward this question, about which I have done much thinking and some writing, is that in the end it is an empirical question, not an a priori question. It is entirely credible that monetary policy may work, in part, through changing the volume of credit supplied by banks in the form of commercial loans, as well as its cost. That way it may have the same effects as acting through market interest rates, but without necessarily producing large movements in interest rates. I think that future research will help in sorting this out. But the answer will not be perpetual, since the answer depends upon the structure of financial intermediaries and the laws regulating them.
Solow: Now one of the questions I've wanted to ask you, which I think you've already now answered, is what does it mean to be a Keynesian today? But I take it that what you just said is the essence of Keynesian economics, and by that definition you would describe yourself as a Keynesian.
Modigliani: Absolutely. I consider myself a Keynesian. Now as I think it over in this light, I consider Keynesian economics to be a great revolution, having a really tremendous impact, with tremendously novel ideas. Again I consider myself a Keynesian in the very fundamental sense that I know the system does not automatically tend to full employment without appropriate policies. Price flexibility will not produce full employment, and therefore unemployment is always due to an insufficiency of real money. But it must be recognized that there are certain circumstances under which the Central Bank may not be able to produce the right real money supply. For instance, the case of Italy was interesting. Unemployment there was due to the fact that real wages were too high, in the sense that they resulted in substantially negative net exports at full employment. Under those circumstances, if the Central Bank expanded the money supply to create more aggregate demand and employment, the balance of trade would run into nonfinancable deficits, and the Central Bank would be forced to contract. So, you're not always able to increase the real money supply. But that's not the case in Europe, where the money supply could be easily increased and the unemployment is largely due to insufficient real money supply.
Solow: You know I rather agree with you about that.
Modigliani: Interest rates are too high. There is not enough real money being supplied. This is not being understood. Keynes is not being understood. That's the main source of European unemployment. Some improvements in the labor market, such as more wage flexibility, could help, but would not get very far without a significant rise in aggregate demand (which at present would not significantly increase the danger of inflation).

Solow: Right, but you cannot get European central bankers to see that.
Modigliani: In Europe they accept the view that long-lasting unemployment contributes to the current high level because it reduces search by the unemployed, *causing long-term unemployment.* No, sir. That's a consequence of the too restrictive policy.
Barnett: You have argued that stock market bubbles sometimes are produced by misinterpreting capital gains as a maintainable component of current returns (a permanent addition to current profits). Do you believe that that phenomenon is going on now, or do you believe that current stock market valuations are consistent with the fundamentals?
Modigliani: I am very much interested and concerned about bubbles, and I believe that bubbles do exist. They are one of the sources of malfunctioning of the market mechanism. The essence of these bubbles is that indeed capital gains get confused with profits, and this results in the stock becoming more attractive, so people bid up the price, which produces more capital gains, and so on. I believe that indeed the stock market in the United States is in the grips of a serious bubble. I think the overvaluation of stocks is probably on the order of 25 percent or so, but, by the nature of the process, it is not possible to predict just when the whole thing will collapse. In my view, there will be a collapse because if there is a marked overvaluation, as I hold, it cannot disappear slowly.
Barnett: How does your research help us understand what has occurred over the past few years in the volatile economies of East Asia?
Modigliani: My view is that what has happened in East Asia is very much in the nature of a bubble, where expected high returns have attracted capital. The attraction of capital has held up exchange rates, permitting large deficits in the balance of trade; the influx of capital has supported the exchange rate making capital investment more attractive. So, you have a spiral until people realize that those returns are really not maintainable. I think it is important for the future of the international situation to set up systems under which bubbles cannot develop or are hard to develop, such as requiring reserves against short-term capital movements.
Solow: Now, I want to ask what's your current belief about wage behavior? How would you today model the behavior of nominal or real wages?
Modigliani: This, I think, is one of the fundamental issues that we face today, because in my model the wage and the price level are exogenous. Why is the price level exogenous? Because prices fundamentally depend upon wages, and wages are not flexible. Wages are certainly not responding mechanically to unemployment. So what do we do about wages? Well, I do think that some of this rigidity of wages is historical. It's very likely that in the nineteenth century the situation was different. In that century there was a greater role for competitive

industries such as agriculture. In any event, the wage is *the* fundamental component of the price level. What's going to determine wages? Well, we've come to a difficult period, mostly since unions in Europe have been very powerful. They've become unreasonable and pushed for higher and higher wages, nominal wages. But my view is that in the long run we'll have to reach the point at which the wage, the nominal wage, is negotiated in a general simultaneous settlement of wages and prices. Now that's what's happened in Italy.

Solow: Say some more about that.

Modigliani: What saved Italy from the tremendously disastrous situation that existed just before devaluation was the fact that workers agreed to fix nominal wages for three years together with a price program, so that real as well as nominal wages were set. To me, that is the future because I do not know what else to say about the price picture.

Solow: What you're saying is that Keynes's remark that labor cannot determine the real wage may turn out to be false because institutions change and permit bargaining over the real wage.

Modigliani: That's right. Yes, yes.

Solow: What's your current feeling about NAIRU, the nonaccelerating-inflation rate of unemployment?

Modigliani: I think it is true that if unemployment gets too low, then you will have accelerating inflation, not just higher inflation but higher rate of change of inflation. But I do not believe that if unemployment gets very high, you'll ever get to falling nominal wages. You may get very low acceleration of wages, or you may get to the point at which wages don't move. But I don't believe that high unemployment will give us negative wage changes.

Solow: We might even get falling wages for a while, but you would surely not get accelerating reductions in wages. No believer in the NAIRU ever wants to speak about that side of the equation.

Modigliani: That's right. So I think that these views are consistent, in the sense that left alone there may be a tendency for the system to be always in inflation. The Central Bank can pursue full employment policy without simultaneously being concerned that it must keep the inflation rate at zero. What I regard as a real tragedy today is the fact that all of a sudden the European banks and many other banks have shifted to the single-minded target of price stability. I think that is one of the sources of the European tragedy, in contrast with the shining performance of the United States. No concern whatever about employment.

Solow: Well, they argue that there is nothing they can do about it, but you and I think that's fundamentally wrong and simply a way of avoiding responsibility.

Modigliani: Exactly. And on the contrary, I think they should say the priority target should be "first unemployment," though price stability is also very important. There are situations in which indeed you may either have to accommodate inflation or stop it at the cost of temporary unemployment. I think *then* I would accept unemployment as a temporary state to stop an inflationary spiral. But to say that price stability is *the* only target, I think is wrong.

Solow: So, you don't believe that the NAIRU in France is 13 percent today?

Modigliani: Absolutely not, absolutely not. Nor do I believe that here in this country it is as low as 4 percent. I have great doubts about the stability of NAIRU but even more about the appropriate way to estimate it.

Barnett: While I was an undergraduate student at MIT, I was permitted to take your graduate course in corporate finance. I shared with the graduate students in the class the view that the Modigliani–Miller work on the cost of capital was dramatically raising the level of sophistication of the field of corporate finance. What motivated you to enter that area of research, and what earlier research inspired you?

Modigliani: Ever since my 1944 article on Keynes, I have become interested in empirical tests of the Keynesian structure. As everybody knows, one of the key components of that structure is the investment function, which explains investment in terms of the interest rate, seen as the cost of capital, the cost of funds invested. I was then under the influence of the views of the corporate finance specialists that the cost of funds depended upon the way in which the firm was financed. If you issued stock, then the cost of that would be the return on equity, which might be 10 percent, but if you used bonds, the cost would be their interest rate, which might be only 5 percent. That sort of answer didn't seem to me to be very convincing. In the end, what was the cost of capital: 5 percent or 10 percent? To an economist it could not be rational to say that the required return was 5 percent if you chose to finance the project by debt and 10 percent if you chose equity. After listening to a paper by David Durand suggesting (and then rejecting) the so-called "entity theory" of valuation, I gradually became convinced of the hypothesis that market value should be independent of the structure of financing, and was able to sketch out a proof of the possibility of arbitraging differences in valuation that are due only to differences in the liability structure. This result later became part of the proof of the Modigliani–Miller theorem. In essence, the market value of liabilities could not depend on its structure, because the investor could readily reproduce any leverage structure through personal lending or borrowing (as long as there was no tax impediment). As a consequence, there was no difference between the use of equity and debt funds. Even though debt had a lower apparent cost, it increased the required return on equity, and the weighted average of the two would be unaffected by the composition. I unveiled my proof in a class in which Miller happened to be an auditor. He was convinced instantly and decided to join me in the crusade to bring the truth to the heathens.

The theorem, which by now is well known, was proven very laboriously in about 30 pages. The reason for the laboriousness was in part because the theorem was so much against the grain of the teachings of corporate finance—the art and science of designing the "optimal capital structure." We were threatening to take the bread away, and so, we felt that we had to give a "laborious" proof to persuade them. Unfortunately, the price was paid by generations of students that had to read the paper; I have met many MBA students that remember that paper as a torture, the most difficult reading in the course. It's too bad because, nowadays, the theorem seems to me to be so obvious that I wonder whether it deserves two Nobel Prizes. All that it really says is that (with well-working markets, rational-return-maximizing behavior for any given risk, and no distorting taxes) the value of a firm—its market capitalization of all liabilities—must be the value of its assets. The composition of the claims can change (equity, debt, preferred, convertible preferred., derivatives, and what not), but the aggregate value of the claims cannot change. It is the value of the assets. Of course, it is true that this conclusion implies that the way that you finance investment is immaterial. It follows that in estimating the required return, the cost of capital, we do not have to bother with the details of the composition of the financing. In that sense, Jorgenson is right.

In later years, the Modigliani–Miller theorem has provided the foundation for the work on derivatives, such as options. All of that work assumes that the underlying value of the firm is independent of its current liability structure. But let me remind you of the assumptions needed to establish the theorem and, in particular, the assumption of no distorting effects of taxation on the net-of-tax amount received by an individual from one dollar of before-interest corporate earnings. If there are such effects, then the situation is more complicated, and in fact in this area there is a disagreement between Miller and me. I believe that taxes can introduce a differential advantage between different kinds of instruments, while Miller thinks not. But I should add that even though, in principle, taxation could affect the comparative advantage of different instruments, Miller and I agree that, with the current system of taxation, the differences are unlikely to be appreciable.

Solow: Now I want to make room here for you to make a brief comment about real business-cycle theory. If you look at macroeconomic theory today, what has replaced the Keynesian economics that you and I both accept is, in the minds of young people, real business-cycle theory.

Modigliani: I have no difficulty in believing that business cycles can exist in the real economy. You don't need money, and I myself built models of that kind, when it was fashionable. There was Hicks's article on the business cycle, and then Sidney Alexander had a very interesting article on the introduction of a bound that can permit you to get a cycle without money. But I think that has little to do with Keynesian unemployment. In the thirties, for instance, there was a tremendous depression that I think was caused by an insufficiency of real money. That was a horrible error made by the Federal Reserve, a point on which Milton Friedman and I agree. There was a serious shortage of real money and irresponsible behavior in letting the money supply shrink. I think that unemployment is mostly due to the rigidity of wages and to the shifting conditions. Therefore, there is the need for adjustment by the Central Bank, and the adjustment must be fast enough.

Solow: What's distinctive about real business-cycle theory is not just that it says that the monetary mechanism has nothing to do with cycles, but that business cycles, as we observe them, are optimal reactions of the economy to unexpected shocks to technology and tastes and things like that.

Modigliani: Yes, yes. Well, of course, much of this goes back to rational expectations, and my attitude toward rational expectations is that it is a wonderful theory. It is indeed the crowning of the classical theory. The classical theory spoke of optimal response to expectations. Lucas and company add optimal, formation of expectations. From that point of view, I am satisfied that that is what economic theory would say; and I am proud because I contributed an important concept, which is, I think, at the essence of rational expectations, namely, the existence of expectations that map into themselves.

Solow: Self-validating.

Modigliani: Self-validating. No, not "self-validating"—"internally consistent"

Solow: That's what I meant by self-validating. There is one set of expectations that is self-validating, not that every set is self-validating.

Modigliani: That's right, because usually self-validating means that it happens because you expect it. This is not the case. In addition, I believe that it is *not* a description of the world. I don't believe that the world is behaving rationally in that extreme sense, and there are many circumstances under which the model will not apply. In particular, I do not believe that that model justifies the conclusion that anything the government does is bad.

Solow: It adds variance and the mean is already right, so discretionary policy is bad.

Modigliani: It creates noise, so therefore whatever government does is bad. Wage rigidity to me is a perfect example contradicting the above conclusion. Nor can you dispose of wage rigidity with the hypothesis of staggered contract. If that contract is rational, then wages *are* rigid and one better take this into account in theory and policy; or the staggered contract is not rational and in a Chicago world, it should have long ago disappeared.

Barnett: Robert Barro, who I understood was a student in some of your classes, advocates a version of Ricardian equivalence that appears to be analogous in governmental finance to the Modigliani–Miller theorem in corporate finance and in some ways to your life-cycle theory of savings with bequests. In fact, he sometimes speaks of one of your classes at MIT that he attended in 1969 as being relevant to his views. But I understand that you do not agree with Barro's views of government finance. Why is that?

Modigliani: Barro's Ricardian equivalence theorem has nothing in common with the Modigliani–Miller proposition, *except the trivial relation that something doesn't matter.* In the Modigliani–Miller theorem, it is capital structure, and in the Barro theorem it is government deficit. In my view, Barro's theorem, despite its elegance, has *no* substance. I don't understand why so many seem to be persuaded by a proposition whose proof rests on the incredible assumption that everybody cares about his heirs as if they were himself. If you drop that assumption, there is no proof based on rational behavior, and the theorem is untenable. But that kind of behavior is very rare and can't be universal. Just ask yourself what would happen with two families, when one family has no children and another family has 10. Under Ricardian equivalence, both families would be indifferent between using taxes or deficit financing. But it is obvious that the no-children family would prefer the deficit, and the other would presumably prefer taxation. Indeed, why should the no-children-family save more, when the government runs a deficit? I am just sorry that any parallel is made between Modigliani–Miller and Ricardian equivalence.

I have in fact offered concrete empirical evidence, and plenty of it, that government debt displaces capital in the portfolio of households and hence in the economy. My paper is a bit old, though it has been replicated in unpublished research. But there is an episode in recent history that provides an excellent opportunity to test Barro's model of no burden against the life-cycle hypothesis measure of burden—the displacement effect. I am referring to the great experiment unwittingly performed by Reagan cutting taxes and increasing expenditure between 1981 (the first Reagan budget) and 1992. The federal debt increased $3\frac{1}{4}$ times or from 7 percent of initial private net worth to about 30 percent. In the same interval, disposable (nominal) personal income grew 117 percent (all data from the *Economic Report of the President*, 1994, Table B-112 and B-28). According to my model, private wealth is roughly proportional to net-of-tax income, and hence it should also have increased by 117 percent, relative to the initial net worth. But net national wealth (net worth less government debt, which represents essentially the stock of productive private capital) should have increased 117 percent minus the growth of debt, or 117 − 23 = 94 percent (of initial net worth). The 23 percent is the crowding-out effect of government debt, according to the life-cycle hypothesis. The actual growth of national wealth turns out to be 88 percent, pretty close to my prediction of 94 percent. On the other hand, if the government debt does not crowd out national wealth, as Barro firmly holds, then the increase in the latter should have been the same as that of income, or 117 percent compared with 88 percent. Similarly for Barro the growth of private net worth should be the growth of income of 117 percent *plus* the 23 percent growth of debt, or 140 percent. The actual growth is 111 percent, very close to my prediction of 117 percent and far from his, and the small deviation is in the direction opposite to that predicted by Barro. Why do so many economists continue to pay so much attention to Barro's model over the life-cycle hypothesis?

Solow: OK, let's move on. I think the next thing we ought to discuss is your Presidential Address to the American Economic Association and how, in your mind, it relates both to the 1944 paper that you've been talking about and your later work.

Modigliani: As I said before about Keynes, I stick completely to my view that to maintain a stable economy you need stabilization policy. Fiscal policy should, first of all, come in as an automatic stabilizer. Secondly, fiscal policy might enter in support of monetary policy in extreme conditions. But normally we should try to maintain full employment with savings used to finance investment, not to finance deficits. We should rely on monetary policy to ensure full employment with a balanced budget. But one thing I'd like to add is that it seems to me that in the battle between my recommendation to make use of discretion (or common sense) and Friedman's recommendation to renounce discretion in favor of blind rules (like 3 percent money growth per year), my prescription has won hands down. There is not a country in the world today that uses a mechanical rule.

Solow: It's hard to imagine in a democratic country.

Modigliani: There is not a country that doesn't use discretion.

Solow: You know, I agree with you there. How would you relate the view of your Presidential Address to monetarism? It was stimulated by monetarism, in a way. How do you look at old monetarism, Milton Friedman's monetarism, now?

Modigliani: If by monetarism one means money matters, I am in agreement. In fact, my present view is that *real* money is the most important variable. But I think that a rigid monetary rule is a mistake. It is quite possible that in a very stable period, that might be a good starting point, but I would certainly not accept the idea that that's the way to conduct an economic policy in general.

Solow: And hasn't Milton sometimes, but not always, floated the idea that he can find no interest elasticity in the demand for money.

Modigliani: I've done several papers on that subject and rejected that claim all over the place. Anybody who wants to find it, finds it strikingly—absolutely no problem.

Solow: You had a major involvement in the development of the Federal Reserve's MPS quarterly macroeconometric model, but not lately. How do you feel about large econometric models now? There was a time when someone like Bob Hall might have thought that that's the future of macroeconomics. There is no room for other approaches. All research will be conducted in the context of his model..

Modigliani: Right. Well, I don't know. I imagine that, first of all, the notion of parsimoniousness is a useful notion, the notion that one should try to construct models that are not too big, models that are more compact in size. I think that at the present time these models are still useful. They still give useful forecasts and especially ways of gauging responses to alternative policies, which is most important. But under some international circumstances, there is no room for domestic monetary policy in some countries. In such a country, an econometric model may not be very helpful. But an econometric model would be somewhat useful in considering different fiscal policies.

Barnett: Has mentoring younger economists been important to you as your fame grew within the profession?

Modigliani: My relation with my students, which by now are legion, has been the best aspect of my life. I like teaching but I especially like working with students and associating them with my work. Paul Samuelson makes jokes about the fact that so many of my articles are coauthored with so many people that he says are unknown—such as Paul Samuelson himself. The reason is that whenever any of my research assistants has developed an interesting idea, I want their names to appear as coauthors. Many of my "children" now occupy very high positions, including Fazio, the Governor of the Bank of Italy, Draghi, the Director General of the Treasury of Italy, Padoa Schioppa, a member of the Directorate of the European Central Bank, and Stan Fischer, Joe Stiglitz, and several past and current members of the Federal Reserve Board. All have been very warm to me, and I have the warmest feeling for them.

Solow: Now if you were giving advice to a young macroeconomist just getting a Ph.D., what would you say is the most fertile soil to cultivate in macroeconomics these days?

Modigliani: I think that these days, in terms of my own shifts of interest, I've been moving toward open-economy macroeconomics and especially international finance. It's a very interesting area, and it's an area where wage rigidity is very important. Now the distinction becomes very sharp between nominal wage rigidity and real wage rigidity.

Solow: Explain that.

Modigliani: With nominal wage rigidity, you will want floating exchange rates. With real rigidity, there's nothing you can do about unemployment. I've been looking at the experiences of countries that tried fixing exchange rates and countries that tried floating exchange rates, and I am finding that both experiences have not been good. Europe has been doing miserably.

Barnett: You have been an important observer of the international monetary system and the role of the United States and Europe in it, and I believe that you have supported the European Monetary Union. Would you comment on the EMS and the future of the international monetary system, in relation to what you think about the recent financial crises and the role that exchange rates have played in them.

Modigliani: Yes, I have been a supporter of the Euro, but to a large extent for its political implications, peace in Europe, over the purely economic ones. However, I have also pointed out the difficulties in a system which will have fixed exchange rates and how, for that to work, it will require a great deal of flexibility in the behavior of wages of individual countries having differential productivity growth and facing external shocks. I have also pointed out that the union was born under unfavorable conditions, as the role of the central bank has been played, not legally but *de facto*, by the Bundesbank, which has pursued consistently a wrong overtight monetary policy resulting in high European unemployment. It has reached 12 percent and sometimes even higher, and that policy is now being pursued to a considerable extent by the European Central Bank, which is making essentially the same errors as the Bundesbank. This does not promise too much for the near future.

Solow: What we're going to do now is switch over to talking about the life-cycle theory of savings, and what I'd like you to do is comment on the simplest life-cycle model, the one that you and Albert Ando used for practical purposes, with no bequests, etc.

Modigliani: Well, let me say that bequests are not to be regarded as an exception. Bequests are part of the life-cycle model. But it is true that you can go very far with assuming no bequests, and therefore it's very interesting to follow that direction. The model in which bequests are unimportant does produce a whole series of consequences which were completely unrecognized before the Modigliani–Brumberg articles. There were revolutionary changes in paradigm stemming from the life-cycle hypothesis. Fundamentally, the traditional theory of saving reduced to: the proportion of income saved rises with income, so rich people (and countries) save; poor people dissave. Why do rich people save? God knows. Maybe to leave bequests. That was the whole story, from which you would get very few implications and, in particular, you got the implication that rich countries save and poor countries dissave, an absurd concept since poor countries cannot dissave forever. No one can. But from the life-cycle hypothesis, you have a rich set of consequences. At the *micro* level, you have all the consequences of "Permanent Income," including the fact that consumption depends upon (is proportional to) *permanent* income, while saving depends basically on transitory income: The high savers are not the rich, but the temporarily rich (i.e., rich relative to their own normal income).

The difference between life-cycle and permanent income is that the latter treats the life span as infinite, while in the life-cycle model, lifetime is finite. For the purpose of analyzing short-term behavior, it makes no difference whether life lasts *50* years or forever. So you do have fundamentally the same story about the great bias that comes from the standard way of relating saving to current family income. But, in fact, in reality it does make a difference what the variability of income is in terms of short term versus long term. The marginal propensity to save of farmers is much higher than that of government employees, not because farmers are great savers, but because their income is very unstable. Other consequences that are very interesting include the fact, found from many famous surveys, that successive generations seem to be less and less thrifty, that is, save less and less at any given level of income. These conclusions all are consequences of the association between current and transitory income.

Then you have consequences in terms of the behavior of saving and wealth over the lifetime, and here is where the difference between life cycle and permanent income become important. With the life-cycle hypothesis, saving behavior varies over the person's finite lifetime, because with finite life comes a life cycle of income and consumption: youth, middle age, children, old age, death, and bequests. That's why there is little saving when your are very young. You have more saving in middle age, and dissaving when you are old. With infinite life, there is no life cycle. Aggregate saving reflects that life cycle and its interaction with demography and productivity growth, causing aggregate saving to rise with growth, as has been shown with overlapping generations models. All that has been shown to receive empirical support.

Solow: Dissaving and old age, as well?

Modigliani: Right. Now let me comment on that. Some people have spent a lot of time trying to show that the life-cycle model is wrong because people don't dissave in old age. That is because the poor guys have just done the thing wrong. They have treated Social Security contribution as if it were a sort of income tax, instead of mandatory saving, and they have treated pension as a handout, rather than a drawing down of accumulated pension claims. If you treat Social Security properly, measuring saving as income earned (net of personal taxes) minus consumption, you will find that people dissave tremendous amounts when they are old; they largely consume their pensions, while having no income.

Solow: They are running down their Social Security assets.

Modigliani: In addition to running down their Social Security assets, they also are running down their own assets, but not very much. Somewhat. But, if you include Social Security, wealth has a tremendous hump. It gets to a peak at the age of around 55–60 and then comes down quickly. All of these things have been completely supported by the evidence. Now, next, you do not need bequests to explain the existence of wealth, and that's another very important concept. Even without bequests, you can explain a large portion of the wealth we have. Now that does not mean there are no bequests. There are. In all my papers on the life-cycle hypothesis, there is always a long footnote that explains how to include bequests.

Solow: How you would include it, yes.

Modigliani: In such a way that it remains true that saving does not depend upon current income, but on life-cycle income. That ensures that the ratio of bequeathed wealth to income tends to remain stable, no matter how much income might rise. It is also important to recognize the macro implications of the life cycle, which are totally absent in the permanent-income hypothesis, namely, that the saving rate depends not on income, but on income growth. The permanent income hypothesis has nothing really to say; in fact, it has led Friedman to advance the wrong conclusion, namely that growth *reduces saving*. Why? Because growth results in expectations that future income will exceed current income. But with finite lifetime, terminating with retirement and dissaving, growth generates saving.

Consider again the simplified case of no bequests. Then each individual saves zero over its life cycle. If there is no growth, the path of saving by age is the same as the path of saving over life: it aggregates to zero. But if, say, population is growing, then there are more young in their saving phase than old in the dissaving mode, and so, the aggregate saving ratio is positive and increasing with growth. The same turns out with productivity growth, because the young enjoy a higher life income than the retired. Quite generally, the life-cycle model implies that aggregate wealth is proportional to aggregate income: hence the rate of growth of wealth, which is saving, tends to be proportional to the rate of growth of income. This in essence is the causal link between growth rate and saving ratio, which is one of the most significant and innovative implications of the life-cycle hypothesis.

Barnett: There has been much research and discussion about possible reforms or changes to the Social Security System. What are your views on that subject?

Modigliani: The problems of the Social Security System are my current highest interest and priority, because I think its importance is enormous; and I think there is a tragedy ahead, although in my view we can solve the problem in a way that is to everyone's advantage. In a word, we need to abandon the pay-as-you-go system, which is a wasteful and inefficient system, and replace it with a fully funded system. If we do, we should be able to reduce the Social Security contribution from the 18 percent that it would have to be by the middle of the next century, to below 6 percent using my approach, and I have worked out the transition. It is possible to go from here to there without any significant sacrifices. In fact, it can be done with no sacrifice, except using the purported surplus to increase national saving rather than consumption. And given the current low private saving rate and huge (unsustainable) capital imports, increasing national saving must be considered as a high priority.

Barnett: Are there any other areas to which you feel you made a relevant contribution that we have left out?

Modigliani: Perhaps that dealing with the effects of inflation. At a time when, under the influence of rational expectations, it was fashionable to claim that inflation had no *real* effects worth mentioning, I have delighted in showing that, in reality, it has extensive and massive real effects; and they are not very transitory. This work includes the paper with Stan Fisher on the effects of inflation, and the paper with Rich Cohen showing that investors are incapable of responding rationally to inflation, basically because of the (understandable) inability to distinguish between nominal and Fisherian real interest rates. For this reason, inflation systematically depresses the value of equities.

I have also shown that inflation reduces saving for the same reason. Both propositions have been supported by many replications. In public finance, the calculation of the debt service using the nominal instead of the real rate leads to grievous overstatement of the deficit-to-income ratio during periods of high inflation, such as the mid-seventies to early eighties in the presence of high debt-to-income ratios. In corporate finance, it understates the profits of highly levered firms.

Barnett: Your public life has been very intense, at least stalling at some point in your life. I presume that you do not agree with Walras, who believed that economists should be technical experts only, and should not be active in the formation of policy. Would you comment on the role of economists as "public servants"?

Modigliani: I believe that economists should recognize that economics has two parts. One is economic theory. One is economic policy. The principles of economic theory are universal, and we all should agree on them, as I think we largely do as economists. On economic policy, we do not necessarily agree, and we should not, because economic policy has to do with value judgments, not about what is true, but about what we like. It has to do with the distribution of income, not just total income. So long as they are careful not to mix the two, economists should be ready to participate in policy, but they should be careful to distinguish what part has to do with their value judgment and what part with knowledge of the working of an economy.

Barnett: You have been repeatedly involved in advocating specific economic policies. Were there instances in which, in your view, your advice had a tangible impact on governments and people?

Modigliani: Yes, I can think of several cases. The first relates to Italy and is a funny one. Through the sixties and seventies, Italian wage contracts had an escalator clause with very high coverage. But in 1975, in the middle of the oil crisis, the unions had the brilliant idea of demanding a new type of escalator clause in which an *x%* increase in prices would entitle a worker to an increase in wages not of *x%* of *his* wage but of *x%* of the *average* wage—the same number of liras for everyone! And the high-wage employers went along with glee! I wrote a couple of indignant articles trying to explain the folly and announcing doomsday. To my surprise, it took quite a while before my Italian colleagues came to my support. In fact, one of those colleagues contributed a "brilliant" article suggesting that the measure had economic justification, for, with the high rate of inflation of the time, all real salaries would soon be roughly the same, at which time it was justified to give everyone the same cost-of-living adjustment! It took several years of economic turmoil before the uniform cost-of-living adjustment was finally abolished and its promoters admitted their mistake. It took until 1993 before the cost-of-living adjustment was abolished all together.

A second example is the recommendations in the 1996 book by two coauthors and me, *Il Miracolo Possibile* [The Achievable Miracle], which helped Italy to satisfy the requirement to enter the Euro. This, at the time, was generally understood to be impossible, because of the huge deficit, way above the permissible 3 percent. We argued that the deficit was a fake, due to the use of inflation-swollen nominal interest rates in the presence of an outlandish debt-to-income ratio ($1\frac{1}{4}$), but the target was achievable through a drastic reduction of inflation and corresponding decline in nominal rates. This could be achieved without significant real costs by programming a minimal wage and price inflation through collaboration of labor, employer, and government. It worked, even beyond the results of the simulations reported in the book! And Italy entered the Euro from the beginning.

A third example is my campaign against European unemployment and the role played by a mistaken monetary policy. "An Economists' Manifesto on Unemployment in the European Union," issued by me and a group of distinguished European and American economists, was published a little over a year ago. Although it is not proving as effective as we had hoped, it is making some progress.

Finally, I hope that our proposed Social Security reform will have a significant impact. Here the stakes are truly enormous for most of the world, but the payoff remains to be seen.

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